think algo

# GRADE 5 MATHS

## ADVANCED

### BOOK 2

Teacher's edition

For Grade 5 students

ThinkAlgo — From Math Thinking to Code

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# 1 Working with Percentages

## ① Writing a number as a percentage – and back

- The sign % means "out of 100": $1\% = \frac{1}{100} = 0.01$. So $a = a \times 100\%$.
- **Number → percentage:** multiply the number by 100 and write the sign % after it.

  Examples: $3 = 3 \times 100\% = 300\%$; $0.425 = 42.5\%$; $\frac{3}{4} = \frac{3}{4} \times 100\% = 75\%$.
- **Percentage → decimal:** divide by 100 (move the decimal point two places to the left).

  Examples: $36\% = 36 : 100 = 0.36$; $4.5\% = 0.045$; $120\% = 1.2$.
- **Percentage → fraction:** write it as a fraction with denominator 100, then simplify.

  Example: $85\% = \frac{85}{100} = \frac{17}{20}$.

*The sign % simply takes the place of "divided by 100".*

## ② Calculating with percentages

- **Adding and subtracting:** add or subtract the numbers and keep the sign %:

  $a\% + b\% = (a + b)\%$; $a\% - b\% = (a - b)\%$.
- **Multiplying or dividing by an ordinary number $k$:** $a\% \times k = (a \times k)\%$; $a\% : k = (a : k)\%$.
- **Percentage times percentage:** $a\% \times b\% = \frac{a}{100} \times \frac{b}{100}$. Change both to decimals first:

  $30\% \times 40\% = 0.3 \times 0.4 = 0.12 = 12\%$ (*not* 1 200%!).
- **Percentage divided by percentage:** the two signs % cancel and the answer is an ordinary number: $a\% : b\% = a : b$. Example: $45\% : 5\% = 9$.

*When in doubt, treat each percentage as a fraction with denominator 100 (or as a decimal) and calculate as usual.*

## ③ Expressions with several kinds of numbers

- An expression may contain decimals, fractions, mixed numbers and percentages together. First change everything into **one kind** of number (all decimals, or all fractions), then follow the usual order of operations: brackets first, then $\times$ and :, then $+$ and $-$.
- To find $x$, work backwards exactly as with whole numbers, after changing each percentage into a decimal or a fraction.

## ④ Percentages in word problems

- "$A$ is $a\%$ of $B$" means $A = \frac{a}{100}$ of $B$. Simplify the fraction to see the parts:

  "$A$ is 40% of $B$" $\Rightarrow$ $A = \frac{2}{5}$ of $B$: $A$ is 2 parts, $B$ is 5 equal parts.
- Then use the familiar methods with a bar model: **sum and ratio**, **difference and ratio**.
- The whole of something is 100%. If parts of it are given as percentages, the part that is left is 100% minus the given percentages.

### Example 1

**a)** Write as percentages: $0.36$; $\frac{7}{20}$; $3$; $1.075$.

**b)** Write as a decimal and as a fraction in simplest form: $45\%$; $6.5\%$; $160\%$.

**Guided solution.**

**a)** $0.36 = 0.36 \times 100\% = 36\%$.

$\frac{7}{20} = \frac{7 \times 5}{20 \times 5} = \frac{35}{100} = 35\%$ or $\frac{7}{20} = 7 : 20 = 0.35 = 35\%$.

$3 = 3 \times 100\% = 300\%$; $1.075 = 107.5\%$.

**b)** $45\% = 45 : 100 = 0.45$; $45\% = \frac{45}{100} = \frac{9}{20}$.

$6.5\% = 6.5 : 100 = 0.065$; $6.5\% = \frac{65}{1\,000} = \frac{13}{200}$.

$160\% = 1.6$; $160\% = \frac{160}{100} = \frac{8}{5}$.

### Example 2

Calculate.

**a)** $37\% + 48\%$ **b)** $126\% - 58\%$ **c)** $15.5\% \times 6$

**d)** $40\% \times 15\%$ **e)** $72\% : 9\%$ **g)** $72\% : 9$

**Guided solution.**

**a)** *Method 1.* $37\% + 48\% = (37 + 48)\% = 85\%$.

*Method 2.* $37\% + 48\% = 0.37 + 0.48 = 0.85 = 85\%$.

**b)** $126\% - 58\% = (126 - 58)\% = 68\%$.

**c)** $15.5\% \times 6 = (15.5 \times 6)\% = 93\%$.

**d)** $40\% \times 15\% = 0.4 \times 0.15 = 0.06 = 6\%$.

**e)** $72\% : 9\% = 0.72 : 0.09 = 8$.

**g)** $72\% : 9 = (72 : 9)\% = 8\%$.

*Be careful:* $40\% \times 15\%$ *is* not $600\%$, *and* $72\% : 9\%$ *is the number* $8$, *not* $8\%$.

## Example 3

Calculate: $A = 4.5 - 1\frac{3}{4} + 63\% : 18\%$.

**Guided solution.**

*Method 1 (change to decimals).*

$A = 4.5 - 1.75 + 0.63 : 0.18 = 4.5 - 1.75 + 3.5 = 2.75 + 3.5 = 6.25$.

*Method 2 (change to fractions).*

$$A = \frac{9}{2} - \frac{7}{4} + \frac{63}{100} : \frac{18}{100} = \frac{9}{2} - \frac{7}{4} + \frac{7}{2} = \frac{18}{4} - \frac{7}{4} + \frac{14}{4} = \frac{25}{4} = 6\frac{1}{4}.$$

*Both answers* $A = 6.25$ *and* $A = 6\frac{1}{4}$ *are correct.*

## Example 4

Find $x$: $x : 25\% + 1.6 = 9 : 0.5$.

**Guided solution.**

Work out the right-hand side: $9 : 0.5 = 18$.

So $x : 25\% + 1.6 = 18$, hence $x : 25\% = 18 - 1.6 = 16.4$.

Then $x = 16.4 \times 25\% = 16.4 \times 0.25 = 4.1$.

*Check:* $4.1 : 0.25 + 1.6 = 16.4 + 1.6 = 18$.

**Answer:** $x = 4.1$.

## Example 5

Lan and Hoa have 72 stickers altogether. Hoa's stickers are 80% of Lan's. How many stickers does each girl have?

**Guided solution.**

$80\% = \frac{80}{100} = \frac{4}{5}$, so Hoa's stickers are $\frac{4}{5}$ of Lan's.




Total number of equal parts: $5 + 4 = 9$.
One part: $72 : 9 = 8$ (stickers).
Lan: $8 \times 5 = 40$ stickers; Hoa: $72 - 40 = 32$ stickers.
**Answer:** Lan 40 stickers, Hoa 32 stickers.

## Practice

### Problem 1

Write as percentages.

**a)** 0.47 **b)** 1.36 **c)** 0.209

**d)** 0.8 **e)** 2.5 **g)** 0.035

### Solution

Multiply by 100 and write the sign %:
**a)** $0.47 = 47\%$; **b)** $1.36 = 136\%$; **c)** $0.209 = 20.9\%$;
**d)** $0.8 = 80\%$; **e)** $2.5 = 250\%$; **g)** $0.035 = 3.5\%$.
**Answer:** 47%; 136%; 20.9%; 80%; 250%; 3.5%.

### Problem 2

Write as percentages.

**a)** $\frac{3}{5}$ **b)** $\frac{9}{20}$ **c)** $\frac{21}{30}$

**d)** $\frac{7}{8}$ **e)** $\frac{13}{4}$ **g)** $\frac{33}{40}$

### Solution

**a)** $\frac{3}{5} = \frac{60}{100} = 60\%$. **b)** $\frac{9}{20} = \frac{45}{100} = 45\%$. **c)** $\frac{21}{30} = \frac{7}{10} = \frac{70}{100} = 70\%$.
**d)** $\frac{7}{8} = 7 : 8 = 0.875 = 87.5\%$. **e)** $\frac{13}{4} = \frac{325}{100} = 325\%$.
**g)** $\frac{33}{40} = 33 : 40 = 0.825 = 82.5\%$.
**Answer:** 60%; 45%; 70%; 87.5%; 325%; 82.5%.

## Problem 3

Write as decimals.

**a)** 38% **b)** 4% **c)** 215%

**d)** 9.6% **e)** 300% **g)** 0.5%

## Solution

Divide by 100:

**a)** $38\% = 0.38$; **b)** $4\% = 0.04$; **c)** $215\% = 2.15$;

**d)** $9.6\% = 0.096$; **e)** $300\% = 3$; **g)** $0.5\% = 0.005$.

**Answer:** 0.38; 0.04; 2.15; 0.096; 3; 0.005.

## Problem 4

Write as fractions in simplest form.

**a)** 40% **b)** 85% **c)** 64%

**d)** 150% **e)** 2.5% **g)** 62.5%

## Solution

**a)** $40\% = \frac{40}{100} = \frac{2}{5}$. **b)** $85\% = \frac{85}{100} = \frac{17}{20}$. **c)** $64\% = \frac{64}{100} = \frac{16}{25}$.

**d)** $150\% = \frac{150}{100} = \frac{3}{2}$. **e)** $2.5\% = \frac{25}{1\,000} = \frac{1}{40}$. **g)** $62.5\% = \frac{625}{1\,000} = \frac{5}{8}$.

**Answer:** $\frac{2}{5}$; $\frac{17}{20}$; $\frac{16}{25}$; $\frac{3}{2}$; $\frac{1}{40}$; $\frac{5}{8}$.

## Problem 5

Compare (write $>$, $<$ or $=$).

**a)** 0.7 ... 69% **b)** $\frac{3}{8}$ ... 37.5%

**c)** 125% ... $1\frac{1}{5}$ **d)** $\frac{2}{3}$ ... 66%

## Solution

**a)** $0.7 = 70\%$ and $70\% > 69\%$, so $0.7 > 69\%$.

**b)** $\frac{3}{8} = 3 : 8 = 0.375 = 37.5\%$, so $\frac{3}{8} = 37.5\%$.

**c)** $125\% = 1.25$ and $1\frac{1}{5} = 1.2$; $1.25 > 1.2$, so $125\% > 1\frac{1}{5}$.

**d)** $66\% = \frac{66}{100} = \frac{33}{50}$, and $\frac{2}{3} = \frac{100}{150} > \frac{99}{150} = \frac{33}{50}$, so $\frac{2}{3} > 66\%$.

**Answer: a)** $>$; **b)** $=$; **c)** $>$; **d)** $>$.

## Problem 6

Write in order from smallest to largest: $0.43$, $\frac{2}{5}$, $41.5\%$, $\frac{9}{20}$, $39\%$.

## Solution

Change all of them into decimals:

$0.43$; $\frac{2}{5} = 0.4$; $41.5\% = 0.415$; $\frac{9}{20} = 0.45$; $39\% = 0.39$.

Order: $0.39 < 0.4 < 0.415 < 0.43 < 0.45$.

**Answer:** $39\%$, $\frac{2}{5}$, $41.5\%$, $0.43$, $\frac{9}{20}$.

## Problem 7

Calculate.

**a)** $28\% + 56\%$

**b)** $143\% - 67\%$

**c)** $12.5\% + 7.25\% + 30.25\%$

**d)** $90\% - 45\% + 15\%$

## Solution

**a)** $28\% + 56\% = (28 + 56)\% = 84\%$.

**b)** $143\% - 67\% = (143 - 67)\% = 76\%$.

**c)** $12.5\% + 7.25\% + 30.25\% = (12.5 + 7.25 + 30.25)\% = 50\%$.

**d)** $90\% - 45\% + 15\% = (90 - 45 + 15)\% = 60\%$.

**Answer:** $84\%$; $76\%$; $50\%$; $60\%$.

## Problem 8

Calculate.

**a)** $23.4\% \times 5$

**b)** $252\% : 12$

**c)** $6.8\% \times 15$

**d)** $99\% : 4.5$

**Solution**

**a)** $23.4\% \times 5 = (23.4 \times 5)\% = 117\%$.
**b)** $252\% : 12 = (252 : 12)\% = 21\%$.
**c)** $6.8\% \times 15 = (6.8 \times 15)\% = 102\%$.
**d)** $99\% : 4.5 = (99 : 4.5)\% = 22\%$.
**Answer:** 117%; 21%; 102%; 22%.

**Problem 9**

Calculate and write the result as a decimal.

**a)** $60\% \times 35\%$

**b)** $140\% \times 25\%$

**c)** $3.5\% \times 80\%$

**d)** $84\% : 12\%$

**e)** $45\% : 0.5\%$

**g)** $7.2\% : 16\%$

**Solution**

**a)** $60\% \times 35\% = 0.6 \times 0.35 = 0.21$.
**b)** $140\% \times 25\% = 1.4 \times 0.25 = 0.35$.
**c)** $3.5\% \times 80\% = 0.035 \times 0.8 = 0.028$.
**d)** $84\% : 12\% = 84 : 12 = 7$.
**e)** $45\% : 0.5\% = 45 : 0.5 = 90$.
**g)** $7.2\% : 16\% = 7.2 : 16 = 0.45$.
**Answer:** 0.21; 0.35; 0.028; 7; 90; 0.45.

**Problem 10**

Calculate.

**a)** $2.4 + 35\% - \frac{3}{4}$

**b)** $1\frac{1}{2} \times 40\% + 2.8 : 70\%$

**Solution**

**a)** $2.4 + 35\% - \frac{3}{4} = 2.4 + 0.35 - 0.75 = 2.75 - 0.75 = 2$.

**b)** $1\frac{1}{2} \times 40\% + 2.8 : 70\% = 1.5 \times 0.4 + 2.8 : 0.7 = 0.6 + 4 = 4.6.$

**Answer: a)** 2; **b)** 4.6.

## Problem 11

Calculate.

**a)** $25\% \times 8 + 13.5 : 4.5 - 2\frac{2}{5}$ **b)** $4.8 - 1\frac{1}{4} + 36\% : 144\%$

## Solution

**a)** $25\% \times 8 + 13.5 : 4.5 - 2\frac{2}{5} = 0.25 \times 8 + 3 - 2.4 = 2 + 3 - 2.4 = 2.6.$

**b)** $4.8 - 1\frac{1}{4} + 36\% : 144\% = 4.8 - 1.25 + 36 : 144 = 4.8 - 1.25 + 0.25 = 3.8.$

**Answer: a)** 2.6; **b)** 3.8.

## Problem 12

Calculate in a clever way.

**a)** $37\% \times 2.5 + 63\% \times 2.5$ **b)** $125\% \times 4.7 \times 80\%$ **c)** $6.4 \times 45\% + 6.4 \times 55\%$

## Solution

**a)** $37\% \times 2.5 + 63\% \times 2.5 = (37\% + 63\%) \times 2.5 = 100\% \times 2.5 = 1 \times 2.5 = 2.5.$

**b)** $125\% \times 4.7 \times 80\% = (1.25 \times 0.8) \times 4.7 = 1 \times 4.7 = 4.7.$

**c)** $6.4 \times 45\% + 6.4 \times 55\% = 6.4 \times (45\% + 55\%) = 6.4 \times 1 = 6.4.$

**Answer:** 2.5; 4.7; 6.4.

## Problem 13

Find $x$.

**a)** $20\% \times x = 1.35 + 0.25$ **b)** $x : 40\% - 2.5 = 7.5 : 1.5$

## Solution

**a)** $0.2 \times x = 1.6$, so $x = 1.6 : 0.2 = 8.$

**b)** $x : 40\% - 2.5 = 5$, so $x : 40\% = 7.5$ and $x = 7.5 \times 0.4 = 3.$

**Answer: a)** $x = 8$; **b)** $x = 3$.

## Problem 14

Find $x$.

**a)** $150\% \times x = 4.2 - 1.8$

**b)** $x : 25\% + 0.75 = 0.75 \times 21$

## Solution

**a)** $1.5 \times x = 2.4$, so $x = 2.4 : 1.5 = 1.6$.

**b)** $x : 25\% + 0.75 = 15.75$, so $x : 25\% = 15$ and $x = 15 \times 0.25 = 3.75$.

**Answer: a)** $x = 1.6$; **b)** $x = 3.75$.

## Problem 15

Find $x$.

**a)** $x + 35\% \times x = 5.4$

**b)** $x - 12\% \times x = 22$

## Solution

**a)** $x = 100\% \times x$, so $x + 35\% \times x = 135\% \times x = 1.35 \times x$.

$1.35 \times x = 5.4$, so $x = 5.4 : 1.35 = 4$.

**b)** $x - 12\% \times x = 88\% \times x = 0.88 \times x = 22$, so $x = 22 : 0.88 = 25$.

**Answer: a)** $x = 4$; **b)** $x = 25$.

## Problem 16

A tap fills an empty tank. In the first hour it fills 30% of the tank; in the second hour it fills 12% of the tank more than in the first hour. After these two hours, what percentage of the tank is still empty?

## Solution

Second hour: $30\% + 12\% = 42\%$ of the tank.

Both hours: $30\% + 42\% = 72\%$ of the tank.

Still empty: $100\% - 72\% = 28\%$ of the tank.

**Answer:** 28% of the tank.

## Problem 17

A school library has three kinds of books: storybooks, science books and textbooks. Storybooks make up 45% of all the books; the number of science books is $\frac{2}{3}$ of the number of storybooks. What percentage of the books are textbooks?

**Solution**

Science books: $45\% \times \frac{2}{3} = 30\%$ of all the books.

Textbooks: $100\% - 45\% - 30\% = 25\%$ of all the books.

**Answer:** 25%.

**Problem 18**

This year Khoa's age is 25% of his father's age, and his grandfather's age is 175% of his father's age. Khoa and his father are 50 years old altogether. How old is each of the three people?

**Solution**

$25\% = \frac{1}{4}$: Khoa's age is 1 part, his father's age is 4 parts.

Total number of parts: $1 + 4 = 5$; one part: $50 : 5 = 10$ (years).

Khoa: 10 years; father: $10 \times 4 = 40$ years.

Grandfather: $40 \times 175\% = 40 \times 1.75 = 70$ (years).

**Answer:** Khoa 10, father 40, grandfather 70 years old.

**Problem 19**

A rectangle has a perimeter of 32 dm. Its width is 60% of its length. Find the area of the rectangle.

**Solution**

Length + width: $32 : 2 = 16$ (dm).

$60\% = \frac{3}{5}$: the width is 3 parts, the length 5 parts.

One part: $16 : (3 + 5) = 2$ (dm). Length: $2 \times 5 = 10$ dm; width: $2 \times 3 = 6$ dm.

Area: $10 \times 6 = 60$ ($dm^2$).

**Answer:** 60 $dm^2$.

**Problem 20**

A farm has 150 rabbits and chickens in all. After 10 rabbits and 20 chickens are sold, the number of rabbits left is 50% of the number of chickens left. How many rabbits and how many chickens were there at first?

**Solution**

Animals left: $150 - 10 - 20 = 120$.

$50\% = \frac{1}{2}$: rabbits left are 1 part, chickens left are 2 parts.

One part: $120 : 3 = 40$. Rabbits left: 40; chickens left: 80.
At first: rabbits $40 + 10 = 50$; chickens $80 + 20 = 100$.
**Answer:** 50 rabbits and 100 chickens.

## Problem 21

Two shelves hold 72 books altogether. If 20% of the books on the first shelf are moved to the second shelf, the two shelves will hold the same number of books. How many books are on each shelf?

### Solution

After the move each shelf holds $72 : 2 = 36$ books.
The first shelf keeps $100\% - 20\% = 80\%$ of its books, so 80% of its books is 36.
First shelf: $36 : 80\% = 36 : 0.8 = 45$ (books).
Second shelf: $72 - 45 = 27$ (books).
*Check:* 20% of 45 is 9; $45 - 9 = 36$ and $27 + 9 = 36$.
**Answer:** 45 books and 27 books.

## Problem 22

Two warehouses A and B store rice. If 2 tonnes are moved from A to B, the two warehouses hold equal amounts. If instead 1 tonne is moved from B to A, the rice in A is 300% of the rice in B. How much rice did each warehouse hold at first?

### Solution

Moving 2 tonnes from A to B makes them equal, so A has $2 \times 2 = 4$ tonnes more than B.
If 1 tonne goes from B to A, the difference becomes $4 + 1 + 1 = 6$ (tonnes).
Then A is $300\% = 3$ times B: B is 1 part, A is 3 parts, the difference is 2 parts.
One part: $6 : 2 = 3$ (tonnes). B then has 3 tonnes, so at first B had $3 + 1 = 4$ tonnes.
At first A had $4 + 4 = 8$ tonnes.
**Answer:** A 8 tonnes, B 4 tonnes.

## Problem 23

At the start of the school year, the number of boys in class 5A was 80% of the number of girls. In the middle of the year, 2 more boys joined the class, and then the number of boys was 90% of the number of girls. How many pupils did class 5A have at the start of the year?

### Solution

The number of girls did not change.
The 2 new boys make up $90\% - 80\% = 10\%$ of the number of girls.

Girls: $2 : 10\% = 2 : 0.1 = 20$.
Boys at the start: $20 \times 80\% = 16$.
Pupils at the start: $20 + 16 = 36$.
**Answer:** 36 pupils.

**Problem 24**

A shop had a stock of fabric. On Monday it sold 18.5% of the stock, on Tuesday 26.5%, and on Wednesday as much as on Monday and Tuesday together. What percentage of the stock was left?

**Solution**

Monday and Tuesday: $18.5\% + 26.5\% = 45\%$ of the stock.
Wednesday: also 45%.
Left: $100\% - 45\% - 45\% = 10\%$ of the stock.
**Answer:** 10%.

**Problem 25**

Find $x$.

**a)** $75\% \times x + x \times \frac{1}{4} = 3.6$ **b)** $x : 20\% - x = 12$

**Solution**

**a)** $\frac{1}{4} = 25\%$, so $75\% \times x + 25\% \times x = 100\% \times x = x$. Hence $x = 3.6$.
**b)** $x : 20\% = x : 0.2 = x \times 5$, so $x \times 5 - x = x \times 4 = 12$ and $x = 3$.
**Answer: a)** $x = 3.6$; **b)** $x = 3$.

**Problem 26**

Calculate: $(45\% + 0.55) \times 3.7 - 75\% \times 4$.

**Solution**

$45\% + 0.55 = 0.45 + 0.55 = 1$, and $75\% \times 4 = 0.75 \times 4 = 3$.
So the expression equals $1 \times 3.7 - 3 = 0.7$.
**Answer:** 0.7.

## Problem 27

In a school, 60% of the pupils are girls, and 25% of the girls sing in the school choir. What percentage of all the pupils are girls who sing in the choir?

### Solution

The choir girls are 25% of 60% of the pupils:
$60\% \times 25\% = 0.6 \times 0.25 = 0.15 = 15\%$.
**Answer:** 15% of the pupils.

## Problem 28

The sum of two numbers is 2.7. The first number is 125% of the second. Find the two numbers.

### Solution

$125\% = \frac{125}{100} = \frac{5}{4}$: the first number is 5 parts, the second 4 parts.
One part: $2.7 : (5 + 4) = 0.3$.
First number: $0.3 \times 5 = 1.5$; second number: $0.3 \times 4 = 1.2$.
**Answer:** 1.5 and 1.2.

## Problem 29

The difference between two numbers is 3.6. The smaller number is 40% of the larger number. Find the two numbers.

### Solution

$40\% = \frac{2}{5}$: the smaller number is 2 parts, the larger 5 parts.
The difference is $5 - 2 = 3$ parts, so one part is $3.6 : 3 = 1.2$.
Smaller: $1.2 \times 2 = 2.4$; larger: $1.2 \times 5 = 6$.
**Answer:** 2.4 and 6.

## Problem 30

Mum cuts 35% off a ribbon, then cuts 40% of what is left. **a)** What percentage of the ribbon is left? **b)** The piece left is 7.8 m long. How long was the ribbon?

### Solution

**a)** After the first cut: $100\% - 35\% = 65\%$ of the ribbon is left.
The second cut leaves $100\% - 40\% = 60\%$ of that: $65\% \times 60\% = 0.65 \times 0.6 = 0.39 = 39\%$.
**b)** 39% of the ribbon is 7.8 m, so the ribbon was $7.8 : 0.39 = 20$ (m).
**Answer: a)** 39%; **b)** 20 m.

## Challenge – Extension

### Problem 31

At a swimming pool, the number of children sitting on the side is 25% of the number of children in the water. Then 4 children from the side jump into the water, and now the number of children on the side is 12.5% of the number in the water. How many children are there in all?

### Solution

$25\% = \frac{1}{4}$: at first the children on the side are $\frac{1}{1+4} = \frac{1}{5}$ of all the children.

$12.5\% = \frac{1}{8}$: afterwards the children on the side are $\frac{1}{1+8} = \frac{1}{9}$ of all the children.

The total does not change. The 4 children who jumped in make up $\frac{1}{5} - \frac{1}{9} = \frac{4}{45}$ of the total.

Total: $4 : \frac{4}{45} = 45$ (children).

*Check:* at first 9 on the side, 36 in the water (9 is 25% of 36); afterwards 5 and 40 (5 is 12.5% of 40).

**Answer:** 45 children.

### Problem 32

An's stamps are 75% of Bình's stamps. If each of them gets 10 more stamps, An's stamps will be 80% of Bình's. How many stamps does each boy have now?

### Solution

$75\% = \frac{3}{4}$: An has 3 parts, Bình 4 parts, so Bình has 1 part more than An.

After each gets 10 stamps, the difference is still 1 part.

$80\% = \frac{4}{5}$: then An has 4 new parts and Bình 5 new parts, a difference of 1 new part.

So 1 old part = 1 new part, and Bình's stamps go from 4 parts to 5 parts: the 10 stamps are 1 part.

Now: An $3 \times 10 = 30$ stamps, Bình $4 \times 10 = 40$ stamps.

*Check:* $40 : 50 = 0.8 = 80\%$.

**Answer:** An 30 stamps, Bình 40 stamps.

### Problem 33

The price of a toy was raised by 20%. Later the new price was lowered by 20%. Compared with the original price, did the final price go up or down, and by what percentage?

**Solution**

After the rise the price is 120% of the original.
After the cut it is 80% of that: $120\% \times 80\% = 1.2 \times 0.8 = 0.96 = 96\%$ of the original.
So the price went down by $100\% - 96\% = 4\%$.
**Answer:** down by 4%.

**Problem 34** *

The length of a rectangle is increased by 25% and its width is decreased by 20%. How does the area change?

**Solution**

New length: 125% of the old length; new width: 80% of the old width.
New area: $125\% \times 80\% = 1.25 \times 0.8 = 1 = 100\%$ of the old area.
**Answer:** the area does not change.

**Problem 35** *

Three numbers add up to 108. The first number is 50% of the second, and the third is 150% of the second. Find the three numbers.

**Solution**

$50\% = \frac{1}{2}$ and $150\% = \frac{3}{2}$. Take the second number as 2 parts: the first is 1 part, the third is 3 parts.
One part: $108 : (1 + 2 + 3) = 18$.
The numbers: 18, $18 \times 2 = 36$, $18 \times 3 = 54$.
**Answer:** 18; 36; 54.

**Problem 36** *

Find $x$: $(x - 15\% \times x) : 1.7 = 30\%$.

**Solution**

$x - 15\% \times x = 85\% \times x = 0.85 \times x$.
$0.85 \times x = 30\% \times 1.7 = 0.3 \times 1.7 = 0.51$.
$x = 0.51 : 0.85 = 0.6$.
**Answer:** $x = 0.6$.

**Problem 37** *

Calculate: $A = 1\% + 2\% + 3\% + \cdots + 98\% + 99\%$. Write the answer as a decimal.

**Solution**

$A = (1 + 2 + 3 + \cdots + 99)\%$.

$1 + 2 + \cdots + 99 = (1 + 99) \times 99 : 2 = 4\,950$.

$A = 4\,950\% = 49.5$.

**Answer:** 49.5.

**Problem 38** *

Calculate in a clever way: $12.5\% \times 0.25 \times 7.9 \times 8 \times 400\%$.

**Solution**

Group the factors: $(12.5\% \times 8) \times (0.25 \times 400\%) \times 7.9$.

$12.5\% \times 8 = 0.125 \times 8 = 1$; $\quad 0.25 \times 400\% = 0.25 \times 4 = 1$.

Result: $1 \times 1 \times 7.9 = 7.9$.

**Answer:** 7.9.

**Problem 39** *

A box contains red and blue balls; the red balls are 60% of all the balls. After 20 blue balls are put in, the red balls are 40% of all the balls. How many red balls are in the box?

**Solution**

The number of red balls does not change.

At first: red 60%, blue 40% of the balls, so blue $= \frac{40}{60} = \frac{2}{3}$ of red.

Afterwards: red 40%, blue 60%, so blue $= \frac{60}{40} = \frac{3}{2}$ of red.

The 20 blue balls added are $\frac{3}{2} - \frac{2}{3} = \frac{5}{6}$ of the red balls.

Red balls: $20 : \frac{5}{6} = 24$.

*Check:* at first 24 red, 16 blue ($24 : 40 = 60\%$); after: 24 red, 36 blue ($24 : 60 = 40\%$).

**Answer:** 24 red balls.

**Problem 40** *

Three taps fill a tank together. Tap A pours in 40% of the tank, tap B pours in 75% of what tap A pours, and tap C pours in the rest. Tap C pours 54 litres less than tap A. How many litres does the tank hold?

**Solution**

Tap B: $40\% \times 75\% = 0.4 \times 0.75 = 0.3 = 30\%$ of the tank.
Tap C: $100\% - 40\% - 30\% = 30\%$ of the tank.
Tap A pours $40\% - 30\% = 10\%$ of the tank more than tap C, and this is 54 litres.
The tank: $54 : 10\% = 54 : 0.1 = 540$ (litres).
**Answer:** 540 litres.

**Problem 41** *

A shop sold 40% of its rice on the first day and 50% of the remaining rice on the second day; 48 kg were left. How many kilograms of rice did the shop have?

**Solution**

*Work backwards.* After the second day, $100\% - 50\% = 50\%$ of the remainder is left, and this is 48 kg.
Remainder after day one: $48 : 50\% = 96$ (kg).
This is $100\% - 40\% = 60\%$ of all the rice, so the shop had $96 : 60\% = 96 : 0.6 = 160$ (kg).
**Answer:** 160 kg.

**Problem 42** *

Find $x$: $x \times 125\% + x : 4 + x \times 0.5 = 3.4$.

**Solution**

$x : 4 = x \times 0.25$, so the left side is $x \times (1.25 + 0.25 + 0.5) = x \times 2$.
$x \times 2 = 3.4$, so $x = 1.7$.
**Answer:** $x = 1.7$.

**Problem 43** *

This year a daughter's age is 25% of her mother's age. In 6 years' time, the daughter's age will be $\frac{1}{3}$ of her mother's age. How old is each of them now?

**Solution**

Now: daughter 1 part, mother 4 parts; the difference is 3 parts.
In 6 years: daughter 1 new part, mother 3 new parts; the difference is 2 new parts.
The age difference never changes, so 3 parts = 2 new parts, so 1 new part = 1.5 parts.
The daughter grows from 1 part to 1.5 parts: 0.5 part = 6 years, so 1 part = 12 years.
Daughter: 12 years; mother: $12 \times 4 = 48$ years.

*Check:* in 6 years, 18 and 54, and $18 = \frac{1}{3}$ of 54.

**Answer:** daughter 12, mother 48 years old.

### Problem 44 *

A jug holds 200 g of salt water in which salt makes up 5%. How many grams of pure water must be added so that salt makes up 4%?

### Solution

Salt in the jug: $200 \times 5\% = 10$ (g). Adding water does not change this.

After adding, 10 g is 4% of the salt water: $10 : 4\% = 10 : 0.04 = 250$ (g).

Water to add: $250 - 200 = 50$ (g).

**Answer:** 50 g of water.

### Problem 45 *

Number A is 25% more than number B. By what percentage is B smaller than A?

### Solution

A is 125% of B, so $A = \frac{5}{4}$ of B: B is 4 parts, A is 5 parts.

B is smaller by 1 part, and $1 : 5 = 0.2 = 20\%$ of A.

**Answer:** 20%.

### Problem 46 *

In a class, 45% of the pupils are boys, and there are 6 more girls than boys. How many pupils are in the class?

### Solution

Girls: $100\% - 45\% = 55\%$ of the class.

The difference: $55\% - 45\% = 10\%$ of the class, which is 6 pupils.

The class: $6 : 10\% = 60$ (pupils).

**Answer:** 60 pupils.

### Problem 47 **

Fresh mushrooms contain 92% water. After drying, the mushrooms contain 20% water. How many kilograms of dried mushrooms do we get from 100 kg of fresh mushrooms?

**Solution**

In 100 kg of fresh mushrooms the dry matter is $100\% - 92\% = 8\%$: $100 \times 8\% = 8$ (kg).
Drying only removes water, so the dried mushrooms still contain 8 kg of dry matter.
In dried mushrooms the dry matter is $100\% - 20\% = 80\%$.
Dried mushrooms: $8 : 80\% = 8 : 0.8 = 10$ (kg).
**Answer:** 10 kg.

**Problem 48** *

The sum of two numbers is 168; 40% of the first number equals 30% of the second number. Find the two numbers.

**Solution**

$\frac{4}{10}$ of the first number $= \frac{3}{10}$ of the second, so 4 times the first $= 3$ times the second.
Hence the first number is 3 parts and the second 4 parts.
One part: $168 : 7 = 24$. The numbers: $24 \times 3 = 72$ and $24 \times 4 = 96$.
*Check:* $40\% \times 72 = 28.8 = 30\% \times 96$.
**Answer:** 72 and 96.

**Problem 49** **

A number is decreased by 25%. By what percentage must the new number be increased to get back to the original number?

**Solution**

Let the original number be 4 parts. Decreasing it by $25\% = \frac{1}{4}$ leaves 3 parts.

To get back we must add 1 part, which is $\frac{1}{3}$ of the new number.

$\frac{1}{3} = 1 : 3 = 33\frac{1}{3}\%$.

**Answer:** $33\frac{1}{3}\%$.

**Problem 50** *

An, Bình and Chi share the cost of a present. An pays 30% of the cost; Bình pays 40% of the remaining amount; Chi pays the rest. Chi pays 14 000 dong more than Bình. How much does each person pay?

**Solution**

After An: $100\% - 30\% = 70\%$ of the cost remains.

Bình: $70\% \times 40\% = 28\%$ of the cost; Chi: $70\% - 28\% = 42\%$ of the cost.

Chi pays $42\% - 28\% = 14\%$ of the cost more than Bình, and this is 14 000 dong.

Cost: $14\,000 : 14\% = 14\,000 : 0.14 = 100\,000$ (dong).

An: 30 000 dong; Bình: 28 000 dong; Chi: 42 000 dong.

**Answer:** An 30 000 dong, Bình 28 000 dong, Chi 42 000 dong.

# 2 Percentage Word Problems

## ① Finding the percentage one number is of another

- To find what percentage $A$ is of $B$: divide $A$ by $B$, then write the quotient as a percentage (multiply by 100 and write %).

  Example: 9 is what percentage of 12? $9 : 12 = 0.75 = 75\%$.

- Read the question carefully: "$A$ compared with $B$" means $B$ is the whole (100%) and we divide by $B$.
- **Above or below plan:** if the plan is 100%, then actual : plan tells what percentage of the plan was done; the part above 100% is the percentage *exceeding* the plan.
- **Percentage increase or decrease:** (change) : (old amount), written as a percentage.

## ② Finding a percentage of a number

- To find $a\%$ of $M$, calculate $M \times a : 100$ or $M : 100 \times a$.

  Example: 15% of 60 kg is $60 \times 15 : 100 = 9$ (kg).

- $M : 100$ is 1% of $M$; multiplying by $a$ gives $a\%$ of $M$. This "go to 1% first" idea also helps to work backwards: if $a\%$ of a number is $N$, then 1% of it is $N : a$ and the number is $N : a \times 100$.

## ③ Percentages in everyday life

- **Discount:** a price cut of $a\%$ means we pay $(100 - a)\%$ of the old price.
- **Profit:** profit = selling price − cost price. "Profit of 20%" usually means 20% of the cost price, so the selling price is 120% of the cost price.
- **Savings interest:** with an interest rate of 0.5% a month, each month the bank adds 0.5% of the money in the account.
- **Salt water:** "5% salt" means the salt weighs 5% of the whole salt water.

*Always ask: "percentage of* what*?" – that quantity is the* 100%.

**Example 1**

Find the percentage: **a)** of 0.6 compared with 2.4; **b)** of 7 compared with 4.

**Guided solution.**

**a)** $0.6 : 2.4 = 0.25 = 25\%$.

**b)** $7 : 4 = 1.75 = 175\%$.

**Answer: a)** 25%; **b)** 175%.

**Example 2**

Class 5B has 18 boys and 22 girls. What percentage of the class are boys? What percentage are girls?

**Guided solution.**

Pupils in the class: $18 + 22 = 40$.

Boys compared with the whole class: $18 : 40 = 0.45 = 45\%$.

Girls compared with the whole class:

*Method 1.* $22 : 40 = 0.55 = 55\%$. *Method 2.* $100\% - 45\% = 55\%$.

**Answer:** boys 45%, girls 55%.

**Example 3**

Find 35% of 240.

**Guided solution.**

*Method 1.* $240 : 100 \times 35 = 2.4 \times 35 = 84$.

*Method 2.* $240 \times 35 : 100 = 8\,400 : 100 = 84$.

**Answer:** 84.

**Example 4**

A youth group planned to plant 250 trees, but actually planted 290 trees. What percentage of the plan did they reach, and by what percentage did they exceed the plan?

**Guided solution.**

Actual compared with plan: $290 : 250 = 1.16 = 116\%$.

They exceeded the plan by: $116\% - 100\% = 16\%$.

**Answer:** 116% of the plan; exceeded by 16%.

**Example 5**

A school bag costs 180 000 dong. During a sale its price is cut by 15%. How much does the bag cost in the sale?

**Guided solution.**

*Method 1.* The price cut: $180\,000 \times 15 : 100 = 27\,000$ (dong).

Sale price: $180\,000 - 27\,000 = 153\,000$ (dong).

*Method 2.* We pay $100\% - 15\% = 85\%$ of the price: $180\,000 \times 85 : 100 = 153\,000$ (dong).

**Answer:** 153 000 dong.

## Practice

### Problem 1

Find the percentage of the first number compared with the second.

**a)** 18 and 24 **b)** 4.5 and 12.5 **c)** 0.9 and 3.6

**d)** 27 and 15 **e)** 3.4 and 8 **g)** $\frac{1}{4}$ and 5

### Solution

**a)** $18 : 24 = 0.75 = 75\%$. **b)** $4.5 : 12.5 = 0.36 = 36\%$. **c)** $0.9 : 3.6 = 0.25 = 25\%$.

**d)** $27 : 15 = 1.8 = 180\%$. **e)** $3.4 : 8 = 0.425 = 42.5\%$. **g)** $\frac{1}{4} : 5 = 0.05 = 5\%$.

**Answer:** 75%; 36%; 25%; 180%; 42.5%; 5%.

### Problem 2

Calculate.

**a)** 15% of 60 kg **b)** 8% of 450 m **c)** 25% of 7.2 l

**d)** 40% of 3.5 tonnes **e)** 120% of 85 cm **g)** 12.5% of 96 $m^2$

### Solution

**a)** $60 \times 15 : 100 = 9$ (kg). **b)** $450 \times 8 : 100 = 36$ (m). **c)** $7.2 \times 25 : 100 = 1.8$ (l).

**d)** $3.5 \times 40 : 100 = 1.4$ (tonnes). **e)** $85 \times 120 : 100 = 102$ (cm).

**g)** $96 \times 12.5 : 100 = 12$ ($m^2$).

**Answer:** 9 kg; 36 m; 1.8 l; 1.4 tonnes; 102 cm; 12 $m^2$.

### Problem 3

A bag holds 48 marbles: 30 are red and the rest are green. **a)** What percentage of the marbles are red? **b)** What is the percentage of green marbles compared with red marbles?

**Solution**

**a)** $30 : 48 = 0.625 = 62.5\%$.

**b)** Green marbles: $48 - 30 = 18$. $18 : 30 = 0.6 = 60\%$.

**Answer: a)** 62.5%; **b)** 60%.

**Problem 4**

In a maths test, 34 of the 40 pupils of class 5C got a good mark. What percentage of the class got a good mark?

**Solution**

$34 : 40 = 0.85 = 85\%$.

**Answer:** 85%.

**Problem 5**

A bicycle workshop planned to make 400 bicycles in a month, but it made 460. What percentage of the plan did it reach? By what percentage did it exceed the plan?

**Solution**

$460 : 400 = 1.15 = 115\%$.

Exceeded by: $115\% - 100\% = 15\%$.

**Answer:** 115%; exceeded by 15%.

**Problem 6**

A village planned to plant 1 250 trees this year. By the end of March it had planted 1 000 trees. What percentage of the plan has been done, and what percentage is still to do?

**Solution**

Done: $1\,000 : 1\,250 = 0.8 = 80\%$ of the plan.

Still to do: $100\% - 80\% = 20\%$ of the plan.

**Answer:** 80% done, 20% still to do.

**Problem 7**

Aunt Lan earns 12 million dong a month and saves 2.7 million dong of it. What percentage of her pay does she save?

**Solution**

$2.7 : 12 = 0.225 = 22.5\%$.
**Answer:** 22.5%.

**Problem 8**

Last year Minh weighed 38 kg. This year his weight has gone up by 10% of last year's weight. How much does Minh weigh now?

**Solution**

Increase: $38 \times 10 : 100 = 3.8$ (kg).
Weight now: $38 + 3.8 = 41.8$ (kg).
**Answer:** 41.8 kg.

**Problem 9**

A rectangular plot of land is 50 m long; its width is 60% of its length. Vegetables are grown on 45% of its area, flowers on 35%, and the rest is paths. Find the area of the paths.

**Solution**

Width: $50 \times 60 : 100 = 30$ (m). Area: $50 \times 30 = 1\,500$ ($m^2$).
Paths: $100\% - 45\% - 35\% = 20\%$ of the area.
Area of the paths: $1\,500 \times 20 : 100 = 300$ ($m^2$).
**Answer:** 300 $m^2$.

**Problem 10**

A shop sold 1 200 kg of rice, of which 35% was sticky rice. How many kilograms of ordinary rice did it sell?

**Solution**

Ordinary rice: $100\% - 35\% = 65\%$ of the rice.
$1\,200 \times 65 : 100 = 780$ (kg).
**Answer:** 780 kg.

**Problem 11**

A school has 640 pupils, and 55% of them are girls. How many boys are there?

**Solution**

Girls: $640 \times 55 : 100 = 352$.
Boys: $640 - 352 = 288$.
**Answer:** 288 boys.

**Problem 12**

A shopkeeper buys a shirt for 150 000 dong and sells it for 180 000 dong. Her profit is what percentage of the cost price?

**Solution**

Profit: $180\,000 - 150\,000 = 30\,000$ (dong).
$30\,000 : 150\,000 = 0.2 = 20\%$.
**Answer:** 20% of the cost price.

**Problem 13**

A storybook costs 60 000 dong. At the book fair it is sold at a 10% discount. How much does it cost at the fair?

**Solution**

We pay $100\% - 10\% = 90\%$ of the price.
$60\,000 \times 90 : 100 = 54\,000$ (dong).
**Answer:** 54 000 dong.

**Problem 14**

Mrs Hạnh puts 20 000 000 dong into a savings account with interest of 0.6% a month. After one month, how much interest does she get, and how much money is in the account?

**Solution**

Interest: $20\,000\,000 \times 0.6 : 100 = 120\,000$ (dong).
Money in the account: $20\,000\,000 + 120\,000 = 20\,120\,000$ (dong).
**Answer:** interest 120 000 dong; total 20 120 000 dong.

**Problem 15**

Seawater contains 3.5% salt. How many kilograms of salt are there in 40 kg of seawater?

**Solution**

$40 \times 3.5 : 100 = 1.4$ (kg).

**Answer:** 1.4 kg.

**Problem 16**

The price of a notebook went up from 8 000 dong to 9 000 dong. By what percentage did the price go up?

**Solution**

Increase: $9\,000 - 8\,000 = 1\,000$ (dong).

Compared with the old price: $1\,000 : 8\,000 = 0.125 = 12.5\%$.

**Answer:** 12.5%.

**Problem 17**

The price of a pair of trainers went down from 250 000 dong to 200 000 dong. By what percentage did the price go down?

**Solution**

Decrease: $250\,000 - 200\,000 = 50\,000$ (dong).

Compared with the old price: $50\,000 : 250\,000 = 0.2 = 20\%$.

**Answer:** 20%.

**Problem 18**

A football team played 25 matches: it won 16, drew 4 and lost the rest. What percentage of the matches did it win, draw and lose?

**Solution**

Lost: $25 - 16 - 4 = 5$ (matches).

Won: $16 : 25 = 0.64 = 64\%$; drew: $4 : 25 = 0.16 = 16\%$; lost: $5 : 25 = 0.2 = 20\%$.

**Answer:** won 64%, drew 16%, lost 20%.

**Problem 19**

Hoa is reading a 240-page book. On the first day she reads 30% of the book, on the second day 45%. How many pages are left?

**Solution**

Pages left: $100\% - 30\% - 45\% = 25\%$ of the book.
$240 \times 25 : 100 = 60$ (pages).
**Answer:** 60 pages.

**Problem 20**

A farmer planted 450 young trees; 18 of them died. What percentage of the trees survived?

**Solution**

Trees that survived: $450 - 18 = 432$.
$432 : 450 = 0.96 = 96\%$.
**Answer:** 96%.

**Problem 21**

Out of 500 bean seeds, 92% sprouted. How many seeds did not sprout?

**Solution**

Seeds that sprouted: $500 \times 92 : 100 = 460$.
Seeds that did not sprout: $500 - 460 = 40$.
**Answer:** 40 seeds.

**Problem 22**

Class 5A has 35 pupils; 40% of them got an excellent report. How many pupils did not get an excellent report?

**Solution**

Excellent: $35 \times 40 : 100 = 14$ (pupils).
Not excellent: $35 - 14 = 21$ (pupils).
**Answer:** 21 pupils.

**Problem 23**

A basket holds 80 apples and pears; the pears make up 40%. Find the percentage of apples compared with pears.

**Solution**

Pears: $80 \times 40 : 100 = 32$. Apples: $80 - 32 = 48$.
Apples compared with pears: $48 : 32 = 1.5 = 150\%$.

**Answer:** 150%.

### Problem 24

For a tree-planting day, 32 of the 40 pupils of class 5A and 27 of the 30 pupils of class 5B took part. Which class took part at the higher rate?

### Solution

Class 5A: $32 : 40 = 0.8 = 80\%$.
Class 5B: $27 : 30 = 0.9 = 90\%$.
$90\% > 80\%$, so class 5B took part at the higher rate.
**Answer:** class 5B (90% against 80%).

### Problem 25

Last year a factory made 2 400 products. This year it made 15% more. How many products did it make this year?

### Solution

This year it made $100\% + 15\% = 115\%$ of last year's number.
$2\,400 \times 115 : 100 = 2\,760$ (products).
**Answer:** 2 760 products.

### Problem 26

A museum ticket costs 40 000 dong. Children get a 25% discount. How much does a family of 2 adults and 3 children pay?

### Solution

A child's ticket: $40\,000 \times (100 - 25) : 100 = 30\,000$ (dong).
The family pays: $40\,000 \times 2 + 30\,000 \times 3 = 80\,000 + 90\,000 = 170\,000$ (dong).
**Answer:** 170 000 dong.

### Problem 27

A rectangular garden is 25 m long and 16 m wide. The length is increased by 20% and the width stays the same. Find the new area. By what percentage did the area increase?

**Solution**

New length: $25 \times 120 : 100 = 30$ (m). New area: $30 \times 16 = 480$ ($m^2$).
Old area: $25 \times 16 = 400$ ($m^2$). Increase: $480 - 400 = 80$ ($m^2$).
$80 : 400 = 0.2 = 20\%$.
**Answer:** 480 $m^2$; the area increased by 20%.

## Problem 28

Fresh milk contains 4% fat. How many grams of fat are there in 3 kg of fresh milk?

**Solution**

3 kg = 3 000 g. Fat: $3\,000 \times 4 : 100 = 120$ (g).
**Answer:** 120 g.

## Problem 29

A lorry has to travel 240 km in three days. On the first day it covers 35% of the distance, on the second day 40%. How many kilometres are left for the third day?

**Solution**

Third day: $100\% - 35\% - 40\% = 25\%$ of the distance.
$240 \times 25 : 100 = 60$ (km).
**Answer:** 60 km.

## Problem 30

A class fund has 1 500 000 dong. The class spends 40% of it on books, then 30% of the remaining money on pens. How much money is left?

**Solution**

Books: $1\,500\,000 \times 40 : 100 = 600\,000$ (dong).
Remaining: $1\,500\,000 - 600\,000 = 900\,000$ (dong).
Pens: $900\,000 \times 30 : 100 = 270\,000$ (dong).
Left: $900\,000 - 270\,000 = 630\,000$ (dong).
**Answer:** 630 000 dong.

## Challenge – Extension

**Problem 31** *

25% of a number is 18. Find the number.

**Solution**

1% of the number: $18 : 25 = 0.72$.
The number (100%): $0.72 \times 100 = 72$.

*Or:* $25\% = \frac{1}{4}$, *so the number is* $18 \times 4 = 72$.

**Answer:** 72.

**Problem 32** *

A factory made 1 380 products and so exceeded its plan by 15%. How many products did the plan ask for?

**Solution**

The factory made $100\% + 15\% = 115\%$ of the plan.
1% of the plan: $1\,380 : 115 = 12$ (products).
The plan: $12 \times 100 = 1\,200$ (products).
**Answer:** 1 200 products.

**Problem 33** *

A shop sells a fan for 240 000 dong and makes a profit of 20% of the cost price. What was the cost price?

**Solution**

Selling price: $100\% + 20\% = 120\%$ of the cost price.
Cost price: $240\,000 : 120 \times 100 = 200\,000$ (dong).
**Answer:** 200 000 dong.

**Problem 34** *

After a 25% discount, Mum paid 360 000 dong for a jacket. What was the price before the discount?

**Solution**

Mum paid $100\% - 25\% = 75\%$ of the price.
Price before the discount: $360\,000 : 75 \times 100 = 480\,000$ (dong).
**Answer:** 480 000 dong.

**Problem 35**

A shop's profit on a toy is 20% of the *selling* price. What percentage of the *cost* price is the profit?

**Solution**

Take the selling price as 100%: the profit is 20% and the cost price is $100\% - 20\% = 80\%$ of the selling price.

Profit compared with cost price: $20 : 80 = 0.25 = 25\%$.

**Answer:** 25%.

**Problem 36**

A shop buys goods at 80% of the listed price and sells them at 5% off the listed price. Its profit is what percentage of the cost price?

**Solution**

Take the listed price as 100%.

Cost price: 80%; selling price: $100\% - 5\% = 95\%$; profit: $95\% - 80\% = 15\%$ of the listed price.

Profit compared with cost price: $15 : 80 = 0.1875 = 18.75\%$.

**Answer:** 18.75%.

**Problem 37**

Mr Dũng saves 10 000 000 dong at an interest of 0.5% a month. The interest of the first month is added to the account and also earns interest in the second month. How much money is in the account after two months?

**Solution**

After month 1: interest $10\,000\,000 \times 0.5 : 100 = 50\,000$ (dong); total 10 050 000 dong.

After month 2: interest $10\,050\,000 \times 0.5 : 100 = 50\,250$ (dong); total $10\,050\,000 + 50\,250 = 10\,100\,250$ (dong).

**Answer:** 10 100 250 dong.

**Problem 38**

A jar holds 300 g of salt water containing 6% salt. How many grams of pure water must be added to get salt water containing 4% salt?

**Solution**

Salt: $300 \times 6 : 100 = 18$ (g); it does not change when water is added.

New salt water: 18 g is 4% of it, so it weighs $18 : 4 \times 100 = 450$ (g).

Water added: $450 - 300 = 150$ (g).

**Answer:** 150 g.

**Problem 39**

Mix 200 g of salt water with 10% salt and 300 g of salt water with 5% salt. What percentage of the mixture is salt?

**Solution**

Salt in the first jar: $200 \times 10 : 100 = 20$ (g); in the second: $300 \times 5 : 100 = 15$ (g).

Mixture: $200 + 300 = 500$ (g), with $20 + 15 = 35$ (g) of salt.

$35 : 500 = 0.07 = 7\%$.

**Answer:** 7%.

**Problem 40**

In a class, the boys make up 40% of the pupils. After 6 girls move to another class, the boys make up 50% of the class. How many pupils did the class have at first?

**Solution**

The number of boys does not change.

At first: girls 60%, boys 40%, so girls $= \frac{60}{40} = \frac{3}{2}$ of the boys.

Afterwards: girls = boys (50% each).

The 6 girls who left are $\frac{3}{2} - 1 = \frac{1}{2}$ of the boys, so there are $6 \times 2 = 12$ boys.

At first the class had $12 : 40 \times 100 = 30$ (pupils).

**Answer:** 30 pupils.

**Problem 41**

The price of a bicycle was raised by 25%. By what percentage must the new price be lowered to get back to the old price?

**Solution**

Take the old price as 4 parts; $25\% = \frac{1}{4}$, so the new price is 5 parts.

To get back we must take off 1 part of the new price: $1 : 5 = 0.2 = 20\%$.

**Answer:** 20%.

### Problem 42 *

Shop A sells a rice cooker for 200 000 dong with 15% off. Shop B sells the same rice cooker for 180 000 dong with 5% off. Which shop is cheaper, and by how much?

### Solution

Shop A: $200\,000 \times 85 : 100 = 170\,000$ (dong).
Shop B: $180\,000 \times 95 : 100 = 171\,000$ (dong).
Shop A is cheaper by $171\,000 - 170\,000 = 1\,000$ (dong).
**Answer:** shop A, cheaper by 1 000 dong.

### Problem 43 **

A trader sells two vases for 240 000 dong each. On one he makes a profit of 20% of its cost; on the other he loses 20% of its cost. Overall, does he gain or lose, and how much?

### Solution

First vase: 240 000 is 120% of its cost, so the cost is $240\,000 : 120 \times 100 = 200\,000$ (dong).
Second vase: 240 000 is $100\% - 20\% = 80\%$ of its cost, so the cost is $240\,000 : 80 \times 100 = 300\,000$ (dong).
Total cost: 500 000 dong; money received: 480 000 dong.
He loses $500\,000 - 480\,000 = 20\,000$ (dong).
**Answer:** he loses 20 000 dong.

### Problem 44 *

A theatre lowered its ticket price by 20%, and the number of people coming went up by 30%. By what percentage did the theatre's ticket money change?

### Solution

New price: 80% of the old price; new number of people: 130% of the old number.
New ticket money: $80\% \times 130\% = 0.8 \times 1.3 = 1.04 = 104\%$ of the old ticket money.
It went up by $104\% - 100\% = 4\%$.
**Answer:** it went up by 4%.

### Problem 45 *

The side of a square is increased by 10%. By what percentage does its area increase?

**Solution**

New side: 110% of the old side.
New area: $110\% \times 110\% = 1.1 \times 1.1 = 1.21 = 121\%$ of the old area.
The area increases by 21%.
**Answer:** 21%.

**Problem 46** *

A village has 5 000 people. Each year the population grows by 2% of the previous year's number. How many people will live in the village after 2 years?

**Solution**

After 1 year: $5\,000 + 5\,000 \times 2 : 100 = 5\,000 + 100 = 5\,100$ (people).
After 2 years: $5\,100 + 5\,100 \times 2 : 100 = 5\,100 + 102 = 5\,202$ (people).
**Answer:** 5 202 people.

**Problem 47** **

A 500 g piece of bronze contains 80% copper; the rest is tin. How many grams of copper must be added so that copper makes up 90%?

**Solution**

Tin: $100\% - 80\% = 20\%$, which is $500 \times 20 : 100 = 100$ (g). Adding copper does not change the tin.
In the new piece, tin is $100\% - 90\% = 10\%$, so the new piece weighs $100 : 10 \times 100 = 1\,000$ (g).
Copper added: $1\,000 - 500 = 500$ (g).
**Answer:** 500 g.

**Problem 48** *

A stationery shop offers: "Buy 4 notebooks, get 1 free." Buying in this way, by what percentage is the price of each notebook reduced?

**Solution**

You pay for 4 notebooks and get 5. The money of 1 notebook is saved out of the price of 5 notebooks.
$1 : 5 = 0.2 = 20\%$.
**Answer:** 20%.

## Problem 49 ⋆

A saleswoman gets a basic pay of 6 000 000 dong a month plus 5% of the value of the goods she sells. Last month she received 8 500 000 dong. What was the value of the goods she sold?

### Solution

Her extra pay: $8\,500\,000 - 6\,000\,000 = 2\,500\,000$ (dong), which is 5% of the goods sold.

Value of the goods: $2\,500\,000 : 5 \times 100 = 50\,000\,000$ (dong).

**Answer:** 50 000 000 dong.

## Problem 50 ⋆⋆

In a maths contest, 60% of the pupils come from the city and the rest from the countryside. 25% of the city pupils and 40% of the countryside pupils win a prize. What percentage of all the pupils win a prize?

### Solution

Countryside pupils: $100\% - 60\% = 40\%$ of all the pupils.

City prize winners: $60\% \times 25\% = 0.6 \times 0.25 = 0.15 = 15\%$ of all the pupils.

Countryside prize winners: $40\% \times 40\% = 0.4 \times 0.4 = 0.16 = 16\%$ of all the pupils.

All prize winners: $15\% + 16\% = 31\%$.

**Answer:** 31%.

# 3 Cuboids and Cubes

## ① Cuboids

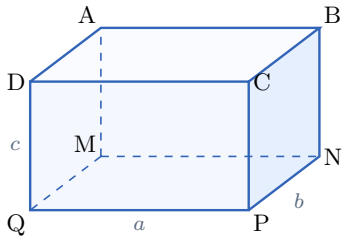
A matchbox, a brick and a shoebox all have the shape of a **cuboid** (a rectangular box).

Cuboid ABCD.MNPQ has:

- 6 **faces**, all rectangles; opposite faces are equal. ABCD and MNPQ are the two **bases**; the other four faces are the **side faces**;
- 12 **edges**, in three groups of 4 equal edges;
- 8 **vertices**: A, B, C, D, M, N, P, Q.

The three **dimensions** are the **length** $a$, the **width** $b$ (the two sides of the base) and the **height** $c$ (a side edge such as $DQ$).

In a drawing, edges we cannot see are drawn *dashed*.

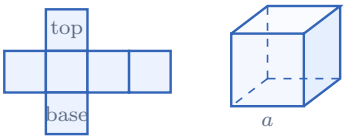



## ② Cubes and nets

A **cube** is a cuboid whose 12 edges are all equal. Its 6 faces are equal squares.

If we cut a box along some edges and unfold it flat, we get a **net** of the box. A net of a cube is made of 6 equal squares; the figure on the right is one of them. Folding the net back up gives the cube again.

Not every set of 6 joined squares is a net: two squares must never land on the same face when we fold.




## ③ Lateral area and total surface area

The **lateral area** $S_{\text{lat}}$ of a cuboid is the total area of its 4 side faces. Unfolded, the four side faces make one long rectangle whose length is the **perimeter of the base** and whose width is the **height**:

$$S_{\text{lat}} = (a + b) \times 2 \times c \qquad \text{(perimeter of base} \times \text{height)}$$

The **total surface area** $S_{\text{tot}}$ is the lateral area plus the areas of the two bases:

$$S_{\text{tot}} = S_{\text{lat}} + a \times b \times 2$$

For a **cube** with edge $a$: $S_{\text{lat}} = a \times a \times 4$, $S_{\text{tot}} = a \times a \times 6$.

*Remember:* all dimensions must be in the same unit. A box *without a lid* has only one base: its area is $S_{\text{lat}} + a \times b$.

## Example 1

A cuboid is 8 cm long, 5 cm wide and 6 cm high. Find:

**a)** its lateral area; **b)** the area of one base; **c)** its total surface area.

### Guided solution.

**a)** Perimeter of the base: $(8 + 5) \times 2 = 26$ cm.

$S_{\text{lat}} = 26 \times 6 = 156$ cm$^2$.

**b)** Area of one base: $8 \times 5 = 40$ cm$^2$.

**c)** $S_{\text{tot}} = 156 + 40 \times 2 = 236$ cm$^2$.

**Answer: a)** 156 cm$^2$; **b)** 40 cm$^2$; **c)** 236 cm$^2$.

## Example 2

A cube has a total surface area of 294 cm$^2$. Find its edge and its lateral area.

### Guided solution.

A cube has 6 equal faces. Area of one face: $294 : 6 = 49$ cm$^2$.

Since $49 = 7 \times 7$, the edge is 7 cm.

Lateral area: $49 \times 4 = 196$ cm$^2$.

**Answer:** Edge 7 cm; 196 cm$^2$.

## Example 3

A cuboid has a base of area 60 cm$^2$. Its length is 3 times its height and is 8 cm more than its height. Find its lateral area and total surface area.

### Guided solution.

The length is 3 times the height, so the difference is $3 - 1 = 2$ times the height.

Height: $8 : 2 = 4$ cm. Length: $4 \times 3 = 12$ cm. Width: $60 : 12 = 5$ cm.

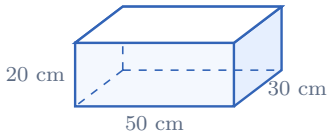
$S_{\text{lat}} = (12 + 5) \times 2 \times 4 = 136$ cm$^2$.

$S_{\text{tot}} = 136 + 60 \times 2 = 256$ cm$^2$.

**Answer:** 136 cm$^2$; 256 cm$^2$.

## Example 4

A tin box *without a lid* is 50 cm long, 30 cm wide and 20 cm high. How much tin is needed to make it? (Ignore the seams.)




### Guided solution.

The box has 4 side faces and only 1 base.

Lateral area: $(50 + 30) \times 2 \times 20 = 3\,200$ cm$^2$.

Base: $50 \times 30 = 1\,500$ cm$^2$.

Tin needed: $3\,200 + 1\,500 = 4\,700$ cm$^2$.

**Answer:** 4 700 cm$^2$.

## Example 5

A room is 6 m long, 4.5 m wide and 3.5 m high. The four walls and the ceiling are to be painted. The doors and windows have a total area of 7.5 m$^2$. Find the area to be painted.

### Guided solution.

Four walls (lateral area): $(6 + 4.5) \times 2 \times 3.5 = 73.5$ m$^2$.

Ceiling: $6 \times 4.5 = 27$ m$^2$.

Area to paint: $73.5 + 27 - 7.5 = 93$ m$^2$.

*The floor is not painted, and the doors and windows are taken away.*

**Answer:** 93 m$^2$.

# Practice

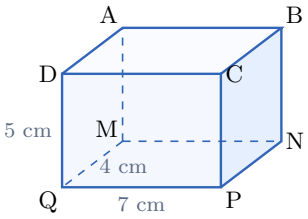
## Problem 1

Cuboid ABCD.MNPQ has $QP = 7$ cm, $QM = 4$ cm and $QD = 5$ cm.

**a)** Find the lengths of the edges AB, BN and NP.

**b)** Find the areas of the base MNPQ and of the faces DCPQ and ADQM.

**c)** Find its lateral area and total surface area.




## Solution

**a)** $AB = QP = 7$ cm; $BN = QD = 5$ cm; $NP = QM = 4$ cm.

**b)** $S_{MNPQ} = 7 \times 4 = 28$ cm$^2$; $S_{DCPQ} = 7 \times 5 = 35$ cm$^2$; $S_{ADQM} = 4 \times 5 = 20$ cm$^2$.

**c)** $S_{\text{lat}} = (7 + 4) \times 2 \times 5 = 110$ cm$^2$.

$S_{\text{tot}} = 110 + 28 \times 2 = 166$ cm$^2$.

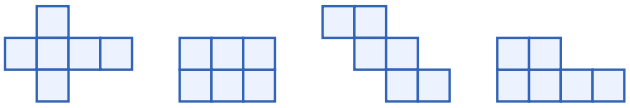
**Answer: a)** 7 cm, 5 cm, 4 cm;

**b)** 28 $\text{cm}^2$, 35 $\text{cm}^2$, 20 $\text{cm}^2$;

**c)** 110 $\text{cm}^2$, 166 $\text{cm}^2$.

## Problem 2

Which of the figures below are nets of a cube?




## Solution

**a)** The row of 4 squares folds into the 4 side faces; the square above is the top and the square below is the base. It is a net.

**b)** Folding the $2 \times 3$ block, the squares overlap and two faces stay open. It is not a net.

**c)** This "staircase" folds into a cube with no overlap. It is a net.

**d)** The two squares on top of the row both become the same end face, and the other end stays open. It is not a net.

**Answer:** Figures **a)** and **c)**.

## Problem 3

Find the lateral area and the total surface area of a cuboid with:

**a)** $a = 9$ cm, $b = 6$ cm, $c = 5$ cm;

**b)** $a = 2.5$ m, $b = 1.5$ m, $c = 1.2$ m;

**c)** $a = \frac{4}{5}$ m, $b = \frac{1}{2}$ m, $c = \frac{1}{3}$ m;

**d)** $a = 1.2$ dm, $b = 8$ cm, $c = 5$ cm.

## Solution

**a)** $S_{\text{lat}} = (9 + 6) \times 2 \times 5 = 150$ $\text{cm}^2$; $S_{\text{tot}} = 150 + 9 \times 6 \times 2 = 258$ $\text{cm}^2$.

**b)** $S_{\text{lat}} = (2.5 + 1.5) \times 2 \times 1.2 = 9.6$ $\text{m}^2$; $S_{\text{tot}} = 9.6 + 2.5 \times 1.5 \times 2 = 17.1$ $\text{m}^2$.

**c)** $S_{\text{lat}} = \left(\frac{4}{5} + \frac{1}{2}\right) \times 2 \times \frac{1}{3} = \frac{13}{10} \times \frac{2}{3} = \frac{13}{15}$ $\text{m}^2$;

$S_{\text{tot}} = \frac{13}{15} + \frac{4}{5} \times \frac{1}{2} \times 2 = \frac{13}{15} + \frac{12}{15} = \frac{5}{3}$ $\text{m}^2$.

**d)** 1.2 dm = 12 cm. $S_{\text{lat}} = (12 + 8) \times 2 \times 5 = 200$ $\text{cm}^2$; $S_{\text{tot}} = 200 + 12 \times 8 \times 2 = 392$ $\text{cm}^2$.

**Answer: a)** 150 $\text{cm}^2$, 258 $\text{cm}^2$;

**b)** 9.6 $\text{m}^2$, 17.1 $\text{m}^2$;

**c)** $\frac{13}{15}$ m$^2$, $\frac{5}{3}$ m$^2$;

**d)** 200 cm$^2$, 392 cm$^2$.

## Problem 4

Find the lateral area and the total surface area of a cube with edge:

**a)** 6 cm; **b)** 2.5 dm; **c)** $\frac{3}{4}$ m.

## Solution

**a)** $S_{\text{lat}} = 6 \times 6 \times 4 = 144$ cm$^2$; $S_{\text{tot}} = 6 \times 6 \times 6 = 216$ cm$^2$.

**b)** $S_{\text{lat}} = 2.5 \times 2.5 \times 4 = 25$ dm$^2$; $S_{\text{tot}} = 2.5 \times 2.5 \times 6 = 37.5$ dm$^2$.

**c)** $S_{\text{lat}} = \frac{3}{4} \times \frac{3}{4} \times 4 = \frac{9}{4}$ m$^2$; $S_{\text{tot}} = \frac{3}{4} \times \frac{3}{4} \times 6 = \frac{27}{8}$ m$^2$.

**Answer: a)** 144 cm$^2$, 216 cm$^2$;

**b)** 25 dm$^2$, 37.5 dm$^2$;

**c)** $\frac{9}{4}$ m$^2$, $\frac{27}{8}$ m$^2$.

## Problem 5

A cuboid is 15 cm long, 10 cm wide and 8 cm high. Find:

**a)** the perimeter of its base; **b)** its lateral area; **c)** its total surface area.

## Solution

**a)** $(15 + 10) \times 2 = 50$ cm.

**b)** $50 \times 8 = 400$ cm$^2$.

**c)** $400 + 15 \times 10 \times 2 = 700$ cm$^2$.

**Answer: a)** 50 cm; **b)** 400 cm$^2$; **c)** 700 cm$^2$.

## Problem 6

Mai wraps a gift box 24 cm long, 16 cm wide and 10 cm high, covering every face. What is the least area of wrapping paper she needs?

**Solution**

The paper must cover all 6 faces, i.e. the total surface area.

$S_{\text{lat}} = (24 + 16) \times 2 \times 10 = 800$ cm$^2$.

$S_{\text{tot}} = 800 + 24 \times 16 \times 2 = 1\,568$ cm$^2$.

**Answer:** 1 568 cm$^2$.

**Problem 7**

A box is 40 cm long, 25 cm wide and 20 cm high. Lan covers its four side faces with blue paper and its two bases with yellow paper (outside only). Which colour of paper is used more, and by how many square centimetres?

**Solution**

Blue paper (lateral area): $(40 + 25) \times 2 \times 20 = 2\,600$ cm$^2$.

Yellow paper (two bases): $40 \times 25 \times 2 = 2\,000$ cm$^2$.

Blue is more by $2\,600 - 2\,000 = 600$ cm$^2$.

**Answer:** Blue paper, by 600 cm$^2$.

**Problem 8**

An iron box without a lid is 60 cm long, 40 cm wide and 30 cm high. How much iron sheet is used to make it? (Ignore the seams.)

**Solution**

Lateral area: $(60 + 40) \times 2 \times 30 = 6\,000$ cm$^2$.

Base: $60 \times 40 = 2\,400$ cm$^2$.

Iron sheet: $6\,000 + 2\,400 = 8\,400$ cm$^2$.

**Answer:** 8 400 cm$^2$.

**Problem 9**

A classroom is 8 m long, 6 m wide and 3.5 m high. The four walls and the ceiling are to be painted; the doors and windows cover 9.5 m$^2$. Find the area to be painted.

**Solution**

Walls: $(8 + 6) \times 2 \times 3.5 = 98$ m$^2$.

Ceiling: $8 \times 6 = 48$ m$^2$.

Area to paint: $98 + 48 - 9.5 = 136.5$ m$^2$.

**Answer:** 136.5 m$^2$.

**Problem 10**

A water tank in the shape of a cube without a lid has edge 1.5 m. Its outside (side faces and bottom) is painted. Find the painted area.

**Solution**

One face: $1.5 \times 1.5 = 2.25$ m$^2$.
A cube without a lid has 5 faces: $2.25 \times 5 = 11.25$ m$^2$.
**Answer:** 11.25 m$^2$.

**Problem 11**

A cube has a lateral area of 100 cm$^2$. Find its edge and its total surface area.

**Solution**

One face: $100 : 4 = 25$ cm$^2$; since $25 = 5 \times 5$, the edge is 5 cm.
$S_{\text{tot}} = 25 \times 6 = 150$ cm$^2$.
**Answer:** 5 cm; 150 cm$^2$.

**Problem 12**

A cube has a total surface area of 600 cm$^2$. Find its lateral area.

**Solution**

One face: $600 : 6 = 100$ cm$^2$.
$S_{\text{lat}} = 100 \times 4 = 400$ cm$^2$.
**Answer:** 400 cm$^2$.

**Problem 13**

A cube has a lateral area of 1.44 m$^2$. Find its edge and its total surface area.

**Solution**

One face: $1.44 : 4 = 0.36$ m$^2$; since $0.36 = 0.6 \times 0.6$, the edge is 0.6 m.
$S_{\text{tot}} = 0.36 \times 6 = 2.16$ m$^2$.
**Answer:** 0.6 m; 2.16 m$^2$.

**Problem 14**

A cuboid has a lateral area of 360 cm$^2$ and a height of 9 cm. Its length is 4 cm more than its width. Find the length and the width.

**Solution**

Perimeter of the base: $360 : 9 = 40$ cm.
Length + width: $40 : 2 = 20$ cm.
Length: $(20 + 4) : 2 = 12$ cm. Width: $20 - 12 = 8$ cm.
**Answer:** Length 12 cm, width 8 cm.

**Problem 15**

A cuboid has a base perimeter of 36 dm and a height of 7 dm. Find its lateral area.

**Solution**

$S_{\text{lat}} = 36 \times 7 = 252$ dm$^2$.
**Answer:** 252 dm$^2$.

**Problem 16**

A cuboid has a lateral area of 480 cm$^2$ and a base perimeter of 48 cm. Find its height.

**Solution**

Height = lateral area : perimeter of base.
$c = 480 : 48 = 10$ cm.
**Answer:** 10 cm.

**Problem 17**

A cuboid is 2.4 m long and 1.6 m wide, and its total surface area is 19.68 m$^2$. Find its height.

**Solution**

Two bases: $2.4 \times 1.6 \times 2 = 7.68$ m$^2$.
Lateral area: $19.68 - 7.68 = 12$ m$^2$.
Perimeter of base: $(2.4 + 1.6) \times 2 = 8$ m.
Height: $12 : 8 = 1.5$ m.
**Answer:** 1.5 m.

**Problem 18**

A wooden crate is 1.2 m long, 0.8 m wide and 0.6 m high. All of its outside faces are varnished, using 0.25 kg of varnish per square metre. How many kilograms of varnish are needed?

**Solution**

$S_{\text{lat}} = (1.2 + 0.8) \times 2 \times 0.6 = 2.4$ m$^2$.

$S_{\text{tot}} = 2.4 + 1.2 \times 0.8 \times 2 = 4.32$ m$^2$.

Varnish: $0.25 \times 4.32 = 1.08$ kg.

**Answer:** 1.08 kg.

**Problem 19**

A swimming pool is 25 m long, 10 m wide and 1.6 m deep. Its bottom and its four walls are covered with tiles measuring 25 cm by 20 cm. How many tiles are needed? (Ignore the joints.)

**Solution**

Walls: $(25 + 10) \times 2 \times 1.6 = 112$ m$^2$. Bottom: $25 \times 10 = 250$ m$^2$.

Area to tile: $112 + 250 = 362$ m$^2$.

One tile: $25 \times 20 = 500$ cm$^2$ $= 0.05$ m$^2$.

Tiles: $362 : 0.05 = 7\,240$ tiles.

**Answer:** 7 240 tiles.

**Problem 20**

A fish pond is 3 m long, 2 m wide and 0.8 m deep. Its bottom and walls are covered with square tiles of side 2 dm. How many tiles are needed?

**Solution**

Walls: $(3 + 2) \times 2 \times 0.8 = 8$ m$^2$. Bottom: $3 \times 2 = 6$ m$^2$.

Area to tile: $8 + 6 = 14$ m$^2$ $= 1\,400$ dm$^2$.

One tile: $2 \times 2 = 4$ dm$^2$. Tiles: $1\,400 : 4 = 350$.

**Answer:** 350 tiles.

**Problem 21**

A cuboid has a base perimeter of 30 cm; its length is 3 cm more than its width, and its height is $\frac{2}{3}$ of its length. Find its lateral area and total surface area.

**Solution**

Length + width: $30 : 2 = 15$ cm.

Length: $(15 + 3) : 2 = 9$ cm. Width: 6 cm. Height: $9 \times \frac{2}{3} = 6$ cm.

$S_{\text{lat}} = 30 \times 6 = 180$ cm$^2$.

$S_{\text{tot}} = 180 + 9 \times 6 \times 2 = 288$ cm$^2$.

**Answer:** 180 $\text{cm}^2$; 288 $\text{cm}^2$.

**Problem 22**

Compare the total surface area of a cube with edge 6 cm and that of a cuboid 8 cm long, 6 cm wide and 4 cm high.

**Solution**

Cube: $6 \times 6 \times 6 = 216$ $\text{cm}^2$.
Cuboid: $(8 + 6) \times 2 \times 4 + 8 \times 6 \times 2 = 112 + 96 = 208$ $\text{cm}^2$.
$216 - 208 = 8$ $\text{cm}^2$.
**Answer:** The cube's is larger, by 8 $\text{cm}^2$.

**Problem 23**

A cuboid has a square base of side 5 cm and a height of 12 cm. Find its lateral area and total surface area.

**Solution**

Perimeter of base: $5 \times 4 = 20$ cm.
$S_{\text{lat}} = 20 \times 12 = 240$ $\text{cm}^2$.
$S_{\text{tot}} = 240 + 5 \times 5 \times 2 = 290$ $\text{cm}^2$.
**Answer:** 240 $\text{cm}^2$; 290 $\text{cm}^2$.

**Problem 24**

A lantern has a frame in the shape of a cuboid with a square base of side 20 cm and a height of 35 cm. Paper is glued on the four sides and the bottom. How many square decimetres of paper are used?

**Solution**

Four sides: $20 \times 4 \times 35 = 2\,800$ $\text{cm}^2$.
Bottom: $20 \times 20 = 400$ $\text{cm}^2$.
Paper: $2\,800 + 400 = 3\,200$ $\text{cm}^2 = 32$ $\text{dm}^2$.
**Answer:** 32 $\text{dm}^2$.

**Problem 25**

A piece of cardboard is just enough to make a cube-shaped box without a lid, with edge 2.5 dm. Find the area of the cardboard. (Ignore the flaps.)

**Solution**

One face: $2.5 \times 2.5 = 6.25$ dm$^2$.
The box has 5 faces: $6.25 \times 5 = 31.25$ dm$^2$.
**Answer:** 31.25 dm$^2$.

**Problem 26**

A cube-shaped tank without a lid has a base perimeter of 4.8 m. Both its inside and its outside are painted, and 1 kg of paint covers 3 m$^2$. How many kilograms of paint are needed?

**Solution**

Edge: $4.8 : 4 = 1.2$ m. One face: $1.2 \times 1.2 = 1.44$ m$^2$.
Without a lid there are 5 faces, painted on both sides: $1.44 \times 5 \times 2 = 14.4$ m$^2$.
Paint: $14.4 : 3 = 4.8$ kg.
**Answer:** 4.8 kg.

**Problem 27**

The 12 edges of a cuboid add up to 96 cm. Its length is 12 cm and its width is 8 cm. Find its total surface area.

**Solution**

A cuboid has 4 lengths, 4 widths and 4 heights.
Length + width + height: $96 : 4 = 24$ cm. Height: $24 - 12 - 8 = 4$ cm.
$S_{\text{lat}} = (12 + 8) \times 2 \times 4 = 160$ cm$^2$.
$S_{\text{tot}} = 160 + 12 \times 8 \times 2 = 352$ cm$^2$.
**Answer:** 352 cm$^2$.

**Problem 28**

The 12 edges of a cube add up to 72 cm. Find its total surface area.

**Solution**

Edge: $72 : 12 = 6$ cm.
$S_{\text{tot}} = 6 \times 6 \times 6 = 216$ cm$^2$.
**Answer:** 216 cm$^2$.

**Problem 29**

A glass aquarium without a lid is 80 cm long, 40 cm wide and 50 cm high. How many square decimetres of glass are used?

**Solution**

Four walls: $(80 + 40) \times 2 \times 50 = 12\,000$ cm$^2$.
Bottom: $80 \times 40 = 3\,200$ cm$^2$.
Glass: $12\,000 + 3\,200 = 15\,200$ cm$^2$ $= 152$ dm$^2$.
**Answer:** 152 dm$^2$.

**Problem 30**

The total surface area of a cube is 50 cm$^2$ more than its lateral area. Find its lateral area.

**Solution**

The difference is the area of the two bases: one face is $50 : 2 = 25$ cm$^2$.
$S_{\text{lat}} = 25 \times 4 = 100$ cm$^2$.
**Answer:** 100 cm$^2$.

## Challenge – Extension

**Problem 31** ⋆

A cuboid has a base of area 120 cm$^2$. Its length is 9 cm more than its height, and its height is $\frac{2}{5}$ of its length. Find its lateral area and total surface area.

**Solution**

Height 2 parts, length 5 parts; the difference is 3 parts = 9 cm, so 1 part is 3 cm.
Height: $3 \times 2 = 6$ cm. Length: $3 \times 5 = 15$ cm. Width: $120 : 15 = 8$ cm.
$S_{\text{lat}} = (15 + 8) \times 2 \times 6 = 276$ cm$^2$.
$S_{\text{tot}} = 276 + 120 \times 2 = 516$ cm$^2$.
**Answer:** 276 cm$^2$; 516 cm$^2$.

**Problem 32** ⋆

A water tank has a lateral area of 10.5 m$^2$ and a height of 1.5 m. Its length is 1 m more than its width. Find its length and width.

**Solution**

Perimeter of base: $10.5 : 1.5 = 7$ m. Length + width: $7 : 2 = 3.5$ m.
Length: $(3.5 + 1) : 2 = 2.25$ m. Width: $3.5 - 2.25 = 1.25$ m.
**Answer:** Length 2.25 m, width 1.25 m.

## Problem 33 *

An iron crate in the shape of a cuboid has a width equal to $\frac{3}{4}$ of its length and 0.5 m less than its length; its height is 1.2 m.

**a)** Find its total surface area.

**b)** The outside is painted completely, using 3 kg of paint for every 4 $m^2$. How many kilograms of paint are needed?

### Solution

**a)** Length 4 parts, width 3 parts; the difference is 1 part = 0.5 m.

Length: $0.5 \times 4 = 2$ m. Width: $0.5 \times 3 = 1.5$ m.

$S_{\text{lat}} = (2 + 1.5) \times 2 \times 1.2 = 8.4$ $m^2$. $S_{\text{tot}} = 8.4 + 2 \times 1.5 \times 2 = 14.4$ $m^2$.

**b)** 1 $m^2$ needs $3 : 4 = 0.75$ kg. Paint: $0.75 \times 14.4 = 10.8$ kg.

**Answer: a)** 14.4 $m^2$; **b)** 10.8 kg.

## Problem 34 * *

If the edge of a cube is increased by 2 cm, its total surface area increases by 120 $cm^2$. Find the original edge.

### Solution

Each of the 6 faces grows by $120 : 6 = 20$ $cm^2$.

A square of side $a$ becomes a square of side $a + 2$: the new part is two strips $a$ by 2 and a corner square 2 by 2.

$a \times 2 \times 2 + 2 \times 2 = 20$, so $a \times 4 = 16$ and $a = 4$ cm.

*Check:* $6 \times 6 \times 6 - 4 \times 4 \times 6 = 216 - 96 = 120$ $cm^2$.

**Answer:** 4 cm.

## Problem 35 *

Two cubes, each with edge 5 cm, are glued face to face to make a cuboid. Find the total surface area of the cuboid. How much less is it than the total of the two cubes' surface areas?

### Solution

The cuboid is 10 cm $\times$ 5 cm $\times$ 5 cm.

$S_{\text{tot}} = (10 + 5) \times 2 \times 5 + 10 \times 5 \times 2 = 150 + 100 = 250$ $cm^2$.

Two cubes: $5 \times 5 \times 6 \times 2 = 300$ $cm^2$.

Less by $300 - 250 = 50$ $cm^2$: the two glued faces ($2 \times 25$ $cm^2$) are hidden.

**Answer:** 250 $cm^2$; 50 $cm^2$ less.

**Problem 36**

Three cubes with edge 4 cm are put in a row and glued into a cuboid. Find the total surface area of that cuboid.

**Solution**

The cuboid is 12 cm $\times$ 4 cm $\times$ 4 cm.

$S_{\text{tot}} = (12 + 4) \times 2 \times 4 + 12 \times 4 \times 2 = 128 + 96 = 224$ cm$^2$.

*Other way:* the 3 cubes have 18 faces; each of the 2 joints hides 2 faces, so $18 - 4 = 14$ faces remain: $14 \times 4 \times 4 = 224$ cm$^2$.

**Answer:** 224 cm$^2$.

**Problem 37**

A cuboid 10 cm long, 6 cm wide and 4 cm high is cut into two equal halves by a cut parallel to its base. Find the total surface area of the original cuboid and the total surface area of the two halves together.

**Solution**

Original: $(10 + 6) \times 2 \times 4 + 10 \times 6 \times 2 = 128 + 120 = 248$ cm$^2$.

The cut makes two new faces, each $10 \times 6 = 60$ cm$^2$.

Two halves together: $248 + 60 \times 2 = 368$ cm$^2$.

**Answer:** 248 cm$^2$; 368 cm$^2$.

**Problem 38**

A cuboid has a square base and a lateral area of 160 cm$^2$. Its height is 2.5 times the side of the base. Find its total surface area.

**Solution**

Each side face is a rectangle (base side) $\times$ (2.5$\times$ base side), so it equals 2.5 squares of the base.

Lateral area $= 4 \times 2.5 = 10$ base squares: one base square is $160 : 10 = 16$ cm$^2$ (side 4 cm, height 10 cm).

$S_{\text{tot}} = 160 + 16 \times 2 = 192$ cm$^2$.

**Answer:** 192 cm$^2$.

**Problem 39**

A cube has a total surface area of 1.5 m$^2$. Find its edge and its lateral area.

**Solution**

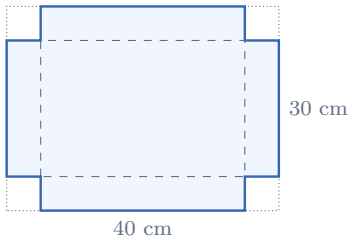
One face: $1.5 : 6 = 0.25$ m$^2$; since $0.25 = 0.5 \times 0.5$, the edge is 0.5 m.

$S_{\text{lat}} = 0.25 \times 4 = 1$ m$^2$.

**Answer:** 0.5 m; 1 m$^2$.

**Problem 40** ★★

From a rectangular sheet of card 40 cm by 30 cm, Nam cuts a square of side 5 cm from each corner and folds up the sides to make an open box. Find the length, width and height of the box and the area of card it uses.




**Solution**

Length: $40 - 5 \times 2 = 30$ cm. Width: $30 - 5 \times 2 = 20$ cm. Height: 5 cm.

Card used: $40 \times 30 - 5 \times 5 \times 4 = 1\,200 - 100 = 1\,100$ cm$^2$.

*Check:* $S_{\text{lat}}$ + base $= (30 + 20) \times 2 \times 5 + 30 \times 20 = 500 + 600 = 1\,100$ cm$^2$.

**Answer:** 30 cm, 20 cm, 5 cm; 1 100 cm$^2$.

**Problem 41** ★

A room is 5 m long, 4 m wide and 3 m high. The inside walls and the ceiling are painted. The door and windows cover 4 m$^2$. Painting costs 25 000 đồng per square metre. Find the cost.

**Solution**

Walls: $(5 + 4) \times 2 \times 3 = 54$ m$^2$. Ceiling: $5 \times 4 = 20$ m$^2$.

Area to paint: $54 + 20 - 4 = 70$ m$^2$.

Cost: $25\,000 \times 70 = 1\,750\,000$ đồng.

**Answer:** 1 750 000 đồng.

**Problem 42** ★

The 12 edges of a cuboid add up to 120 cm. Its length is 3 times its width and its height is 2 times its width. Find its total surface area.

**Solution**

Length + width + height: $120 : 4 = 30$ cm $= 3 + 1 + 2 = 6$ widths.

Width: $30 : 6 = 5$ cm. Length: 15 cm. Height: 10 cm.

$S_{\text{tot}} = (15+5) \times 2 \times 10 + 15 \times 5 \times 2 = 400 + 150 = 550$ cm$^2$.
**Answer:** 550 cm$^2$.

**Problem 43** ⋆⋆

Three different faces of a cuboid have areas 12 cm$^2$, 15 cm$^2$ and 20 cm$^2$. Find its total surface area, and find its three dimensions (whole numbers of centimetres).

**Solution**

Each face appears twice: $S_{\text{tot}} = (12+15+20) \times 2 = 94$ cm$^2$.
Try whole numbers: the edge shared by the faces 12 and 15 divides both, so it is 1 or 3.
If it is 3: the other edges are $12 : 3 = 4$ and $15 : 3 = 5$, and $4 \times 5 = 20$. ✓
(If it were 1: $12 \times 15 = 180 \neq 20$.)
**Answer:** 94 cm$^2$; 3 cm, 4 cm, 5 cm.

**Problem 44** ⋆

A cube-shaped box without a lid has edge 15 cm. Coloured paper is glued on both the inside and the outside. Find the area of paper used.

**Solution**

One face: $15 \times 15 = 225$ cm$^2$.
5 faces, both sides: $225 \times 5 \times 2 = 2\,250$ cm$^2$.
**Answer:** 2 250 cm$^2$.

**Problem 45** ⋆

A cuboid has a square base of side 6 cm. Its lateral area is 3 times the total area of its two bases. Find its height and total surface area.

**Solution**

Two bases: $6 \times 6 \times 2 = 72$ cm$^2$.
$S_{\text{lat}} = 72 \times 3 = 216$ cm$^2$. Height: $216 : (6 \times 4) = 9$ cm.
$S_{\text{tot}} = 216 + 72 = 288$ cm$^2$.
**Answer:** 9 cm; 288 cm$^2$.

**Problem 46** ⋆

Two identical cuboids, each 8 cm × 5 cm × 3 cm, are glued together along their largest faces. Find the total surface area of the new cuboid.

**Solution**

The largest face is $8 \times 5 = 40$ cm$^2$. The new cuboid is 8 cm $\times$ 5 cm $\times$ 6 cm.

$S_{\text{tot}} = (8 + 5) \times 2 \times 6 + 8 \times 5 \times 2 = 156 + 80 = 236$ cm$^2$.

*Check:* one cuboid has $(8 + 5) \times 2 \times 3 + 80 = 158$ cm$^2$; $158 \times 2 - 40 \times 2 = 236$ cm$^2$.

**Answer:** 236 cm$^2$.

**Problem 47** ★★

A pool is 12.5 m long, 6 m wide and 1.4 m deep. Its bottom and walls are tiled with tiles of 25 cm $\times$ 20 cm. Tiles are sold in boxes of 12. How many boxes must be bought?

**Solution**

Walls: $(12.5 + 6) \times 2 \times 1.4 = 51.8$ m$^2$. Bottom: $12.5 \times 6 = 75$ m$^2$.

Area: $51.8 + 75 = 126.8$ m$^2$. One tile: 0.05 m$^2$.

Tiles: $126.8 : 0.05 = 2\,536$.

$2\,536 : 12 = 211$ remainder 4, so 211 boxes are not enough; we need 212 boxes.

**Answer:** 212 boxes.

**Problem 48** ★

A cube has the same total surface area as a cuboid 12 cm long, 9 cm wide and 4 cm high. Find the edge of the cube.

**Solution**

Cuboid: $(12 + 9) \times 2 \times 4 + 12 \times 9 \times 2 = 168 + 216 = 384$ cm$^2$.

One face of the cube: $384 : 6 = 64$ cm$^2$ $= 8 \times 8$.

**Answer:** 8 cm.

**Problem 49** ★★

A cuboid is 5 cm high. If its length is decreased by 3 cm and its width increased by 3 cm, its base becomes a square of side 8 cm. Compare the total surface areas of the old and the new cuboid.

**Solution**

Old length: $8 + 3 = 11$ cm. Old width: $8 - 3 = 5$ cm.

Base perimeter does not change (32 cm), so the lateral area is the same: $32 \times 5 = 160$ cm$^2$.

Old: $160 + 11 \times 5 \times 2 = 270$ cm$^2$. New: $160 + 8 \times 8 \times 2 = 288$ cm$^2$.

The new one is larger by 18 cm$^2$.

**Answer:** The new cuboid's is larger, by 18 cm$^2$.

## Problem 50 **

A cube of edge 6 cm is cut into 8 equal small cubes. Find the total surface area of the 8 small cubes and compare it with that of the big cube.

## Solution

Each small cube has edge $6 : 2 = 3$ cm.

One small cube: $3 \times 3 \times 6 = 54$ cm$^2$. 8 cubes: $54 \times 8 = 432$ cm$^2$.

Big cube: $6 \times 6 \times 6 = 216$ cm$^2$. $432 : 216 = 2$.

**Answer:** 432 cm$^2$, twice as much.

# 4 Volume

## ① Volume and units of volume

The **volume** of a solid is the amount of space it takes up. To measure volume we count how many *unit cubes* fit into the solid.

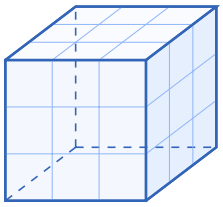
A **cubic centimetre** ($cm^3$) is the volume of a cube with edge 1 cm. Likewise, 1 $dm^3$ is a cube with edge 1 dm and 1 $m^3$ is a cube with edge 1 m.

$$1\ m^3 = 1\,000\ dm^3 \qquad 1\ dm^3 = 1\,000\ cm^3$$

Each unit of volume is 1 000 times the next smaller unit, so when converting we move the decimal point **three** places.

For liquids:

$$1\ l = 1\ dm^3 \qquad 1\ ml = 1\ cm^3 \qquad 1\ m^3 = 1\,000\ l$$



1 dm = 10 cm

1 $dm^3$: 10 layers of $10 \times 10$ cubes = 1 000 $cm^3$ (drawn coarser)

## ② Volume of a cuboid and of a cube

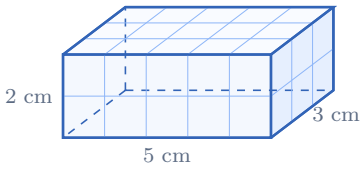
A cuboid 5 cm long, 3 cm wide and 2 cm high holds 2 layers of $5 \times 3 = 15$ unit cubes, i.e. $5 \times 3 \times 2 = 30\ cm^3$.

The **volume of a cuboid** is length × width × height (same unit):

$$V = a \times b \times c = S_{\text{base}} \times c$$

The **volume of a cube** with edge $a$: $V = a \times a \times a$.

Working backwards: $c = V : (a \times b)$, $S_{\text{base}} = V : c$.




## ③ Volume of an object in water

A stone or a piece of coral has no simple shape, so we cannot use a formula for it. Put it into a tank of water so that it is **completely under water**: the water rises because the stone takes up space.

- *Method 1:* volume of the stone = volume of (water + stone) − volume of the water.

- *Method 2:* volume of the stone = base area of the tank × the rise in water level.

If the tank is already full, the volume of water that **overflows** equals the volume of the object.

**Example 1**

Fill in the blanks.

**a)** $3.5\ \text{m}^3 = \ldots\ \text{dm}^3$

**b)** $2\,400\ \text{cm}^3 = \ldots\ \text{dm}^3$

**c)** $0.75\ \text{dm}^3 = \ldots\ \text{cm}^3 = \ldots\ \text{ml}$

**d)** $1.2\ \text{m}^3 = \ldots\ \text{l}$

**Guided solution.**

**a)** $1\ \text{m}^3 = 1\,000\ \text{dm}^3$, so $3.5\ \text{m}^3 = 3.5 \times 1\,000 = 3\,500\ \text{dm}^3$.

**b)** $2\,400\ \text{cm}^3 = 2\,400 : 1\,000 = 2.4\ \text{dm}^3$.

**c)** $0.75\ \text{dm}^3 = 750\ \text{cm}^3 = 750\ \text{ml}$.

**d)** $1.2\ \text{m}^3 = 1\,200\ \text{dm}^3 = 1\,200\ \text{l}$.

**Answer: a)** 3 500; **b)** 2.4; **c)** 750, 750; **d)** 1 200.

**Example 2**

Find the volume of:

**a)** a cuboid 1.2 m long, 5 dm wide and 8 dm high;

**b)** a cube with edge 2.5 cm.

**Guided solution.**

**a)** Change to the same unit: 1.2 m = 12 dm.

$V = 12 \times 5 \times 8 = 480\ \text{dm}^3$.

**b)** $V = 2.5 \times 2.5 \times 2.5 = 15.625\ \text{cm}^3$.

**Answer: a)** $480\ \text{dm}^3$; **b)** $15.625\ \text{cm}^3$.

**Example 3**

A water tank is 1.5 m long, 0.8 m wide and 1 m high.

**a)** How many litres does it hold when full?

**b)** At the moment the tank is $\frac{3}{4}$ full. How many litres of water are in it?

**Guided solution.**

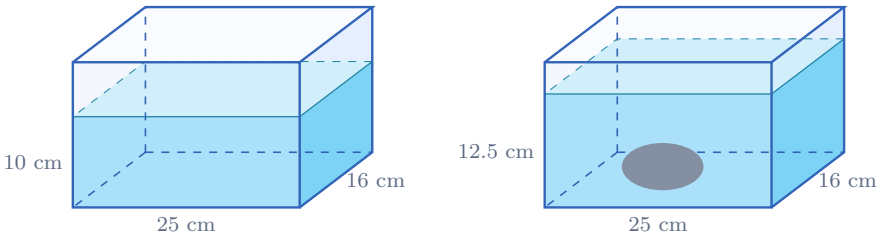
**a)** $V = 1.5 \times 0.8 \times 1 = 1.2\ \text{m}^3 = 1\,200\ \text{dm}^3 = 1\,200\ \text{l}$.

**b)** $1\,200 \times \frac{3}{4} = 900\ \text{l}$.

**Answer: a)** 1 200 l; **b)** 900 l.

## Example 4

A glass tank has a base 25 cm long and 16 cm wide. The water is 10 cm deep. After a stone is dropped in (and is completely under water), the water is 12.5 cm deep. Find the volume of the stone.




**Guided solution.**

*Method 1.* Water: $25 \times 16 \times 10 = 4\,000$ cm$^3$.

Water and stone: $25 \times 16 \times 12.5 = 5\,000$ cm$^3$.

Stone: $5\,000 - 4\,000 = 1\,000$ cm$^3$.

*Method 2.* The water rises $12.5 - 10 = 2.5$ cm.

Stone: $25 \times 16 \times 2.5 = 1\,000$ cm$^3$.

**Answer:** $1\,000$ cm$^3$ $(= 1$ dm$^3)$.

## Example 5

A cube has a total surface area of 864 cm$^2$. Find its volume.

**Guided solution.**

One face: $864 : 6 = 144$ cm$^2$. Since $144 = 12 \times 12$, the edge is 12 cm.

$V = 12 \times 12 \times 12 = 1\,728$ cm$^3$.

**Answer:** $1\,728$ cm$^3$.

# Practice

## Problem 1

Fill in the blanks.

**a)** $5$ m$^3 = \ldots$ dm$^3$ **b)** $3.2$ dm$^3 = \ldots$ cm$^3$ **c)** $4\,500$ cm$^3 = \ldots$ dm$^3$

**d)** $750$ dm$^3 = \ldots$ m$^3$ **e)** $2.05$ m$^3 = \ldots$ dm$^3$ **f)** $\frac{1}{4}$ m$^3 = \ldots$ dm$^3$

## Solution

**a)** $5 \times 1\,000 = 5\,000$. **b)** $3.2 \times 1\,000 = 3\,200$. **c)** $4\,500 : 1\,000 = 4.5$.

**d)** $750 : 1\,000 = 0.75$. **e)** $2.05 \times 1\,000 = 2\,050$. **f)** $1\,000 : 4 = 250$.

**Answer:** 5 000; 3 200; 4.5; 0.75; 2 050; 250.

## Problem 2

Fill in the blanks.

**a)** $3.6\ \text{m}^3 = \ldots\ \text{l}$

**b)** $450\ \text{l} = \ldots\ \text{m}^3$

**c)** $2.5\ \text{l} = \ldots\ \text{cm}^3$

**d)** $750\ \text{ml} = \ldots\ \text{dm}^3$

## Solution

**a)** $3.6\ \text{m}^3 = 3\,600\ \text{dm}^3 = 3\,600\ \text{l}$.

**b)** $450\ \text{l} = 450\ \text{dm}^3 = 0.45\ \text{m}^3$.

**c)** $2.5\ \text{l} = 2.5\ \text{dm}^3 = 2\,500\ \text{cm}^3$.

**d)** $750\ \text{ml} = 750\ \text{cm}^3 = 0.75\ \text{dm}^3$.

**Answer:** 3 600; 0.45; 2 500; 0.75.

## Problem 3

Compare (write $>$, $<$ or $=$).

**a)** $2.5\ \text{m}^3 \ \ldots\ 2\,500\ \text{dm}^3$ **b)** $3\ \text{dm}^3\ 40\ \text{cm}^3 \ \ldots\ 3.4\ \text{dm}^3$ **c)** $0.9\ \text{m}^3 \ \ldots\ 90\,000\ \text{cm}^3$

## Solution

**a)** $2.5\ \text{m}^3 = 2\,500\ \text{dm}^3$, so they are equal.

**b)** $3\ \text{dm}^3\ 40\ \text{cm}^3 = 3\,040\ \text{cm}^3$ and $3.4\ \text{dm}^3 = 3\,400\ \text{cm}^3$; $3\,040 < 3\,400$.

**c)** $0.9\ \text{m}^3 = 900\ \text{dm}^3 = 900\,000\ \text{cm}^3 > 90\,000\ \text{cm}^3$.

**Answer: a)** $=$; **b)** $<$; **c)** $>$.

## Problem 4

Find the volume of a cuboid with length $a$, width $b$ and height $c$:

**a)** $a = 8$ cm, $b = 5$ cm, $c = 4$ cm;

**b)** $a = 1.5$ m, $b = 0.8$ m, $c = 1.2$ m;

**c)** $a = \frac{2}{3}$ m, $b = \frac{3}{5}$ m, $c = \frac{1}{2}$ m;

**d)** $a = 2$ dm, $b = 15$ cm, $c = 8$ cm.

**Solution**

**a)** $8 \times 5 \times 4 = 160$ cm$^3$.

**b)** $1.5 \times 0.8 \times 1.2 = 1.44$ m$^3$.

**c)** $\frac{2}{3} \times \frac{3}{5} \times \frac{1}{2} = \frac{1}{5}$ m$^3$.

**d)** $2$ dm $= 20$ cm; $20 \times 15 \times 8 = 2\,400$ cm$^3$.

**Answer:** 160 cm$^3$; 1.44 m$^3$; $\frac{1}{5}$ m$^3$; 2 400 cm$^3$.

**Problem 5**

Find the volume of a cube with edge:

**a)** 7 cm; **b)** 1.2 m; **c)** $\frac{3}{4}$ dm.

**Solution**

**a)** $7 \times 7 \times 7 = 343$ cm$^3$.

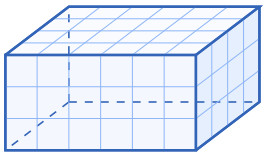
**b)** $1.2 \times 1.2 \times 1.2 = 1.728$ m$^3$.

**c)** $\frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{27}{64}$ dm$^3$.

**Answer:** 343 cm$^3$; 1.728 m$^3$; $\frac{27}{64}$ dm$^3$.

**Problem 6**

The box on the right is built from cubes of edge 1 cm, with no gaps. How many small cubes are there? What is the volume of the box?



**Solution**

One layer has $6 \times 4 = 24$ cubes, and there are 3 layers.

Number of cubes: $24 \times 3 = 72$.

Volume: 72 cm$^3$ (also $6 \times 4 \times 3 = 72$ cm$^3$).

**Answer:** 72 cubes; 72 cm$^3$.

**Problem 7**

A cuboid has a volume of 360 cm$^3$, a length of 9 cm and a width of 8 cm. Find its height.

**Solution**

Base area: $9 \times 8 = 72$ cm$^2$.
Height: $360 : 72 = 5$ cm.
**Answer:** 5 cm.

**Problem 8**

A cuboid has a volume of 1.8 m$^3$ and a base area of 1.2 m$^2$. Find its height.

**Solution**

$c = V : S_{\text{base}} = 1.8 : 1.2 = 1.5$ m.
**Answer:** 1.5 m.

**Problem 9**

A cube has a volume of 512 cm$^3$. Find its edge.

**Solution**

We need a number that multiplied by itself three times gives 512.
$7 \times 7 \times 7 = 343$, $8 \times 8 \times 8 = 512$. The edge is 8 cm.
**Answer:** 8 cm.

**Problem 10**

A water tank is 2 m long, 1.5 m wide and 1.2 m high. How many litres of water does it hold when full?

**Solution**

$V = 2 \times 1.5 \times 1.2 = 3.6$ m$^3$ $= 3\,600$ dm$^3$ $= 3\,600$ l.
**Answer:** 3 600 l.

**Problem 11**

An aquarium is 60 cm long, 30 cm wide and 40 cm high. The water reaches $\frac{3}{4}$ of its height. How many litres of water are in it?

**Solution**

Water depth: $40 \times \frac{3}{4} = 30$ cm.

Water: $60 \times 30 \times 30 = 54\,000$ cm$^3$ $= 54$ dm$^3$ $= 54$ l.
**Answer:** 54 l.

## Problem 12

A tank is 1.2 m long, 0.9 m wide and 1 m high. The water in it fills 60% of the tank. How many litres of water are in the tank?

### Solution

Tank: $1.2 \times 0.9 \times 1 = 1.08$ m$^3$ $= 1\,080$ l.
Water: $1\,080 \times 60 : 100 = 648$ l.
**Answer:** 648 l.

## Problem 13

A tank has a base 40 cm long and 25 cm wide. When a stone is put in, completely under water, the water rises by 3 cm. Find the volume of the stone in cubic decimetres.

### Solution

Stone: $40 \times 25 \times 3 = 3\,000$ cm$^3$ $= 3$ dm$^3$.
**Answer:** 3 dm$^3$.

## Problem 14

A cube-shaped glass tank has edge 5 dm; the water in it is 3 dm deep. After a stone is put in (completely under water), the water is 3.4 dm deep. Find the volume of the stone.

### Solution

Rise: $3.4 - 3 = 0.4$ dm.
Stone: $5 \times 5 \times 0.4 = 10$ dm$^3$.
**Answer:** 10 dm$^3$.

## Problem 15

A tank has a base 50 cm long and 30 cm wide; the water is 20 cm deep. A stone of volume 1.8 dm$^3$ is put in and is completely under water. How deep is the water now?

### Solution

1.8 dm$^3$ $= 1\,800$ cm$^3$. Base area: $50 \times 30 = 1\,500$ cm$^2$.
Rise: $1\,800 : 1\,500 = 1.2$ cm.
New depth: $20 + 1.2 = 21.2$ cm.
**Answer:** 21.2 cm.

## Problem 16

A metal block in the shape of a cube has edge $\frac{1}{5}$ m. Each cubic centimetre of the metal weighs 7.8 g. How many kilograms does the block weigh?

### Solution

$\frac{1}{5}$ m $= 20$ cm. Volume: $20 \times 20 \times 20 = 8\,000$ cm$^3$.

Mass: $7.8 \times 8\,000 = 62\,400$ g $= 62.4$ kg.

**Answer:** 62.4 kg.

## Problem 17

A cuboid is 10 cm long, 8 cm wide and 6 cm high. A cube has an edge equal to the average of the three dimensions of the cuboid. Which has the larger volume, and by how much?

### Solution

Cuboid: $10 \times 8 \times 6 = 480$ cm$^3$.

Edge of the cube: $(10 + 8 + 6) : 3 = 8$ cm. Cube: $8 \times 8 \times 8 = 512$ cm$^3$.

$512 - 480 = 32$ cm$^3$.

**Answer:** The cube, by 32 cm$^3$.

## Problem 18

The edge of cube B is 2 times the edge of cube A. How many times the volume of cube A is the volume of cube B?

### Solution

Let the edge of A be $a$; the edge of B is $a \times 2$.

$V_B = (a \times 2) \times (a \times 2) \times (a \times 2) = a \times a \times a \times 8 = V_A \times 8$.

**Answer:** 8 times.

## Problem 19

An empty tank is 1.6 m long, 1 m wide and 1.2 m high. A tap pours in 40 l of water per minute. How long does it take to fill the tank?

### Solution

Tank: $1.6 \times 1 \times 1.2 = 1.92$ m$^3 = 1\,920$ l.

Time: $1\,920 : 40 = 48$ minutes.

**Answer:** 48 minutes.

## Problem 20

A full water tank is 3 m long, 2 m wide and 1.5 m high. A family uses 1 800 l of water a day. How many days does the water last?

### Solution

Water: $3 \times 2 \times 1.5 = 9$ m$^3$ $= 9\,000$ l.
Days: $9\,000 : 1\,800 = 5$ days.
**Answer:** 5 days.

## Problem 21

Each brick is 20 cm long, 10 cm wide and 5 cm high. Bricks are stacked tightly into a block 2 m long, 1 m wide and 0.5 m high. How many bricks are in the block?

### Solution

One brick: $20 \times 10 \times 5 = 1\,000$ cm$^3$ $= 1$ dm$^3$.
Block: $2 \times 1 \times 0.5 = 1$ m$^3$ $= 1\,000$ dm$^3$.
Bricks: $1\,000 : 1 = 1\,000$ bricks.
**Answer:** 1 000 bricks.

## Problem 22

A carton is 20 cm long, 12 cm wide and 8 cm high. Small boxes of chalk, each 4 cm $\times$ 3 cm $\times$ 2 cm, are packed into it to fill it exactly. How many boxes of chalk fit?

### Solution

Along the length: $20 : 4 = 5$; along the width: $12 : 3 = 4$; up the height: $8 : 2 = 4$.
Boxes: $5 \times 4 \times 4 = 80$.
*Check:* $20 \times 12 \times 8 = 1\,920$ cm$^3$, $4 \times 3 \times 2 = 24$ cm$^3$, $1\,920 : 24 = 80$.
**Answer:** 80 boxes.

## Problem 23

A cuboid has a lateral area of 480 cm$^2$ and a height of 8 cm; its length is 6 cm more than its width. Find its volume.

### Solution

Perimeter of base: $480 : 8 = 60$ cm. Length + width: 30 cm.
Length: $(30 + 6) : 2 = 18$ cm. Width: 12 cm.
$V = 18 \times 12 \times 8 = 1\,728$ cm$^3$.
**Answer:** 1 728 cm$^3$.

## Problem 24

A tank has a base perimeter of 26 dm and a height of 4 dm. Its length is 3 dm more than its width. How many litres does it hold?

### Solution

Length + width: $26 : 2 = 13$ dm. Length: $(13 + 3) : 2 = 8$ dm. Width: 5 dm.

$V = 8 \times 5 \times 4 = 160$ dm$^3$ $= 160$ l.

**Answer:** 160 l.

## Problem 25

A cube-shaped glass tank has edge 8 dm; the water is 5 dm deep. After a piece of coral is put in (completely under water), the water is 5.5 dm deep. Find the volume of the coral.

### Solution

Rise: $5.5 - 5 = 0.5$ dm.

Coral: $8 \times 8 \times 0.5 = 32$ dm$^3$.

**Answer:** 32 dm$^3$.

## Problem 26

A tank is 2.5 m long, 1.6 m wide and 1.2 m high. The water in it is 0.8 m deep.

**a)** How many litres of water are in the tank?

**b)** How many more litres are needed to fill it?

### Solution

**a)** $2.5 \times 1.6 \times 0.8 = 3.2$ m$^3$ $= 3\,200$ l.

**b)** Empty part: $1.2 - 0.8 = 0.4$ m high; $2.5 \times 1.6 \times 0.4 = 1.6$ m$^3$ $= 1\,600$ l.

**Answer: a)** 3 200 l; **b)** 1 600 l.

## Problem 27

A tank is 2 m long, 1.5 m wide and 1.4 m high. The water now fills 30% of the tank. How many litres must be added so that the water fills 80% of the tank?

### Solution

Tank: $2 \times 1.5 \times 1.4 = 4.2$ m$^3$ $= 4\,200$ l.

Water to add: $80\% - 30\% = 50\%$ of the tank.

$4\,200 \times 50 : 100 = 2\,100$ l.

**Answer:** 2 100 l.

### Problem 28

A fish tank has a base 50 cm long and 40 cm wide. Four identical stones are put in, completely under water, and the water rises by 1.5 cm. Find the volume of one stone.

**Solution**

Four stones: $50 \times 40 \times 1.5 = 3\,000$ cm$^3$.
One stone: $3\,000 : 4 = 750$ cm$^3$.
**Answer:** 750 cm$^3$.

### Problem 29

A tank is full of water. Three identical iron cubes are lowered into it and 24 l of water overflow. Find the edge of one iron cube.

**Solution**

Three cubes: $24$ l $= 24$ dm$^3$. One cube: $24 : 3 = 8$ dm$^3$.
$2 \times 2 \times 2 = 8$, so the edge is 2 dm.
**Answer:** 2 dm.

### Problem 30

A milk carton is 6 cm long, 4 cm wide and 10 cm high and is filled exactly. How many such cartons can be filled from 4.8 l of milk?

**Solution**

One carton: $6 \times 4 \times 10 = 240$ cm$^3$ $= 240$ ml.
$4.8$ l $= 4\,800$ ml. Cartons: $4\,800 : 240 = 20$.
**Answer:** 20 cartons.

## Challenge – Extension

### Problem 31 ★

A fish tank is 60 cm long, 40 cm wide and 45 cm high. At first the water reaches $\frac{3}{5}$ of the height. After a stone is put in (completely under water), the water reaches $\frac{2}{3}$ of the height. Find the volume of the stone.

**Solution**

Depth before: $45 \times \frac{3}{5} = 27$ cm. Depth after: $45 \times \frac{2}{3} = 30$ cm.

Rise: $30 - 27 = 3$ cm.
Stone: $60 \times 40 \times 3 = 7\,200$ cm$^3$.
**Answer:** $7\,200$ cm$^3$.

### Problem 32 *

A glass fish tank without a lid is 80 cm long, 50 cm wide and 50 cm high; the water is 30 cm deep.

**a)** Find the area of glass used to make the tank.

**b)** A stone of volume 16 dm$^3$ is put in, completely under water. How deep is the water now?

### Solution

**a)** Walls: $(80 + 50) \times 2 \times 50 = 13\,000$ cm$^2$; bottom: $80 \times 50 = 4\,000$ cm$^2$.
Glass: $13\,000 + 4\,000 = 17\,000$ cm$^2$ $= 170$ dm$^2$.
**b)** $16$ dm$^3$ $= 16\,000$ cm$^3$. Rise: $16\,000 : 4\,000 = 4$ cm. Depth: $30 + 4 = 34$ cm.
**Answer: a)** $170$ dm$^2$; **b)** $34$ cm.

### Problem 33 **

A tank is 60 cm long, 40 cm wide and 50 cm high; the water is 45 cm deep. A stone of volume 15 dm$^3$ is lowered into it. How many litres of water overflow?

### Solution

Space left above the water: $60 \times 40 \times (50 - 45) = 12\,000$ cm$^3$ $= 12$ dm$^3$.
The stone ($15$ dm$^3$) is larger than the space left, so water overflows.
Overflow: $15 - 12 = 3$ dm$^3$ $= 3$ l.
**Answer:** 3 l.

### Problem 34 *

Tank A is 1.5 m long, 1 m wide and 0.8 m high and is full of water. All of its water is poured into empty tank B, whose base is 2 m long and 1.2 m wide. How deep is the water in tank B?

### Solution

Water: $1.5 \times 1 \times 0.8 = 1.2$ m$^3$.
Base of B: $2 \times 1.2 = 2.4$ m$^2$.
Depth: $1.2 : 2.4 = 0.5$ m.
**Answer:** 0.5 m.

**Problem 35** *

The total surface area of a cube is 72 $cm^2$ more than its lateral area. Find its volume.

**Solution**

The difference is the two bases: one face is $72 : 2 = 36$ $cm^2 = 6 \times 6$.
Edge 6 cm; $V = 6 \times 6 \times 6 = 216$ $cm^3$.
**Answer:** 216 $cm^3$.

**Problem 36** *

A cuboid has a volume of 120 $cm^3$. Its length is doubled, its width is tripled and its height is halved. Find the new volume.

**Solution**

The volume is multiplied by $2 \times 3 = 6$ and then divided by 2, i.e. multiplied by 3.
New volume: $120 \times 3 = 360$ $cm^3$.
**Answer:** 360 $cm^3$.

**Problem 37** **

The edge of a 10 cm cube is increased by 20%. By how many cubic centimetres, and by what percentage, does its volume increase?

**Solution**

New edge: $10 + 10 \times 20 : 100 = 12$ cm.
Volumes: $10 \times 10 \times 10 = 1\,000$ $cm^3$ and $12 \times 12 \times 12 = 1\,728$ $cm^3$.
Increase: $1\,728 - 1\,000 = 728$ $cm^3$, i.e. $728 : 1\,000 = 0.728 = 72.8\%$.
**Answer:** 728 $cm^3$; 72.8%.

**Problem 38** **

Three different faces of a cuboid have areas 24 $cm^2$, 30 $cm^2$ and 20 $cm^2$, and its edges are whole numbers of centimetres. Find its volume.

**Solution**

The edge shared by the faces 24 and 30 divides both: it can be 1, 2, 3 or 6.
Try 6: the other edges are $24 : 6 = 4$ and $30 : 6 = 5$, and $4 \times 5 = 20$. ✓
(With 1, 2 or 3 the third face would be 720, 180 or 80, not 20.)
$V = 6 \times 4 \times 5 = 120$ $cm^3$.
**Answer:** 120 $cm^3$.

## Problem 39 ⋆⋆

A cuboid's length is 3 times its height and its width is 2 times its height. Its volume is 384 $cm^3$. Find its dimensions.

### Solution

Cut the cuboid into cubes with edge equal to the height: there are $3 \times 2 \times 1 = 6$ such cubes.
One cube: $384 : 6 = 64$ $cm^3 = 4 \times 4 \times 4$, so the height is 4 cm.
Length: 12 cm; width: 8 cm.
**Answer:** 12 cm, 8 cm, 4 cm.

## Problem 40 ⋆⋆

An empty tank is 1.5 m long, 1.2 m wide and 1 m high. Tap A pours in 30 l per minute while tap B lets out 12 l per minute. Both are opened at the same time. How long does it take to fill the tank?

### Solution

Tank: $1.5 \times 1.2 \times 1 = 1.8$ $m^3 = 1\,800$ l.
Each minute the water increases by $30 - 12 = 18$ l.
Time: $1\,800 : 18 = 100$ minutes $= 1$ hour 40 minutes.
**Answer:** 1 hour 40 minutes.

## Problem 41 ⋆

A swimming pool is 20 m long and 10 m wide. A pump puts in 60 $m^3$ of water per hour. How long does it take to fill the empty pool to a depth of 1.5 m?

### Solution

Water needed: $20 \times 10 \times 1.5 = 300$ $m^3$.
Time: $300 : 60 = 5$ hours.
**Answer:** 5 hours.

## Problem 42 ⋆⋆

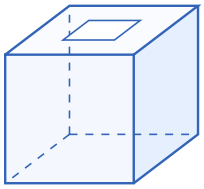
In one storm, 15 mm of rain fell on a flat roof 8 m long and 6 m wide. All of the rain ran into a tank. How many litres of water went into the tank?

### Solution

15 mm = 0.015 m.
Rain on the roof: $8 \times 6 \times 0.015 = 0.72$ $m^3 = 720$ l.
**Answer:** 720 l.

## Problem 43 *

A wooden cube has edge 10 cm. A square hole 4 cm by 4 cm is cut straight through it, from the top face to the bottom face. Find the volume of the wood that is left.




### Solution

Cube: $10 \times 10 \times 10 = 1\,000$ cm$^3$.
The hole is a cuboid $4 \times 4 \times 10 = 160$ cm$^3$.
Wood left: $1\,000 - 160 = 840$ cm$^3$.
**Answer:** 840 cm$^3$.

## Problem 44 * *

When water freezes, its volume increases by $\frac{1}{11}$. An ice tray holds water 22 cm long, 10 cm wide and 3 cm deep. What is the volume of the ice after the water freezes?

### Solution

Water: $22 \times 10 \times 3 = 660$ cm$^3$.
Increase: $660 \times \frac{1}{11} = 60$ cm$^3$.
Ice: $660 + 60 = 720$ cm$^3$.
**Answer:** 720 cm$^3$.

## Problem 45 *

A tank has a base 2 m long and 1.2 m wide. In the morning the water is 1 m deep; by evening it is 0.65 m deep. How many litres of water were used during the day?

### Solution

The level fell by $1 - 0.65 = 0.35$ m.
Water used: $2 \times 1.2 \times 0.35 = 0.84$ m$^3$ $= 840$ l.
**Answer:** 840 l.

## Problem 46 * *

How many cubes of edge 3 cm can be packed into a box 16 cm long, 12 cm wide and 9 cm high (the cubes sit upright, side by side)? What volume of the box is left empty?

**Solution**

Along the length: $16 : 3 = 5$ (remainder 1), so 5 cubes; along the width: $12 : 3 = 4$; up the height: $9 : 3 = 3$.
Cubes: $5 \times 4 \times 3 = 60$.
Box: $16 \times 12 \times 9 = 1\,728$ cm$^3$; cubes: $60 \times 27 = 1\,620$ cm$^3$.
Empty: $1\,728 - 1\,620 = 108$ cm$^3$.
**Answer:** 60 cubes; 108 cm$^3$.

## Problem 47 *

A cuboid has a volume of 1.2 m$^3$ and a base area of 0.8 m$^2$. Its length is 1 m. Find its height and its lateral area.

**Solution**

Height: $1.2 : 0.8 = 1.5$ m. Width: $0.8 : 1 = 0.8$ m.
$S_{\text{lat}} = (1 + 0.8) \times 2 \times 1.5 = 5.4$ m$^2$.
**Answer:** 1.5 m; 5.4 m$^2$.

## Problem 48 *

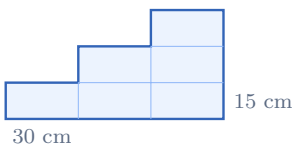
All the water from a full cube-shaped tank of edge 6 dm is poured into an empty tank whose base is 9 dm long and 8 dm wide. How deep is the water?

**Solution**

Water: $6 \times 6 \times 6 = 216$ dm$^3$.
Base: $9 \times 8 = 72$ dm$^2$.
Depth: $216 : 72 = 3$ dm.
**Answer:** 3 dm.

## Problem 49 * *

A flight of 3 concrete steps is 1.5 m wide. Each step is 30 cm deep and 15 cm high (see the side view). Find the volume of concrete.




**Solution**

Cut the steps into three layers, each 15 cm = 0.15 m high:
bottom layer 0.9 m deep, middle layer 0.6 m deep, top layer 0.3 m deep.
Volume: $1.5 \times 0.15 \times (0.9 + 0.6 + 0.3) = 1.5 \times 0.15 \times 1.8 = 0.405$ m$^3$.

**Answer:** 0.405 $m^3$.

## Problem 50

A tank is 1.5 m long. The water in it is 60 cm deep. After 240 l of water are added, it is 80 cm deep. Find the width of the tank.

## Solution

240 l = 240 $dm^3$; the rise is 80 − 60 = 20 cm = 2 dm.

Base area: 240 : 2 = 120 $dm^2$ = 1.2 $m^2$.

Width: 1.2 : 1.5 = 0.8 m.

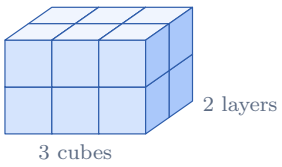
**Answer:** 0.8 m.

# 5 Cube Stacks and Painted Faces

## ① Building a box from small blocks

When equal small blocks (unit cubes, small cubes, bricks) are stacked tightly into a new solid, the **volume of the new solid** is the sum of the volumes of the blocks:

$$V_{\text{big}} = V_{\text{one block}} \times \text{number of blocks.}$$




$3 \times 2 \times 2 = 12$ cubes

To find the lateral area, total surface area or volume of a box built from equal blocks:

**Step 1.** Find the dimensions of the big box:
length = length of a block × number of blocks along the length;
width = width of a block × number of blocks along the width;
height = height of a block × number of layers.

**Step 2.** Use the formulas for the cuboid or cube.

## ② How many cubes fit in a box?

To pack cubes of edge $a$ into a box (all lengths in the same unit):

– along each edge of the box, count how many times $a$ fits (the quotient; any remainder is wasted space);
– number of cubes = (number along the length) × (number along the width) × (number along the height).

If a big cube is built from small cubes, the edge of the big cube is a whole number of small edges, so

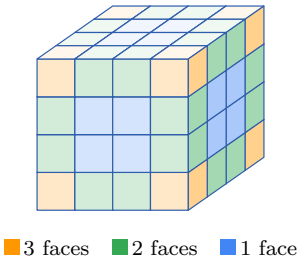
$$\text{number of small cubes} = n \times n \times n, \qquad n = \text{big edge} : \text{small edge}.$$

If we know the total surface area of the big cube, one face is $S_{\text{total}} : 6$, and the edge is the number that multiplied by itself gives this area.

## ③ Painting the outside of a solid made of small cubes

A big cube with $n$ small cubes along each edge ($n \geq 3$) is painted on all 6 faces, then taken apart:

– **painted on 3 faces**: the cubes at the 8 corners: 8 cubes;
– **painted on exactly 2 faces**: on the 12 edges, not at corners: $12 \times (n-2)$;
– **painted on exactly 1 face**: on the 6 faces, not on edges: $6 \times (n-2) \times (n-2)$;
– **not painted**: the hidden core, a cube of edge $n-2$: $(n-2) \times (n-2) \times (n-2)$.




For a box with $a$, $b$, $c$ small cubes along its edges (all $\geq 3$):
3 faces: 8; exactly 2 faces: $4 \times [(a-2)+(b-2)+(c-2)]$;
exactly 1 face: $2 \times [(a-2)(b-2)+(b-2)(c-2)+(a-2)(c-2)]$;
none: $(a-2)(b-2)(c-2)$.

*Check:* the four groups add up to all the small cubes. If an edge has fewer than 3 small cubes, count directly.

## ④ Counting cubes in a stack. Views

To count the cubes in a stack, count **column by column** (or layer by layer). Unless the problem says otherwise, every cube rests on the table or on another cube, so hidden cubes are under the visible ones.

A **top view with numbers** shows the base squares; the number in each square is how many cubes are stacked there. The number of cubes is the sum of these numbers.

In the **front view**, each column is as tall as the tallest stack in that column.

top view

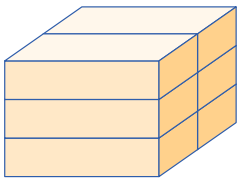
| | | |
|---|---|---|
| **2** | **3** | **1** |
| **1** | **2** | **1** |

front

$2+3+1+1+2+1 = 10$ cubes

### Example 1

A brick is a cuboid 24 cm long, 12 cm wide and 6 cm high. A builder stacks 6 such bricks into a block shaped like a cuboid: 2 bricks side by side in each layer, 3 layers (see the figure).




**a)** Find the lateral area and the total surface area of the block.

**b)** Find the volume of the block.

**Guided solution.**

**a)** Length of the block: 24 cm.

Width of the block: $12 \times 2 = 24$ cm.
Height of the block: $6 \times 3 = 18$ cm.
Lateral area: $(24 + 24) \times 2 \times 18 = 1\,728$ cm$^2$.
Area of the base: $24 \times 24 = 576$ cm$^2$.
Total surface area: $1\,728 + 576 \times 2 = 2\,880$ cm$^2$.
**b)** *Method 1.* $V = 24 \times 24 \times 18 = 10\,368$ cm$^3$.
*Method 2.* One brick: $24 \times 12 \times 6 = 1\,728$ cm$^3$; 6 bricks: $1\,728 \times 6 = 10\,368$ cm$^3$.
**Answer: a)** 1 728 cm$^2$; 2 880 cm$^2$;
**b)** 10 368 cm$^3$.

## Example 2

Small cubes of edge 2 cm are stacked into a big cube whose total surface area is 864 cm$^2$. How many small cubes are used?

### Guided solution.

*Method 1.* One face of the big cube: $864 : 6 = 144$ cm$^2$.
$144 = 12 \times 12$, so the edge of the big cube is 12 cm.
Along each edge there are $12 : 2 = 6$ small cubes.
Number of small cubes: $6 \times 6 \times 6 = 216$.
*Method 2.* The edge of the big cube is 12 cm (as above).
Volume of the big cube: $12 \times 12 \times 12 = 1\,728$ cm$^3$.
Volume of a small cube: $2 \times 2 \times 2 = 8$ cm$^3$.
Number of small cubes: $1\,728 : 8 = 216$.
**Answer:** 216 small cubes.

## Example 3

A crate (a cuboid) is 1.2 m long, 0.8 m wide and 0.5 m high.

**a)** At most how many cube-shaped toy blocks of edge 1 dm fit in the crate?

**b)** At most how many cube-shaped gift boxes of edge 3 dm fit in the crate?

### Guided solution.

Change units: 1.2 m = 12 dm; 0.8 m = 8 dm; 0.5 m = 5 dm.
**a)** Along the length: $12 : 1 = 12$ blocks; along the width: 8 blocks; along the height: 5 blocks.
Number of blocks: $12 \times 8 \times 5 = 480$.
**b)** Along the length: $12 : 3 = 4$ boxes.
Along the width: $8 : 3 = 2$ (remainder 2 dm), so 2 boxes.
Along the height: $5 : 3 = 1$ (remainder 2 dm), so 1 box.
Number of gift boxes: $4 \times 2 \times 1 = 8$.
**Answer: a)** 480 blocks;
**b)** 8 boxes.

## Example 4

Mai glues 1 cm cubes into a big cube of edge 5 cm and paints all 6 outside faces.

**a)** How many small cubes did she use?

**b)** How many small cubes are painted on 3 faces? On exactly 2 faces? On exactly 1 face? On no face?

### Guided solution.

**a)** Each edge has $5 : 1 = 5$ small cubes.

Number of small cubes: $5 \times 5 \times 5 = 125$.

**b)** Painted on 3 faces: the 8 corner cubes.

Each edge has $5 - 2 = 3$ cubes that are not corners.

Exactly 2 faces: $12 \times 3 = 36$.

Each face has $3 \times 3 = 9$ cubes not on an edge.

Exactly 1 face: $6 \times 9 = 54$.

No face: the core cube of edge 3: $3 \times 3 \times 3 = 27$.

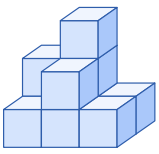
*Check:* $8 + 36 + 54 + 27 = 125$.

**Answer: a)** 125;

**b)** 8; 36; 54; 27.

## Example 5

Nam builds the stack of cubes in the figure.



front

**a)** How many cubes are in the stack?

**b)** Draw the front view.

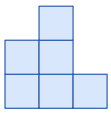
**c)** How many more cubes does Nam need to turn the stack into a cuboid 3 cubes long, 2 cubes wide and 3 cubes high?

### Guided solution.

**a)** Write the number of cubes on each base square (top view):

| | | |
|---|---|---|
| **2** | **3** | **1** |
| **1** | **2** | **1** |

front



front view

Back row: $2 + 3 + 1 = 6$ cubes; front row: $1 + 2 + 1 = 4$ cubes.

Number of cubes: $6 + 4 = 10$.

**b)** The columns of the front view are as tall as the tallest stacks: 2, 3, 1 (figure above).

**c)** The cuboid has $3 \times 2 \times 3 = 18$ cubes.

Cubes still needed: $18 - 10 = 8$.

**Answer: a)** 10 cubes;

**c)** 8 cubes.

## Practice

**Problem 1**

How many 1 cm cubes are needed to build a cube of edge 6 cm?

**Solution**

Each edge has $6 : 1 = 6$ small cubes.
Number of small cubes: $6 \times 6 \times 6 = 216$.
**Answer:** 216 cubes.

**Problem 2**

How many cubes of edge 3 cm are needed to build a cube of edge 12 cm?

**Solution**

Each edge has $12 : 3 = 4$ small cubes.
Number of small cubes: $4 \times 4 \times 4 = 64$.
**Answer:** 64 cubes.

**Problem 3**

How many cubes of edge 2 cm are needed to fill a box 10 cm long, 8 cm wide and 6 cm high?

**Solution**

Along the length: $10 : 2 = 5$; along the width: $8 : 2 = 4$; along the height: $6 : 2 = 3$.
Number of cubes: $5 \times 4 \times 3 = 60$.
**Answer:** 60 cubes.

**Problem 4**

A toy chest is 1.2 m long, 0.6 m wide and 0.4 m high. How many cube-shaped blocks of edge 2 dm fill it exactly?

**Solution**

1.2 m = 12 dm; 0.6 m = 6 dm; 0.4 m = 4 dm.
Along the length: $12 : 2 = 6$; along the width: $6 : 2 = 3$; along the height: $4 : 2 = 2$.
Number of blocks: $6 \times 3 \times 2 = 36$.
**Answer:** 36 blocks.

## Problem 5

64 small cubes, each with volume 1 $cm^3$, are stacked into a big cube. Find the lateral area, the total surface area and the volume of the big cube.

### Solution

A small cube of volume 1 $cm^3$ has edge 1 cm.
$64 = 4 \times 4 \times 4$, so each edge of the big cube has 4 small cubes: the edge is 4 cm.
Lateral area: $4 \times 4 \times 4 = 64$ $cm^2$.
Total surface area: $4 \times 4 \times 6 = 96$ $cm^2$.
Volume: $64 \times 1 = 64$ $cm^3$.
**Answer:** 64 $cm^2$; 96 $cm^2$; 64 $cm^3$.

## Problem 6

125 small cubes, each with volume 27 $cm^3$, are stacked into a big cube. Find the total surface area and the volume of the big cube.

### Solution

$27 = 3 \times 3 \times 3$, so a small cube has edge 3 cm.
$125 = 5 \times 5 \times 5$, so the big cube has 5 small cubes along each edge.
Edge of the big cube: $3 \times 5 = 15$ cm.
Total surface area: $15 \times 15 \times 6 = 1\,350$ $cm^2$.
Volume: $27 \times 125 = 3\,375$ $cm^3$.
**Answer:** 1 350 $cm^2$; 3 375 $cm^3$.

## Problem 7

A big cube with total surface area 216 $cm^2$ is built from 1 cm cubes. How many small cubes are used?

### Solution

One face: $216 : 6 = 36$ $cm^2$.
$36 = 6 \times 6$, so the edge is 6 cm.
Number of small cubes: $6 \times 6 \times 6 = 216$.
**Answer:** 216 cubes.

## Problem 8

A big cube with lateral area 400 $cm^2$ is built from cubes of edge 2 cm. How many small cubes are used?

**Solution**

One face: $400 : 4 = 100$ cm$^2$.
$100 = 10 \times 10$, so the edge is 10 cm.
Along each edge: $10 : 2 = 5$ small cubes.
Number of small cubes: $5 \times 5 \times 5 = 125$.
**Answer:** 125 cubes.

**Problem 9**

Five bricks, each 20 cm long, 10 cm wide and 5 cm high, are laid flat one on top of another. Find the lateral area, the total surface area and the volume of the pile.

**Solution**

The pile is a cuboid 20 cm long, 10 cm wide and $5 \times 5 = 25$ cm high.
Lateral area: $(20 + 10) \times 2 \times 25 = 1\,500$ cm$^2$.
Total surface area: $1\,500 + 20 \times 10 \times 2 = 1\,900$ cm$^2$.
Volume: $20 \times 10 \times 25 = 5\,000$ cm$^3$.
**Answer:** 1 500 cm$^2$; 1 900 cm$^2$; 5 000 cm$^3$.

**Problem 10**

Two equal cubes are stacked to make a cuboid 16 cm high. Find the lateral area and the total surface area of the cuboid.

**Solution**

Edge of each cube: $16 : 2 = 8$ cm.
The cuboid is 8 cm long, 8 cm wide and 16 cm high.
Lateral area: $(8 + 8) \times 2 \times 16 = 512$ cm$^2$.
Total surface area: $512 + 8 \times 8 \times 2 = 640$ cm$^2$.
**Answer:** 512 cm$^2$; 640 cm$^2$.

**Problem 11**

Three equal cubes are put in a row to make a cuboid 21 cm long. Find the total surface area of the cuboid.

**Solution**

Edge of each cube: $21 : 3 = 7$ cm. One face: $7 \times 7 = 49$ cm$^2$.
The four long sides of the cuboid consist of $3 \times 4 = 12$ cube faces; the two ends are 2 more faces.
Total surface area: $49 \times 14 = 686$ cm$^2$.
*Check:* $2 \times (21 \times 7 + 21 \times 7 + 7 \times 7) = 686$ cm$^2$.

**Answer:** 686 $cm^2$.

### Problem 12

A carton is 60 cm long, 40 cm wide and 30 cm high. How many cube-shaped boxes of edge 10 cm fill it exactly?

### Solution

Along the length: $60 : 10 = 6$; along the width: $40 : 10 = 4$; along the height: $30 : 10 = 3$.
Number of boxes: $6 \times 4 \times 3 = 72$.
**Answer:** 72 boxes.

### Problem 13

A drawer is 50 cm long, 35 cm wide and 22 cm high. At most how many cube-shaped boxes of edge 10 cm fit in it?

### Solution

Along the length: $50 : 10 = 5$.
Along the width: $35 : 10 = 3$ (remainder 5 cm), so 3 boxes.
Along the height: $22 : 10 = 2$ (remainder 2 cm), so 2 boxes.
Number of boxes: $5 \times 3 \times 2 = 30$.
**Answer:** 30 boxes.

### Problem 14

A crate is 9 dm long, 6 dm wide and 4 dm high. Cube-shaped boxes of edge 3 dm are packed in it.

**a)** At most how many boxes fit?

**b)** With that many boxes inside, how much space in the crate is left empty?

### Solution

**a)** Along the length: $9 : 3 = 3$; along the width: $6 : 3 = 2$; along the height: $4 : 3 = 1$ (remainder 1 dm).
Number of boxes: $3 \times 2 \times 1 = 6$.

**b)** Volume of the crate: $9 \times 6 \times 4 = 216$ $dm^3$.
Volume of 6 boxes: $3 \times 3 \times 3 \times 6 = 162$ $dm^3$.
Empty space: $216 - 162 = 54$ $dm^3$.

**Answer: a)** 6 boxes;
**b)** 54 $dm^3$.

**Problem 15**

A cube of edge 3 cm made of 1 cm cubes is painted on all its faces. How many small cubes are painted on 3 faces, on exactly 2 faces, on exactly 1 face, on no face?

**Solution**

Each edge has 3 small cubes; $3 - 2 = 1$ of them is not a corner.

3 faces: 8.

Exactly 2 faces: $12 \times 1 = 12$.

Exactly 1 face: $6 \times 1 \times 1 = 6$.

No face: $1 \times 1 \times 1 = 1$ (the cube in the very middle).

*Check:* $8 + 12 + 6 + 1 = 27 = 3 \times 3 \times 3$.

**Answer:** 8; 12; 6; 1.

**Problem 16**

A big cube is made of 216 equal small cubes and then painted on all its faces. How many small cubes are painted on exactly 2 faces? On exactly 1 face?

**Solution**

$216 = 6 \times 6 \times 6$, so each edge has 6 small cubes.

Exactly 2 faces: $12 \times (6 - 2) = 48$.

Exactly 1 face: $6 \times 4 \times 4 = 96$.

**Answer:** 48; 96.

**Problem 17**

A big cube is made of 343 equal small cubes and then painted on all its faces. How many small cubes have no paint at all?

**Solution**

$343 = 7 \times 7 \times 7$, so each edge has 7 small cubes.

The unpainted cubes form a core cube with $7 - 2 = 5$ cubes along each edge.

Number of unpainted cubes: $5 \times 5 \times 5 = 125$.

**Answer:** 125 cubes.

**Problem 18**

A cube of edge 10 cm is built from 1 cm cubes and painted on all its faces. How many small cubes are painted on at least 2 faces?

**Solution**

Each edge has 10 small cubes.
Painted on 3 faces: 8.
Painted on exactly 2 faces: $12 \times (10 - 2) = 96$.
At least 2 faces: $8 + 96 = 104$.
**Answer:** 104 cubes.

**Problem 19**

A cuboid 5 cm long, 4 cm wide and 3 cm high is built from 1 cm cubes and painted on all its faces. How many small cubes are painted on 3 faces, on exactly 2 faces, on exactly 1 face, on no face?

**Solution**

Along the edges there are 5, 4, 3 cubes; without the corners: 3, 2, 1.
3 faces: 8.
Exactly 2 faces: $4 \times (3 + 2 + 1) = 24$.
Exactly 1 face: $2 \times (3 \times 2 + 2 \times 1 + 3 \times 1) = 22$.
No face: $3 \times 2 \times 1 = 6$.
*Check:* $8 + 24 + 22 + 6 = 60 = 5 \times 4 \times 3$.
**Answer:** 8; 24; 22; 6.

**Problem 20**

A cuboid 14 cm long, 10 cm wide and 6 cm high is built from cubes of edge 2 cm and painted on all its faces. How many small cubes are painted on exactly 2 faces, on exactly 1 face, on no face?

**Solution**

Along the edges: $14 : 2 = 7$, $10 : 2 = 5$, $6 : 2 = 3$ cubes; without the corners: 5, 3, 1.
Exactly 2 faces: $4 \times (5 + 3 + 1) = 36$.
Exactly 1 face: $2 \times (5 \times 3 + 3 \times 1 + 5 \times 1) = 46$.
No face: $5 \times 3 \times 1 = 15$.
**Answer:** 36; 46; 15.

**Problem 21**

A big cube made of equal small cubes is painted on all its faces. Exactly 64 small cubes have no paint. How many small cubes are there in all, and how many are painted on exactly 1 face?

**Solution**

The unpainted core is a cube: $64 = 4 \times 4 \times 4$, so it has 4 cubes along each edge.
The big cube has $4 + 2 = 6$ cubes along each edge.
In all: $6 \times 6 \times 6 = 216$ small cubes.

Exactly 1 face: $6 \times 4 \times 4 = 96$.
**Answer:** 216 cubes; 96 cubes.

**Problem 22**

A big cube made of equal small cubes is painted on all its faces. Exactly 150 small cubes are painted on just 1 face. How many small cubes are there in all?

**Solution**

On each face: $150 : 6 = 25$ such cubes; $25 = 5 \times 5$.
So $n - 2 = 5$, and each edge has 7 small cubes.
In all: $7 \times 7 \times 7 = 343$.
**Answer:** 343 cubes.

**Problem 23**

A big cube is built from 1 cm cubes and painted on all its faces. Exactly 84 small cubes are painted on 2 faces. Find the volume of the big cube.

**Solution**

On each edge: $84 : 12 = 7$ such cubes.
Each edge has $7 + 2 = 9$ small cubes, so the edge is 9 cm.
Volume: $9 \times 9 \times 9 = 729$ cm$^3$.
**Answer:** 729 cm$^3$.

**Problem 24**

A cube of edge 5 cm is built from 1 cm cubes. The top layer is then lifted off. How many small cubes are left, and what is the total surface area of the remaining solid?

**Solution**

One layer has $5 \times 5 = 25$ cubes, so $125 - 25 = 100$ cubes are left.
The remaining solid is a cuboid 5 cm long, 5 cm wide and 4 cm high.
Lateral area: $(5 + 5) \times 2 \times 4 = 80$ cm$^2$.
Total surface area: $80 + 5 \times 5 \times 2 = 130$ cm$^2$.
**Answer:** 100 cubes; 130 cm$^2$.

**Problem 25**

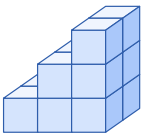
A cube of edge 4 cm is built from 1 cm cubes. One corner cube is removed. Find the total surface area of the remaining solid.

**Solution**

Total surface area at first: $4 \times 4 \times 6 = 96$ cm$^2$.
Removing the corner cube takes away 3 outside squares of 1 cm$^2$, but uncovers 3 new squares of 1 cm$^2$ inside the gap.
The area does not change: $96 - 3 + 3 = 96$ cm$^2$.
**Answer:** 96 cm$^2$.

## Problem 26

How many cubes are there in this staircase? (No cube is missing from the back.)



**Solution**

The staircase is 2 cubes deep.
Seen from the front, the columns have 1, 2 and 3 cubes: $1 + 2 + 3 = 6$ cubes in one row.
Number of cubes: $6 \times 2 = 12$.
**Answer:** 12 cubes.

## Problem 27

The figure is the top view of a stack of cubes; each number tells how many cubes stand on that square.

**a)** How many cubes are in the stack?

**b)** How tall is each column of the front view?

| | | |
|---|---|---|
| **3** | **1** | **2** |
| **2** | **2** | **1** |

front

**Solution**

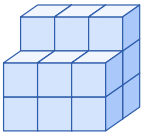
**a)** $3 + 1 + 2 + 2 + 2 + 1 = 11$ cubes.

**b)** From left to right, the tallest stack in each column has 3, 2 and 2 cubes.

**Answer: a)** 11 cubes;
**b)** 3, 2, 2.

## Problem 28

How many cubes are in the stack? How many more cubes are needed to have enough for a cube with 3 small cubes along each edge?



**Solution**

Front row: $3 \times 2 = 6$ cubes; back row: $3 \times 3 = 9$ cubes.
Number of cubes: $6 + 9 = 15$.
A cube with 3 small cubes along each edge has $3 \times 3 \times 3 = 27$ cubes.
More cubes needed: $27 - 15 = 12$.
**Answer:** 15 cubes; 12 more cubes.

**Problem 29**

Khoa stacks 64 dice, each a cube of edge 2 cm, into one big cube. Find the total surface area and the volume of the big cube.

**Solution**

$64 = 4 \times 4 \times 4$, so each edge has 4 dice.
Edge of the big cube: $2 \times 4 = 8$ cm.
Total surface area: $8 \times 8 \times 6 = 384$ cm$^2$.
Volume: $8 \times 8 \times 8 = 512$ cm$^3$.
**Answer:** 384 cm$^2$; 512 cm$^3$.

**Problem 30**

A wooden block 12 dm long, 8 dm wide and 6 dm high is painted on all its faces and then sawn into cubes of edge 2 dm. How many cubes are there? How many have no paint? How many are painted on exactly 1 face?

**Solution**

Along the edges: $12 : 2 = 6$, $8 : 2 = 4$, $6 : 2 = 3$ cubes.
Number of cubes: $6 \times 4 \times 3 = 72$.
Without the corners: 4, 2, 1 cubes along the edges.
No paint: $4 \times 2 \times 1 = 8$.
Exactly 1 face: $2 \times (4 \times 2 + 2 \times 1 + 4 \times 1) = 28$.
**Answer:** 72 cubes; 8 cubes; 28 cubes.

## Challenge – Extension

**Problem 31** ★

A big cube made of small cubes is painted on all its faces. The number of small cubes painted on exactly 1 face equals the number painted on exactly 2 faces. How many small cubes make up the big cube?

**Solution**

Let each edge have $n$ small cubes and put $k = n - 2$.
Exactly 1 face: $6 \times k \times k$; exactly 2 faces: $12 \times k$.
$6 \times k \times k = 12 \times k$ gives $k = 2$, so $n = 4$.
Number of small cubes: $4 \times 4 \times 4 = 64$.
**Answer:** 64 cubes.

**Problem 32** ★★

A big cube made of small cubes is painted on all its faces. The number of unpainted small cubes equals the number of small cubes painted on exactly 1 face. How many small cubes are there?

**Solution**

Let $k = n - 2$. Unpainted: $k \times k \times k$; exactly 1 face: $6 \times k \times k$.
$k \times k \times k = 6 \times k \times k$ gives $k = 6$, so $n = 8$.
Number of small cubes: $8 \times 8 \times 8 = 512$.
*Check:* unpainted $6 \times 6 \times 6 = 216$; one face $6 \times 36 = 216$.
**Answer:** 512 cubes.

**Problem 33** ★

A cube of edge 4 cm made of 1 cm cubes is painted on all its faces and taken apart. What fraction of all the faces of the small cubes is painted?

**Solution**

Painted squares: the surface of the big cube, $4 \times 4 \times 6 = 96$ squares.
All faces of the small cubes: $64 \times 6 = 384$ squares.
Fraction painted: $\frac{96}{384} = \frac{1}{4}$.
**Answer:** $\frac{1}{4}$.

**Problem 34** ★

A cube of edge 5 cm is built from 1 cm cubes. Only its 4 side faces are painted (not the top, not the bottom). How many small cubes are painted on exactly 2 faces, on exactly 1 face, on no face?

**Solution**

Each of the 5 layers is a $5 \times 5$ square whose border is painted.
In one layer: 4 corner cubes with 2 faces; $4 \times 3 = 12$ cubes with 1 face; $3 \times 3 = 9$ unpainted.
Exactly 2 faces: $4 \times 5 = 20$.
Exactly 1 face: $12 \times 5 = 60$.

No face: $9 \times 5 = 45$.
*Check:* $20 + 60 + 45 = 125$.
**Answer:** 20; 60; 45.

**Problem 35** ★★

A cube of edge 6 cm is built from 1 cm cubes. The 4 side faces are painted blue; the top and the bottom are painted red. How many small cubes carry both colours?

**Solution**

A cube with red paint lies in the top layer or the bottom layer.
It also has blue paint if it lies on the border of that layer.
Border of a $6 \times 6$ layer: $6 \times 6 - 4 \times 4 = 20$ cubes.
Both colours: $20 \times 2 = 40$ cubes.
**Answer:** 40 cubes.

**Problem 36** ★

343 cubes of edge 2 cm are stacked into a big cube. The top layer is then lifted off. Find the total surface area of the remaining solid.

**Solution**

$343 = 7 \times 7 \times 7$, so the big cube is $2 \times 7 = 14$ cm along each edge.
Without the top layer it is a cuboid 14 cm by 14 cm by $14 - 2 = 12$ cm.
Lateral area: $(14 + 14) \times 2 \times 12 = 672$ cm$^2$.
Total surface area: $672 + 14 \times 14 \times 2 = 1\,064$ cm$^2$.
**Answer:** $1\,064$ cm$^2$.

**Problem 37** ★★

A cube of edge 5 cm is built from 1 cm cubes. Then a straight tunnel 1 cm by 1 cm is cut right through the middle, from the centre of the front face to the centre of the back face, by taking out a row of small cubes. How many small cubes are left? What is the total surface area of the solid now (inside of the tunnel included)?

**Solution**

The tunnel takes out 5 small cubes: $125 - 5 = 120$ cubes are left.
Outside surface at first: $5 \times 5 \times 6 = 150$ cm$^2$; the two openings remove 2 cm$^2$.
The tunnel has 4 walls, each 5 cm by 1 cm: $4 \times 5 = 20$ cm$^2$.
Total surface area: $150 - 2 + 20 = 168$ cm$^2$.

**Answer:** 120 cubes; 168 $cm^2$.

**Problem 38**

A box is 20 cm long, 12 cm wide and 8 cm high. It is filled exactly with equal cubes whose edge is a whole number of centimetres. What is the smallest possible number of cubes?

**Solution**

Fewest cubes means the biggest possible edge.
The edge must divide 20, 12 and 8 exactly; the largest such number is 4.
With edge 4 cm: $20 : 4 = 5$, $12 : 4 = 3$, $8 : 4 = 2$.
Number of cubes: $5 \times 3 \times 2 = 30$.
**Answer:** 30 cubes.

**Problem 39**

A crate is 6 dm long, 4 dm wide and 2 dm high. It is filled exactly with equal cubes whose edge is a whole number of decimetres. What are the largest and the smallest possible numbers of cubes?

**Solution**

Largest number: edge 1 dm, $6 \times 4 \times 2 = 48$ cubes.
Smallest number: the edge must divide 6, 4 and 2; the largest such number is 2.
With edge 2 dm: $3 \times 2 \times 1 = 6$ cubes.
**Answer:** Largest 48, smallest 6.

**Problem 40**

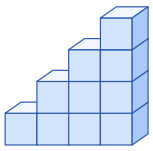
1 cm cubes are glued face to face in a straight row. A row of 10 cubes has total surface area 42 $cm^2$. How many cubes are in a row whose total surface area is 82 $cm^2$?

**Solution**

Each cube shows 4 faces along the sides of the row; the 2 ends show 2 more faces.
A row of $n$ cubes has area $4 \times n + 2$ ($cm^2$). Check: $4 \times 10 + 2 = 42$.
$4 \times n + 2 = 82$, so $4 \times n = 80$ and $n = 20$.
**Answer:** 20 cubes.

## Problem 41

A staircase is glued from 1 cm cubes (one cube deep). Find the number of cubes and the total surface area of the staircase, bottom included.



### Solution

Cubes: $1 + 2 + 3 + 4 = 10$.

Front and back: $10 + 10 = 20$ cm$^2$.

Bottom: $4$ cm$^2$; from above we see $4$ cm$^2$ (one square on each column).

Facing left: $1 + 1 + 1 + 1 = 4$ cm$^2$ (the left end and the three risers); facing right: $4$ cm$^2$.

Total surface area: $20 + 4 + 4 + 4 + 4 = 36$ cm$^2$.

**Answer:** 10 cubes; 36 cm$^2$.

## Problem 42

The figure is the top view of a stack, with the number of cubes on each square. How many cubes are in the stack? How many more cubes are needed to make a cube with 3 small cubes along each edge?

| | | |
|---|---|---|
| **3** | **2** | **1** |
| **2** | **2** | **1** |
| **1** | **1** | **1** |

front

### Solution

Number of cubes: $(3 + 2 + 1) + (2 + 2 + 1) + (1 + 1 + 1) = 6 + 5 + 3 = 14$.

The full cube has 27 cubes.

More cubes needed: $27 - 14 = 13$.

**Answer:** 14 cubes; 13 more cubes.

## Problem 43

A cube of edge 6 cm made of 1 cm cubes is painted on all its faces and then taken apart. Find the total painted area and the total unpainted area of all the small cubes.

### Solution

The painted area is the surface of the big cube: $6 \times 6 \times 6 = 216$ cm$^2$.

All the small cubes: 216 cubes, each with 6 faces of 1 cm$^2$: $216 \times 6 = 1\,296$ cm$^2$.

Unpainted area: $1\,296 - 216 = 1\,080$ cm$^2$.

**Answer:** Painted 216 cm$^2$; unpainted 1 080 cm$^2$.

## Problem 44

A cuboid made of 1 cm cubes has a base 6 cm by 5 cm. It is painted on all its faces, and 24 small cubes have no paint. Find the height of the cuboid and the number of small cubes painted on exactly 1 face.

### Solution

The unpainted core has $6-2=4$ by $5-2=3$ cubes in each layer: 12 cubes per layer.
Layers in the core: $24:12=2$, so the height is $2+2=4$ cm.
Without corners the edges have 4, 3, 2 cubes.
Exactly 1 face: $2\times(4\times3+3\times2+4\times2)=2\times26=52$.
**Answer:** Height 4 cm; 52 cubes.

## Problem 45

Bricks 12 cm long, 8 cm wide and 6 cm high are all laid the same way, side by side and in layers, to build a solid cube. What is the smallest number of bricks needed?

### Solution

The edge of the cube must be a whole number of brick lengths, of brick widths and of brick heights.
The smallest number divisible by 12, 8 and 6 is 24, so the cube has edge 24 cm.
Along the length: $24:12=2$; along the width: $24:8=3$; in height: $24:6=4$.
Number of bricks: $2\times3\times4=24$.
**Answer:** 24 bricks.

## Problem 46

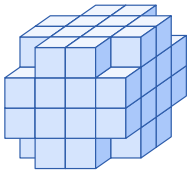
A big cube made of small cubes is painted on all its faces. Exactly 98 small cubes have paint on them. How many small cubes are there along each edge?

### Solution

Painted cubes = all cubes − the core.
Try $n=4$: $64-8=56$. Try $n=5$: $125-27=98$. ✓
So each edge has 5 small cubes.
**Answer:** 5 cubes.

## Problem 47 ★★

From a cube of edge 4 cm made of 1 cm cubes, all 8 corner cubes are removed. How many small cubes are left? What is the total surface area of the solid?



### Solution

Cubes left: $64 - 8 = 56$.
Each removed corner takes away 3 outside squares and uncovers 3 new squares, so the area does not change.
Total surface area: $4 \times 4 \times 6 = 96$ cm$^2$.
**Answer:** 56 cubes; 96 cm$^2$.

## Problem 48 ★★

A cuboid 8 cm long, 6 cm wide and 4 cm high is built from 1 cm cubes and stands on a table. All its faces are painted except the bottom. How many small cubes are painted on exactly 1 face?

### Solution

*Top face:* cubes not on its border: $(8 - 2) \times (6 - 2) = 24$.
*Front face and back face:* cubes not on the left or right edge (6 in a row) and not in the top row (3 rows; the bottom row has no paint underneath): $6 \times 3 = 18$ on each, 36 in all.
*Left face and right face:* $(6 - 2) \times 3 = 12$ on each, 24 in all.
Exactly 1 face: $24 + 36 + 24 = 84$.
**Answer:** 84 cubes.

## Problem 49 ★★

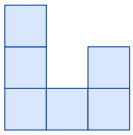
A cube is built from 27 small cubes (3 along each edge). How many cubes of all sizes can be seen in it (cubes of 1, 2 or 3 small cubes along an edge)?

### Solution

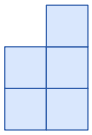
Cubes of edge 1: 27.
Cubes of edge 2: along each direction there are 2 positions, so $2 \times 2 \times 2 = 8$.
Cubes of edge 3: 1.
In all: $27 + 8 + 1 = 36$.
**Answer:** 36 cubes.

## Problem 50 ★★

A stack of cubes stands on a base of $3 \times 2$ squares; every base square has at least one cube. Its front view and its view from the right are shown. What are the largest and the smallest possible numbers of cubes in the stack?



front view



right view

(front | back)

## Solution

Front view: the three columns (left to right) are 3, 1, 2 tall. Right view: the front row is 2 tall, the back row 3 tall.

*Largest:* each stack can be as tall as both views allow, i.e. the smaller of its column height and its row height.

Front row: $2 + 1 + 2 = 5$; back row: $3 + 1 + 2 = 6$. Largest: $5 + 6 = 11$ cubes.

*Smallest:* put 3 cubes on the left square of the back row (this gives both the 3 of the left column and the 3 of the back row) and 2 cubes on the right square of the front row; every other square gets 1 cube.

Smallest: $3 + 2 + 1 + 1 + 1 + 1 = 9$ cubes.

**Answer:** Largest 11, smallest 9.

# 6 Uniform Motion

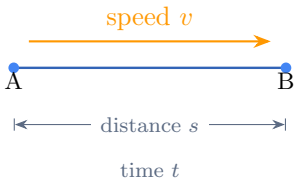
## ① Speed, distance and time

A **motion problem** is about a person or a vehicle moving from a place A to a place B in a certain amount of time.

– The length AB is the **distance**, written $s$ (usually in km).

– The time taken to go from A to B is the **time**, written $t$ (usually in hours).

– The distance covered in one unit of time is the **speed**, written $v$.




For example, An cycles 36 km in 3 hours, so his speed is $36 : 3 = 12$ km/h.

In **uniform motion** (steady speed):

$$s = v \times t \qquad v = s : t \qquad t = s : v$$

*Units:* distance in km and time in hours give speed in km/h; metres and minutes give m/min; metres and seconds give m/s. Always put the quantities in matching units before calculating.

## ② Units of time and of speed

1 day = 24 h; 1 h = 60 min; 1 min = 60 s; 1 h = 3 600 s.

– Hours to minutes: multiply by 60. $0.4 \text{ h} = 0.4 \times 60 \text{ min} = 24 \text{ min}$.

– Minutes to hours: divide by 60. $1 \text{ h } 15 \text{ min} = 1\frac{15}{60} \text{ h} = 1.25 \text{ h}$.

**Changing units of speed:**

– 54 km/h: in 1 hour 54 000 m, so in 1 second $54\,000 : 3\,600 = 15$ m. So 54 km/h = 15 m/s.

– In short: km/h → m/s: divide by 3.6; m/s → km/h: multiply by 3.6.

– 600 m/min: in 1 hour $600 \times 60 = 36\,000$ m = 36 km. So 600 m/min = 36 km/h.

## ③ Working with times of day

When a journey has a departure time, an arrival time and a stop:

travel time = arrival time − departure time − stopping time;

arrival time = departure time + travel time + stopping time.

When adding or subtracting times, work with hours and minutes separately; 60 min make 1 h.

To use $s = v \times t$, write the travel time in hours (as a decimal or a fraction).




travel time = 9 h 20 min − 7 h − 20 min = 2 h

## Example 1

Fill in the blanks.

**a)** 0.75 h = ... min; **b)** 2 h 24 min = ... h; **c)** 3 min 12 s = ... s.

### Guided solution.

**a)** 0.75 h = 0.75 × 60 min = 45 min.

**b)** 24 min = 24 : 60 h = 0.4 h, so 2 h 24 min = 2.4 h.

**c)** 3 min = 3 × 60 = 180 s, so 3 min 12 s = 180 + 12 = 192 s.

**Answer: a)** 45 min;

**b)** 2.4 h;

**c)** 192 s.

## Example 2

Fill in the blanks.

**a)** 72 km/h = ... m/s; **b)** 15 m/s = ... km/h; **c)** 800 m/min = ... km/h.

### Guided solution.

**a)** 72 km = 72 000 m; 1 h = 3 600 s.

72 000 : 3 600 = 20, so 72 km/h = 20 m/s.

**b)** In 1 hour: 15 × 3 600 = 54 000 m = 54 km. So 15 m/s = 54 km/h.

**c)** In 1 hour: 800 × 60 = 48 000 m = 48 km. So 800 m/min = 48 km/h.

**Answer: a)** 20 m/s;

**b)** 54 km/h;

**c)** 48 km/h.

## Example 3

A coach leaves town A at 6:45 and reaches town B, 110 km away, at 9:15. On the way it stops for 30 min. Find the speed of the coach.

### Guided solution.

From 6:45 to 9:15: 9 h 15 min − 6 h 45 min = 2 h 30 min.
Travel time: 2 h 30 min − 30 min = 2 h.
Speed: 110 : 2 = 55 km/h.

**Answer:** 55 km/h.

## Example 4

Hùng cycles from home at 7:20 at 14 km/h and arrives at his grandparents' house at 9:05 without stopping. How far is it from his home to his grandparents' house?

### Guided solution.

Travel time: 9 h 5 min − 7 h 20 min = 1 h 45 min.
1 h 45 min = $1\frac{45}{60}$ h = 1.75 h.
Distance: 14 × 1.75 = 24.5 km.
**Answer:** 24.5 km.

## Example 5

At 6:30 a lorry leaves A for B, 135 km away, at 45 km/h. At B it stops for 40 min to unload, then drives back to A at 54 km/h. At what time does it get back to A?




### Guided solution.

Time from A to B: 135 : 45 = 3 h.
Time from B back to A: 135 : 54 = 2.5 h = 2 h 30 min.
Back at A: 6:30 + 3 h + 40 min + 2 h 30 min = 12:40.
**Answer:** 12:40.

# Practice

## Problem 1

Fill in the blanks.

**a)** 0.4 h = . . . min;

**b)** 1.25 h = . . . min;

**c)** 2 h 18 min = . . . h;

**d)** 45 min = . . . h.

**Solution**

**a)** $0.4 \times 60 = 24$ min.
**b)** $1.25 \times 60 = 75$ min.
**c)** $18 : 60 = 0.3$, so 2.3 h.
**d)** $45 : 60 = 0.75$ h.
**Answer:** 24 min; 75 min; 2.3 h; 0.75 h.

## Problem 2

Fill in the blanks.

**a)** 3.5 min = ... s;

**b)** 2 min 15 s = ... s;

**c)** 1.5 days = ... h;

**d)** 150 s = ... min.

**Solution**

**a)** $3.5 \times 60 = 210$ s.
**b)** $2 \times 60 + 15 = 135$ s.
**c)** $1.5 \times 24 = 36$ h.
**d)** $150 : 60 = 2.5$ min.
**Answer:** 210 s; 135 s; 36 h; 2.5 min.

## Problem 3

Fill in the blanks.

**a)** 36 km/h = ... m/s;

**b)** 25 m/s = ... km/h;

**c)** 54 km/h = ... m/min;

**d)** 750 m/min = ... km/h.

**Solution**

**a)** $36\,000 : 3\,600 = 10$ m/s.
**b)** $25 \times 3\,600 = 90\,000$ m $= 90$ km, so 90 km/h.
**c)** $54\,000 : 60 = 900$ m/min.
**d)** $750 \times 60 = 45\,000$ m $= 45$ km, so 45 km/h.
**Answer:** 10 m/s; 90 km/h; 900 m/min; 45 km/h.

## Problem 4

Work out.

**a)** 3 h 45 min + 2 h 30 min;

**b)** 8 h 10 min − 5 h 35 min;

**c)** 1 h 25 min × 3;

**d)** 7 h 30 min : 6.

## Solution

**a)** 5 h 75 min = 6 h 15 min.

**b)** 8 h 10 min = 7 h 70 min; 7 h 70 min − 5 h 35 min = 2 h 35 min.

**c)** 3 h 75 min = 4 h 15 min.

**d)** 7 h 30 min = 450 min; 450 : 6 = 75 min = 1 h 15 min.

**Answer:** 6 h 15 min; 2 h 35 min; 4 h 15 min; 1 h 15 min.

## Problem 5

A car travels 150 km in 3 hours. Find its speed.

## Solution

Speed: 150 : 3 = 50 km/h.

**Answer:** 50 km/h.

## Problem 6

A cyclist rides 31.5 km in 2 h 15 min. Find the cyclist's speed.

## Solution

2 h 15 min = 2.25 h.

Speed: 31.5 : 2.25 = 14 km/h.

**Answer:** 14 km/h.

## Problem 7

Lan runs one lap of a 400 m track in 50 seconds. Find her speed in m/s.

## Solution

Speed: 400 : 50 = 8 m/s.

**Answer:** 8 m/s.

**Problem 8**

A car travels at 48 km/h for 2.5 hours. How far does it go?

**Solution**

Distance: $48 \times 2.5 = 120$ km.
**Answer:** 120 km.

**Problem 9**

Grandma walks at 4.5 km/h for 1 h 20 min. How far does she walk?

**Solution**

1 h 20 min $= 1\frac{20}{60}$ h $= \frac{4}{3}$ h.

Distance: $4.5 \times \frac{4}{3} = 6$ km.

**Answer:** 6 km.

**Problem 10**

A motorbike travels 84 km at 35 km/h. How long does the journey take?

**Solution**

Time: $84 : 35 = 2.4$ h.
0.4 h $= 0.4 \times 60 = 24$ min, so 2.4 h = 2 h 24 min.
**Answer:** 2 h 24 min.

**Problem 11**

A swallow flies at 108 km/h. How many minutes does it need to fly 27 km?

**Solution**

Time: $27 : 108 = 0.25$ h.
$0.25 \text{ h} = 0.25 \times 60 = 15$ min.
**Answer:** 15 min.

**Problem 12**

A coach leaves at 7:15 and arrives at 9:45 after travelling 125 km without stopping. Find its speed.

**Solution**

Travel time: 9 h 45 min − 7 h 15 min = 2 h 30 min = 2.5 h.
Speed: 125 : 2.5 = 50 km/h.
**Answer:** 50 km/h.

**Problem 13**

A motorbike leaves at 13:40 and arrives at 15:10, riding at 42 km/h without stopping. How long is the journey?

**Solution**

Travel time: 15 h 10 min − 13 h 40 min = 1 h 30 min = 1.5 h.
Distance: 42 × 1.5 = 63 km.
**Answer:** 63 km.

**Problem 14**

A van leaves at 6:50 to travel 96 km at 40 km/h without stopping. At what time does it arrive?

**Solution**

Travel time: 96 : 40 = 2.4 h = 2 h 24 min.
Arrival: 6:50 + 2 h 24 min = 8 h 74 min = 9:14.
**Answer:** 9:14.

**Problem 15**

Mr Tâm must reach a city 150 km away at 11:00. His car goes at 60 km/h and he will not stop. At what time must he set off?

**Solution**

Travel time: 150 : 60 = 2.5 h = 2 h 30 min.
Departure: 11 h − 2 h 30 min = 8:30.
**Answer:** 8:30.

**Problem 16**

A car leaves A at 7:00 and arrives at B, 162 km away, at 10:20. On the way it stops for 20 min. Find its speed.

**Solution**

From 7:00 to 10:20: 3 h 20 min.
Travel time: 3 h 20 min − 20 min = 3 h.

Speed: $162 : 3 = 54$ km/h.
**Answer:** 54 km/h.

**Problem 17**

A motorbike leaves at 8:15 at 36 km/h and arrives at 11:30. On the way the rider stops for 45 min. How long is the journey?

**Solution**

From 8:15 to 11:30: 3 h 15 min.
Travel time: 3 h 15 min − 45 min = 2 h 30 min = 2.5 h.
Distance: $36 \times 2.5 = 90$ km.
**Answer:** 90 km.

**Problem 18**

A motorcyclist rides 37.5 km in 1 h 15 min. At the same speed, how long will 51 km take?

**Solution**

1 h 15 min = 1.25 h.
Speed: $37.5 : 1.25 = 30$ km/h.
Time for 51 km: $51 : 30 = 1.7$ h = 1 h 42 min.
**Answer:** 1 h 42 min.

**Problem 19**

A bus at 40 km/h covers a road in 3 h 30 min. How long does a car at 56 km/h take on the same road?

**Solution**

3 h 30 min = 3.5 h.
Length of the road: $40 \times 3.5 = 140$ km.
Time for the car: $140 : 56 = 2.5$ h = 2 h 30 min.
**Answer:** 2 h 30 min.

**Problem 20**

Mai walks 1.2 km from home to school in 15 min. Find her speed in km/h and in m/min.

**Solution**

15 min = 0.25 h.
Speed: $1.2 : 0.25 = 4.8$ km/h.
1.2 km = 1 200 m; speed: $1\,200 : 15 = 80$ m/min.
**Answer:** 4.8 km/h; 80 m/min.

**Problem 21**

Hà cycles to school at 12 km/h and needs 15 min. On the same road, what speed would get her there in 10 min?

**Solution**

15 min = 0.25 h. Distance: $12 \times 0.25 = 3$ km.
10 min = $\frac{10}{60}$ h = $\frac{1}{6}$ h.
Speed needed: $3 : \frac{1}{6} = 18$ km/h.
**Answer:** 18 km/h.

**Problem 22**

Sound travels through air at 340 m/s. Bình hears the thunder 5 seconds after he sees the lightning. About how far away is the storm? (Light arrives almost at once.)

**Solution**

Distance: $340 \times 5 = 1\,700$ m = 1.7 km.
**Answer:** About 1.7 km.

**Problem 23**

A night train leaves at 21:30 and runs 360 km at 72 km/h without stopping. At what time does it arrive?

**Solution**

Travel time: $360 : 72 = 5$ h.
21 h 30 min + 5 h = 26 h 30 min, that is 2:30 the next morning.
**Answer:** 2:30 the next day.

**Problem 24**

Quân cycles for 40 min at 15 km/h, then walks for 18 min at 5 km/h. How far does he go altogether?

**Solution**

Cycling: 40 min $= \frac{2}{3}$ h; $15 \times \frac{2}{3} = 10$ km.
Walking: 18 min $= 0.3$ h; $5 \times 0.3 = 1.5$ km.
Altogether: $10 + 1.5 = 11.5$ km.
**Answer:** 11.5 km.

## Problem 25

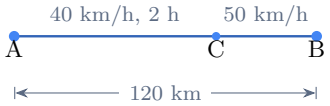
A taxi leaves A at 7:00 for B, 90 km away, at 45 km/h. It waits at B for 30 min, then returns to A at 60 km/h. At what time is it back at A?

**Solution**

A to B: $90 : 45 = 2$ h. B to A: $90 : 60 = 1.5$ h $= 1$ h 30 min.
Back at A: 7:00 + 2 h + 30 min + 1 h 30 min = 11:00.
**Answer:** 11:00.

## Problem 26

The road AB is 120 km long. At 7:30 a car leaves A at 40 km/h. After 2 hours it stops for 15 min to change a tyre, then covers the rest of the road at 50 km/h. At what time does it reach B?




**Solution**

Distance in the first 2 h: $40 \times 2 = 80$ km.
Rest of the road: $120 - 80 = 40$ km.
Time for it: $40 : 50 = 0.8$ h $= 48$ min.
Arrival: 7:30 + 2 h + 15 min + 48 min = 10:33.
**Answer:** 10:33.

## Problem 27

At 6:30 Mr Phúc walks from his house at 5 km/h. At 7:06 he gets on a bus, which goes at 40 km/h and arrives at the market town at 7:51. How far is the market town from his house?

**Solution**

Walking time: 7 h 6 min − 6 h 30 min = 36 min = 0.6 h; walking: $5 \times 0.6 = 3$ km.
Bus time: 7 h 51 min − 7 h 6 min = 45 min = 0.75 h; bus: $40 \times 0.75 = 30$ km.
Distance: $3 + 30 = 33$ km.
**Answer:** 33 km.

### Problem 28

A car covers 180 km in 2 h 30 min. Find its speed in km/h and in m/s.

### Solution

2 h 30 min = 2.5 h.
Speed: 180 : 2.5 = 72 km/h.
72 km/h = 72 000 : 3 600 = 20 m/s.
**Answer:** 72 km/h; 20 m/s.

### Problem 29

A snail crawls at 15 cm/min. How many minutes does it need to cross a path 1.2 m wide?

### Solution

1.2 m = 120 cm.
Time: 120 : 15 = 8 min.
**Answer:** 8 min.

### Problem 30

A ferry leaves at 22:45 and arrives at 4:15 the next morning without stopping, after sailing 132 km. Find its speed.

### Solution

From 22:45 to 24:00: 1 h 15 min; from 0:00 to 4:15: 4 h 15 min.
Travel time: 1 h 15 min + 4 h 15 min = 5 h 30 min = 5.5 h.
Speed: 132 : 5.5 = 24 km/h.
**Answer:** 24 km/h.

## Challenge – Extension

### Problem 31 *

The road AB is 150 km long. At 7:30 a car leaves A. It covers the first $\frac{1}{3}$ of the road at 40 km/h and the rest at 50 km/h. At what time does it reach B?

### Solution

First part: 150 : 3 = 50 km; time 50 : 40 = 1.25 h = 1 h 15 min.
Rest: 150 − 50 = 100 km; time 100 : 50 = 2 h.
Arrival: 7:30 + 1 h 15 min + 2 h = 10:45.

**Answer:** 10:45.

### Problem 32

A cyclist rides for 2 h 15 min at 16 km/h, then for 1 h 45 min at 12 km/h. How far does she ride in all? On average, how many kilometres does she ride per hour?

### Solution

First part: $16 \times 2.25 = 36$ km. Second part: $12 \times 1.75 = 21$ km.
In all: $36 + 21 = 57$ km in $2.25 + 1.75 = 4$ h.
On average: $57 : 4 = 14.25$ km per hour.
**Answer:** 57 km; 14.25 km/h.

### Problem 33

A road has an uphill part of 9 km and a downhill part of 15 km. A cyclist goes uphill at 12 km/h and downhill at 30 km/h. How long does the whole road take? What is the average speed over the whole road?

### Solution

Uphill: $9 : 12 = 0.75$ h. Downhill: $15 : 30 = 0.5$ h.
Whole road: $0.75 + 0.5 = 1.25$ h $= 1$ h 15 min.
Average speed: $(9 + 15) : 1.25 = 19.2$ km/h.
**Answer:** 1 h 15 min; 19.2 km/h.

### Problem 34

A train leaves at 23:40 and runs for 2 h 50 min at 60 km/h. It stops at a station for 20 min, then runs another 75 km at 50 km/h to the last station. At what time does it arrive, and how long is the whole route?

### Solution

First part: 2 h 50 min $= 2\frac{5}{6}$ h; $60 \times 2\frac{5}{6} = 170$ km.
Second part: $75 : 50 = 1.5$ h $= 1$ h 30 min.
Arrival: 23:40 + 2 h 50 min + 20 min + 1 h 30 min = 28 h 20 min, that is 4:20 the next day.
Whole route: $170 + 75 = 245$ km.
**Answer:** 4:20 the next day; 245 km.

### Problem 35

Hoa jogs 3 laps of a 400 m track in 6 min. Find her speed in m/min and in km/h.

**Solution**

Distance: $400 \times 3 = 1\,200$ m.
Speed: $1\,200 : 6 = 200$ m/min.
In 1 hour: $200 \times 60 = 12\,000$ m $= 12$ km, so 12 km/h.
**Answer:** 200 m/min; 12 km/h.

**Problem 36**

A car moves at 15 m/s. How many kilometres does it cover in 1 h 20 min?

**Solution**

15 m/s $= 15 \times 3.6 = 54$ km/h.
1 h 20 min $= \frac{4}{3}$ h.
Distance: $54 \times \frac{4}{3} = 72$ km.
**Answer:** 72 km.

**Problem 37**

At 8:15 a motorbike leaves A at 36 km/h. At 9:35 the rider speeds up by 9 km/h and keeps that speed until reaching B at 10:55. How far is it from A to B?

**Solution**

From 8:15 to 9:35: 1 h 20 min $= \frac{4}{3}$ h; distance $36 \times \frac{4}{3} = 48$ km.
New speed: $36 + 9 = 45$ km/h.
From 9:35 to 10:55: 1 h 20 min $= \frac{4}{3}$ h; distance $45 \times \frac{4}{3} = 60$ km.
AB: $48 + 60 = 108$ km.
**Answer:** 108 km.

**Problem 38**

The road AB is 108 km long. A lorry leaves A for B at 7:00 at 36 km/h. A car leaves A at 7:30 at 54 km/h. Which arrives at B first, and by how many minutes?

**Solution**

Lorry: $108 : 36 = 3$ h, arriving at 10:00.
Car: $108 : 54 = 2$ h, arriving at 7:30 + 2 h = 9:30.
The car arrives first, 30 min before the lorry.
**Answer:** The car, 30 min earlier.

## Problem 39

An runs 100 m in 16 s. Bình runs 1 km in 3 min 20 s. Cường cycles at 21.6 km/h. Order the three boys from fastest to slowest.

### Solution

An: $100 : 16 = 6.25$ m/s.

Bình: 3 min 20 s = 200 s; $1\,000 : 200 = 5$ m/s.

Cường: 21.6 km/h $= 21.6 : 3.6 = 6$ m/s.

$6.25 > 6 > 5$.

**Answer:** An, Cường, Bình.

## Problem 40

Tuấn's school is 4 km from home and lessons start at 7:15.

**a)** He leaves at 6:40 and cycles at 12 km/h. How many minutes early is he?

**b)** One day he leaves at 6:59. What speed must he ride at to arrive exactly at 7:15?

### Solution

**a)** Time: $4 : 12 = \frac{1}{3}$ h = 20 min; he arrives at 7:00, that is 15 min early.

**b)** He has 16 min $= \frac{16}{60}$ h $= \frac{4}{15}$ h.

Speed: $4 : \frac{4}{15} = 15$ km/h.

**Answer: a)** 15 min;

**b)** 15 km/h.

## Problem 41

A car leaves A at 7:05 and reaches B at 9:41. On the way it stops for 18 min. Its speed is 45 km/h. Find the distance AB.

### Solution

From 7:05 to 9:41: 2 h 36 min.

Travel time: 2 h 36 min − 18 min = 2 h 18 min = 2.3 h.

AB: $45 \times 2.3 = 103.5$ km.

**Answer:** 103.5 km.

## Problem 42

A plane flies at 840 km/h. It takes off at 6:50 for a city 1 260 km away. At what time does it land?

**Solution**

Flying time: $1\,260 : 840 = 1.5$ h $= 1$ h 30 min.
Landing: 6:50 + 1 h 30 min = 8:20.
**Answer:** 8:20.

**Problem 43** ⋆

A runner covers 3 km in 12 min 30 s. Find the speed in m/s and in km/h.

**Solution**

12 min 30 s = 750 s; 3 km = 3 000 m.
Speed: $3\,000 : 750 = 4$ m/s.
$4$ m/s $= 4 \times 3.6 = 14.4$ km/h.
**Answer:** 4 m/s; 14.4 km/h.

**Problem 44** ⋆

Bình cycles for 45 min at 16 km/h; then his bike breaks and he walks the remaining 2.4 km in 36 min. How long is the whole road, and how fast did he walk?

**Solution**

Cycling: 45 min $= 0.75$ h; $16 \times 0.75 = 12$ km.
Whole road: $12 + 2.4 = 14.4$ km.
Walking: 36 min $= 0.6$ h; speed $2.4 : 0.6 = 4$ km/h.
**Answer:** 14.4 km; 4 km/h.

**Problem 45** ⋆⋆

The road AB is 60 km long. A car spends half of its travel time at 40 km/h and the other half at 60 km/h. How long does it take from A to B?

**Solution**

In each hour of the first half it covers 40 km, in each hour of the second half 60 km.
One hour of the first half together with one hour of the second half covers $40 + 60 = 100$ km.
So each half lasts $60 : 100 = 0.6$ h.
Total time: $0.6 \times 2 = 1.2$ h $= 1$ h 12 min.
*Check:* $40 \times 0.6 + 60 \times 0.6 = 24 + 36 = 60$ km.
**Answer:** 1 h 12 min.

**Problem 46** ★★

The road AB is 60 km long. A car covers the first half of the road at 30 km/h and the second half at 60 km/h. Find the total time and the average speed. Is the average speed 45 km/h?

**Solution**

Each half: 30 km.
First half: $30 : 30 = 1$ h; second half: $30 : 60 = 0.5$ h.
Total time: 1.5 h.
Average speed: $60 : 1.5 = 40$ km/h, not 45 km/h (the car spends longer at the slower speed).
**Answer:** 1.5 h; 40 km/h (no).

**Problem 47** ★

On a motorway, Dad's car passes one kilometre post every 48 seconds. What is its speed in km/h?

**Solution**

1 h = 3 600 s. In 1 hour it passes $3\,600 : 48 = 75$ posts.
Speed: 75 km/h.
**Answer:** 75 km/h.

**Problem 48** ★★

A tortoise and a hare race over 120 m. The tortoise crawls steadily at 0.2 m/s. The hare runs at 6 m/s, stops after 90 m to sleep, then wakes up and runs the last 30 m at the same speed. At most how long can the hare sleep and still not lose?

**Solution**

Tortoise: $120 : 0.2 = 600$ s.
Hare running: $120 : 6 = 20$ s in all.
Longest sleep: $600 - 20 = 580$ s = 9 min 40 s.
**Answer:** 9 min 40 s.

**Problem 49** ★★

A train leaves station A at 19:20 and reaches station B, 570 km away, at 6:05 the next day. Its stops at stations add up to 1 h 15 min. Find the speed of the train while it is moving.

**Solution**

From 19:20 to 6:05 the next day: 4 h 40 min + 6 h 5 min = 10 h 45 min.
Moving time: 10 h 45 min − 1 h 15 min = 9 h 30 min = 9.5 h.
Speed: $570 : 9.5 = 60$ km/h.

**Answer:** 60 km/h.

## Problem 50 ★★

The road AB is 115 km long. A car leaves A at 7:00 and must reach B at 9:30. For the first 1 h 15 min it goes at 40 km/h. At what speed must it cover the rest of the road?

## Solution

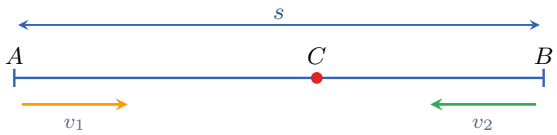
Distance so far: $40 \times 1.25 = 50$ km; still to go: $115 - 50 = 65$ km.
Time left: 9 h 30 min − 7 h − 1 h 15 min = 1 h 15 min = 1.25 h.
Speed needed: $65 : 1.25 = 52$ km/h.

**Answer:** 52 km/h.

# 7 Meeting and Catching Up

## ① Two objects moving towards each other

Two objects start *at the same time* from $A$ and $B$, a distance $s$ apart, and move towards each other with speeds $v_1$ and $v_2$.




- Every hour the gap between them shrinks by $v_1 + v_2$ (the *closing speed*).
- Time until they meet: $t = s : (v_1 + v_2)$.
- The meeting point $C$ is $v_1 \times t$ from $A$ and $v_2 \times t$ from $B$.
- Clock time of meeting = starting time + $t$.
- If one object sets off earlier, first find how far it has gone by the time the other one starts; the *remaining* distance is then shared as above.

## ② Two objects moving in the same direction

Two objects start *at the same time* from $A$ and $B$ ($s$ apart) and move in the same direction $A \to B$; the one behind is faster: $v_1 > v_2$.



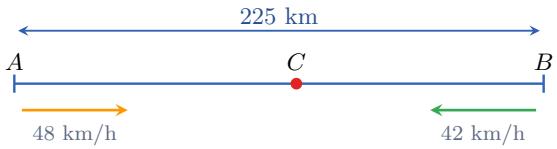

- Every hour the faster one gains $v_1 - v_2$ on the slower one.
- Time to catch up: $t = s : (v_1 - v_2)$; the catching point is $v_1 \times t$ from $A$.
- If both start from the *same place* at *different times*, the head start is $s = v_2 \times$ (difference in starting times).

**Note:** use the same units throughout (km, h, km/h or m, min, m/min). Two objects moving *away* from each other (opposite directions from the same point) separate by $v_1 + v_2$ every hour.

## Example 1

Two towns $A$ and $B$ are 225 km apart. At 6:45 a car leaves $A$ for $B$ at 48 km/h; at the same time a coach leaves $B$ for $A$ at 42 km/h. At what time do they meet, and how far from $A$ is the meeting point?

**Guided solution.**




Every hour the two vehicles close up: $48 + 42 = 90$ (km).
Time until they meet: $225 : 90 = 2.5$ (h) $= 2$ h 30 min.
They meet at: 6:45 + 2 h 30 min = 9:15.
The meeting point is $48 \times 2.5 = 120$ (km) from $A$.

**Answer: 9:15; 120 km from $A$.**

## Example 2

At 7:10 Lan leaves school by bicycle at 12 km/h. At 7:40 her brother leaves the school on a motorbike at 36 km/h and rides after her along the same road. At what time does he catch up with her, and how far from school?

**Guided solution.**

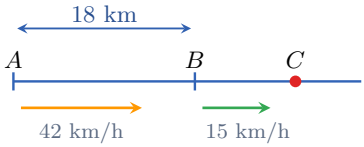
Lan's head start in time: 7:40 − 7:10 = 30 min = 0.5 h.
Distance Lan has covered by 7:40: $12 \times 0.5 = 6$ (km).
Every hour the motorbike gains: $36 - 12 = 24$ (km).
Time to catch up: $6 : 24 = 0.25$ (h) $= 15$ min. They meet at 7:40 + 15 min = 7:55.
Distance from school: $36 \times 0.25 = 9$ (km).

**Answer: 7:55; 9 km from school.**

## Example 3

Points $A$ and $B$ are 18 km apart. At 6:30 a motorbike leaves $A$ at 42 km/h and, at the same time, a cyclist leaves $B$ at 15 km/h; both travel in the direction from $A$ to $B$. At what time does the motorbike catch the cyclist, and how far from $A$?

**Guided solution.**




Every hour the motorbike gains: $42 - 15 = 27$ (km).

Time to catch up: $18 : 27 = \frac{2}{3}$ (h) $= 40$ min. It catches up at 6:30 + 40 min = 7:10.

Distance from $A$: $42 \times \frac{2}{3} = 28$ (km).

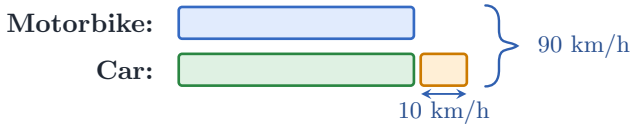
**Answer: 7:10; 28 km from $A$.**

## Example 4

Two towns are 180 km apart. A car and a motorbike set off from the two towns at the same time towards each other and meet after 2 hours. The car is 10 km/h faster than the motorbike. Find the speed of each.

### Guided solution.

Sum of the two speeds (the closing speed): $180 : 2 = 90$ (km/h).




Speed of the motorbike: $(90 - 10) : 2 = 40$ (km/h).

Speed of the car: $40 + 10 = 50$ (km/h).

**Answer: car 50 km/h, motorbike 40 km/h.**

## Example 5

Two places are 150 km apart. At 8:00 two cars leave them and drive towards each other at 35 km/h and 40 km/h. At what time are the two cars 30 km apart?

### Guided solution.

Every hour the cars close up by $35 + 40 = 75$ (km). There are *two* moments when they are 30 km apart.

**Before they meet:** together they have driven $150 - 30 = 120$ (km); time: $120 : 75 = 1.6$ (h) $= 1$ h 36 min, so at 9:36.

**After they meet:** together they have driven $150 + 30 = 180$ (km); time: $180 : 75 = 2.4$ (h) $= 2$ h 24 min, so at 10:24.

**Answer: at 9:36 and at 10:24.**

## Practice

**Problem 1**

Two villages are 36 km apart. Two cyclists leave the villages at the same time and ride towards each other at 13 km/h and 11 km/h. After how long do they meet?

**Solution**

Closing speed: $13 + 11 = 24$ (km/h).

Time: $36 : 24 = 1.5$ (h) $= 1$ h 30 min.

**Answer:** 1 h 30 min.

**Problem 2**

Road $AB$ is 279 km long. At 7:20 a car leaves $A$ for $B$ at 68 km/h; at the same time a coach leaves $B$ for $A$ at 56 km/h. At what time do they meet?

**Solution**

Closing speed: $68 + 56 = 124$ (km/h).

Time: $279 : 124 = 2.25$ (h) $= 2$ h 15 min.

They meet at 7:20 + 2 h 15 min = 9:35.

**Answer:** 9:35.

**Problem 3**

Two cities $A$ and $B$ are 168 km apart. At the same moment a motorbike leaves $A$ at 38 km/h and a car leaves $B$ at 46 km/h, heading towards each other. After how long do they meet, and how far from $A$?

**Solution**

Closing speed: $38 + 46 = 84$ (km/h).

Time: $168 : 84 = 2$ (h).

Distance from $A$: $38 \times 2 = 76$ (km).

**Answer:** after 2 h, 76 km from $A$.

**Problem 4**

Two towns $A$ and $B$ are 210 km apart. At 6:30 two cars leave $A$ and $B$ and drive towards each other, the car from $A$ at 45 km/h and the car from $B$ at 39 km/h. At what time do they meet, and how far from $A$ is the meeting point?

**Solution**

Closing speed: $45 + 39 = 84$ (km/h).

Time: $210 : 84 = 2.5$ (h) $= 2$ h 30 min; they meet at 9:00.

Distance from $A$: $45 \times 2.5 = 112.5$ (km).

**Answer:** 9:00; 112.5 km from $A$.

**Problem 5**

Two towns are 196 km apart. A car and a motorbike leave the two towns at the same time towards each other and meet after 2 h 20 min. The car is 12 km/h faster than the motorbike. Find the speed of each.

**Solution**

2 h 20 min $= 2\frac{1}{3}$ h $= \frac{7}{3}$ h.

Sum of the speeds: $196 : \frac{7}{3} = 84$ (km/h).

Motorbike: $(84 - 12) : 2 = 36$ (km/h); car: $36 + 12 = 48$ (km/h).

**Answer:** car 48 km/h, motorbike 36 km/h.

**Problem 6**

Two towns $A$ and $B$ are 245 km apart. Two cars leave $A$ and $B$ at the same time towards each other and meet after 2.5 h at a point $C$ that is 105 km from $A$. Find the speed of each car.

**Solution**

Car from $A$: $105 : 2.5 = 42$ (km/h).

Distance $CB$: $245 - 105 = 140$ (km).

Car from $B$: $140 : 2.5 = 56$ (km/h).

**Answer:** 42 km/h and 56 km/h.

**Problem 7**

Road $AB$ is 288 km long. A car and a motorbike leave $A$ and $B$ at the same time towards each other and meet after 3 h 12 min. The car's speed is 1.5 times the motorbike's. Find the speed of each.

**Solution**

3 h 12 min $= 3.2$ h. Sum of the speeds: $288 : 3.2 = 90$ (km/h).

$1.5 = \frac{3}{2}$: motorbike 2 parts, car 3 parts; 5 parts $= 90$, one part $= 18$ km/h.

Motorbike: $18 \times 2 = 36$ (km/h); car: $18 \times 3 = 54$ (km/h).
**Answer:** car 54 km/h, motorbike 36 km/h.

### Problem 8

Road $AB$ is 270 km long. At 7:00 a lorry leaves $A$ for $B$ at 40 km/h. At 8:15 a car leaves $B$ for $A$ at 60 km/h. At what time do they meet, and how far from $A$?

### Solution

From 7:00 to 8:15 is 1 h 15 min = 1.25 h; the lorry has gone $40 \times 1.25 = 50$ (km).
Remaining distance: $270 - 50 = 220$ (km). Closing speed: $40 + 60 = 100$ (km/h).
Time: $220 : 100 = 2.2$ (h) = 2 h 12 min; they meet at 8:15 + 2 h 12 min = 10:27.
Distance from $A$: $50 + 40 \times 2.2 = 138$ (km).
**Answer:** 10:27; 138 km from $A$.

### Problem 9

At 6:30 a coach leaves $A$ for $B$ at 50 km/h. Half an hour later a second coach leaves $B$ for $A$ at 45 km/h. The two coaches meet at 9:00. How long is road $AB$?

### Solution

First coach drives from 6:30 to 9:00: 2.5 h, covering $50 \times 2.5 = 125$ (km).
Second coach starts at 7:00 and drives 2 h, covering $45 \times 2 = 90$ (km).
$AB = 125 + 90 = 215$ (km).
**Answer:** 215 km.

### Problem 10

Two places are 160 km apart. Two cars leave them at the same time and drive towards each other at 38 km/h and 42 km/h. How far apart are they after 1 h 30 min?

### Solution

Closing speed: $38 + 42 = 80$ (km/h).
In 1.5 h they close up $80 \times 1.5 = 120$ (km).
Still apart: $160 - 120 = 40$ (km).
**Answer:** 40 km.

## Problem 11

Two cars leave $A$ and $B$ at the same time and drive towards each other. The first car needs 4 h to drive the whole road $AB$; the second needs 6 h. After 2 h the cars are still 30 km apart. How long is road $AB$?

### Solution

In 1 h the first car covers $\frac{1}{4}$ of $AB$ and the second $\frac{1}{6}$ of $AB$; together $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ of $AB$.

In 2 h together: $\frac{5}{12} \times 2 = \frac{5}{6}$ of $AB$. Remaining: $1 - \frac{5}{6} = \frac{1}{6}$ of $AB$, which is 30 km.

$AB = 30 \times 6 = 180$ (km).

**Answer:** 180 km.

## Problem 12

A motorbike needs 3 h to ride from $A$ to $B$; a bicycle needs 6 h to ride from $B$ to $A$. They set off at the same time towards each other. After how long do they meet?

### Solution

In 1 h together they cover $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$ of $AB$.

Time to cover the whole road together: $1 : \frac{1}{2} = 2$ (h).

**Answer:** 2 h.

## Problem 13

Lan's house and Mai's house are 2.4 km apart. At 16:40 the two girls leave home and walk towards each other, Lan at 4.5 km/h and Mai at 3.5 km/h. After how many minutes do they meet, and at what time?

### Solution

Closing speed: $4.5 + 3.5 = 8$ (km/h).

Time: $2.4 : 8 = 0.3$ (h) $= 18$ min.

They meet at 16:40 + 18 min = 16:58.

**Answer:** after 18 min, at 16:58.

## Problem 14

$A$ and $B$ are 24 km apart. A walker leaves $A$ at 4 km/h and, at the same time, a cyclist leaves $B$ at 12 km/h; they go towards each other. How far from $A$ do they meet?

**Solution**

Closing speed: $4 + 12 = 16$ (km/h).

Time: $24 : 16 = 1.5$ (h).

Distance from $A$: $4 \times 1.5 = 6$ (km).

**Answer:** 6 km.

**Problem 15**

Road $AB$ is 132 km long. A car leaves $A$ at 52 km/h and, at the same time, a motorbike leaves $B$; they travel towards each other and meet after 1 h 30 min. Find the speed of the motorbike.

**Solution**

Sum of the speeds: $132 : 1.5 = 88$ (km/h).

Motorbike: $88 - 52 = 36$ (km/h).

**Answer:** 36 km/h.

**Problem 16**

$A$ and $B$ are 48 km apart. At the same time a car leaves $A$ at 58 km/h and a motorbike leaves $B$ at 42 km/h, both going in the direction from $A$ to $B$. After how long does the car catch the motorbike, and how far from $A$?

**Solution**

The car gains $58 - 42 = 16$ (km) every hour.

Time: $48 : 16 = 3$ (h).

Distance from $A$: $58 \times 3 = 174$ (km).

**Answer:** after 3 h, 174 km from $A$.

**Problem 17**

At 6:40 a cyclist leaves a town at 14 km/h. At 7:10 a motorbike leaves the same place at 35 km/h and follows the same road. At what time does the motorbike catch the cyclist?

**Solution**

Head start in time: 30 min $= 0.5$ h; head start in distance: $14 \times 0.5 = 7$ (km).

The motorbike gains $35 - 14 = 21$ (km) every hour.

Time: $7 : 21 = \frac{1}{3}$ (h) $= 20$ min; it catches up at 7:30.

**Answer:** 7:30.

## Problem 18

A car leaves $A$ and a bicycle leaves $B$ at the same time, both going in the direction from $A$ to $B$. The car drives at 52 km/h and the bicycle at 16 km/h. The car catches the bicycle after 1 h 45 min. How far apart are $A$ and $B$?

### Solution

The car gains $52 - 16 = 36$ (km) every hour.

1 h 45 min $= 1.75$ h. $AB = 36 \times 1.75 = 63$ (km).

**Answer:** 63 km.

## Problem 19

A cyclist riding at 15 km/h has already covered 15 km when a motorbike sets off after him from the same starting point at 40 km/h. After how many minutes does the motorbike catch the cyclist?

### Solution

The motorbike gains $40 - 15 = 25$ (km) every hour.

Time: $15 : 25 = 0.6$ (h) $= 36$ min.

**Answer:** 36 min.

## Problem 20

At 8:00 a lorry leaves $A$ at 45 km/h. At 9:20 a car leaves $A$ at 75 km/h along the same road. At what time does the car catch the lorry, and how far from $A$?

### Solution

From 8:00 to 9:20 is 1 h 20 min $= \frac{4}{3}$ h; the lorry is $45 \times \frac{4}{3} = 60$ (km) ahead.

The car gains $75 - 45 = 30$ (km) every hour. Time: $60 : 30 = 2$ (h); it catches up at 11:20.

Distance from $A$: $75 \times 2 = 150$ (km).

**Answer:** 11:20; 150 km from $A$.

## Problem 21

$A$ and $B$ are 36 km apart. Two cars leave $A$ and $B$ at the same time in the same direction (from $A$ to $B$). After 3 h the car from $A$ catches the car from $B$. The sum of their speeds is 84 km/h. Find the speed of each car.

**Solution**

Difference of the speeds: $36 : 3 = 12$ (km/h).

Car from $A$: $(84 + 12) : 2 = 48$ (km/h); car from $B$: $48 - 12 = 36$ (km/h).

**Answer:** 48 km/h and 36 km/h.

**Problem 22**

A motorbike is 30 km behind a lorry; both drive in the same direction, the motorbike at 50 km/h and the lorry at 35 km/h. How far apart are they after 1 h 20 min?

**Solution**

The gap shrinks by $50 - 35 = 15$ (km) every hour.

In 1 h 20 min $= \frac{4}{3}$ h it shrinks by $15 \times \frac{4}{3} = 20$ (km).

Still apart: $30 - 20 = 10$ (km).

**Answer:** 10 km.

**Problem 23**

Nam leaves home by bicycle at 250 m/min. 6 minutes later his mum sets off after him on a motorbike at 550 m/min. After how many minutes does Mum catch Nam, and how far from home?

**Solution**

Nam's head start: $250 \times 6 = 1\,500$ (m).

Mum gains $550 - 250 = 300$ (m) every minute.

Time: $1\,500 : 300 = 5$ (min). Distance from home: $550 \times 5 = 2\,750$ (m) $= 2.75$ km.

**Answer:** 5 min; 2.75 km.

**Problem 24**

Two cars leave the same crossroads at the same time and drive in opposite directions, at 45 km/h and 55 km/h. After how long are they 250 km apart?

**Solution**

They move apart by $45 + 55 = 100$ (km) every hour.

Time: $250 : 100 = 2.5$ (h) $= 2$ h 30 min.

**Answer:** 2 h 30 min.

## Problem 25

At 7:00 two cars leave the same place and drive along the same road in the same direction, at 42 km/h and 54 km/h. At what time are they 15 km apart?

### Solution

The faster car pulls away by $54 - 42 = 12$ (km) every hour.

Time: $15 : 12 = 1.25$ (h) $= 1$ h 15 min; at 8:15.

**Answer:** 8:15.

## Problem 26

At 7:00 a cyclist leaves $A$ at 12 km/h. At 8:00 a motorbike leaves $A$ along the same road and catches the cyclist at 8:30. Find the speed of the motorbike.

### Solution

By 8:30 the cyclist has ridden 1.5 h: $12 \times 1.5 = 18$ (km).

The motorbike covers these 18 km in 30 min $= 0.5$ h.

Speed: $18 : 0.5 = 36$ (km/h).

**Answer:** 36 km/h.

## Problem 27

Road $AB$ is 150 km long. At 7:00 a car leaves $A$ at 50 km/h and a motorbike leaves $B$ at 40 km/h, towards each other. Before they meet, the motorbike rider stops to rest for 18 minutes. At what time do they meet, and how far from $A$?

### Solution

During the 18 min $= 0.3$ h rest only the car moves: $50 \times 0.3 = 15$ (km).

The rest of the distance, $150 - 15 = 135$ (km), is closed at $50 + 40 = 90$ (km/h): $135 : 90 = 1.5$ (h).

Total time: $1.5 + 0.3 = 1.8$ (h) $= 1$ h 48 min; they meet at 8:48.

Distance from $A$: $50 \times 1.8 = 90$ (km).

**Answer:** 8:48; 90 km from $A$.

## Problem 28

Two towns are 120 km apart. At 7:00 a coach and a car leave the two towns towards each other at 42 km/h and 53 km/h. At what time, before they meet, are they 25 km apart?

**Solution**

They must close up $120 - 25 = 95$ (km) at $42 + 53 = 95$ (km/h).
Time: $95 : 95 = 1$ (h); at 8:00.
**Answer:** 8:00.

**Problem 29**

At 6:00 a motorbike leaves $A$ for $B$ at 40 km/h. At 7:00 a car leaves $A$ at 60 km/h on the same road and catches the motorbike exactly at $B$. How long is road $AB$?

**Solution**

Head start of the motorbike: $40 \times 1 = 40$ (km).
The car gains $60 - 40 = 20$ (km) every hour; time: $40 : 20 = 2$ (h).
$AB = 60 \times 2 = 120$ (km).
**Answer:** 120 km.

**Problem 30**

Lan's house and Hoa's house are 4.2 km apart. At 6:45 Lan walks from her house at 80 m/min and Hoa cycles from her house at 200 m/min, towards each other. At what time do they meet, and how far from Lan's house?

**Solution**

4.2 km $= 4\,200$ m. Closing speed: $80 + 200 = 280$ (m/min).
Time: $4\,200 : 280 = 15$ (min); they meet at 7:00.
Distance from Lan's house: $80 \times 15 = 1\,200$ (m) $= 1.2$ km.
**Answer:** 7:00; 1.2 km from Lan's house.

## Challenge – Extension

**Problem 31** ★★

Road $AB$ is 240 km long. At 7:00 a car leaves $A$ for $B$ at 80 km/h; at the same time a motorbike and a lorry leave $B$ for $A$ at 30 km/h and 50 km/h. At what time is the car exactly halfway between the motorbike and the lorry?

**Solution**

Think of a point $M$ always halfway between the motorbike and the lorry. It starts at $B$ and moves towards $A$ at the average speed $(30 + 50) : 2 = 40$ (km/h).
The car is halfway when it meets $M$: closing speed $80 + 40 = 120$ (km/h); time $240 : 120 = 2$ (h);

at 9:00.
*Check:* at 9:00 the car is 160 km from $A$, the motorbike $240 - 60 = 180$ km and the lorry $240 - 100 = 140$ km from $A$; 160 is halfway between 140 and 180.
**Answer:** 9:00.

**Problem 32** *

Road $AB$ is 150 km long. Two cars start from $A$ and $B$ at the same time. If they drive towards each other, they meet after 1 h 15 min. If they drive in the same direction (from $A$ towards $B$), the car from $A$ catches the other after 7 h 30 min. Find the speed of each car.

**Solution**

Sum of the speeds: $150 : 1.25 = 120$ (km/h). Difference: $150 : 7.5 = 20$ (km/h).
Car from $A$: $(120 + 20) : 2 = 70$ (km/h); car from $B$: $70 - 20 = 50$ (km/h).
**Answer:** 70 km/h and 50 km/h.

**Problem 33** *

$A$ and $B$ are 90 km apart. Two riders start from $A$ and $B$ at the same time. Riding towards each other they meet after 1.5 h; riding in the same direction, the rider from $A$ catches the other after 4.5 h. Find their speeds.

**Solution**

Sum: $90 : 1.5 = 60$ (km/h). Difference: $90 : 4.5 = 20$ (km/h).
Rider from $A$: $(60 + 20) : 2 = 40$ (km/h); rider from $B$: 20 km/h.
**Answer:** 40 km/h and 20 km/h.

**Problem 34** *

Two cars at $A$ and $B$, 60 km apart, set off at the same time in the same direction towards $C$ ($B$ lies between $A$ and $C$). After 2.5 h the car from $A$ catches the car from $B$. The sum of their speeds is 104 km/h. Find the speed of each.

**Solution**

Difference of the speeds: $60 : 2.5 = 24$ (km/h).
Car from $A$: $(104 + 24) : 2 = 64$ (km/h); car from $B$: $64 - 24 = 40$ (km/h).
**Answer:** 64 km/h and 40 km/h.

## Problem 35

Two cars leave $A$ and $B$ at the same time towards each other and first meet 40 km from $A$. Each car drives on to the other end, turns back at once and they meet a second time 20 km from $B$. How long is road $AB$?

### Solution

At the first meeting the two cars together have covered $AB$ once; the car from $A$ has covered 40 km.

At the second meeting together they have covered $AB$ three times, so the car from $A$ has covered $40 \times 3 = 120$ (km).

That car has gone the whole of $AB$ and then 20 km back: $AB = 120 - 20 = 100$ (km).

**Answer:** 100 km.

## Problem 36

Road $AB$ is 210 km. Two cars leave $A$ and $B$ at the same time towards each other and meet after 2 h. At that moment the car from $A$ has gone 30 km more than the car from $B$. Find their speeds.

### Solution

Sum of the speeds: $210 : 2 = 105$ (km/h). Difference: $30 : 2 = 15$ (km/h).

Car from $A$: $(105 + 15) : 2 = 60$ (km/h); car from $B$: 45 km/h.

**Answer:** 60 km/h and 45 km/h.

## Problem 37

At 6:00 a cyclist leaves $A$ at 15 km/h. When he reaches the 15-km mark he rests there for 30 minutes, then rides on. At 7:00 a motorbike leaves $A$ at 45 km/h along the same road. At what time does the motorbike catch the cyclist?

### Solution

The cyclist reaches the 15-km mark at $15 : 15 = 1$ h after 6:00, i.e. at 7:00, and rests there until 7:30.

The motorbike reaches that mark after $15 : 45 = \frac{1}{3}$ h $= 20$ min, at 7:20 — while the cyclist is still resting.

So it catches him at 7:20 (at the 15-km mark).

**Answer:** 7:20.

**Problem 38** *

Two friends are 6 km apart and walk towards each other at 4 km/h and 5 km/h. A dog starts with the first friend and runs at 15 km/h to the second, then back to the first, and so on until the friends meet. How far does the dog run?

**Solution**

The friends meet after $6 : (4 + 5) = \frac{2}{3}$ (h).

The dog runs all that time: $15 \times \frac{2}{3} = 10$ (km).

**Answer:** 10 km.

**Problem 39** *

At 7:00 a lorry leaves $A$ at 40 km/h. At 8:00 a car leaves $A$ at 60 km/h on the same road. After the car has started, at what times are the two vehicles 10 km apart?

**Solution**

At 8:00 the lorry is 40 km ahead; the car gains $60 - 40 = 20$ (km) every hour.
Before catching up: gain $40 - 10 = 30$ km, time $30 : 20 = 1.5$ h; at 9:30.
After catching up: gain $40 + 10 = 50$ km, time $50 : 20 = 2.5$ h; at 10:30.
**Answer:** 9:30 and 10:30.

**Problem 40** *

At 8:00 a motorbike leaves $A$ for $B$ at 36 km/h; $AB = 108$ km. A car drives at 54 km/h. At what time must the car leave $A$ to catch the motorbike exactly at $B$?

**Solution**

The motorbike needs $108 : 36 = 3$ (h); it reaches $B$ at 11:00.
The car needs $108 : 54 = 2$ (h), so it must leave at 9:00.
**Answer:** 9:00.

**Problem 41** **

A car leaves $A$ at 45 km/h. Half an hour later a second car leaves $B$ at 60 km/h and drives towards $A$. They meet exactly at the midpoint of $AB$. How long is $AB$?

**Solution**

Each car covers half of $AB$. To cover 1 km the first car needs $\frac{1}{45}$ h and the second $\frac{1}{60}$ h; the

difference is $\frac{1}{45} - \frac{1}{60} = \frac{1}{180}$ (h).

Over half of $AB$ the difference is 0.5 h, so half of $AB$ is $0.5 : \frac{1}{180} = 90$ (km).

$AB = 90 \times 2 = 180$ (km). (*Check:* $90 : 45 = 2$ h, $90 : 60 = 1.5$ h.)

**Answer:** 180 km.

## Problem 42 *

Two cars leave $A$ and $B$ at the same time towards each other at 50 km/h and 62 km/h. They meet at a point 18 km from the midpoint of $AB$. How long is $AB$?

### Solution

At the meeting point the faster car has gone $18 \times 2 = 36$ (km) more than the slower one.

It gains $62 - 50 = 12$ (km) every hour, so they meet after $36 : 12 = 3$ (h).

$AB = (50 + 62) \times 3 = 336$ (km).

**Answer:** 336 km.

## Problem 43 *

Two cyclists leave $A$ and $B$ at the same time towards each other at 15 km/h and 10 km/h. After meeting, the faster cyclist needs 40 more minutes to reach $B$. After how long did they meet, and how long is $AB$?

### Solution

After meeting, the faster cyclist rides the part that the slower one rode before meeting: $15 \times \frac{2}{3} = 10$ (km).

The slower cyclist rode those 10 km in $10 : 10 = 1$ (h), so they met after 1 h.

$AB = (15 + 10) \times 1 = 25$ (km).

**Answer:** after 1 h; 25 km.

## Problem 44 **

$A$ and $B$ are 90 km apart. If a cyclist from $A$ and a motorbike from $B$ start at the same time towards each other, they meet after 2 h. If the cyclist starts 1 h earlier, they meet 1 h 36 min after the motorbike starts. Find the speed of each.

### Solution

Sum of the speeds: $90 : 2 = 45$ (km/h).

Second case: in 1 h 36 min = 1.6 h after the motorbike starts, the two close up $45 \times 1.6 = 72$

(km).
So in the first hour alone the cyclist rode $90 - 72 = 18$ (km): his speed is 18 km/h.
Motorbike: $45 - 18 = 27$ (km/h).
**Answer:** cyclist 18 km/h, motorbike 27 km/h.

**Problem 45** ★★

At exactly 3:00 the minute hand of a clock points to 12 and the hour hand to 3. After how many minutes does the minute hand first catch up with the hour hand?

**Solution**

Measure in minute marks: at 3:00 the hour hand is 15 marks ahead.
The minute hand moves 1 mark per minute; the hour hand 5 marks per hour, i.e. $\frac{1}{12}$ mark per minute.
Gain per minute: $1 - \frac{1}{12} = \frac{11}{12}$ (mark). Time: $15 : \frac{11}{12} = \frac{180}{11} = 16\frac{4}{11}$ (min).
**Answer:** $16\frac{4}{11}$ min.

**Problem 46** ★

Road $AB$ is 240 km. At 7:00 two cars leave $A$ and $B$ towards each other and meet at 9:00. The speed of the car from $A$ is $\frac{3}{5}$ of the speed of the car from $B$. Find the speeds and the distance from $A$ to the meeting point.

**Solution**

Time: 2 h; sum of the speeds: $240 : 2 = 120$ (km/h).
$3 + 5 = 8$ parts; one part $120 : 8 = 15$ (km/h).
Car from $A$: 45 km/h; car from $B$: 75 km/h.
Distance from $A$: $45 \times 2 = 90$ (km).
**Answer:** 45 km/h, 75 km/h; 90 km from $A$.

**Problem 47** ★

At 7:00 car $X$ is 40 km behind car $Y$ on the same road; both drive forward, $X$ at 60 km/h and $Y$ at 40 km/h. At 8:00 car $Y$ speeds up to 50 km/h. At what time does $X$ catch $Y$?

**Solution**

From 7:00 to 8:00 the gap shrinks by $60 - 40 = 20$ (km): at 8:00 it is 20 km.

After 8:00 it shrinks by $60 - 50 = 10$ (km) each hour: $20 : 10 = 2$ (h). $X$ catches $Y$ at 10:00.

**Answer:** 10:00.

## Problem 48 ★★

At 6:00 a walker leaves $A$ for $B$ at 5 km/h. At 7:00 a cyclist leaves $A$ at 15 km/h along the same road. The cyclist overtakes the walker, rides on and reaches $B$ 1 hour before the walker. How long is $AB$?

**Solution**

Head start: 5 km; gain 10 km/h; the cyclist overtakes after 0.5 h, at 7:30, $15 \times 0.5 = 7.5$ km from $A$.

From there, for each km the walker needs $\frac{1}{5}$ h and the cyclist $\frac{1}{15}$ h: a difference of $\frac{1}{5} - \frac{1}{15} = \frac{2}{15}$ (h).

The difference is 1 h, so the remaining distance is $1 : \frac{2}{15} = 7.5$ (km).

$AB = 7.5 + 7.5 = 15$ (km).

**Answer:** 15 km.

## Problem 49 ★

Hoa's house and Lan's house are 3.4 km apart. At 7:00 Hoa walks towards Lan's house at 80 m/min; at 7:05 Lan walks towards Hoa's house at 70 m/min. At what time do they meet, and how far from Hoa's house?

**Solution**

By 7:05 Hoa has walked $80 \times 5 = 400$ (m); remaining $3\,400 - 400 = 3\,000$ (m).

Closing speed $80 + 70 = 150$ (m/min); time $3\,000 : 150 = 20$ (min); they meet at 7:25.

Distance from Hoa's house: $80 \times 25 = 2\,000$ (m) $= 2$ km.

**Answer:** 7:25; 2 km from Hoa's house.

## Problem 50 ★★

Car $X$ drives the whole road $AB$ in 6 h; car $Y$ drives it in 4 h. Car $X$ leaves $A$ for $B$; 1 hour later car $Y$ leaves $B$ for $A$. How long after car $Y$ starts do they meet?

**Solution**

In its first hour car $X$ covers $\frac{1}{6}$ of $AB$; $\frac{5}{6}$ of $AB$ remains.

Together they cover $\frac{1}{6} + \frac{1}{4} = \frac{5}{12}$ of $AB$ per hour.

Time: $\frac{5}{6} : \frac{5}{12} = 2$ (h).

**Answer:** 2 h.

# 8 Speed Ratios and Average Speed

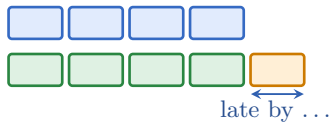
## ① Speed, distance and time depend on each other

$$s = v \times t; \qquad v = s : t; \qquad t = s : v.$$

- **Same distance:** the faster you go, the less time you need. If the speed becomes $k$ times as big, the time becomes $k$ times as small. We say speed and time are *inversely proportional*: the ratio of the times is the ratio of the speeds turned upside down.
- **Same speed:** a longer distance needs a longer time; distance and time are *directly proportional*.
- **Same time:** the faster object goes further; distance and speed are *directly proportional*.

## ② Using ratios with bar models

On the same road, if the planned speed and the actual speed are in the ratio $\frac{a}{b}$, then the planned time and the actual time are in the ratio $\frac{b}{a}$. Draw the two times as bars made of equal parts: the *difference in time* (late, early, longer, shorter) matches the *difference in parts*. From one part we get the time, then the distance.




## ③ Average speed

- **Average speed = total distance : total time.**
- In general the average speed is *not* the average of the speeds. For example, going half the way at 12 km/h and half the way at 4 km/h does *not* give 8 km/h.
- When the distance is not given ("half the way", "there and back"), choose a convenient length for it, usually a number divisible by all the speeds (*reducing to a unit*), and work out the times.

## Example 1

Huy cycles 18 km in 1 h 30 min. At the same speed, how long does he need to cycle 30 km?

**Guided solution.**

The speed is the same, so the time grows in the same ratio as the distance.

$30 : 18 = \frac{5}{3}$, so the distance is $\frac{5}{3}$ times as long.

Time: $1.5 \times \frac{5}{3} = 2.5$ (h) $= 2$ h 30 min.

*Another way: speed* $18 : 1.5 = 12$ *(km/h); time* $30 : 12 = 2.5$ *(h).*

**Answer: 2 h 30 min.**

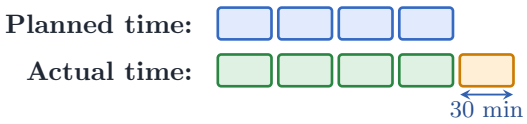
## Example 2

A car planned to drive from $A$ to $B$ at 60 km/h. Because of road works it could only drive at 48 km/h and arrived 30 minutes late. How long is road $AB$?

**Guided solution.**

Ratio of planned speed to actual speed: $\frac{60}{48} = \frac{5}{4}$.

On the same road speed and time are inversely proportional, so planned time : actual time $= \frac{4}{5}$.




Difference: $5 - 4 = 1$ part $= 30$ min. Actual time: $30 \times 5 = 150$ (min) $= 2.5$ h.

$AB = 48 \times 2.5 = 120$ (km).

**Answer: 120 km.**

## Example 3

A car drives road $AB$ in 4 h; a motorbike drives the same road in 6 h. The car is 15 km/h faster than the motorbike. Find both speeds and the length of $AB$.

**Guided solution.**

Time of car : time of motorbike $= \frac{4}{6} = \frac{2}{3}$; so speed of car : speed of motorbike $= \frac{3}{2}$.




Difference: $3 - 2 = 1$ part $= 15$ km/h.

Car: $15 \times 3 = 45$ (km/h); motorbike: $45 - 15 = 30$ (km/h).

$AB = 45 \times 4 = 180$ (km).

**Answer: car 45 km/h, motorbike 30 km/h; $AB = 180$ km.**

## Example 4

Thu cycles from $A$ to $B$ at 12 km/h. On the way back, she rides the first 6 km at 12 km/h, then a strong wind slows her to 10 km/h for the rest of the way. So the way back takes 12 minutes longer than the way there. How long is $AB$?

### Guided solution.

Let $C$ be the point 6 km from $B$. Both trips take the same time on $BC$; the extra 12 min comes from the part $CA$.




On $CA$: planned speed : actual speed $= \frac{12}{10} = \frac{6}{5}$, so planned time : actual time $= \frac{5}{6}$.

$6 - 5 = 1$ part $= 12$ min; actual time on $CA$: $12 \times 6 = 72$ (min) $= 1.2$ h.

$CA = 10 \times 1.2 = 12$ (km); $AB = 12 + 6 = 18$ (km).

**Answer: 18 km.**

## Example 5

**a)** A motorbike rides for 2 h at 45 km/h and then for 1 h at 60 km/h. Find its average speed.

**b)** Khoa cycles the first half of a road at 12 km/h, then his bike breaks and he walks the second half at 4 km/h. Find his average speed over the whole road.

### Guided solution.

**a)** Total distance: $45 \times 2 + 60 \times 1 = 150$ (km); total time: $2 + 1 = 3$ (h).

Average speed: $150 : 3 = 50$ (km/h).

**b)** 12 is divisible by both 12 and 4; suppose half the road is 12 km (a unit chosen for convenience).

First half: $12 : 12 = 1$ (unit of time); second half: $12 : 4 = 3$ (units).

Whole road: 24 km in $1 + 3 = 4$ units; average speed: $24 : 4 = 6$ (km/h).

*It is not* $(12 + 4) : 2 = 8$ *km/h: Khoa spends much longer walking than cycling.*

**Answer: a) 50 km/h; b) 6 km/h.**

## Practice

## Problem 1

A motorbike rides 96 km in 2 h 40 min. At the same speed, how far does it ride in 4 h?

### Solution

2 h 40 min $= \frac{8}{3}$ h. Speed: $96 : \frac{8}{3} = 36$ (km/h).

Distance in 4 h: $36 \times 4 = 144$ (km).

**Answer:** 144 km.

## Problem 2

A car drives 150 km in 2.5 h. At the same speed, how long does it need for 210 km?

### Solution

Speed: $150 : 2.5 = 60$ (km/h).

Time: $210 : 60 = 3.5$ (h) $= 3$ h 30 min.

**Answer:** 3 h 30 min.

## Problem 3

A cyclist and a motorbike leave a town at the same time. The motorbike's speed is 2.5 times the cyclist's. When the cyclist has ridden 14 km, how far has the motorbike gone?

### Solution

In the same time the distance is proportional to the speed.

Motorbike: $14 \times 2.5 = 35$ (km).

**Answer:** 35 km.

## Problem 4

Towns $X$ and $Y$ are 150 km apart. A car drives from $X$ to $Y$ at 60 km/h; at the same time a motorbike rides from $Y$ to $X$ at 40 km/h. When the car reaches $Y$, how far is the motorbike from $X$?

### Solution

The car needs $150 : 60 = 2.5$ (h).

In 2.5 h the motorbike rides $40 \times 2.5 = 100$ (km).

Distance to $X$: $150 - 100 = 50$ (km).

**Answer:** 50 km.

## Problem 5

A car drives the 96 km from $A$ to $B$ at 48 km/h. During the same time a motorbike leaves $A$ along the same road at a speed equal to $\frac{3}{4}$ of the car's speed. When the car reaches $B$, how far is the motorbike from $B$?

### Solution

In the same time the motorbike covers $\frac{3}{4}$ of the car's distance: $96 \times \frac{3}{4} = 72$ (km).

Distance to $B$: $96 - 72 = 24$ (km).

**Answer:** 24 km.

## Problem 6

A motorbike at 40 km/h needs 3 h for a road. A car drives 1.5 times as fast. How long does the car need for the same road?

### Solution

Same road: the speed is 1.5 times as big, so the time is 1.5 times as small.

Time: $3 : 1.5 = 2$ (h).

**Answer:** 2 h.

## Problem 7

A motorbike at 42 km/h covers a road in 2 h 30 min. A car drives the same road at 60 km/h and leaves at 6:50. At what time does the car arrive?

### Solution

Length of the road: $42 \times 2.5 = 105$ (km).

Car's time: $105 : 60 = 1.75$ (h) $= 1$ h 45 min.

Arrival: 6:50 + 1 h 45 min = 8:35.

**Answer:** 8:35.

## Problem 8

A motorbike planned to ride from $A$ to $B$ in 4 h, but it actually took 3 h 12 min because it rode 10 km/h faster. Find its actual speed.

### Solution

3 h 12 min = 3.2 h. Planned time : actual time $= 4 : 3.2 = \frac{5}{4}$.

So planned speed : actual speed $= \frac{4}{5}$; the difference 1 part $= 10$ km/h.

Actual speed: $10 \times 5 = 50$ (km/h).

**Answer:** 50 km/h.

**Problem 9**

A coach planned to drive at 50 km/h but, because of heavy traffic, could only drive at 40 km/h and arrived 36 minutes late. How long was the journey?

**Solution**

Planned speed : actual speed $= \frac{50}{40} = \frac{5}{4}$; planned time : actual time $= \frac{4}{5}$.

1 part $= 36$ min; actual time: $36 \times 5 = 180$ (min) $= 3$ h.

Distance: $40 \times 3 = 120$ (km).

**Answer:** 120 km.

**Problem 10**

A car planned to drive at 45 km/h but actually drove at 54 km/h and arrived 20 minutes early. How long was the road?

**Solution**

Planned speed : actual speed $= \frac{45}{54} = \frac{5}{6}$; planned time : actual time $= \frac{6}{5}$.

1 part $= 20$ min; actual time: $20 \times 5 = 100$ (min) $= \frac{5}{3}$ h.

Road: $54 \times \frac{5}{3} = 90$ (km).

**Answer:** 90 km.

**Problem 11**

Bình cycles home from school at 15 km/h. Halfway, his tyre goes flat and he walks the rest of the way at 5 km/h. The whole trip takes 40 minutes. How far is Bình's home from school?

**Solution**

For each km of half the road: cycling takes $\frac{1}{15}$ h, walking $\frac{1}{5}$ h; together $\frac{1}{15} + \frac{1}{5} = \frac{4}{15}$ (h).

40 min $= \frac{2}{3}$ h. Half the road: $\frac{2}{3} : \frac{4}{15} = 2.5$ (km).

Whole road: $2.5 \times 2 = 5$ (km).

**Answer:** 5 km.

**Problem 12**

A coach drives the first half of a road at 60 km/h; then rain slows it to 40 km/h for the second half. The whole trip takes 5 h. How long is the road?

**Solution**

For each km of half the road the coach needs $\frac{1}{60} + \frac{1}{40} = \frac{1}{24}$ (h).

Half the road: $5 : \frac{1}{24} = 120$ (km); whole road: 240 km.

*Check:* $120 : 60 + 120 : 40 = 2 + 3 = 5$ (h).

**Answer:** 240 km.

**Problem 13**

A delivery van leaves $A$ at 7:15, drives to $B$ at 40 km/h, spends 45 minutes unloading, then drives straight back at 60 km/h and arrives at $A$ at 11:30. How long is road $AB$?

**Solution**

From 7:15 to 11:30: 4 h 15 min; driving time: 4 h 15 min − 45 min = 3.5 h.

For each km of $AB$ the van needs $\frac{1}{40} + \frac{1}{60} = \frac{1}{24}$ (h) there and back.

$AB = 3.5 : \frac{1}{24} = 84$ (km).

**Answer:** 84 km.

**Problem 14**

Mr Tâm rides from $A$ to $B$ in 2 h. On the way back he rides 12 km/h slower, and the trip takes 30 minutes longer. How long is $AB$?

**Solution**

Time there : time back $= 2 : 2.5 = \frac{4}{5}$; so speed there : speed back $= \frac{5}{4}$.

1 part = 12 km/h; speed there: $12 \times 5 = 60$ (km/h).

$AB = 60 \times 2 = 120$ (km).

**Answer:** 120 km.

## Problem 15

A driver works out that at 50 km/h he will arrive at 10:12, and at 60 km/h he will arrive at 9:42. How long is the journey?

## Solution

The difference in arrival times is 30 min.

Speeds $\frac{50}{60} = \frac{5}{6}$, so times $\frac{6}{5}$; 1 part = 30 min.

Time at 60 km/h: $30 \times 5 = 150$ (min) $= 2.5$ h. Distance: $60 \times 2.5 = 150$ (km).

**Answer:** 150 km.

## Problem 16

A car must get from $A$ to $B$ in a fixed time. At 40 km/h it would be 20 min late; at 50 km/h it would be 16 min early. Find the length of $AB$ and the fixed time.

## Solution

The two times differ by $20 + 16 = 36$ (min).

Speeds $\frac{4}{5}$, so times $\frac{5}{4}$; 1 part = 36 min.

Time at 50 km/h: $36 \times 4 = 144$ (min) $= 2.4$ h; $AB = 50 \times 2.4 = 120$ (km).

Fixed time: $144 + 16 = 160$ (min) $= 2$ h 40 min.

**Answer:** 120 km; 2 h 40 min.

## Problem 17

Brothers Quân and Khoa ride from home to their grandmother's house. Quân rides at 45 km/h. Khoa leaves 9 minutes after Quân, rides at 54 km/h and arrives 3 minutes before him. How far is it to Grandma's house?

## Solution

Khoa's time is $9 + 3 = 12$ (min) shorter.

Speeds $\frac{45}{54} = \frac{5}{6}$, so times $\frac{6}{5}$; 1 part = 12 min.

Quân's time: $12 \times 6 = 72$ (min) $= 1.2$ h. Distance: $45 \times 1.2 = 54$ (km).

**Answer:** 54 km.

## Problem 18

Minh cycles from home to school at 12 km/h. On the way back he rides the first 2 km at 12 km/h, then speeds up to 16 km/h. The way back takes 5 minutes less. How far is school from home?

**Solution**

Only the part after the first 2 km matters.

Speeds there $\frac{12}{16} = \frac{3}{4}$, so times $\frac{4}{3}$; 1 part = 5 min.

Time at 16 km/h: $5 \times 3 = 15$ (min) $= 0.25$ h; length $16 \times 0.25 = 4$ (km).

Home to school: $2 + 4 = 6$ (km).

**Answer:** 6 km.

**Problem 19**

A coach leaves the city at 7:00 for a town 150 km away, planning to drive the whole way at 50 km/h. On the way it stops for 12 minutes; then it drives on at 60 km/h and arrives 3 minutes earlier than planned. How far from the city did it stop?

**Solution**

On the part after the stop the coach saved $12 + 3 = 15$ (min).

Speeds $\frac{50}{60} = \frac{5}{6}$, so times $\frac{6}{5}$; 1 part = 15 min.

Planned time for that part: $15 \times 6 = 90$ (min) $= 1.5$ h; its length: $50 \times 1.5 = 75$ (km).

Stop: $150 - 75 = 75$ (km) from the city.

**Answer:** 75 km.

**Problem 20**

Two cars leave at the same time. In the same time car $X$ goes 150 km and car $Y$ goes 120 km. Car $X$ is 12 km/h faster. Find both speeds.

**Solution**

Same time: speeds are in the ratio of the distances, $\frac{150}{120} = \frac{5}{4}$.

1 part = 12 km/h.

$X$: 60 km/h; $Y$: 48 km/h.

**Answer:** 60 km/h and 48 km/h.

**Problem 21**

A car drives road $AB$ in 3 h; a motorbike needs 5 h. The car is 24 km/h faster. Find both speeds and the length of $AB$.

**Solution**

Times $\frac{3}{5}$, so speeds $\frac{5}{3}$; difference 2 parts = 24 km/h, 1 part = 12 km/h.

Car: 60 km/h; motorbike: 36 km/h. $AB = 60 \times 3 = 180$ (km).

**Answer:** 60 km/h, 36 km/h; 180 km.

**Problem 22**

A lorry drives for 1.5 h at 40 km/h and then for 0.5 h at 60 km/h. Find its average speed.

**Solution**

Distance: $40 \times 1.5 + 60 \times 0.5 = 90$ (km); time: 2 h.

Average speed: $90 : 2 = 45$ (km/h).

**Answer:** 45 km/h.

**Problem 23**

Grandpa walks 2 km in 30 min, then 3 km in 45 min. Find his average speed in km/h.

**Solution**

Distance: 5 km; time: 75 min = 1.25 h.

Average speed: $5 : 1.25 = 4$ (km/h).

**Answer:** 4 km/h.

**Problem 24**

A car drives a road of 160 km: the first half at 60 km/h and the second half at 40 km/h. Find its average speed.

**Solution**

Times: $80 : 60 = \frac{4}{3}$ (h) and $80 : 40 = 2$ (h); total $\frac{10}{3}$ h.

Average speed: $160 : \frac{10}{3} = 48$ (km/h).

**Answer:** 48 km/h.

**Problem 25**

A motorbike rides from $A$ to $B$ at 30 km/h and returns along the same road at 20 km/h. Find its average speed for the round trip.

**Solution**

Choose $AB = 60$ km (divisible by 30 and 20).

Times: $60 : 30 = 2$ and $60 : 20 = 3$ (units); total 5.

Average speed: $120 : 5 = 24$ (km/h).

**Answer:** 24 km/h.

**Problem 26**

A car drives for 3 h at an average speed of 50 km/h. In the first 2 h it drove at 45 km/h. What was its speed in the third hour?

**Solution**

Total distance: $50 \times 3 = 150$ (km).

First 2 h: $45 \times 2 = 90$ (km).

Third hour: $150 - 90 = 60$ (km), so 60 km/h.

**Answer:** 60 km/h.

**Problem 27**

A hiker climbs a hill at 3 km/h and comes down the same path at 6 km/h. Find the average speed for the whole walk.

**Solution**

Choose the path to be 6 km.

Up: $6 : 3 = 2$ (h); down: $6 : 6 = 1$ (h).

Average: $12 : 3 = 4$ (km/h).

**Answer:** 4 km/h.

**Problem 28**

A road is divided into three equal parts. A cyclist rides them at 30 km/h, 20 km/h and 60 km/h. Find the average speed over the whole road.

**Solution**

Choose each part to be 60 km.

Times: 2 h, 3 h, 1 h; total 6 h for 180 km.

Average: $180 : 6 = 30$ (km/h).

**Answer:** 30 km/h.

### Problem 29

In the time Lan walks 600 m, Mai cycles 1.5 km. Lan walks at 4 km/h. Find Mai's speed.

### Solution

Same time: speeds are proportional to distances; $1\,500 : 600 = 2.5$.

Mai: $4 \times 2.5 = 10$ (km/h).

**Answer:** 10 km/h.

### Problem 30

A car usually needs 2 h for a road. Today it drives at $\frac{4}{5}$ of its usual speed. How many minutes late is it?

### Solution

Speed $\times\frac{4}{5}$, so time $\times\frac{5}{4}$: $2 \times \frac{5}{4} = 2.5$ (h).

Late by 0.5 h = 30 min.

**Answer:** 30 min.

## Challenge – Extension

### Problem 31 ⋆

Road $AB$ consists of an uphill part and a downhill part. Vy walks from $A$ to $B$ and back, going uphill at 4 km/h and downhill at 6 km/h. The round trip takes 5 h. How long is $AB$?

### Solution

Over the round trip each km of the road is walked once uphill and once downhill: $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ (h) per km.

$AB = 5 : \frac{5}{12} = 12$ (km).

**Answer:** 12 km.

### Problem 32 ⋆⋆

Road $AB$ has an uphill part followed by a downhill part. Uphill speed 4 km/h, downhill speed 6 km/h. Going from $A$ to $B$ takes 2 h 20 min; returning takes 2 h 40 min. How long is $AB$, and how long is the uphill part on the way from $A$?

**Solution**

Round trip: 5 h; as in Problem 31, $AB = 5 : \frac{5}{12} = 12$ (km).

If all 12 km from $A$ were downhill, the trip would take $12 : 6 = 2$ h; it takes $\frac{1}{3}$ h more.

Each uphill km takes $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ h longer, so the uphill part is $\frac{1}{3} : \frac{1}{12} = 4$ (km).

**Answer:** $AB = 12$ km; uphill 4 km.

**Problem 33** *

A car drives the first half of a road at 40 km/h. How fast must it go on the second half for the average speed to be 48 km/h?

**Solution**

Choose half the road = 240 km. Whole road 480 km at 48 km/h: 10 h.

First half: $240 : 40 = 6$ (h); second half must take $10 - 6 = 4$ (h).

Speed: $240 : 4 = 60$ (km/h).

**Answer:** 60 km/h.

**Problem 34** **

Lâm cycles the first half of a road at 30 km/h. Can he ride the second half fast enough to make his average speed 60 km/h?

**Solution**

Choose half the road = 30 km. Average 60 km/h on 60 km needs 1 h in total.

But the first half alone already takes $30 : 30 = 1$ h, leaving no time for the second half.

So it is impossible.

**Answer:** No, it is impossible.

**Problem 35** *

A cyclist leaves at 7:00 and arrives at 8:30, 18 km away, after a 15-minute rest on the way. Find **a)** the average speed for the whole trip; **b)** the speed while riding.

**Solution**

**a)** Total time 1.5 h: $18 : 1.5 = 12$ (km/h).

**b)** Riding time $1.5 - 0.25 = 1.25$ (h): $18 : 1.25 = 14.4$ (km/h).

**Answer:** a) 12 km/h; b) 14.4 km/h.

## Problem 36 *

A driver works out that at 45 km/h he arrives at 10:00, and at 60 km/h at 9:15. How long is the journey, and at what time does he set off?

### Solution

Difference: 45 min. Speeds $\frac{3}{4}$, so times $\frac{4}{3}$; 1 part = 45 min.

Time at 60 km/h: $45 \times 3 = 135$ (min) $= 2.25$ h; distance $60 \times 2.25 = 135$ (km).

Departure: 9:15 − 2 h 15 min = 7:00.

**Answer:** 135 km; 7:00.

## Problem 37 *

Road $AB$ is 120 km. A car plans to drive at 40 km/h. After 1 h it speeds up by 10 km/h. How many minutes early does it arrive?

### Solution

Planned time: $120 : 40 = 3$ (h).

Actual: 1 h for 40 km, then $80 : 50 = 1.6$ (h); total 2.6 h.

Early by 0.4 h = 24 min.

**Answer:** 24 min.

## Problem 38 **

A car planned to drive at 50 km/h. After 1 h it had to slow down to 40 km/h, and it arrived 24 minutes late. How long was the road?

### Solution

Only the part after the first hour matters. Speeds $\frac{5}{4}$, so times $\frac{4}{5}$; 1 part = 24 min.

Actual time on that part: $24 \times 5 = 120$ (min) $= 2$ h; its length $40 \times 2 = 80$ (km).

Road: $50 \times 1 + 80 = 130$ (km).

**Answer:** 130 km.

## Problem 39 *

A car drives from $A$ to $B$ in 2 h at 45 km/h and returns along the same road in 3 h. Find the average speed for the round trip.

**Solution**

$AB = 45 \times 2 = 90$ (km). Round trip: 180 km in 5 h.

Average: $180 : 5 = 36$ (km/h).

**Answer:** 36 km/h.

**Problem 40** *

**a)** A car drives half of the *time* at 40 km/h and the other half of the time at 60 km/h. **b)** Another car drives half of the *distance* at 40 km/h and the other half at 60 km/h. Find each average speed.

**Solution**

**a)** Take 2 h: $40 + 60 = 100$ km in 2 h: 50 km/h.

**b)** Take each half = 120 km: times 3 h and 2 h; 240 km in 5 h: 48 km/h.

**Answer:** a) 50 km/h; b) 48 km/h.

**Problem 41** **

If a car increases its speed by a quarter, it arrives 30 minutes earlier. How long does the trip take at the usual speed?

**Solution**

New speed $= \frac{5}{4}$ of the usual, so new time $= \frac{4}{5}$ of the usual.

Saved: $\frac{1}{5}$ of the usual time = 30 min. Usual time: $30 \times 5 = 150$ (min) = 2 h 30 min.

**Answer:** 2 h 30 min.

**Problem 42** *

Two cyclists leave the same place at the same time in the same direction. After 1.5 h the first is 6 km ahead of the second. The first rides at $\frac{5}{4}$ of the second's speed. Find their speeds.

**Solution**

Difference of the speeds: $6 : 1.5 = 4$ (km/h) = 1 part.

Second: $4 \times 4 = 16$ (km/h); first: $4 \times 5 = 20$ (km/h).

**Answer:** 20 km/h and 16 km/h.

## Problem 43 *

Phong cycles from $A$ to $B$ at 18 km/h. Coming back he rides the first 6 km at 18 km/h, then 15 km/h, and the way back takes 8 minutes longer. How long is $AB$?

### Solution

On the part after the first 6 km: speeds $\frac{18}{15} = \frac{6}{5}$, so times $\frac{5}{6}$; 1 part = 8 min.

Time at 15 km/h: $8 \times 6 = 48$ (min) $= 0.8$ h; length $15 \times 0.8 = 12$ (km).

$AB = 6 + 12 = 18$ (km).

**Answer:** 18 km.

## Problem 44 *

A car drives from $A$ to $B$ at 36 km/h and back at 45 km/h. Find the average speed for the round trip.

### Solution

Choose $AB = 180$ km (divisible by 36 and 45).

Times: 5 h and 4 h; 360 km in 9 h.

Average: 40 km/h.

**Answer:** 40 km/h.

## Problem 45 *

Ngân walks to school at 5 km/h and walks home along the same road at 3 km/h. Altogether she walks for 48 minutes. How far is school?

### Solution

Per km: $\frac{1}{5} + \frac{1}{3} = \frac{8}{15}$ (h) there and back.

48 min $= 0.8$ h. Distance: $0.8 : \frac{8}{15} = 1.5$ (km).

**Answer:** 1.5 km.

## Problem 46 **

A motorbike leaves $A$ at 7:00 and a car leaves $A$ at 8:00 on the same road; they both reach $B$ at the same moment. The car is 1.5 times as fast as the motorbike, and $AB = 120$ km. At what time do they reach $B$, and what are their speeds?

**Solution**

Speeds $\frac{3}{2}$ (car : motorbike), so times $\frac{2}{3}$; the difference 1 part = 1 h.

Motorbike's time: 3 h, so they reach $B$ at 10:00.

Motorbike: $120 : 3 = 40$ (km/h); car: $120 : 2 = 60$ (km/h).

**Answer:** 10:00; motorbike 40 km/h, car 60 km/h.

**Problem 47** *

In 2 h car $X$ goes 30 km further than car $Y$. Their speeds are in the ratio 7 : 5. Find both speeds.

**Solution**

Difference of speeds: $30 : 2 = 15$ (km/h) $= 7 - 5 = 2$ parts; 1 part = 7.5 km/h.

$X$: 52.5 km/h; $Y$: 37.5 km/h.

**Answer:** 52.5 km/h and 37.5 km/h.

**Problem 48** *

Hà must cover 32 km in 2 h. She rides the first 1 h 20 min at 15 km/h. At what speed must she ride the rest?

**Solution**

First part: $15 \times \frac{4}{3} = 20$ (km). Left: 12 km in 40 min $= \frac{2}{3}$ h.

Speed: $12 : \frac{2}{3} = 18$ (km/h).

**Answer:** 18 km/h.

**Problem 49** *

Anh walks 2.4 km in 30 min; Bảo walks 3 km in 40 min. Each walks 6 km. How many minutes longer does the slower one take?

**Solution**

Anh: $2.4 : 0.5 = 4.8$ (km/h); Bảo: $3 : \frac{2}{3} = 4.5$ (km/h). Bảo is slower.

Times for 6 km: Anh $6 : 4.8 = 1.25$ h $= 75$ min; Bảo $6 : 4.5 = \frac{4}{3}$ h $= 80$ min.

Difference: 5 min.

**Answer:** Bảo, 5 min longer.

## Problem 50 ★★

A car drives for 2 h at 60 km/h and then for a while at 40 km/h. Its average speed for the whole trip is 48 km/h. How long did it drive at 40 km/h?

## Solution

In the first 2 h the car is $(60 - 48) \times 2 = 24$ (km) *ahead* of a car going steadily at 48 km/h.
At 40 km/h it loses $48 - 40 = 8$ (km) every hour against that car.
To lose the 24 km: $24 : 8 = 3$ (h).
**Answer:** 3 h.

# 9 River Currents and Long Objects

## ① Boats on a river

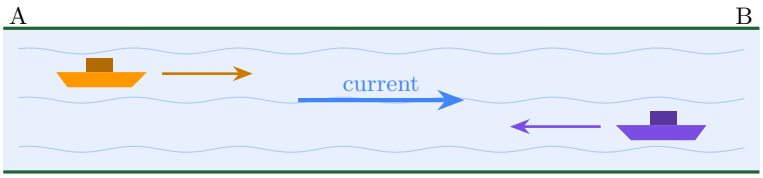
- The water of a river flows in one direction. A boat going *with* the current (downstream) is helped by the water; a boat going *against* the current (upstream) is held back by it.
- Let $v_{\text{boat}}$ be the speed of the boat in still water and $v_{\text{current}}$ the speed of the current. Then

$$v_{\text{down}} = v_{\text{boat}} + v_{\text{current}}, \qquad v_{\text{up}} = v_{\text{boat}} - v_{\text{current}}.$$

- Hence $v_{\text{down}} - v_{\text{up}} = 2 \times v_{\text{current}}$, so

$$v_{\text{current}} = (v_{\text{down}} - v_{\text{up}}) : 2, \qquad v_{\text{boat}} = (v_{\text{down}} + v_{\text{up}}) : 2.$$

- A raft, a leaf or a log floating on the water (no engine, no oars) moves at exactly the speed of the current.




downstream: $v_{\text{boat}} + v_{\text{current}}$

upstream: $v_{\text{boat}} - v_{\text{current}}$

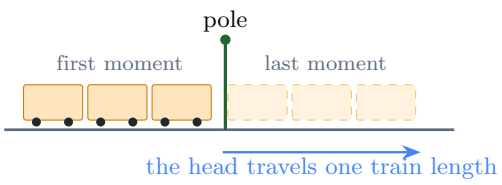
**Note:** when the same stretch AB is travelled both ways, the times are in the *inverse* ratio of the speeds: the faster trip takes less time.

## ② Trains: moving objects that have a length

A train is long, so "the train has passed" something only when its *tail* has gone by. We follow the *head* of the train and ask how far it travels.

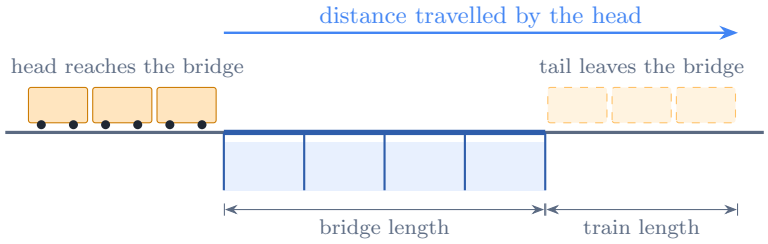
**a) Passing a pole** (a tree, a person standing still): from the moment the head reaches the pole until the tail leaves it, the head travels *one train length*.

time to pass a pole = train length : train speed.



**b) Crossing a bridge** (or a tunnel, a platform): from the moment the head reaches the bridge until the tail leaves it, the head travels the *bridge length plus the train length*.

$$\text{time to cross a bridge} = (\text{train length} + \text{bridge length}) : \text{train speed}.$$




**Units:** 1 m/s = 3.6 km/h. For example 54 km/h = 54 000 m : 3 600 s = 15 m/s.

## ③ A train and another moving object

- **Two trains in opposite directions** (on parallel tracks): from "the heads meet" to "the tails part", together they cover the sum of the two lengths.

$$\text{time} = (\text{length}_1 + \text{length}_2) : (v_1 + v_2).$$

- **One train overtaking another** (same direction): the faster train must gain the sum of the two lengths.

$$\text{time} = (\text{length}_1 + \text{length}_2) : (v_1 - v_2).$$

- **A train and a cyclist, walker or car** (a short object, its length ignored): replace the sum of the lengths by the train length only; opposite directions: time = train length : $(v_{\text{train}} + v)$; same direction: time = train length : $(v_{\text{train}} - v)$.
- A passenger *sitting* in one train who watches another train go past sees only the length of the *other* train.

**Example 1**

The river stretch AB is 36 km long and the current flows from A to B at 3 km/h. At 6:30 a boat whose speed in still water is 15 km/h leaves A for B, rests 30 minutes at B and then returns to A. At what time does it get back to A?

**Guided solution.**

Downstream speed: $15 + 3 = 18$ (km/h); time from A to B: $36 : 18 = 2$ (h).
Upstream speed: $15 - 3 = 12$ (km/h); time from B to A: $36 : 12 = 3$ (h).
Back at A: 6 h 30 min + 2 h + 30 min + 3 h = 12 h 00 min.

**Answer:** at 12:00 (noon).

## Example 2

A canoe goes downstream from A to B in 4 hours and comes back upstream from B to A in 6 hours. How long does a raft take to drift from A to B?

### Guided solution.

*Method 1 (choose a length).* Let AB be 12 equal units (12 is divisible by both 4 and 6).
Downstream speed: $12 : 4 = 3$ (units per hour); upstream speed: $12 : 6 = 2$ (units per hour).
Speed of the current (= speed of the raft): $(3 - 2) : 2 = 0.5$ (units per hour).
Time for the raft: $12 : 0.5 = 24$ (h).

*Method 2 (fractions of AB).* In 1 hour the canoe covers $\frac{1}{4}$ of AB downstream and $\frac{1}{6}$ of AB upstream.

In 1 hour the current carries the raft $\left(\frac{1}{4} - \frac{1}{6}\right) : 2 = \frac{1}{24}$ of AB, so the raft needs 24 hours.

**Answer:** 24 hours.

## Example 3

A train travelling at 54 km/h passes an electricity pole in 8 seconds.

**a)** How long is the train?

**b)** At the same speed, how long does it take to cross a bridge 330 m long?

### Guided solution.

54 km/h = 54 000 m : 3 600 s = 15 m/s.
**a)** Passing the pole, the train travels its own length: $15 \times 8 = 120$ (m).
**b)** Crossing the bridge, it travels $120 + 330 = 450$ (m); time: $450 : 15 = 30$ (s).
**Answer: a)** 120 m; **b)** 30 seconds.

## Example 4

Train P is 160 m long and runs at 20 m/s. Train Q is 140 m long and runs at 10 m/s on a parallel track. How long does it take them to pass each other

**a)** if they run in opposite directions?

**b)** if P overtakes Q (same direction)?

### Guided solution.

In both cases the distance to be covered is the sum of the lengths: $160 + 140 = 300$ (m).
**a)** Opposite directions: they close up $20 + 10 = 30$ m each second; time: $300 : 30 = 10$ (s).

**b)** Same direction: P gains $20 - 10 = 10$ m each second; time: $300 : 10 = 30$ (s).

**Answer: a)** 10 seconds; **b)** 30 seconds.

**Example 5**

A train 150 m long crosses a bridge 210 m long in 24 seconds. The same train overtakes a cyclist riding in the same direction in 12 seconds. Find the cyclist's speed.

**Guided solution.**

Speed of the train: $(150 + 210) : 24 = 15$ (m/s).

*Method 1.* In 12 s the head of the train travels $15 \times 12 = 180$ (m). It started level with the cyclist and finished one train length ahead, so the cyclist rode $180 - 150 = 30$ (m).

Cyclist's speed: $30 : 12 = 2.5$ (m/s).

*Method 2.* The train gains 150 m in 12 s, so the difference of the speeds is $150 : 12 = 12.5$ (m/s).

Cyclist's speed: $15 - 12.5 = 2.5$ (m/s) $= 9$ km/h.

**Answer:** 2.5 m/s (that is, 9 km/h).

## Practice

**Problem 1**

A boat's speed in still water is 16 km/h and the current flows at 2 km/h. Find the boat's speed downstream and its speed upstream.

**Solution**

Downstream: $16 + 2 = 18$ (km/h).

Upstream: $16 - 2 = 14$ (km/h).

**Answer:** 18 km/h downstream, 14 km/h upstream.

**Problem 2**

At 7:15 a boat leaves A and goes downstream, arriving at B at 9:45. Its speed in still water is 21 km/h and the current is 3 km/h. How long is the river stretch AB?

**Solution**

Travelling time: 9 h 45 min $-$ 7 h 15 min $=$ 2 h 30 min $=$ 2.5 h.

Downstream speed: $21 + 3 = 24$ (km/h).

AB: $24 \times 2.5 = 60$ (km).

**Answer:** 60 km.

## Problem 3

A canoe goes 45 km downstream in 1 hour 30 minutes and comes back the same 45 km upstream in 2 hours 15 minutes. Find the speed of the canoe in still water and the speed of the current.

### Solution

1 h 30 min = 1.5 h; 2 h 15 min = 2.25 h.

Downstream speed: $45 : 1.5 = 30$ (km/h); upstream speed: $45 : 2.25 = 20$ (km/h).

Still water: $(30 + 20) : 2 = 25$ (km/h); current: $(30 - 20) : 2 = 5$ (km/h).

**Answer:** 25 km/h and 5 km/h.

## Problem 4

A canoe goes downstream from A to B, a distance of 48 km, in 3 hours, and back upstream in 4 hours. How long does a leaf take to float from A to B?

### Solution

Downstream speed: $48 : 3 = 16$ (km/h); upstream speed: $48 : 4 = 12$ (km/h).

Current (= the leaf's speed): $(16 - 12) : 2 = 2$ (km/h).

Time for the leaf: $48 : 2 = 24$ (h).

**Answer:** 24 hours.

## Problem 5

A ferry goes downstream at 22 km/h and upstream at 16 km/h. Find its speed in still water and the speed of the current.

### Solution

Still water: $(22 + 16) : 2 = 19$ (km/h).

Current: $(22 - 16) : 2 = 3$ (km/h).

**Answer:** 19 km/h and 3 km/h.

## Problem 6

A boat's speed in still water is 12 km/h and the current flows at 2.5 km/h. How long does the boat take to go 38 km upstream?

### Solution

Upstream speed: $12 - 2.5 = 9.5$ (km/h).

Time: $38 : 9.5 = 4$ (h).

**Answer:** 4 hours.

## Problem 7

A boat with speed 18 km/h in still water goes downstream from A to B in 2 hours. The current is 3 km/h. How long does it take to come back from B to A?

### Solution

Downstream speed: $18 + 3 = 21$ (km/h); AB: $21 \times 2 = 42$ (km).

Upstream speed: $18 - 3 = 15$ (km/h); time: $42 : 15 = 2.8$ (h) $= 2$ h 48 min.

**Answer:** 2 hours 48 minutes.

## Problem 8

A raft drifts 7.5 km in 2.5 hours on a river. On the same river a boat whose speed in still water is 15 km/h goes 36 km upstream. How long does the boat take?

### Solution

Current: $7.5 : 2.5 = 3$ (km/h).

Upstream speed of the boat: $15 - 3 = 12$ (km/h).

Time: $36 : 12 = 3$ (h).

**Answer:** 3 hours.

## Problem 9

A motorboat with speed 20 km/h in still water goes downstream from A to B and immediately comes back to A. AB is 48 km and the current is 4 km/h. How long is the whole trip?

### Solution

Downstream: $48 : (20 + 4) = 2$ (h).

Upstream: $48 : (20 - 4) = 3$ (h).

Whole trip: $2 + 3 = 5$ (h).

**Answer:** 5 hours.

## Problem 10

A boat's downstream speed is 1.5 times its upstream speed. The current flows at 2 km/h. Find the speed of the boat in still water.

### Solution

Downstream speed − upstream speed $= 2 \times 2 = 4$ (km/h).

Upstream speed 1 part, downstream speed 1.5 parts; the difference 0.5 part is 4 km/h.

Upstream speed: $4 : 0.5 = 8$ (km/h); downstream speed: $8 \times 1.5 = 12$ (km/h).

Still water: $(12 + 8) : 2 = 10$ (km/h).

**Answer:** 10 km/h.

### Problem 11

A boat goes downstream from A to B in 3 hours and back upstream in 4 hours. The current is 2 km/h. How long is AB?

### Solution

Same distance, so the speeds are in the inverse ratio of the times: downstream : upstream $= 4 : 3$.

Their difference is $2 \times 2 = 4$ (km/h), matching $4 - 3 = 1$ part.

Downstream speed: $4 \times 4 = 16$ (km/h); AB: $16 \times 3 = 48$ (km).

**Answer:** 48 km.

### Problem 12

A canoe goes downstream from C to D in 2 hours and back upstream in 3 hours. The current is 2.5 km/h. Find the speed of the canoe in still water.

### Solution

Downstream : upstream speed $= 3 : 2$; the difference $2 \times 2.5 = 5$ (km/h) is 1 part.

Downstream speed: 15 km/h; upstream speed: 10 km/h.

Still water: $(15 + 10) : 2 = 12.5$ (km/h).

**Answer:** 12.5 km/h.

### Problem 13

A barge takes 5 hours to go downstream from A to B and 7 hours to go back. How long does a log take to drift from A to B?

### Solution

In 1 hour the barge covers $\frac{1}{5}$ of AB downstream and $\frac{1}{7}$ of AB upstream.

In 1 hour the current carries the log $\left(\frac{1}{5} - \frac{1}{7}\right) : 2 = \frac{2}{35} : 2 = \frac{1}{35}$ of AB.

So the log needs 35 hours.

**Answer:** 35 hours.

## Problem 14

A boat with speed 15 km/h in still water goes downstream from A to B and at once returns to A. The current is 3 km/h and the whole trip takes 5 hours. How long is AB?

### Solution

Downstream speed 18 km/h, upstream speed 12 km/h.
Times are in the inverse ratio: downstream : upstream $= 12 : 18 = 2 : 3$; 5 parts = 5 h, so 1 part = 1 h.
Downstream time 2 h; AB: $18 \times 2 = 36$ (km). (*Check:* $36 : 12 = 3$ h upstream.)
**Answer:** 36 km.

## Problem 15

Nam rows at 4 km/h in still water. On a river with a current of 1 km/h he rows 6 km upstream and then rows back to the start. How long does he take altogether?

### Solution

Upstream: $6 : (4 - 1) = 2$ (h).
Downstream: $6 : (4 + 1) = 1.2$ (h).
Altogether: 3.2 h = 3 h 12 min.
**Answer:** 3 hours 12 minutes.

## Problem 16

A train travelling at 72 km/h passes a pole in 9 seconds. How long is the train?

### Solution

72 km/h = 72 000 m : 3 600 s = 20 m/s.
Train length: $20 \times 9 = 180$ (m).
**Answer:** 180 m.

## Problem 17

A train 240 m long goes past a man standing beside the track in 16 seconds. Find its speed in m/s and in km/h.

### Solution

Speed: $240 : 16 = 15$ (m/s).
$15 \text{ m/s} = 15 \times 3.6 = 54$ km/h.
**Answer:** 15 m/s = 54 km/h.

## Problem 18

A train 150 m long runs at 15 m/s. How long does it take to cross a bridge 300 m long?

### Solution

Distance travelled by the head: $150 + 300 = 450$ (m).

Time: $450 : 15 = 30$ (s).

**Answer:** 30 seconds.

## Problem 19

A train running at 54 km/h crosses a bridge 0.6 km long in 48 seconds. How long is the train?

### Solution

54 km/h = 15 m/s; 0.6 km = 600 m.

Distance in 48 s: $15 \times 48 = 720$ (m).

Train length: $720 - 600 = 120$ (m).

**Answer:** 120 m.

## Problem 20

A train passes a pole in 10 seconds and crosses a bridge 360 m long in 34 seconds. Find the speed and the length of the train.

### Solution

The extra $34 - 10 = 24$ (s) is the time to travel the bridge length 360 m.

Speed: $360 : 24 = 15$ (m/s).

Length: $15 \times 10 = 150$ (m).

**Answer:** 15 m/s and 150 m.

## Problem 21

A train 200 m long runs at 20 m/s through a tunnel 1 km long.

**a)** How long is it from the moment the head enters the tunnel until the tail leaves it?

**b)** For how long is the whole train inside the tunnel?

### Solution

1 km = 1 000 m.

**a)** The head travels $1\,000 + 200 = 1\,200$ (m): $1\,200 : 20 = 60$ (s).

**b)** The whole train is inside from "tail enters" to "head leaves"; the head travels $1\,000 - 200 = 800$

(m): $800 : 20 = 40$ (s).

**Answer: a)** 60 s; **b)** 40 s.

### Problem 22

A train 180 m long passes a cyclist riding towards it at 18 km/h in 9 seconds. Find the speed of the train in km/h.

### Solution

18 km/h = 5 m/s.

Sum of the speeds: $180 : 9 = 20$ (m/s).

Train: $20 - 5 = 15$ (m/s) = 54 km/h.

**Answer:** 54 km/h.

### Problem 23

A train 200 m long overtakes a jogger running at 2 m/s in the same direction. It takes 25 seconds. Find the speed of the train in km/h.

### Solution

Difference of the speeds: $200 : 25 = 8$ (m/s).

Train: $8 + 2 = 10$ (m/s) = 36 km/h.

**Answer:** 36 km/h.

### Problem 24

A train 120 m long at 54 km/h meets a train 180 m long at 36 km/h coming the other way on a parallel track. How long do they take to pass each other?

### Solution

54 km/h = 15 m/s; 36 km/h = 10 m/s.

Sum of lengths: 300 m; sum of speeds: 25 m/s.

Time: $300 : 25 = 12$ (s).

**Answer:** 12 seconds.

### Problem 25

An express train 160 m long at 20 m/s overtakes a goods train 140 m long at 15 m/s on a parallel track. How long does the overtaking take?

**Solution**

Sum of lengths: $160 + 140 = 300$ (m).
Difference of speeds: $20 - 15 = 5$ (m/s).
Time: $300 : 5 = 60$ (s) $= 1$ min.
**Answer:** 1 minute.

**Problem 26**

A car at 72 km/h drives alongside a train going the same way at 54 km/h. It takes the car 40 seconds to get from the tail to the head of the train. How long is the train?

**Solution**

72 km/h $= 20$ m/s; 54 km/h $= 15$ m/s.
The car gains $20 - 15 = 5$ m each second.
Train length: $5 \times 40 = 200$ (m).
**Answer:** 200 m.

**Problem 27**

Lan sits by the window of a train running at 15 m/s. A train on the next track, running the other way at 10 m/s, takes 9 seconds to go past her window. How long is that train?

**Solution**

Lan sees only the other train's length go by, at $15 + 10 = 25$ (m/s).
Length: $25 \times 9 = 225$ (m).
**Answer:** 225 m.

**Problem 28**

A train passes a signal post in 10 seconds and passes along a platform 150 m long in 16 seconds. Find the length and the speed of the train.

**Solution**

The platform needs $16 - 10 = 6$ (s) more, so the speed is $150 : 6 = 25$ (m/s).
Length: $25 \times 10 = 250$ (m).
**Answer:** 250 m and 25 m/s.

**Problem 29**

A train 150 m long runs at 15 m/s and crosses a bridge in 40 seconds. How long is the bridge?

**Solution**

Distance travelled: $15 \times 40 = 600$ (m).

Bridge: $600 - 150 = 450$ (m).

**Answer:** 450 m.

**Problem 30**

A train crosses a bridge 240 m long in 25 seconds and a bridge 420 m long in 37 seconds, at the same speed. Find the speed and the length of the train.

**Solution**

The second bridge is $420 - 240 = 180$ (m) longer and takes $37 - 25 = 12$ (s) longer.

Speed: $180 : 12 = 15$ (m/s).

Distance for the first bridge: $15 \times 25 = 375$ (m); train: $375 - 240 = 135$ (m).

**Answer:** 15 m/s and 135 m.

## Challenge – Extension

**Problem 31** *

A boat with speed 18 km/h in still water goes downstream from A to B and at once comes back. The current is 2 km/h and the round trip takes 4 hours 30 minutes. How long is AB?

**Solution**

Downstream speed 20 km/h, upstream speed 16 km/h.

Times: downstream : upstream $= 16 : 20 = 4 : 5$.

9 parts = 4.5 h, so 1 part = 0.5 h; downstream time 2 h.

AB: $20 \times 2 = 40$ (km).

**Answer:** 40 km.

**Problem 32** *

Ports A and B are 60 km apart and the river flows from A to B at 3 km/h. At the same moment a boat leaves A for B and another boat leaves B for A; both have speed 20 km/h in still water. After how long do they meet, and how far from A?

**Solution**

Speeds: $20 + 3 = 23$ (km/h) and $20 - 3 = 17$ (km/h).

Sum: $23 + 17 = 40$ (km/h) (the current helps one boat as much as it slows the other).

Time: $60 : 40 = 1.5$ (h); distance from A: $23 \times 1.5 = 34.5$ (km).

**Answer:** after 1 h 30 min, 34.5 km from A.

### Problem 33 **

As a boat leaves A downstream, a raft is let go at A. The boat (12 km/h in still water, current 3 km/h) reaches B, 30 km from A, and immediately turns back. How long after leaving A does it meet the raft?

### Solution

Boat downstream: $30 : 15 = 2$ (h). Meanwhile the raft drifts $3 \times 2 = 6$ (km).

Gap when the boat turns: $30 - 6 = 24$ (km). The boat goes up at 9 km/h, the raft comes down at 3 km/h: they close 12 km each hour.

They meet after $24 : 12 = 2$ (h) more, i.e. $2 + 2 = 4$ (h) after leaving A.

*Seen from the water, the boat moves away from the raft for 2 h and comes back for 2 h: the current does not matter.*

**Answer:** 4 hours.

### Problem 34 **

Rowing upstream, Bình's hat falls into the water. He notices 20 minutes later, turns round at once and picks up his hat 2 km downstream of the place where it fell. Find the speed of the current.

### Solution

Relative to the water, the hat stays still and Bình rows away from it for 20 minutes, so he also needs 20 minutes to come back to it.

So the hat drifted for $20 + 20 = 40$ (min) $= \frac{2}{3}$ h, travelling 2 km.

Current: $2 : \frac{2}{3} = 3$ (km/h).

**Answer:** 3 km/h.

### Problem 35 *

A ferry goes downstream at 20 km/h and upstream at 12 km/h. It goes from A to B, rests 30 minutes at B and comes back; the whole trip takes 4 hours 30 minutes. How long is AB?

### Solution

Moving time: 4 h 30 min − 30 min = 4 h.

Times: downstream : upstream = 12 : 20 = 3 : 5; 8 parts = 4 h, 1 part = 0.5 h.

Downstream time 1.5 h; AB: $20 \times 1.5 = 30$ (km).

**Answer:** 30 km.

## Problem 36 *

A boat's speed in still water is 4 times the speed of the current. It goes downstream from A to B in 2 hours. How long does it take to go back?

### Solution

Let the current be 1 part: still water 4 parts, downstream 5 parts, upstream 3 parts.

Times are in the inverse ratio: upstream time $= 2 \times 5 : 3 = \dfrac{10}{3}$ (h) $= 3$ h 20 min.

**Answer:** 3 hours 20 minutes.

## Problem 37 *

A canoe goes downstream from A to B in 3 hours and back in 4 hours 30 minutes. If its engine were switched off, how long would it take to drift from A to B?

### Solution

In 1 hour: downstream $\dfrac{1}{3}$ of AB, upstream $1 : 4.5 = \dfrac{2}{9}$ of AB.

The current carries the canoe $\left(\dfrac{1}{3} - \dfrac{2}{9}\right) : 2 = \dfrac{1}{18}$ of AB per hour.

Drifting time: 18 hours.

**Answer:** 18 hours.

## Problem 38 *

At 7:00 a boat leaves B upstream and reaches A at 10:00. It rests, leaves A at 10:30 and gets back to B at 12:30. The current is 2 km/h. How far is it from A to B?

### Solution

Upstream 3 h, downstream 2 h; speeds upstream : downstream $= 2 : 3$.

The difference $2 \times 2 = 4$ (km/h) is 1 part; upstream speed 8 km/h.

AB: $8 \times 3 = 24$ (km).

**Answer:** 24 km.

## Problem 39 **

At 6:00 a raft starts drifting from A. At 8:00 a boat leaves A downstream (9 km/h in still water). The current is 3 km/h. At what time does the boat catch the raft, and how far from A?

**Solution**

At 8:00 the raft is $3 \times 2 = 6$ (km) ahead.

Boat downstream 12 km/h, raft 3 km/h: the boat gains 9 km/h (its still-water speed).

Time: $6 : 9 = \frac{2}{3}$ (h) $= 40$ min, at 8:40; distance from A: $12 \times \frac{2}{3} = 8$ (km).

**Answer:** at 8:40, 8 km from A.

## Problem 40 ⋆⋆

A train overtakes a man walking in the same direction at 1 m/s in 22 seconds, and passes a man walking towards it at 1 m/s in 18 seconds. Find the speed and the length of the train.

**Solution**

In both cases the train covers its own length: once at speed $v - 1$ (in 22 s), once at speed $v + 1$ (in 18 s).

Same length, so $(v - 1) : (v + 1) = 18 : 22 = 9 : 11$; their difference 2 m/s is 2 parts, so 1 part $= 1$ m/s.

$v - 1 = 9$, so $v = 10$ m/s; length: $9 \times 22 = 198$ (m).

**Answer:** 10 m/s and 198 m.

## Problem 41 ⋆⋆

A train goes through a tunnel 1.5 km long. From the moment its head enters until its tail leaves takes 90 seconds; the whole train is inside the tunnel for 60 seconds. Find the speed and the length of the train.

**Solution**

In 90 s the head travels tunnel + train; in 60 s it travels tunnel − train.

Together: in 150 s it travels 2 tunnels $= 3\,000$ m; speed: $3\,000 : 150 = 20$ (m/s).

Tunnel + train $= 20 \times 90 = 1\,800$ (m); train: $1\,800 - 1\,500 = 300$ (m).

**Answer:** 20 m/s and 300 m.

## Problem 42 ⋆⋆

An express train 180 m long runs at 24 m/s. It takes 50 seconds to overtake a slower train on a parallel track; if the two trains ran towards each other they would pass in 10 seconds. Find the speed and the length of the slower train.

**Solution**

The same distance (the two lengths) takes 50 s with the difference of speeds and 10 s with the sum, so sum : difference = 5 : 1.

Sum 5 parts, difference 1 part: express = 3 parts, slow = 2 parts; 3 parts = 24 m/s.

Slow train: 16 m/s. Two lengths: $(24 + 16) \times 10 = 400$ (m); slow train: $400 - 180 = 220$ (m).

**Answer:** 16 m/s and 220 m.

## Problem 43

A passenger on a train running at 54 km/h sees a train 300 m long, coming the other way, go past in 12 seconds. Find the speed of that train in km/h.

**Solution**

Sum of speeds: $300 : 12 = 25$ (m/s).

Other train: $25 - 15 = 10$ (m/s) = 36 km/h.

**Answer:** 36 km/h.

## Problem 44

A bridge is 4 times as long as a train. The train, at 20 m/s, takes 25 seconds from the moment its head reaches the bridge until its tail leaves it. Find the length of the train and of the bridge.

**Solution**

Distance: $20 \times 25 = 500$ (m) = bridge + train = 5 train lengths.

Train: $500 : 5 = 100$ (m); bridge: 400 m.

**Answer:** train 100 m, bridge 400 m.

## Problem 45

Two trains, each 150 m long, pass each other going in opposite directions in 10 seconds. One train is 1.5 times as fast as the other. Find their speeds.

**Solution**

Sum of speeds: $300 : 10 = 30$ (m/s).

Slower 1 part, faster 1.5 parts: $30 : 2.5 = 12$ (m/s).

Faster: 18 m/s.

**Answer:** 12 m/s and 18 m/s.

### Problem 46 *

Train A, 240 m long, at 72 km/h overtakes train B going the same way at 54 km/h. The overtaking takes 1 minute 30 seconds. How long is train B?

### Solution

Difference of speeds: $20 - 15 = 5$ (m/s); 1 min 30 s = 90 s.
Sum of lengths: $5 \times 90 = 450$ (m).
Train B: $450 - 240 = 210$ (m).
**Answer:** 210 m.

### Problem 47 **

On a river with a current of 1 m/s, a cargo ship 80 m long sails downstream (6 m/s in still water) and meets a ship 70 m long sailing upstream (4 m/s in still water). How long do they take to pass each other?

### Solution

Speeds: $6 + 1 = 7$ (m/s) and $4 - 1 = 3$ (m/s); sum 10 m/s (equal to the sum of the still-water speeds).
Sum of lengths: 150 m. Time: $150 : 10 = 15$ (s).
**Answer:** 15 seconds.

### Problem 48 **

A line of pupils 300 m long walks at 1.5 m/s. A cyclist rides at 4.5 m/s from the back of the line to the front and immediately back to the end. How long does this take?

### Solution

Back to front (same direction): he gains $4.5 - 1.5 = 3$ (m/s): $300 : 3 = 100$ (s).
Front to back (towards the line): $4.5 + 1.5 = 6$ (m/s): $300 : 6 = 50$ (s).
Total: 150 s = 2 min 30 s.
**Answer:** 2 minutes 30 seconds.

### Problem 49 *

A train 250 m long runs at 72 km/h. Its head enters a tunnel at 9:00:00 and its tail leaves at 9:01:40. How long is the tunnel?

**Solution**

Time: 1 min 40 s = 100 s; speed 20 m/s.

Distance: $20 \times 100 = 2\,000$ (m).

Tunnel: $2\,000 - 250 = 1\,750$ (m).

**Answer:** 1 750 m.

**Problem 50** ★★

A boat's upstream speed is $\frac{2}{3}$ of its downstream speed. A raft drifts from A to B in 30 hours. How long does the boat take from A to B downstream, and from B to A upstream?

**Solution**

Downstream 3 parts, upstream 2 parts; difference 1 part = 2× current, so the current is $\frac{1}{2}$ part.

Downstream speed is $3 : \frac{1}{2} = 6$ times the current: $30 : 6 = 5$ (h).

Upstream speed is 4 times the current: $30 : 4 = 7.5$ (h) = 7 h 30 min.

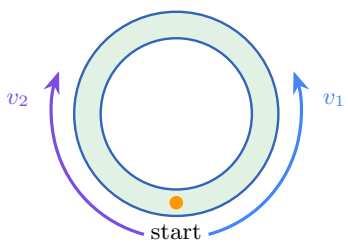
**Answer:** 5 hours downstream, 7 hours 30 minutes upstream.

# 10 Motion around a Circular Track

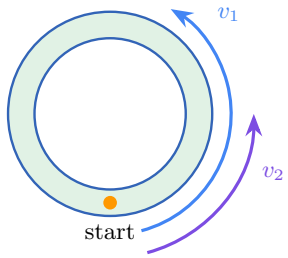
## ① Moving round a closed track

- Runners on a running track, cyclists round a lake, the hands of a clock… all move round a closed path. After each full lap the mover is back exactly where it started, and it can go round as many times as it likes.
- If one lap is $C$ and the distance travelled is $s$, then $s : C$ gives the number of whole laps, and the remainder tells us how far past the start the mover is. For a circular track, $C = \text{diameter} \times 3.14$.
- **Careful:** "how far from the start" along the way ahead is *one lap minus the remainder*, not the remainder itself.

## ② Two movers starting together from the same point




**opposite directions**
time to meet $= C : (v_1 + v_2)$




**same direction** $(v_1 > v_2)$
time to meet $= C : (v_1 - v_2)$

- **Opposite directions:** they meet each time that together they have covered one lap.
- **Same direction:** they meet each time the faster one has gained one whole lap on the slower one (it "laps" the slower one).
- After the first meeting the situation repeats: the 2nd, 3rd, … meetings come after 2, 3, … times that time.
- If they do not start together, use the gap between them in place of $C$ for the first meeting.

**Clock hands** move round the clock face, which has 60 minute-divisions: the minute hand moves 1 division per minute and the hour hand $\frac{1}{12}$ division per minute.

## ③ Going back and forth on a road

# Motion around a Circular Track

Two people start at the same time from the two ends A and B of a road, walk towards each other, and each turns back at once on reaching the other end.




- By the first meeting (at C) together they have covered *one* length AB.
- By the second meeting (at D, after both have turned back) together they have covered *three* lengths AB, so it takes 3 times as long, and each person has walked 3 times as far as before the first meeting.

*This holds when the faster person is less than twice as fast as the slower one, so that nobody is caught from behind before the second meeting.*

## Example 1

Minh and Hà cycle round a lake whose shore path is 1.5 km long. Minh rides at 12 km/h and Hà at 15 km/h. They start together from the same point and go the same way. After how long do they meet for the first time (not counting the start)?

### Guided solution.

To meet again, Hà must gain one whole lap, 1.5 km, on Minh.
Difference of speeds: $15 - 12 = 3$ (km/h).
Time: $1.5 : 3 = 0.5$ (h) $= 30$ min.
**Answer:** 30 minutes.

## Example 2

On a 400 m running track, Tú runs at 3 m/s and Quân at 5 m/s. They start together from the same point and run in opposite directions.

**a)** After how long do they first meet?

**b)** How far from the start is that, along Tú's way?

**c)** How many times do they meet in 10 minutes?

### Guided solution.

**a)** Together they cover one lap: $400 : (3 + 5) = 50$ (s).

**b)** In 50 s Tú runs $3 \times 50 = 150$ (m): they meet 150 m from the start along Tú's way (and 250 m along Quân's way).

**c)** 10 min $= 600$ s; they meet every 50 s, so $600 : 50 = 12$ times.

**Answer: a)** 50 s; **b)** 150 m; **c)** 12 times.

## Example 3

Dũng jogs from his house round a block whose loop is 1.8 km long, at 8 km/h. After 45 minutes, how much farther must he run, keeping the same way, to get back home?

**Guided solution.**

45 min = 0.75 h; distance run: $8 \times 0.75 = 6$ (km).

$6 : 1.8 = 3$ laps, remainder 0.6 km ($6 = 3 \times 1.8 + 0.6$): Dũng is 0.6 km past his house.

To get home going forward he still needs $1.8 - 0.6 = 1.2$ (km).

*Do not answer* 0.6 *km: that is how far he has gone* past *the house, not how far he still has to go.*

**Answer:** 1.2 km.

## Example 4

A circular running track has diameter 100 m. Two pupils start together from the same point and run the same way at 4.5 m/s and 2.5 m/s. After how long does the faster one first catch the slower one?

**Guided solution.**

Length of one lap: $100 \times 3.14 = 314$ (m).

Difference of speeds: $4.5 - 2.5 = 2$ (m/s).

Time: $314 : 2 = 157$ (s) = 2 min 37 s.

**Answer:** 2 minutes 37 seconds.

## Example 5

Two friends set off at the same time, one from A towards B and one from B towards A. They meet first at C, 14 km from A. Each goes on to the other end and turns back at once; they meet a second time at D, 8 km from B. How long is the road AB?

**Guided solution.**




By the first meeting the two together have covered one AB; the friend from A has walked 14 km. By the second meeting they together have covered three AB, so the friend from A has walked $14 \times 3 = 42$ (km).

He walked from A to B and then 8 km back: $42 = AB + 8$, so $AB = 42 - 8 = 34$ (km).

*Check: the speeds are in the ratio* 14 : 20*; the friend from B walks* $20 \times 3 = 60 = 34 + 26$ *km, i.e. to A and* 26 *km back, which is* 8 *km from B.*

**Answer:** 34 km.

## Practice

## Problem 1

On a 400 m track two runners start together from the same point and run in opposite directions at 4 m/s and 6 m/s. After how long do they first meet?

### Solution

Together they cover one lap.
Sum of speeds: $4 + 6 = 10$ (m/s).
Time: $400 : 10 = 40$ (s).
**Answer:** 40 seconds.

## Problem 2

The same two runners (at 4 m/s and 6 m/s) start together from the same point on the 400 m track but run the same way. After how long do they first meet?

### Solution

The faster runner must gain one lap.
Difference of speeds: $6 - 4 = 2$ (m/s).
Time: $400 : 2 = 200$ (s) $= 3$ min 20 s.
**Answer:** 3 minutes 20 seconds.

## Problem 3

A lake path is 3 km round. Two cyclists start together from the same point in opposite directions at 12 km/h and 18 km/h. After how many minutes do they meet?

### Solution

Sum of speeds: $12 + 18 = 30$ (km/h).
Time: $3 : 30 = 0.1$ (h) $= 6$ min.
**Answer:** 6 minutes.

## Problem 4

The two cyclists of Problem 3 now ride the same way round the 3 km lake. After how long does the faster one first catch up with the other?

### Solution

Difference of speeds: $18 - 12 = 6$ (km/h).
Time: $3 : 6 = 0.5$ (h) $= 30$ min.
**Answer:** 30 minutes.

**Problem 5**

A park path is a loop of 900 m. Lan walks at 60 m/min and Mai at 90 m/min; they start together from the gate and walk the same way. After how long does Mai first catch up with Lan, and how many laps has each walked by then?

**Solution**

Difference of speeds: $90 - 60 = 30$ (m/min).

Time: $900 : 30 = 30$ (min).

Lan: $60 \times 30 = 1\,800$ m $= 2$ laps; Mai: $90 \times 30 = 2\,700$ m $= 3$ laps.

**Answer:** after 30 minutes; Lan 2 laps, Mai 3 laps.

**Problem 6**

An runs round an 800 m track at 150 m/min for 14 minutes. How many full laps has he run, and how many more metres must he run to finish the lap he is on?

**Solution**

Distance: $150 \times 14 = 2\,100$ (m).

$2\,100 = 2 \times 800 + 500$: 2 full laps and 500 m more.

To finish the third lap: $800 - 500 = 300$ (m).

**Answer:** 2 full laps; 300 m more.

**Problem 7**

A circular cycle track has diameter 50 m. A child rides 5 laps. How far does the child ride?

**Solution**

One lap: $50 \times 3.14 = 157$ (m).

5 laps: $157 \times 5 = 785$ (m).

**Answer:** 785 m.

**Problem 8**

A circular track has diameter 100 m. Two pupils start together from the same point and run in opposite directions at 3 m/s and 5 m/s. After how long do they meet?

**Solution**

One lap: $100 \times 3.14 = 314$ (m).

Sum of speeds: $3 + 5 = 8$ (m/s).

Time: $314 : 8 = 39.25$ (s).

**Answer:** 39.25 seconds.

### Problem 9

Two cyclists ride the same way round a 2 km circuit at 20 km/h and 25 km/h, starting together. How many times does the faster one overtake the slower one in 2 hours?

### Solution

In 2 hours the faster one gains $(25 - 20) \times 2 = 10$ (km).
Each overtaking is a gain of one lap: $10 : 2 = 5$ (times).
**Answer:** 5 times.

### Problem 10

Two runners start together from the same point of a 500 m track and run in opposite directions at 3 m/s and 2 m/s. How many times do they meet in 10 minutes?

### Solution

They meet every $500 : (3 + 2) = 100$ (s).
10 min = 600 s; $600 : 100 = 6$ (times).
**Answer:** 6 times.

### Problem 11

Two friends run the same way round a 1 km loop, starting together. The faster one runs at 250 m/min and catches the other for the first time after 10 minutes. Find the speed of the slower one.

### Solution

Difference of speeds: $1\,000 : 10 = 100$ (m/min).
Slower one: $250 - 100 = 150$ (m/min).
**Answer:** 150 m/min.

### Problem 12

Two friends run in opposite directions round a 600 m loop, starting together, and meet after 2 minutes. One runs at 140 m/min. Find the speed of the other.

### Solution

Sum of speeds: $600 : 2 = 300$ (m/min).
The other: $300 - 140 = 160$ (m/min).

**Answer:** 160 m/min.

### Problem 13

On a 400 m track, Bảo starts 100 m ahead of Khoa, and both run the same way. Khoa runs at 5 m/s, Bảo at 3 m/s. After how long does Khoa catch Bảo?

### Solution

Khoa must gain 100 m.

Difference of speeds: $5 - 3 = 2$ (m/s).

Time: $100 : 2 = 50$ (s).

**Answer:** 50 seconds.

### Problem 14

Hùng runs one lap of a track in 4 minutes, Việt in 6 minutes. They start together from the same point and run the same way. After how long does Hùng first catch Việt?

### Solution

Each minute Hùng runs $\frac{1}{4}$ lap and Việt $\frac{1}{6}$ lap; Hùng gains $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ lap.

To gain one lap: $1 : \frac{1}{12} = 12$ (min).

**Answer:** 12 minutes.

### Problem 15

Hùng and Việt (one lap in 4 min and in 6 min) now start together and run in opposite directions. After how long do they meet?

### Solution

Each minute together they run $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ lap.

Time: $1 : \frac{5}{12} = \frac{12}{5} = 2.4$ (min) $= 2$ min 24 s.

**Answer:** 2 minutes 24 seconds.

### Problem 16

One runner takes 3 minutes per lap and another 5 minutes per lap. They start together from the start line. After how long are both at the start line together again for the first time?

**Solution**

The first runner is at the start line after 3, 6, 9, 12, 15, ... minutes; the second after 5, 10, 15, ... minutes.
The smallest common number: 15.
**Answer:** 15 minutes.

## Problem 17

A runner runs 5 laps of a 400 m track in 10 minutes. Find the speed in m/min and in km/h.

**Solution**

Distance: $400 \times 5 = 2\,000$ (m).

Speed: $2\,000 : 10 = 200$ (m/min).

$200$ m/min $= 12\,000$ m/h $= 12$ km/h.

**Answer:** 200 m/min = 12 km/h.

## Problem 18

A road AB is 30 km long. A cyclist leaves A for B at 12 km/h, and at the same time a motorcyclist leaves B for A at 18 km/h. Each turns back at once on reaching the other end.

**a)** How far from A do they meet the first time?

**b)** How far from A do they meet the second time?

**Solution**

**a)** Time: $30 : (12 + 18) = 1$ (h); distance from A: $12 \times 1 = 12$ (km).

**b)** By the second meeting they together cover $3 \times 30 = 90$ (km), which takes 3 h.

The cyclist rides $12 \times 3 = 36$ (km) $= 30 + 6$: to B and 6 km back, which is $30 - 6 = 24$ (km) from A.

**Answer: a)** 12 km; **b)** 24 km.

## Problem 19

Two people set off at the same time from A and from B towards each other. They meet first 9 km from A. Each goes on to the other end and turns back at once; they meet again 5 km from B. How long is AB?

**Solution**

By the second meeting the person from A has walked $9 \times 3 = 27$ (km).

That is AB plus 5 km back from B: AB $= 27 - 5 = 22$ (km).

**Answer:** 22 km.

### Problem 20

Hoa walks from A and Quế from B towards each other, then back and forth between A and B without stopping. Their speeds are in the ratio 2 : 3. The first two meeting points are 24 m apart. How long is AB?

### Solution

First meeting: Hoa has walked $\frac{2}{5}$ of AB.

Second meeting: Hoa has walked $3 \times \frac{2}{5} = \frac{6}{5}$ of AB, i.e. to B and $\frac{1}{5}$ AB back: she is $\frac{4}{5}$ of AB from A.

Distance between the meeting points: $\frac{4}{5} - \frac{2}{5} = \frac{2}{5}$ of AB $= 24$ m; AB $= 24 : \frac{2}{5} = 60$ (m).

**Answer:** 60 m.

### Problem 21

A bicycle wheel has diameter 0.6 m. How many turns does it make while the bicycle travels 942 m?

### Solution

One turn: $0.6 \times 3.14 = 1.884$ (m).

Turns: $942 : 1.884 = 500$.

**Answer:** 500 turns.

### Problem 22

The minute hand of a clock is 10 cm long. How far does its tip travel in 1 hour? In 3 hours?

### Solution

In 1 hour the tip goes once round a circle of radius 10 cm: $10 \times 2 \times 3.14 = 62.8$ (cm).

In 3 hours: $62.8 \times 3 = 188.4$ (cm).

**Answer:** 62.8 cm; 188.4 cm.

### Problem 23

On a 400 m track, A runs at 6 m/s and B at 4 m/s, the same way, starting together. When A first laps B, how many metres and how many laps has A run?

**Solution**

Time: $400 : (6 - 4) = 200$ (s).

A runs $6 \times 200 = 1\,200$ (m) $= 3$ laps.

**Answer:** 1 200 m, 3 laps.

**Problem 24**

Phúc runs one lap of a track in 80 seconds and Long in 120 seconds. They start together and run in opposite directions. After how long do they meet?

**Solution**

Each second together they run $\frac{1}{80} + \frac{1}{120} = \frac{3}{240} + \frac{2}{240} = \frac{1}{48}$ lap.

Time: 48 s.

**Answer:** 48 seconds.

**Problem 25**

Two horses gallop the same way round a 1.2 km racecourse at 15 m/s and 12 m/s, starting together. After how long does the faster one first catch the other?

**Solution**

1.2 km $= 1\,200$ m.

Difference: 3 m/s.

Time: $1\,200 : 3 = 400$ (s) $= 6$ min 40 s.

**Answer:** 6 minutes 40 seconds.

**Problem 26**

Two toy trains run in opposite directions round a circular track 3 m long, at 0.25 m/s and 0.35 m/s, starting from the same point. After how long do they meet?

**Solution**

Sum of speeds: $0.25 + 0.35 = 0.6$ (m/s).

Time: $3 : 0.6 = 5$ (s).

**Answer:** 5 seconds.

**Problem 27**

Two runners start together and run in opposite directions at 4 m/s and 6 m/s round a track. They meet after 40 seconds. How long is the track?

**Solution**

In 40 s together they run one lap: $(4 + 6) \times 40 = 400$ (m).

**Answer:** 400 m.

**Problem 28**

Two cyclists start together and ride the same way at 260 m/min and 180 m/min round a loop. The faster first catches the slower after 6 minutes. How long is the loop?

**Solution**

In 6 min the faster gains one lap: $(260 - 180) \times 6 = 480$ (m).

**Answer:** 480 m.

**Problem 29**

Ngân starts from home and runs round a 2.5 km loop at 9 km/h for 1 hour. How much farther must she run, keeping the same way, to get back home?

**Solution**

Distance: 9 km $= 3 \times 2.5 + 1.5$: 3 laps and 1.5 km more.

Still to run: $2.5 - 1.5 = 1$ (km).

**Answer:** 1 km.

**Problem 30**

Two swimmers start at the same time from the two ends of a 50 m pool and swim back and forth at 1 m/s and 1.5 m/s, turning at once at each end. After how long do they meet the first time? The second time?

**Solution**

First meeting: together one length: $50 : (1 + 1.5) = 20$ (s).

Second meeting: together three lengths: $20 \times 3 = 60$ (s).

**Answer:** 20 s and 60 s.

## Challenge – Extension

**Problem 31** *

Three children run the same way round a 300 m track at 3 m/s, 4 m/s and 5 m/s, starting together from the same point. After how long are all three together again for the first time?

**Solution**

The 4 m/s and 3 m/s runners meet every $300 : 1 = 300$ (s); the 5 m/s and 4 m/s runners also every 300 s; the 5 and 3 m/s runners every $300 : 2 = 150$ (s).
All three are together when every pair is together: first at 300 s = 5 min.
(*Check:* they have run 900, 1 200, 1 500 m, i.e. 3, 4, 5 whole laps.)
**Answer:** 5 minutes.

**Problem 32** *

On a 400 m track two runners start together the same way at 4 m/s and 5 m/s. After how long are they both at the start line together for the first time? Is that also the first time they meet?

**Solution**

Lap times: $400 : 4 = 100$ (s) and $400 : 5 = 80$ (s). Both at the start: smallest common multiple of 100 and 80, i.e. 400 s.
They first meet when the faster gains a lap: $400 : (5 - 4) = 400$ (s), so yes, the first meeting happens at the start line.
**Answer:** 400 s = 6 min 40 s; yes.

**Problem 33** **

On a 400 m track two runners start together from point S and run in opposite directions at 3 m/s and 5 m/s. At which meeting do they meet at S again for the first time, and when?

**Solution**

They meet every $400 : 8 = 50$ (s); between meetings the slower one runs 150 m.
Meeting points, measured from S along the slower runner's way: 150, 300, 50, 200, 350, 100, 250, 0 (m).
So the 8th meeting is at S, after $8 \times 50 = 400$ (s). (The slower runner has then run 1 200 m = 3 laps.)
**Answer:** the 8th meeting, after 400 s.

**Problem 34** **

At exactly 3 o'clock, after how many minutes are the hour hand and the minute hand of a clock on top of each other for the first time?

**Solution**

At 3:00 the minute hand is 15 divisions behind the hour hand.

Each minute the minute hand gains $1 - \frac{1}{12} = \frac{11}{12}$ division.

Time: $15 : \frac{11}{12} = \frac{180}{11} = 16\frac{4}{11}$ (min).

**Answer:** $16\frac{4}{11}$ minutes (at about 3:16).

**Problem 35** ★★

How many times are the two hands of a clock on top of each other from just after 12 noon until 12 midnight (including midnight)?

**Solution**

The minute hand gains one full turn (60 divisions) on the hour hand every $60 : \frac{11}{12} = \frac{720}{11}$ min $= \frac{12}{11}$ h.

In 12 hours: $12 : \frac{12}{11} = 11$ (times).

**Answer:** 11 times.

**Problem 36** ★

On a 600 m track, B starts 200 m ahead of A and both run the same way, A at 5 m/s and B at 3 m/s. When does A catch B the first time? The second time?

**Solution**

First: A gains 200 m: $200 : 2 = 100$ (s).

Then A must gain a whole lap more: $600 : 2 = 300$ (s); second time: $100 + 300 = 400$ (s).

**Answer:** after 100 s and after 400 s.

**Problem 37** ★

On a 400 m track, two runners starting together meet after 40 s if they run in opposite directions, and after 200 s if they run the same way. Find their speeds.

**Solution**

Sum of speeds: $400 : 40 = 10$ (m/s); difference: $400 : 200 = 2$ (m/s).

Faster: $(10 + 2) : 2 = 6$ (m/s); slower: $10 - 6 = 4$ (m/s).

**Answer:** 6 m/s and 4 m/s.

## Problem 38 ★★

Yến runs one lap of a track in 6 minutes. Running in opposite directions from the same start, she and Xuân meet every 2 minutes 24 seconds. How long does Xuân take for one lap?

### Solution

2 min 24 s = 2.4 min; together they run $1 : 2.4 = \frac{5}{12}$ lap per minute.

Xuân: $\frac{5}{12} - \frac{1}{6} = \frac{3}{12} = \frac{1}{4}$ lap per minute, so one lap takes 4 min.

**Answer:** 4 minutes.

## Problem 39 ★★

Two friends 6 km apart walk towards each other at 3 km/h and 5 km/h. A dog starts with the first friend and runs at 12 km/h back and forth between them until they meet. How far does the dog run?

### Solution

The friends meet after $6 : (3 + 5) = 0.75$ (h).

The dog runs the whole time: $12 \times 0.75 = 9$ (km).

**Answer:** 9 km.

## Problem 40 ★★

Hà walks back and forth between A and B, starting from A; Tú does the same starting from B at the same moment. Hà's speed is 80% of Tú's. The first two meeting points are 40 m apart. How long is AB?

### Solution

Speeds Hà : Tú = 4 : 5. First meeting: Hà has walked $\frac{4}{9}$ of AB.

Second meeting: Hà has walked $3 \times \frac{4}{9} = \frac{12}{9}$ of AB: to B and $\frac{3}{9}$ back, so she is $\frac{6}{9}$ of AB from A.

Gap: $\frac{6}{9} - \frac{4}{9} = \frac{2}{9}$ of AB = 40 m; AB = $40 : \frac{2}{9} = 180$ (m).

**Answer:** 180 m.

## Problem 41 ★★

Two cars leave A and B at the same time towards each other. They meet first 40 km from A. Each drives on to the other end and turns back at once; they meet the second time 20 km from A. How long is AB?

**Solution**

By the second meeting the car from A has driven $40 \times 3 = 120$ (km).
It drove A to B and then back to 20 km from A: $120 = AB + (AB - 20)$.
So $2 \times AB = 140$, $AB = 70$ km.
**Answer:** 70 km.

**Problem 42** *

On a 500 m track, Kiên (at 3 m/s) and Lâm (at 2 m/s) start together from S and run in opposite directions. When do they meet the third time, and how far from S is that along Kiên's way?

**Solution**

They meet every $500 : 5 = 100$ (s); third time after 300 s.
Kiên has run $3 \times 300 = 900$ (m) $= 500 + 400$: 400 m from S along his way.
**Answer:** after 300 s (5 min), 400 m from S.

**Problem 43** *

On a 400 m track, A runs at 5 m/s and B more slowly, the same way, starting together. A first laps B at the moment A finishes his 5th lap. Find B's speed.

**Solution**

A runs $5 \times 400 = 2\,000$ (m) in $2\,000 : 5 = 400$ (s).
B has run one lap less: 1 600 m; speed $1\,600 : 400 = 4$ (m/s).
**Answer:** 4 m/s.

**Problem 44** *

On a 400 m track, two runners start together from S in opposite directions and meet at a point 160 m from S along the first runner's way. The first runner runs at 4 m/s. Find the speed of the second.

**Solution**

Time: $160 : 4 = 40$ (s). The second runner covers the rest of the lap: $400 - 160 = 240$ (m).
Speed: $240 : 40 = 6$ (m/s).
**Answer:** 6 m/s.

**Problem 45** *

A Ferris wheel has diameter 40 m and turns once every 8 minutes. How fast does a cabin on its rim move (in m/min)?

**Solution**

One turn: $40 \times 3.14 = 125.6$ (m).

Speed: $125.6 : 8 = 15.7$ (m/min).

**Answer:** 15.7 m/min.

**Problem 46** *

A bicycle wheel of diameter 0.5 m turns 200 times a minute. Find the speed of the bicycle in km/h.

**Solution**

One turn: $0.5 \times 3.14 = 1.57$ (m); per minute: $1.57 \times 200 = 314$ (m).

Per hour: $314 \times 60 = 18\,840$ (m) $= 18.84$ km.

**Answer:** 18.84 km/h.

**Problem 47** *

Two runners start at the same moment from two opposite points of a 400 m track (half a lap apart) and run the same way at 6 m/s and 4 m/s. After how long does the faster one catch the slower one?

**Solution**

The faster one is $400 : 2 = 200$ (m) behind.

Time: $200 : (6 - 4) = 100$ (s).

**Answer:** 100 seconds.

**Problem 48** **

Two runners start at the same moment from two opposite points of a 400 m track and run towards each other at 3 m/s and 5 m/s. When do they meet the third time?

**Solution**

First meeting: they close the 200 m gap: $200 : 8 = 25$ (s).

After that they meet every $400 : 8 = 50$ (s).

Third time: $25 + 50 \times 2 = 125$ (s).

**Answer:** after 125 s = 2 min 5 s.

## Problem 49 *

Two friends walk from the two ends of a 1.2 km path towards each other at 70 m/min and 50 m/min, then go on back and forth without stopping. How long after the start do they meet the second time?

### Solution

By the second meeting together they walk $3 \times 1\,200 = 3\,600$ (m).

Time: $3\,600 : (70 + 50) = 30$ (min).

**Answer:** 30 minutes.

## Problem 50 * *

Two runners start together from the same point of a track. Running in opposite directions they meet every 30 s; running the same way, the faster one laps the slower every 2 min 30 s. How long does each take for one lap?

### Solution

2 min 30 s = 150 s. Per second: sum of their parts of a lap $= \frac{1}{30} = \frac{5}{150}$, difference $= \frac{1}{150}$.

Faster: $\left(\frac{5}{150} + \frac{1}{150}\right) : 2 = \frac{3}{150} = \frac{1}{50}$ lap per second, so 50 s per lap.

Slower: $\frac{5}{150} - \frac{3}{150} = \frac{2}{150} = \frac{1}{75}$ lap per second, so 75 s per lap.

**Answer:** 50 s and 75 s.

# 11 Statistics and Probability

## ① Collecting and presenting data

To compare, judge and decide, people **collect information** (data) and then **organise** it. Sometimes we ask everybody; sometimes we ask only a typical group.

Data are usually shown as:

- a **table** – each value is written in its own cell;
- a **pictograph** – each picture stands for the same number of things;
- a **bar chart** – the height of each bar, read on the vertical scale, gives the value;
- a **pie chart** – the whole circle is the whole total (100%); each slice is a percentage of the total. Half the circle is 50%, a quarter is 25%.

From a pie chart: **part** = **total** : 100× **percentage**; **total** = **part** : **percentage** ×100.

## ② The mean (average)

**Mean** = (sum of the values) : (number of values). **Sum** = mean × number of values.

- The mean always lies between the smallest and the largest value.
- To find a *missing value*: work out the sum from the mean, then subtract the values you know.
- If one value grows by some amount, the sum grows by that amount and the mean grows by (that amount) : (number of values).

## ③ Chance and counting outcomes

When we repeat an experiment (tossing a coin, rolling a dice, spinning a spinner, picking a ball without looking), we record how many times each outcome happens.

$$\textbf{ratio describing how often an outcome happened} = \frac{\text{number of times it happened}}{\text{total number of trials}}$$

For example, a dice is rolled 20 times and the face with 6 dots comes up 3 times: the ratio is $\frac{3}{20}$.

If we roll more times, this ratio may change.

**Counting outcomes.** List every possible result in an orderly way (a list, a table or a tree). A

coin has 2 outcomes, a dice has 6; a coin *and* a dice together have $2 \times 6 = 12$.
When all outcomes are *equally likely*, the **chance** of an event = (number of outcomes that give the event) : (number of all outcomes). An event that always happens is *certain*; one that can never happen is *impossible*.

## Example 1

Lan writes down how many pages she reads each day of one week:

| **Day** | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| **Pages** | 18 | 25 | 20 | 32 | 15 | 30 | 28 |

**a)** On which day did she read most? Least?

**b)** How many pages did she read per day on average?

**c)** On how many days did she read more than her average?

**Guided solution.**

**a)** Most on Thursday (32 pages); least on Friday (15 pages).

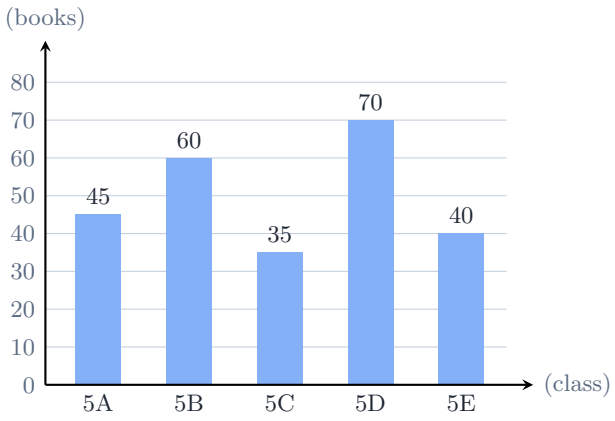
**b)** Pages in the week: $18 + 25 + 20 + 32 + 15 + 30 + 28 = 168$.
Mean per day: $168 : 7 = 24$ pages.

**c)** More than 24: Tuesday, Thursday, Saturday and Sunday, i.e. 4 days.

**Answer: a)** Thursday; Friday. **b)** 24 pages. **c)** 4 days.

## Example 2

The bar chart shows how many books five classes gave to the school library.



**a)** How many books did the five classes give? **b)** How many books did a class give on average?

**c)** How many more books did the class that gave most give than the class that gave least?

**Guided solution.**

**a)** $45 + 60 + 35 + 70 + 40 = 250$ books.

**b)** $250 : 5 = 50$ books.

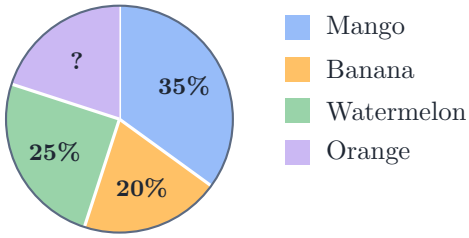
**c)** Most: 5D (70); least: 5C (35).

Difference: $70 - 35 = 35$ books.

**Answer: a)** 250 books. **b)** 50 books. **c)** 35 books.

## Example 3

400 pupils were asked about their favourite fruit. The results are shown in the pie chart.




**a)** What percentage of the pupils chose oranges?

**b)** How many pupils chose mangoes?

**c)** How many more pupils chose mangoes than bananas?

**Guided solution.**

**a)** The whole circle is 100%: $100\% - 35\% - 20\% - 25\% = 20\%$.

**b)** $400 : 100 \times 35 = 140$ pupils.

**c)** *Method 1.* Bananas: $400 : 100 \times 20 = 80$ pupils; $140 - 80 = 60$ pupils.

*Method 2.* Mangoes are $35\% - 20\% = 15\%$ more; $400 : 100 \times 15 = 60$ pupils.

**Answer: a)** 20%. **b)** 140 pupils. **c)** 60 pupils.

## Example 4

Minh rolls a dice 40 times and records the results.

| **Outcome** | 1 dot | 2 dots | 3 dots | 4 dots | 5 dots | 6 dots |
|---|---|---|---|---|---|---|
| **Times** | 6 | 8 | 5 | 7 | 9 | 5 |

Write the ratio describing how often each event happened, compared with the number of rolls:

**a)** 6 dots; **b)** an even number of dots; **c)** at least 5 dots.

**Guided solution.**

**a)** 6 dots came up 5 times: $\frac{5}{40} = \frac{1}{8}$.

**b)** Even (2, 4, 6 dots): $8 + 7 + 5 = 20$ times; $\frac{20}{40} = \frac{1}{2}$.

**c)** At least 5 dots (5 or 6 dots): $9 + 5 = 14$ times; $\frac{14}{40} = \frac{7}{20}$.

*This does not mean the same thing will happen next time: with more rolls the ratios change.*

**Answer: a)** $\frac{1}{8}$. **b)** $\frac{1}{2}$. **c)** $\frac{7}{20}$.

## Example 5

Box I holds three cards numbered 1, 2, 3; box II holds two cards numbered 4, 5. Without looking, Hoa takes one card from each box and adds the two numbers.

**a)** List all possible outcomes. **b)** What is the chance that the sum is 7?

**Guided solution.**

**a)** Make a table (row: box I, column: box II):

| + | 4 | 5 |
|---|---|---|
| 1 | 5 | 6 |
| 2 | 6 | 7 |
| 3 | 7 | 8 |

There are $3 \times 2 = 6$ equally likely outcomes: $1 + 4,\ 1 + 5,\ 2 + 4,\ 2 + 5,\ 3 + 4,\ 3 + 5$.

**b)** The sum is 7 in 2 outcomes ($2 + 5$ and $3 + 4$).

Chance: $\frac{2}{6} = \frac{1}{3}$.

**Answer: a)** 6 outcomes. **b)** $\frac{1}{3}$.

# Practice

## Problem 1

The noon temperature in a town on five days was:

| **Day** | Mon | Tue | Wed | Thu | Fri |
|---|---|---|---|---|---|
| **Temperature (°C)** | 31 | 29 | 33 | 30 | 32 |

**a)** Which day was hottest, which coolest? **b)** Find the mean noon temperature.

**Solution**

**a)** Hottest: Wednesday (33°C); coolest: Tuesday (29°C).

**b)** $(31 + 29 + 33 + 30 + 32) : 5 = 155 : 5 = 31$°C.

**Answer: a)** Wednesday; Tuesday. **b)** 31°C.

## Problem 2

Nam's marks in six maths tests were 7, 9, 8, 10, 6, 8. What is his mean mark?

**Solution**

Sum: $7 + 9 + 8 + 10 + 6 + 8 = 48$.

Mean: $48 : 6 = 8$.

**Answer:** 8.

## Problem 3

Four friends are 135 cm, 142 cm, 138 cm and 145 cm tall. Find their mean height.

**Solution**

$(135 + 142 + 138 + 145) : 4 = 560 : 4 = 140$ cm.

**Answer:** 140 cm.

## Problem 4

The pictograph shows how many books four groups read in May.

| | |
|---|---|
| Group 1 | 📘📘📘📘📘 |
| Group 2 | 📘📘📘 |
| Group 3 | 📘📘📘📘📘📘 |
| Group 4 | 📘📘📘📘 |

Each 📘 stands for 4 books.

**a)** How many books did the four groups read? **b)** How many books did a group read on average?

**Solution**

Number of pictures: $5 + 3 + 6 + 4 = 18$.

**a)** Books: $4 \times 18 = 72$.

**b)** Mean: $72 : 4 = 18$ books.

**Answer: a)** 72 books. **b)** 18 books.

## Problem 5

A shop sold 24 bicycles in January, 30 in February, 27 in March and 39 in April.

**a)** How many bicycles were sold in the four months? **b)** How many a month on average?

**c)** In how many months did the shop sell more than its average?

## Solution

**a)** $24 + 30 + 27 + 39 = 120$ bicycles.

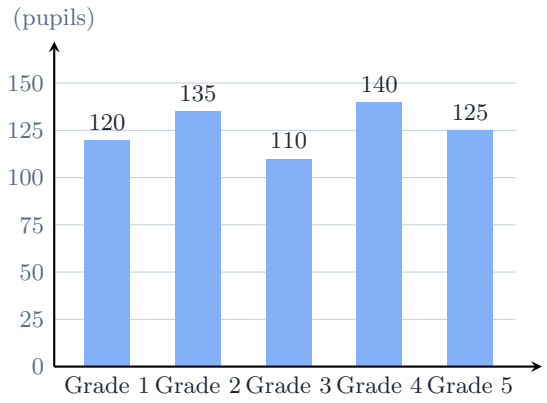
**b)** $120 : 4 = 30$ bicycles.

**c)** Only April ($39 > 30$; February's 30 is equal to the mean, not more): 1 month.

**Answer: a)** 120. **b)** 30. **c)** 1 month.

## Problem 6

The chart shows the number of pupils in each grade of a primary school.




**a)** Which grade has the most pupils? The fewest?

**b)** How many pupils does the school have?

**c)** How many more pupils does Grade 4 have than Grade 3?

## Solution

**a)** Most: Grade 4 (140); fewest: Grade 3 (110).

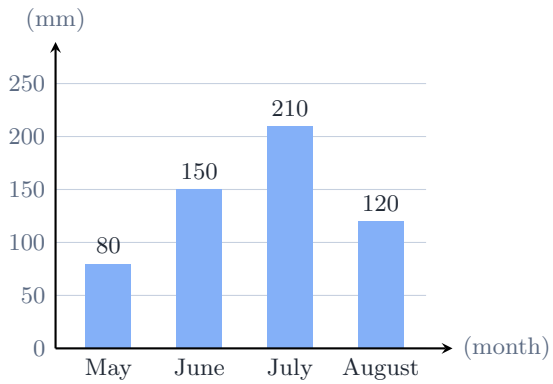
**b)** $120 + 135 + 110 + 140 + 125 = 630$ pupils.

**c)** $140 - 110 = 30$ pupils.

**Answer: a)** Grade 4; Grade 3. **b)** 630 pupils. **c)** 30 pupils.

## Problem 7

The chart shows the rainfall in a town from May to August.



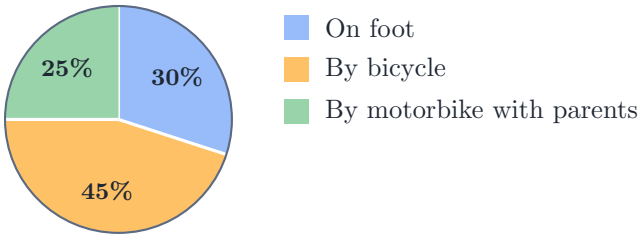

Find the total rainfall and the mean rainfall per month.

## Solution

Total: $80 + 150 + 210 + 120 = 560$ mm.
Mean: $560 : 4 = 140$ mm.
**Answer:** 560 mm; 140 mm.

## Problem 8

Class 5B has 40 pupils. The pie chart shows how they come to school.




**a)** How many pupils walk to school?

**b)** How many more pupils come by bicycle than by motorbike?

## Solution

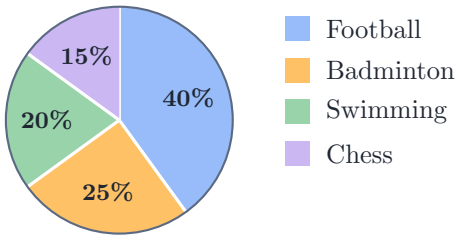
**a)** $40 : 100 \times 30 = 12$ pupils.

**b)** By bicycle: $40 : 100 \times 45 = 18$; by motorbike: $40 : 100 \times 25 = 10$.
Difference: $18 - 10 = 8$ pupils.

**Answer: a)** 12 pupils. **b)** 8 pupils.

## Problem 9

200 pupils chose their favourite sport.




**a)** How many pupils chose football? **b)** chess?

**c)** How many pupils chose football or swimming?

## Solution

**a)** $200 : 100 \times 40 = 80$ pupils.

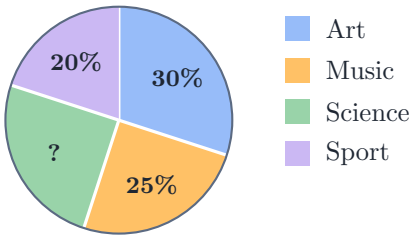
**b)** $200 : 100 \times 15 = 30$ pupils.

**c)** $40\% + 20\% = 60\%$; $200 : 100 \times 60 = 120$ pupils.

**Answer: a)** 80. **b)** 30. **c)** 120 pupils.

## Problem 10

300 pupils each joined one after-school club.




What percentage of the pupils joined the Science club, and how many pupils is that?

## Solution

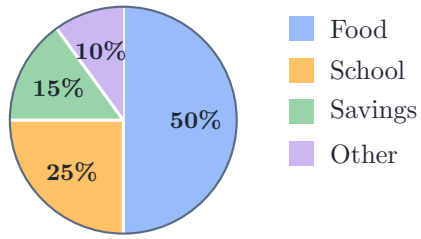
Science: $100\% - 30\% - 25\% - 20\% = 25\%$.

Pupils: $300 : 100 \times 25 = 75$.

**Answer:** 25%; 75 pupils.

## Problem 11

A family spends 12 000 000 dong a month as the chart shows.




How much money goes on food? How much is saved?

### Solution

Food is half: $12\,000\,000 : 2 = 6\,000\,000$ dong.
Savings: $12\,000\,000 : 100 \times 15 = 1\,800\,000$ dong.
**Answer:** Food 6 000 000 dong; savings 1 800 000 dong.

## Problem 12

In a survey, 36 pupils chose mango as their favourite fruit. They are 30% of all the pupils asked. How many pupils were asked?

### Solution

30% of the pupils are 36 pupils, so all the pupils (100%): $36 : 30 \times 100 = 120$.
**Answer:** 120 pupils.

## Problem 13

The mean of five numbers is 18. Four of them are 15, 20, 17 and 22. Find the fifth number.

### Solution

Sum of the five numbers: $18 \times 5 = 90$.
Sum of the four: $15 + 20 + 17 + 22 = 74$.
Fifth number: $90 - 74 = 16$.
**Answer:** 16.

## Problem 14

An's mean mark in three tests is 8. After a fourth test his mean mark is 8.5. What did he get in the fourth test?

**Solution**

Sum of three marks: $8 \times 3 = 24$.
Sum of four marks: $8.5 \times 4 = 34$.
Fourth mark: $34 - 24 = 10$.
**Answer:** 10.

**Problem 15**

Three bags of rice weigh 25 kg on average. A fourth bag of 33 kg is added. What is the mean weight of the four bags?

**Solution**

Three bags: $25 \times 3 = 75$ kg.
Four bags: $75 + 33 = 108$ kg.
Mean: $108 : 4 = 27$ kg.
**Answer:** 27 kg.

**Problem 16**

Tú tosses a coin 50 times; it lands heads up 28 times. Write the ratio describing how often it landed heads, and how often tails, compared with the number of tosses.

**Solution**

Heads: $\frac{28}{50} = \frac{14}{25}$.

Tails: $50 - 28 = 22$ times, $\frac{22}{50} = \frac{11}{25}$.

**Answer:** Heads $\frac{14}{25}$; tails $\frac{11}{25}$.

**Problem 17**

A spinner with red, blue and yellow parts is spun 60 times: red 21 times, blue 15 times, yellow 24 times.
**a)** Write the ratio for each colour. **b)** Write the ratio for "not red".

**Solution**

**a)** Red: $\frac{21}{60} = \frac{7}{20}$; blue: $\frac{15}{60} = \frac{1}{4}$; yellow: $\frac{24}{60} = \frac{2}{5}$.

**b)** Not red: $15 + 24 = 39$ times; $\frac{39}{60} = \frac{13}{20}$.

**Answer: a)** $\frac{7}{20}$; $\frac{1}{4}$; $\frac{2}{5}$. **b)** $\frac{13}{20}$.

## Problem 18

A dice is rolled 30 times:

| **Outcome** | 1 dot | 2 dots | 3 dots | 4 dots | 5 dots | 6 dots |
|---|---|---|---|---|---|---|
| **Times** | 4 | 6 | 5 | 3 | 7 | 5 |

Write the ratio for: **a)** an odd number of dots; **b)** more than 4 dots.

### Solution

**a)** Odd (1, 3, 5): $4 + 5 + 7 = 16$ times; $\frac{16}{30} = \frac{8}{15}$.

**b)** More than 4 (5, 6): $7 + 5 = 12$ times; $\frac{12}{30} = \frac{2}{5}$.

**Answer: a)** $\frac{8}{15}$. **b)** $\frac{2}{5}$.

## Problem 19

Bình rolls a dice many times. The face with 6 dots comes up 9 times, and the ratio for 6 dots is $\frac{3}{20}$. How many times did Bình roll the dice? How many rolls did not show 6 dots?

### Solution

9 times is 3 parts out of 20, so one part is $9 : 3 = 3$ rolls.

Rolls: $3 \times 20 = 60$.

Not 6 dots: $60 - 9 = 51$.

**Answer:** 60 rolls; 51 rolls.

## Problem 20

A bag holds red and blue balls. Mai takes a ball without looking, writes its colour and puts it back. She does this 25 times, and the ratio for red is $\frac{2}{5}$. How many times did she get red, and how many times blue?

### Solution

Red: $25 \times \frac{2}{5} = 10$ times.

Blue: $25 - 10 = 15$ times.

**Answer:** Red 10 times, blue 15 times.

## Problem 21

Two coins are tossed at the same time.

**a)** List all possible outcomes. **b)** In how many outcomes is there one head and one tail?

### Solution

Write H for heads and T for tails (first coin, then second coin).

**a)** HH, HT, TH, TT: 4 outcomes.

**b)** HT and TH: 2 outcomes.

**Answer: a)** 4 outcomes. **b)** 2 outcomes.

## Problem 22

A school canteen offers 3 main dishes (rice, noodles, bread) and 2 drinks (milk, juice). A lunch is one main dish and one drink. How many different lunches are there? List them.

### Solution

Each dish goes with 2 drinks: $3 \times 2 = 6$ lunches.

Rice–milk, rice–juice, noodles–milk, noodles–juice, bread–milk, bread–juice.

**Answer:** 6 lunches.

## Problem 23

Using the digits 1, 4, 7, how many two-digit numbers with two different digits can be written? How many of them are even?

### Solution

Numbers: 14, 17, 41, 47, 71, 74: 6 numbers (3 choices for the tens digit, then 2 for the ones).

Even numbers end in 4: 14 and 74, i.e. 2 numbers.

**Answer:** 6 numbers; 2 even.

## Problem 24

A bag holds 3 red, 5 blue and 2 yellow balls of the same size. One ball is taken without looking.

**a)** Which colour is most likely? **b)** What is the chance of a blue ball?

**c)** Is a green ball certain, possible or impossible?

### Solution

**a)** Blue: there are most blue balls (5).

**b)** 10 balls, 5 blue: $\frac{5}{10} = \frac{1}{2}$.

**c)** There are no green balls: impossible.

**Answer: a)** Blue. **b)** $\frac{1}{2}$. **c)** Impossible.

## Problem 25

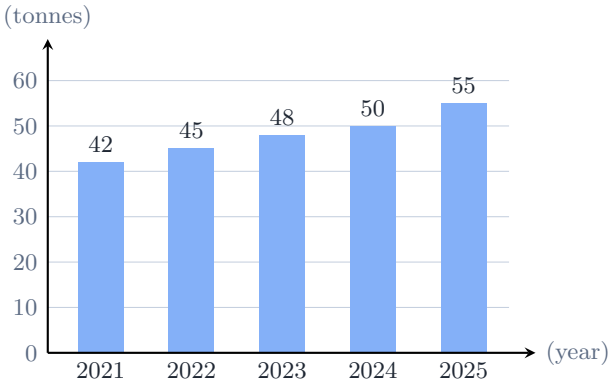
A football team scored 2, 0, 3, 1, 4, 2, 1, 3 goals in eight matches. Find the mean number of goals per match. In how many matches did the team score more than the mean?

## Solution

Goals: $2 + 0 + 3 + 1 + 4 + 2 + 1 + 3 = 16$.
Mean: $16 : 8 = 2$ goals.
More than 2: the matches with 3, 4, 3 goals, i.e. 3 matches.
**Answer:** 2 goals; 3 matches.

## Problem 26

The chart shows the rice harvest of a farm (in tonnes).



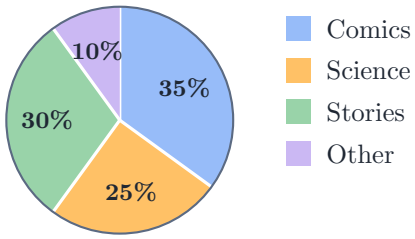

**a)** Find the mean harvest per year. **b)** How much more was harvested in 2025 than in 2021?

## Solution

**a)** $(42 + 45 + 48 + 50 + 55) : 5 = 240 : 5 = 48$ tonnes.
**b)** $55 - 42 = 13$ tonnes.
**Answer: a)** 48 tonnes. **b)** 13 tonnes.

## Problem 27

A library has 1 200 books.




**a)** How many science books are there? **b)** How many more story books than "other" books?

## Solution

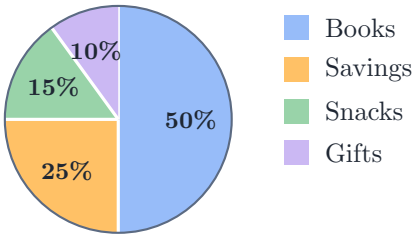
**a)** $1\,200 : 100 \times 25 = 300$ books.

**b)** Stories are $30\% - 10\% = 20\%$ more: $1\,200 : 100 \times 20 = 240$ books.

**Answer: a)** 300 books. **b)** 240 books.

## Problem 28

Hoa spent her 160 000 dong of pocket money as shown.




**a)** How much did she save? **b)** How much more did she spend on books than on snacks?

## Solution

**a)** A quarter: $160\,000 : 4 = 40\,000$ dong.

**b)** Books: 80 000 dong; snacks: $160\,000 : 100 \times 15 = 24\,000$ dong.

Difference: $80\,000 - 24\,000 = 56\,000$ dong.

**Answer: a)** 40 000 dong. **b)** 56 000 dong.

## Problem 29

Five girls weigh 32 kg on average and three boys weigh 36 kg on average. Find the mean weight of all eight children.

**Solution**

Girls: $32 \times 5 = 160$ kg.
Boys: $36 \times 3 = 108$ kg.
Mean: $(160 + 108) : 8 = 268 : 8 = 33.5$ kg.
**Answer:** 33.5 kg.

**Problem 30**

Lan tosses a coin several times. It lands tails up 14 times, and the ratio for heads is $\frac{3}{5}$. How many times did she toss the coin, and how many heads were there?

**Solution**

Ratio for tails: $1 - \frac{3}{5} = \frac{2}{5}$.
So 14 times is 2 parts out of 5: one part is 7 tosses.
Tosses: $7 \times 5 = 35$; heads: $35 - 14 = 21$.
**Answer:** 35 tosses; 21 heads.

## Challenge – Extension

**Problem 31** *

The mean of six numbers is 15. When one number is removed, the mean of the other five is 14. Which number was removed?

**Solution**

Six numbers: $15 \times 6 = 90$.
Five numbers: $14 \times 5 = 70$.
Removed: $90 - 70 = 20$.
**Answer:** 20.

**Problem 32** *

The mean age of 10 players in a team is 30. When the coach is counted too, the mean age is 31. How old is the coach?

**Solution**

Players: $30 \times 10 = 300$ years.
With the coach (11 people): $31 \times 11 = 341$ years.
Coach: $341 - 300 = 41$.
**Answer:** 41 years old.

### Problem 33

Hùng's mean mark in four tests is 7.5. What must he get in the fifth test so that his mean mark becomes 8?

### Solution

Four tests: $7.5 \times 4 = 30$.
Five tests: $8 \times 5 = 40$.
Fifth test: $40 - 30 = 10$.
**Answer:** 10.

### Problem 34

The mean of three numbers is 35. The first is twice the second, and the second is twice the third. Find the three numbers.

### Solution

Sum: $35 \times 3 = 105$.
If the third number is 1 part, the second is 2 parts and the first is 4 parts: 7 parts.
One part: $105 : 7 = 15$.
Numbers: 60, 30, 15.
**Answer:** 60, 30, 15.

### Problem 35

The mean of five consecutive even numbers is 36. Find the largest of them.

### Solution

For equally spaced numbers the mean is the middle number: 36.
The numbers are 32, 34, 36, 38, 40.
**Answer:** 40.

### Problem 36

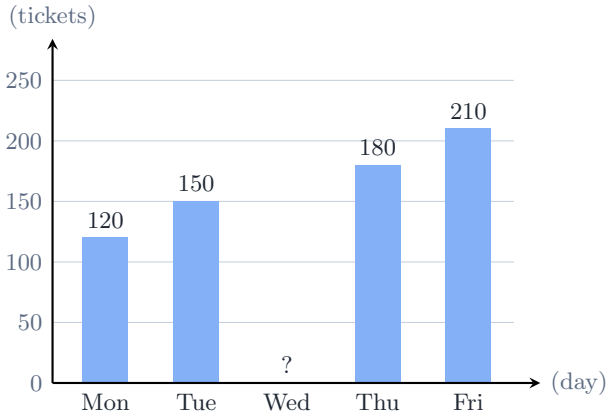
The mean of four marks was worked out as 8.25, but one mark was copied as 6 instead of 9. What is the correct mean?

### Solution

Wrong sum: $8.25 \times 4 = 33$.
Correct sum: $33 - 6 + 9 = 36$.
Correct mean: $36 : 4 = 9$.
**Answer:** 9.

## Problem 37

The chart shows tickets sold at a museum from Monday to Friday; the bar for Wednesday is missing. The mean is 160 tickets a day. How many tickets were sold on Wednesday?

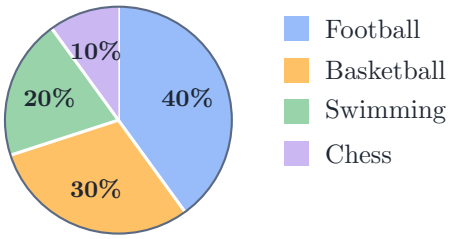



### Solution

Five days: $160 \times 5 = 800$ tickets.
The other four days: $120 + 150 + 180 + 210 = 660$.
Wednesday: $800 - 660 = 140$ tickets.
**Answer:** 140 tickets.

## Problem 38

Each pupil of class 5A chose one sport. 12 pupils chose basketball.




**a)** How many pupils are there in class 5A? **b)** How many chose football? Chess?

### Solution

**a)** 12 pupils are 30%: $12 : 30 \times 100 = 40$ pupils.
**b)** Football: $40 : 100 \times 40 = 16$; chess: $40 : 100 \times 10 = 4$.
**Answer: a)** 40 pupils. **b)** 16; 4.

## Problem 39

160 pupils named their favourite pet: dog 40%, rabbit 15%, and the rest chose cat or fish. Twice as many pupils chose cats as chose fish. How many pupils chose cats?

### Solution

Cat and fish: $100\% - 40\% - 15\% = 45\%$.
Fish 1 part, cat 2 parts: one part is $45\% : 3 = 15\%$.
Cat: 30%, i.e. $160 : 100 \times 30 = 48$ pupils.
**Answer:** 48 pupils.

## Problem 40

In class 5A (40 pupils), 35% like reading best. In class 5B (36 pupils), 50% like reading best. Which class has more pupils who like reading best, and by how many?

### Solution

5A: $40 : 100 \times 35 = 14$ pupils.
5B: $36 : 2 = 18$ pupils.
5B has $18 - 14 = 4$ more.
**Answer:** Class 5B, by 4 pupils.

## Problem 41

Using the digits 0, 3, 5, 8, how many two-digit numbers with different digits can be written? How many of them are divisible by 5?

### Solution

Tens digit: 3 choices (3, 5, 8; not 0); ones digit: 3 choices (any other digit).
Numbers: $3 \times 3 = 9$.
Divisible by 5 means ending in 0 or 5: 30, 50, 80, 35, 85: 5 numbers.
**Answer:** 9 numbers; 5 of them.

## Problem 42

Six friends meet, and every two of them shake hands once. How many handshakes are there?

### Solution

Each person shakes 5 hands: $6 \times 5 = 30$, but each handshake is counted twice.
Handshakes: $30 : 2 = 15$.
**Answer:** 15 handshakes.

## Problem 43

Two dice are rolled together and the dots are added.

**a)** How many outcomes are there? **b)** In how many outcomes is the sum 7? What is the chance of a sum of 7?

**c)** What is the chance of a sum of 12?

## Solution

**a)** Each of the 6 faces of the first dice goes with 6 faces of the second: $6 \times 6 = 36$ outcomes.

**b)** Sum 7: $1+6,\ 2+5,\ 3+4,\ 4+3,\ 5+2,\ 6+1$: 6 outcomes. Chance: $\frac{6}{36} = \frac{1}{6}$.

**c)** Only $6+6$: chance $\frac{1}{36}$.

**Answer: a)** 36. **b)** 6; $\frac{1}{6}$. **c)** $\frac{1}{36}$.

## Problem 44

A box holds 4 red and 6 blue balls. **a)** What is the chance of taking a red ball without looking?

**b)** How many red balls must be added so that the chance of red becomes $\frac{1}{2}$?

## Solution

**a)** $\frac{4}{10} = \frac{2}{5}$.

**b)** The chance is $\frac{1}{2}$ when there are as many red as blue balls: 6 red.

Add $6 - 4 = 2$ red balls.

**Answer: a)** $\frac{2}{5}$. **b)** 2 red balls.

## Problem 45

A bag holds 20 red, blue and green balls. The chance of a red ball is $\frac{1}{4}$ and the chance of a blue ball is $\frac{2}{5}$. How many balls of each colour are there?

## Solution

Red: $20 \times \frac{1}{4} = 5$.

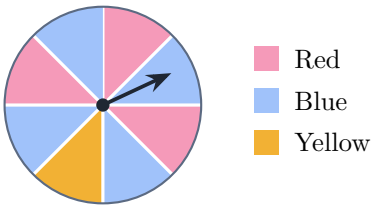
Blue: $20 \times \frac{2}{5} = 8$.

Green: $20 - 5 - 8 = 7$.

**Answer:** 5 red, 8 blue, 7 green.

## Problem 46 *

A spinner is divided into 8 equal parts: 3 red, 4 blue and 1 yellow.




**a)** What is the chance that the arrow stops on blue? **b)** About how many times would you expect blue in 80 spins?

### Solution

**a)** 4 of the 8 equal parts are blue: $\frac{4}{8} = \frac{1}{2}$.

**b)** About half of the spins: $80 \times \frac{1}{2} = 40$ times.

**Answer: a)** $\frac{1}{2}$. **b)** About 40 times.

## Problem 47 *

The mean age of a family of four is 24 years now. What was their mean age 4 years ago (all four were already born then)?

### Solution

Total age now: $24 \times 4 = 96$ years.
4 years ago each person was 4 years younger: $96 - 4 \times 4 = 80$ years.
Mean: $80 : 4 = 20$ years.
**Answer:** 20 years.

## Problem 48 * *

In a test, the 32 pupils of a class have a mean mark of 8. The 20 girls have a mean mark of 8.3. Find the mean mark of the boys.

### Solution

Whole class: $8 \times 32 = 256$.
Girls: $8.3 \times 20 = 166$.
Boys: $32 - 20 = 12$ pupils with $256 - 166 = 90$ marks.
Mean: $90 : 12 = 7.5$.
**Answer:** 7.5.

## Problem 49

A box holds 12 balls, some red and some white. Nam takes a ball, notes its colour and puts it back, 60 times; red comes up 45 times. About how many red balls are there in the box?

### Solution

Ratio for red: $\frac{45}{60} = \frac{3}{4}$.

So about $\frac{3}{4}$ of the balls are red: $12 \times \frac{3}{4} = 9$ balls.

**Answer:** About 9 red balls.

## Problem 50

An, Bình and Chi stand in a row for a photo.

**a)** In how many different orders can they stand? **b)** In how many of them is An in the middle?

### Solution

**a)** First place: 3 choices, second: 2, third: 1; $3 \times 2 \times 1 = 6$ orders:

ABC, ACB, BAC, BCA, CAB, CBA (A = An, B = Bình, C = Chi).

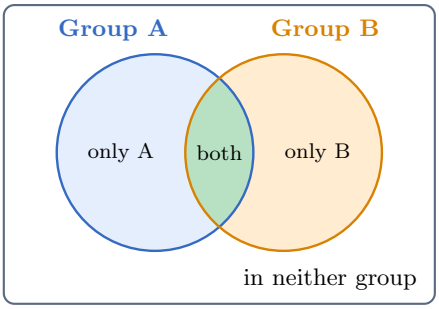
**b)** An in the middle: BAC and CAB, i.e. 2 orders.

**Answer: a)** 6. **b)** 2.

# 12 Venn Diagrams, Working Back

## ① Venn diagrams with two circles

A **Venn diagram** shows groups as closed curves (circles) inside a rectangle that stands for everybody.




- The *overlap* of the two circles holds those who belong to **both** groups.
- The rest of each circle holds those who belong to **only** that group.
- Inside the rectangle but outside both circles are those who belong to **neither** group.

When we add A + B, the members of both groups are counted *twice*. So:

$$\begin{aligned} \textbf{at least one group} &= \text{A} + \text{B} - \text{both} \\ \textbf{both} &= \text{A} + \text{B} - (\text{at least one group}) \\ \textbf{only A} &= \text{A} - \text{both} \\ \textbf{neither} &= \text{everybody} - (\text{at least one group}) \end{aligned}$$

## ② Venn diagrams with three circles

Three circles A, B, C make 7 regions inside them (plus the region outside). Fill in the diagram **from the centre outwards**:

1. the centre: those in all three groups;
2. each overlap of two circles: (in both groups) − (in all three);
3. each "only" region: (whole circle) − (the three numbers already written in it).

Adding all the regions gives the number in at least one group. In one line:

$$\text{at least one} = \text{A} + \text{B} + \text{C} - (\text{A and B}) - (\text{B and C}) - (\text{C and A}) + (\text{all three}).$$

## ③ Working backwards with a table

When things are passed from one person (or box) to another several times and we know the **final** result, we work **backwards** from the end:

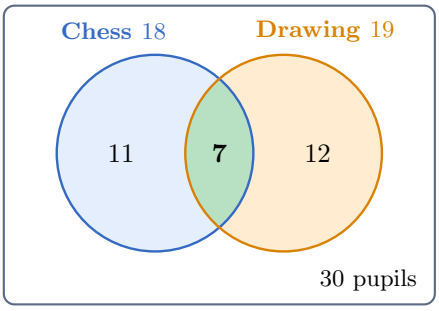
- Passing things around does *not* change the total. If at the end everyone has the same, each has total : number of people.
- Undo the steps in reverse order: whoever *gave* gets the amount back (+), whoever *received* gives it back (−).
- "Gives as many as the other already has" makes the receiver's amount *double*; going back we *halve* it.
- A **table** with one row for each moment keeps the work tidy. Check by working forwards.

### Example 1

A group of 30 pupils each joins at least one of two clubs: 18 join the Chess club and 19 join the Drawing club. How many pupils join both clubs? How many join only one club?

**Guided solution.**

Adding $18 + 19 = 37$ counts the pupils in both clubs twice, so
both clubs: $37 - 30 = 7$ pupils.
Only Chess: $18 - 7 = 11$; only Drawing: $19 - 7 = 12$; only one club: $11 + 12 = 23$ pupils.



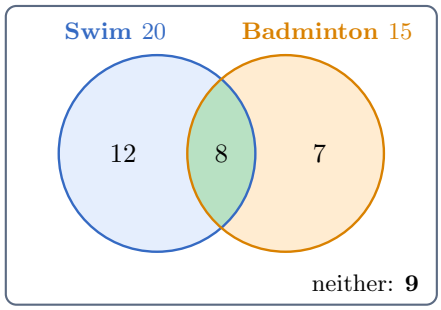

**Answer:** 7 pupils; 23 pupils.

### Example 2

Class 5C has 36 pupils. 20 pupils can swim, 15 can play badminton and 8 can do both. How many pupils can do neither?

**Guided solution.**

*Method 1.* At least one: $20 + 15 - 8 = 27$ pupils.
Neither: $36 - 27 = 9$ pupils.
*Method 2.* Only swim: $20 - 8 = 12$; only badminton: $15 - 8 = 7$.
At least one: $12 + 7 + 8 = 27$; neither: $36 - 27 = 9$.




**Answer:** 9 pupils.

## Example 3

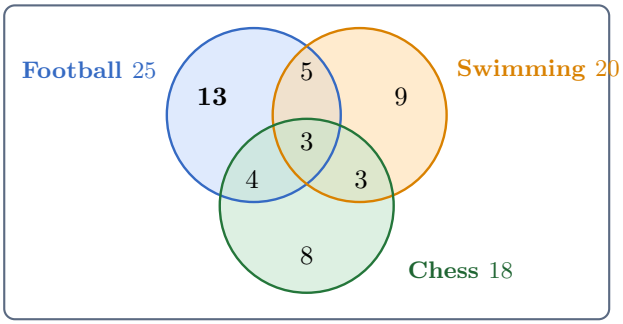
Every pupil in a group likes at least one of football, swimming and chess. 25 like football, 20 like swimming and 18 like chess; 8 like football and swimming, 6 like swimming and chess, 7 like football and chess, and 3 like all three.

**a)** How many like only football? **b)** How many pupils are in the group?

### Guided solution.

Fill in from the centre: all three: 3.
Football and swimming only: $8 - 3 = 5$; swimming and chess only: $6 - 3 = 3$; football and chess only: $7 - 3 = 4$.

**a)** Only football: $25 - 5 - 4 - 3 = 13$.
Only swimming: $20 - 5 - 3 - 3 = 9$; only chess: $18 - 4 - 3 - 3 = 8$.




**b)** $13 + 9 + 8 + 5 + 4 + 3 + 3 = 45$ pupils.
(Check: $25 + 20 + 18 - 8 - 6 - 7 + 3 = 45$.)

**Answer: a)** 13 pupils. **b)** 45 pupils.

## Example 4

An, Bình and Chi have 75 stickers altogether. An gives Bình 6 stickers, then Bình gives Chi 9 stickers, then Chi gives An 4 stickers. Now the three have the same number. How many stickers did each have at first?

### Guided solution.

The total stays 75, so at the end each has $75 : 3 = 25$ stickers. Work backwards, one row per step:

| | An | Bình | Chi |
|---|---|---|---|
| At the end | 25 | 25 | 25 |
| Before "Chi gives An 4" | $25 - 4 = 21$ | 25 | $25 + 4 = 29$ |
| Before "Bình gives Chi 9" | 21 | $25 + 9 = 34$ | $29 - 9 = 20$ |
| Before "An gives Bình 6" (at first) | $21 + 6 = \mathbf{27}$ | $34 - 6 = \mathbf{28}$ | **20** |

Check: $27 + 28 + 20 = 75$.

**Answer:** An 27, Bình 28, Chi 20 stickers.

## Example 5

Lan, Mai and Cúc have 120 shells. First Lan gives Mai and Cúc each as many shells as each of them already has. Then Mai does the same for Lan and Cúc, and finally Cúc does the same for Lan and Mai. Now all three have the same number. How many shells did each have at first?

### Guided solution.

At the end each has $120 : 3 = 40$. Each time, the two receivers *double*; going back we halve them, and the giver gets back what they received.

| | Lan | Mai | Cúc |
|---|---|---|---|
| At the end | 40 | 40 | 40 |
| Before Cúc gives | 20 | 20 | $40 + 20 + 20 = 80$ |
| Before Mai gives | 10 | $20 + 10 + 40 = 70$ | 40 |
| At first (before Lan gives) | $10 + 35 + 20 = \mathbf{65}$ | **35** | **20** |

Check forwards: $(65, 35, 20) \to (10, 70, 40) \to (20, 20, 80) \to (40, 40, 40)$.

**Answer:** Lan 65, Mai 35, Cúc 20 shells.

# Practice

## Problem 1

Every pupil of class 5A got at least one mark 10 in Maths or Vietnamese. 21 pupils got a 10 in Maths, 17 in Vietnamese, and 6 got a 10 in both. How many pupils are there in class 5A?

## Solution

$21 + 17 - 6 = 32$ pupils.

**Answer:** 32 pupils.

## Problem 2

Each of 40 pupils likes reading or drawing (or both). 28 like reading and 22 like drawing. How many like both?

### Solution

$28 + 22 = 50$ counts those who like both twice.
Both: $50 - 40 = 10$ pupils.
**Answer:** 10 pupils.

## Problem 3

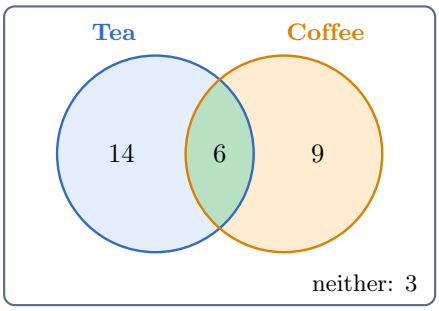
In a group of 45 tourists, 30 speak English, 20 speak French and 5 speak neither language.
**a)** How many speak both? **b)** How many speak only English? **c)** only French?

### Solution

At least one language: $45 - 5 = 40$.
**a)** Both: $30 + 20 - 40 = 10$.
**b)** Only English: $30 - 10 = 20$.
**c)** Only French: $20 - 10 = 10$.
**Answer: a)** 10. **b)** 20. **c)** 10.

## Problem 4

The Venn diagram shows what the people at a party drink.



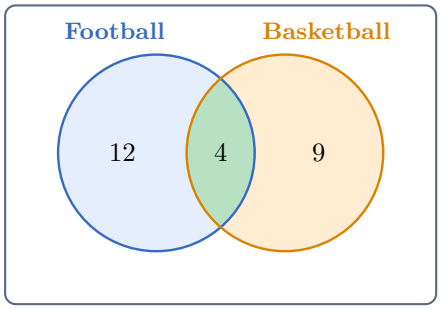

**a)** How many drink tea? **b)** How many drink coffee? **c)** How many people are at the party?

### Solution

**a)** Tea: $14 + 6 = 20$.
**b)** Coffee: $6 + 9 = 15$.
**c)** $14 + 6 + 9 + 3 = 32$ people.
**Answer: a)** 20. **b)** 15. **c)** 32 people.

## Problem 5

Each member of a sports team plays football or basketball (or both), as the diagram shows.




**a)** How many members does the team have? **b)** How many play basketball?

### Solution

**a)** $12 + 4 + 9 = 25$ members.

**b)** $4 + 9 = 13$ members.

**Answer: a)** 25. **b)** 13.

## Problem 6

A maths team of 20 pupils is given two hard problems; everybody solves at least one. 3 pupils cannot solve problem 1 and 5 cannot solve problem 2. How many solve both problems?

### Solution

The 3 who cannot solve problem 1 solve only problem 2.
The 5 who cannot solve problem 2 solve only problem 1.
Both: $20 - 3 - 5 = 12$ pupils.
**Answer:** 12 pupils.

## Problem 7

Class 5D has 38 pupils. 15 sign up for swimming, 20 for football and 7 for both. How many sign up for neither?

### Solution

At least one: $15 + 20 - 7 = 28$.
Neither: $38 - 28 = 10$ pupils.
**Answer:** 10 pupils.

## Problem 8

In a class of 35 pupils, 14 have a dog, 11 have a cat and 5 have both. How many pupils have neither a dog nor a cat?

### Solution

At least one pet: $14 + 11 - 5 = 20$.
Neither: $35 - 20 = 15$ pupils.
**Answer:** 15 pupils.

## Problem 9

In a village, 60 families were asked: 42 have a television, 35 have a computer, and every family has at least one of them.
**a)** How many have both? **b)** only a television? **c)** only a computer?

### Solution

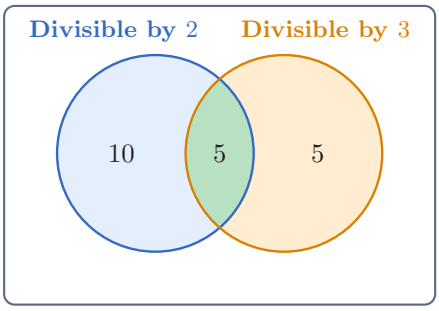
**a)** $42 + 35 - 60 = 17$ families.
**b)** $42 - 17 = 25$ families.
**c)** $35 - 17 = 18$ families.
**Answer: a)** 17. **b)** 25. **c)** 18.

## Problem 10

How many of the numbers from 1 to 30 are divisible by 2 or by 3?

### Solution

Divisible by 2: $30 : 2 = 15$ numbers. Divisible by 3: $30 : 3 = 10$ numbers.
Divisible by both, i.e. by 6: 6, 12, 18, 24, 30: 5 numbers.




By 2 or by 3: $15 + 10 - 5 = 20$ numbers.
**Answer:** 20 numbers.

## Problem 11

How many of the numbers from 1 to 50 are divisible neither by 2 nor by 5?

### Solution

By 2: 25 numbers; by 5: 10 numbers; by both (by 10): 5 numbers.
By 2 or 5: $25 + 10 - 5 = 30$.
Neither: $50 - 30 = 20$ numbers.
**Answer:** 20 numbers.

## Problem 12

In a music group, 16 pupils play the piano, 10 play the guitar, and 22 pupils play at least one of them. How many play both? How many play only the piano?

### Solution

Both: $16 + 10 - 22 = 4$ pupils.
Only piano: $16 - 4 = 12$ pupils.
**Answer:** 4 pupils; 12 pupils.

## Problem 13

18 pupils of a class entered the singing contest, 13 entered the drawing contest, and 25 pupils entered at least one contest. How many pupils entered exactly one contest?

### Solution

Both contests: $18 + 13 - 25 = 6$ pupils.
Exactly one: $25 - 6 = 19$ pupils.
**Answer:** 19 pupils.

## Problem 14

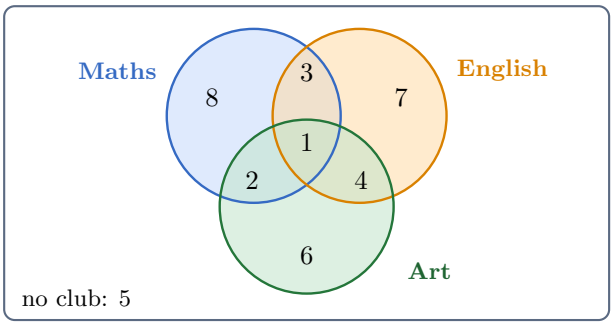
100 pupils were asked about breakfast: 58 like phở, 47 like bánh mì and 15 like neither. How many like both?

### Solution

At least one: $100 - 15 = 85$.
Both: $58 + 47 - 85 = 20$ pupils.
**Answer:** 20 pupils.

## Problem 15

The diagram shows how many pupils of a class are in the Maths, English and Art clubs.




**a)** How many pupils are in the Maths club? **b)** How many are in exactly one club?

**c)** How many pupils are there in the class?

## Solution

**a)** Every number inside the Maths circle: $8 + 3 + 2 + 1 = 14$.

**b)** $8 + 7 + 6 = 21$.

**c)** $8 + 7 + 6 + 3 + 2 + 4 + 1 + 5 = 36$.

**Answer: a)** 14. **b)** 21. **c)** 36 pupils.

## Problem 16

Every pupil of a contest team takes at least one of Maths, English and Art: 25 take Maths, 20 take English and 15 take Art. 6 take Maths and English, 4 take English and Art, 5 take Maths and Art, and nobody takes all three. How many pupils are in the team?

## Solution

Adding $25 + 20 + 15 = 60$ counts each pupil who takes two subjects twice.

Team: $60 - 6 - 4 - 5 = 45$ pupils.

**Answer:** 45 pupils.

## Problem 17

40 pupils were asked about fruit: 20 like apples, 18 bananas, 16 oranges; 7 like apples and bananas, 6 bananas and oranges, 5 apples and oranges, and 2 like all three. How many pupils like none of the three?

## Solution

At least one: $20 + 18 + 16 - 7 - 6 - 5 + 2 = 38$.

None: $40 - 38 = 2$ pupils.

**Answer:** 2 pupils.

## Problem 18

In a class of 30, 15 like singing, 12 dancing and 10 drawing; 5 like singing and dancing, 4 dancing and drawing, 3 singing and drawing, and 1 likes all three.

**a)** How many like only singing? **b)** How many like none of them?

### Solution

**a)** Singing and dancing only: $5 - 1 = 4$; singing and drawing only: $3 - 1 = 2$.

Only singing: $15 - 4 - 2 - 1 = 8$.

**b)** At least one: $15 + 12 + 10 - 5 - 4 - 3 + 1 = 26$; none: $30 - 26 = 4$.

**Answer: a)** 8. **b)** 4.

## Problem 19

In a group, twice as many pupils like football as like volleyball. 6 pupils like both, and 30 pupils like at least one of the two games. How many like football and how many like volleyball?

### Solution

Football + volleyball $= 30 + 6 = 36$.

Volleyball 1 part, football 2 parts: 3 parts $= 36$, one part $= 12$.

Volleyball: 12; football: 24.

**Answer:** Football 24, volleyball 12.

## Problem 20

An, Bình and Cường have 60 marbles. An gives Bình 5 marbles, Bình gives Cường 8, then Cường gives An 3. Now they all have the same number. How many did each have at first?

### Solution

At the end each has $60 : 3 = 20$.

An: $20 - 3 + 5 = 22$.

Bình: $20 + 8 - 5 = 23$.

Cường: $20 - 8 + 3 = 15$.

Check: $22 + 23 + 15 = 60$.

**Answer:** An 22, Bình 23, Cường 15.

## Problem 21

Two boxes hold 48 pens. After 7 pens are moved from box A to box B, the two boxes hold the same number. How many pens were in each box at first?

**Solution**

At the end each box holds $48 : 2 = 24$.
Box A: $24 + 7 = 31$; box B: $24 - 7 = 17$.
**Answer:** Box A 31, box B 17.

**Problem 22**

Three shelves hold 90 books. 8 books are moved from shelf I to shelf II, then 12 from shelf II to shelf III, then 5 from shelf III to shelf I. Now the shelves hold the same number. How many books were on each shelf at first?

**Solution**

At the end: $90 : 3 = 30$ each.
Shelf I: $30 - 5 + 8 = 33$.
Shelf II: $30 + 12 - 8 = 34$.
Shelf III: $30 - 12 + 5 = 23$.
**Answer:** 33, 34, 23 books.

**Problem 23**

Hà gives Lan 4 stickers, and then Lan gives Hà 10 stickers. Now Hà has 26 and Lan has 18. How many did each have at first?

**Solution**

Before Lan gives: Hà $26 - 10 = 16$, Lan $18 + 10 = 28$.
Before Hà gives (at first): Hà $16 + 4 = 20$, Lan $28 - 4 = 24$.
**Answer:** Hà 20, Lan 24.

**Problem 24**

From basket A, Mum moves into basket B as many oranges as basket B already has. Now each basket holds 30 oranges. How many were in each basket at first?

**Solution**

Basket B doubled, so at first it held $30 : 2 = 15$.
Basket A gave 15: at first $30 + 15 = 45$.
**Answer:** Basket A 45, basket B 15.

## Problem 25

Two tanks hold 80 litres of water. Tank A pours into tank B as much water as tank B has; then tank B pours into tank A as much as tank A then has. Now the tanks hold the same amount. How much did each tank hold at first?

### Solution

At the end: 40 litres each.
Before step 2: A $40 : 2 = 20$, B $40 + 20 = 60$.
Before step 1: B $60 : 2 = 30$, A $20 + 30 = 50$.
**Answer:** Tank A 50 litres, tank B 30 litres.

## Problem 26

Hùng puts 36 marbles into three boxes. He moves from box 1 into box 2 as many marbles as box 2 has, then from box 2 into box 3 as many as box 3 has. Now the three boxes hold the same number. How many marbles were in each box at first?

### Solution

At the end: $36 : 3 = 12$ each.
Before step 2: box 3 had $12 : 2 = 6$, box 2 had $12 + 6 = 18$, box 1 had 12.
Before step 1: box 2 had $18 : 2 = 9$, box 1 had $12 + 9 = 21$.
**Answer:** 21, 9, 6 marbles.

## Problem 27

Hòa, Bình and Việt have 70 cards. Hòa gives Bình 5 cards, Bình gives Việt 7, then Việt gives Hòa 9. Now Hòa and Bình have the same number, and each has twice as many as Việt. How many cards did each have at first?

### Solution

At the end: Việt 1 part, Hòa 2 parts, Bình 2 parts; 5 parts = 70, so one part = 14.
End: Hòa 28, Bình 28, Việt 14.
At first: Hòa $28 - 9 + 5 = 24$; Bình $28 + 7 - 5 = 30$; Việt $14 - 7 + 9 = 16$.
**Answer:** Hòa 24, Bình 30, Việt 16.

## Problem 28

A woman sells half of her mangoes and 2 more; then she sells half of the rest and 3 more. She has 5 mangoes left. How many mangoes did she have?

**Solution**

Before the second sale: $(5+3)\times 2 = 16$ mangoes.

At first: $(16+2)\times 2 = 36$ mangoes.

**Answer:** 36 mangoes.

## Problem 29

Ân and Bảo have 64 marbles. Ân gives Bảo as many as Bảo has; then Bảo gives Ân as many as Ân has; then Ân again gives Bảo as many as Bảo has. Now they have the same number. How many did each have at first?

**Solution**

At the end: 32 each.

Before step 3: Bảo 16, Ân 48.

Before step 2: Ân 24, Bảo 40.

At first: Bảo 20, Ân 44.

**Answer:** Ân 44, Bảo 20.

## Problem 30

On the first day $\frac{1}{4}$ of the water in a tank is used; on the second day $\frac{1}{3}$ of what is left is used. 60 litres remain. How much water was in the tank at first?

**Solution**

60 litres is $1-\frac{1}{3}=\frac{2}{3}$ of what was left after day one: $60:\frac{2}{3}=90$ litres.

90 litres is $1-\frac{1}{4}=\frac{3}{4}$ of the tank: $90:\frac{3}{4}=120$ litres.

**Answer:** 120 litres.

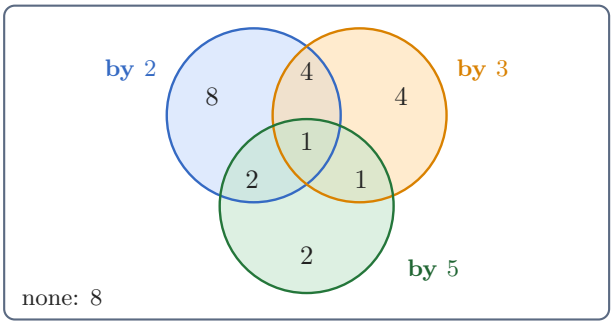
# Challenge – Extension

## Problem 31 ★★

How many numbers from 1 to 30 are divisible by 2, by 3 or by 5? How many are divisible by none of them?

**Solution**

By 2: 15; by 3: 10; by 5: 6.
By 6: 5; by 10: 3; by 15: 2; by 30 (all three): 1.
Fill in from the centre:




At least one: $15 + 10 + 6 - 5 - 3 - 2 + 1 = 22$.
None: $30 - 22 = 8$.
**Answer:** 22 numbers; 8 numbers.

**Problem 32** *

In a class of 40, 25 pupils like Maths and 28 like Literature. What is the least number of pupils who can like both? The greatest?

**Solution**

At most 40 pupils like at least one, so both $\geq 25 + 28 - 40 = 13$ (when everybody likes at least one).
Both cannot be more than the smaller group: at most 25 (when every Maths lover also likes Literature).
**Answer:** Least 13, greatest 25.

**Problem 33** *

45 pupils each belong to at least one of three clubs A, B, C. Club A has 24 members, B has 22, C has 20; 9 pupils are in A and B, 7 in B and C, 8 in C and A. How many are in all three clubs?

**Solution**

$45 = 24 + 22 + 20 - 9 - 7 - 8 + (\text{all three}) = 42 + (\text{all three})$.
All three: $45 - 42 = 3$ pupils.
**Answer:** 3 pupils.

## Problem 34 **

In a school contest team, 30 pupils do Maths, 24 do English and 20 do Science. 46 pupils do Maths or English, 38 do English or Science, 42 do Science or Maths, and nobody does all three.

**a)** How many do Maths and English? **b)** English and Science? **c)** Science and Maths?

**d)** How many pupils are in the team?

### Solution

**a)** $30 + 24 - 46 = 8$.

**b)** $24 + 20 - 38 = 6$.

**c)** $20 + 30 - 42 = 8$.

**d)** $30 + 24 + 20 - 8 - 6 - 8 = 52$ pupils.

**Answer: a)** 8. **b)** 6. **c)** 8. **d)** 52 pupils.

## Problem 35 *

In a class, 8 pupils like both dogs and cats, 5 like neither, and 14 like dogs. The number who like only cats is twice the number who like only dogs. How many pupils are in the class, and how many like cats?

### Solution

Only dogs: $14 - 8 = 6$.

Only cats: $6 \times 2 = 12$.

Class: $6 + 12 + 8 + 5 = 31$ pupils.

Cats: $12 + 8 = 20$ pupils.

**Answer:** 31 pupils; 20 like cats.

## Problem 36 *

How many numbers from 1 to 100 are divisible by 3 but not by 4?

### Solution

By 3: 33 numbers (since $100 = 3 \times 33 + 1$).

By both 3 and 4, i.e. by 12: 8 numbers (12, 24, ..., 96).

By 3 but not 4: $33 - 8 = 25$.

**Answer:** 25 numbers.

## Problem 37 **

Each pupil in a group does at least one of three sports A, B, C: 55 do A, 45 do B and 40 do C. 20 pupils do exactly two sports and 5 do all three. How many pupils are in the group?

**Solution**

In $55 + 45 + 40 = 140$, a pupil with one sport is counted once, with exactly two sports twice, with three sports three times.
Pupils with exactly one sport: $140 - 2 \times 20 - 3 \times 5 = 85$.
Group: $85 + 20 + 5 = 110$ pupils.
**Answer:** 110 pupils.

## Problem 38 *

Of 50 pupils, 35 can swim and 30 can play chess.
**a)** At least how many pupils can do both? **b)** If exactly 4 pupils can do neither, how many can do both?

**Solution**

**a)** At most 50 can do at least one, so both $\geq 35 + 30 - 50 = 15$.
**b)** At least one: $50 - 4 = 46$; both: $35 + 30 - 46 = 19$.
**Answer: a)** 15. **b)** 19.

## Problem 39 **

Quân, Sơn and Tùng have 48 marbles. Quân takes half of his marbles and shares them equally between the other two; then Sơn does the same with half of the marbles he then has; finally Tùng does the same. Now the three have the same number. How many did each have at first?

**Solution**

At the end: 16 each. A giver keeps half; each of the others receives a quarter of what the giver had.

| | Quân | Sơn | Tùng |
|---|---|---|---|
| At the end | 16 | 16 | 16 |
| Before Tùng gives | $16 - 8 = 8$ | $16 - 8 = 8$ | $16 \times 2 = 32$ |
| Before Sơn gives | $8 - 4 = 4$ | $8 \times 2 = 16$ | $32 - 4 = 28$ |
| At first | $4 \times 2 = \mathbf{8}$ | $16 - 2 = \mathbf{14}$ | $28 - 2 = \mathbf{26}$ |

**Answer:** Quân 8, Sơn 14, Tùng 26.

## Problem 40 **

There are three drums of oil A, B, C. The oil in drum A is split into 3 equal parts: one part is poured into B and one into C. Then the oil now in B is split into 4 equal parts: one part is poured into A and one into C. Then the oil now in C is split into 3 equal parts: one into A and one into

B. Now A holds 40 litres, B 46 litres and C 18 litres. How much oil was in each drum at first?

**Solution**

Before C pours: C kept 1 part of 3, so C had $18 \times 3 = 54$; A and B each got 18: A $40 - 18 = 22$, B $46 - 18 = 28$.
Before B pours: B kept 2 parts of 4, so B had $28 \times 2 = 56$; A and C each got $56 : 4 = 14$: A $22 - 14 = 8$, C $54 - 14 = 40$.
Before A pours: A kept 1 part of 3, so A had $8 \times 3 = 24$; B and C each got 8: B $56 - 8 = 48$, C $40 - 8 = 32$.
**Answer:** A 24 litres, B 48 litres, C 32 litres.

**Problem 41** *

Lan reads a book. On day one she reads $\frac{1}{3}$ of the book and 4 more pages; on day two, half of the remaining pages and 6 more; on day three, the last 24 pages. How many pages has the book?

**Solution**

On day two, half of the remaining pages and 6 more leave 24 pages, so before day two there were $(24 + 6) \times 2 = 60$ pages left.

Day one: $60 + 4 = 64$ pages is $1 - \frac{1}{3} = \frac{2}{3}$ of the book.

Book: $64 : \frac{2}{3} = 96$ pages.

**Answer:** 96 pages.

**Problem 42** **

Three boxes I, II, III hold some sweets. First, box I gives boxes II and III each as many sweets as each already holds. Then box II gives boxes I and III each as many as each then holds. Now box I has 20 sweets, box II 30 and box III 40. How many sweets did each box hold at first?

**Solution**

Before box II gives: I $20 : 2 = 10$, III $40 : 2 = 20$, II $30 + 10 + 20 = 60$.
Before box I gives: II $60 : 2 = 30$, III $20 : 2 = 10$, I $10 + 30 + 10 = 50$.
Check: $(50, 30, 10) \to (10, 60, 20) \to (20, 30, 40)$.
**Answer:** I 50, II 30, III 10 sweets.

**Problem 43** *

Bình spends $\frac{1}{4}$ of his money on books, then $\frac{2}{5}$ of the remaining money on a bag, then 15 000 dong

on snacks. He has 30 000 dong left. How much money did he have at first?

**Solution**

Before the snacks: $30\,000 + 15\,000 = 45\,000$ dong.

This is $1 - \frac{2}{5} = \frac{3}{5}$ of the money left after the books: $45\,000 : \frac{3}{5} = 75\,000$ dong.

This is $\frac{3}{4}$ of his money: $75\,000 : \frac{3}{4} = 100\,000$ dong.

**Answer:** 100 000 dong.

**Problem 44** ★

Mai says: "Multiply my age by 3, add 12, divide by 5 and take away 4: you get 5." How old is Mai?

**Solution**

$5 + 4 = 9$; $9 \times 5 = 45$; $45 - 12 = 33$; $33 : 3 = 11$.

Check: $11 \times 3 + 12 = 45$, $45 : 5 = 9$, $9 - 4 = 5$.

**Answer:** 11 years old.

**Problem 45** ★★

40 pupils each join at least one of three clubs: Reading 22, Science 18, Dance 20. 3 pupils join all three. How many pupils join exactly two clubs? Exactly one?

**Solution**

In $22 + 18 + 20 = 60$: one club counted once, two clubs twice, three clubs three times.

Without the 3 pupils in all three clubs: 37 pupils, counted $60 - 9 = 51$ times.

Each pupil in exactly two clubs adds 1 extra: exactly two $= 51 - 37 = 14$.

Exactly one: $37 - 14 = 23$.

**Answer:** Exactly two: 14; exactly one: 23.

**Problem 46** ★★

In a class, $\frac{3}{5}$ of the pupils like Maths, $\frac{1}{2}$ like English, $\frac{1}{4}$ like both, and 6 pupils like neither. How many pupils are in the class?

**Solution**

At least one: $\frac{3}{5} + \frac{1}{2} - \frac{1}{4} = \frac{17}{20}$ of the class.

Neither: $1 - \frac{17}{20} = \frac{3}{20}$ of the class $= 6$ pupils.

Class: $6 : \frac{3}{20} = 40$ pupils.

**Answer:** 40 pupils.

### Problem 47 *

A shop sells 40% of its rice in the morning and 50% of the rest in the afternoon. 66 kg are left. How much rice did the shop have?

### Solution

After the morning, 60% is left; the afternoon leaves half of that: 30% of the rice.

30% is 66 kg, so the rice: $66 : 30 \times 100 = 220$ kg.

**Answer:** 220 kg.

### Problem 48 *

An, Bình, Chi and Dũng have 100 cards. An gives Bình 3, Bình gives Chi 5, Chi gives Dũng 7, and Dũng gives An 9. Now all four have the same number. How many did each have at first?

### Solution

At the end: $100 : 4 = 25$ each.

An: $25 - 9 + 3 = 19$; Bình: $25 + 5 - 3 = 27$; Chi: $25 + 7 - 5 = 27$; Dũng: $25 - 7 + 9 = 27$.

**Answer:** An 19, Bình 27, Chi 27, Dũng 27.

### Problem 49 *

How many two-digit numbers contain at least one digit 5?

### Solution

Tens digit 5: 50 to 59, 10 numbers.

Ones digit 5: 15, 25, ..., 95, 9 numbers.

Both: only 55.

At least one digit 5: $10 + 9 - 1 = 18$.

**Answer:** 18 numbers.

### Problem 50 **

An, Bình and Chi play three games. The loser of each game gives each of the other two as many

marbles as that player already has. An loses game 1, Bình loses game 2 and Chi loses game 3. At the end An has 16, Bình 24 and Chi 32 marbles. How many did each have at first?

**Solution**

Going back, the two winners halve and the loser gets it back.

| | An | Bình | Chi |
|---|---|---|---|
| At the end | 16 | 24 | 32 |
| Before game 3 | 8 | 12 | $32 + 8 + 12 = 52$ |
| Before game 2 | 4 | $12 + 4 + 26 = 42$ | 26 |
| At first | $4 + 21 + 13 = \mathbf{38}$ | **21** | **13** |

**Answer:** An 38, Bình 21, Chi 13 marbles.

# 13 The Supposition Method

## ① The idea of the supposition method

Many problems talk about **two kinds** of things mixed together: chickens and rabbits, adult and child tickets, notes of two values, big and small boats,... We know

- the **total number** of things, and
- the **total of some other quantity** (legs, wheels, money, seats,...), where each thing of the first kind has one value and each thing of the second kind has another.

**The supposition method** ("suppose for a moment"):

1. *Suppose* that all the things are of **one kind**, and work out the total in that case.
2. Compare with the real total: how much **too few** (or **too many**) is it?
3. Swapping *one* thing of that kind for *one* thing of the other kind changes the total by the **difference of the two values**.
4. Number of swaps = (amount too few or too many) : (change per swap). The number of swaps is the number of things of the *other* kind.

The supposition is *not* true: it is only a temporary step that makes the problem easy. The answer does not depend on which kind we suppose, so the problem can be solved in two ways. Always **check** the answer against both conditions.

## ② Problems with penalty points

In a test or a game, a right answer **earns** $a$ points and a wrong answer **loses** $b$ points. Suppose every answer is right. Each time a right answer is swapped for a wrong one, the score falls by

$$a + b \text{ points} \quad \text{(it loses the } a \text{ points it would have earned } \textit{and} \text{ the } b \text{ penalty points).}$$

A question left unanswered (worth 0) makes the score fall by only $a$. The same idea fits wages with fines, deliveries with breakages, and so on.

## ③ When a difference is given instead of a sum

Sometimes the problem gives the **difference** between the totals of the two kinds (for example: *the chickens have* 6 *more legs than the rabbits*). Suppose all the things are of the first kind; then the difference is as large as possible. Swapping one thing of the first kind for one of the second kind makes the first total smaller by $a$ *and* the second total larger by $b$, so the difference falls by

$$a + b.$$

## ④ Three kinds, or an extra condition

- If two kinds are tied together (e.g. *as many bicycles as tricycles*), put them in **groups** and treat each group as one "thing".
- If there are three kinds, use one condition to separate one kind from the other two, then use the supposition method again on the other two.

**Example 1**

A farm has chickens and rabbits: 32 heads and 94 legs in all. How many chickens and how many rabbits are there?

**Guided solution.**

*Method 1.* Suppose all 32 animals were chickens. There would be $32 \times 2 = 64$ legs.
That is $94 - 64 = 30$ legs too few.
Swapping one chicken for one rabbit adds $4 - 2 = 2$ legs.
Number of rabbits: $30 : 2 = 15$.
Number of chickens: $32 - 15 = 17$.
*Method 2.* Suppose all 32 animals were rabbits: $32 \times 4 = 128$ legs, which is $128 - 94 = 34$ legs too many.
Swapping one rabbit for one chicken takes away 2 legs, so chickens: $34 : 2 = 17$; rabbits: $32 - 17 = 15$.
*Check:* $17 \times 2 + 15 \times 4 = 34 + 60 = 94$.
**Answer:** 17 chickens, 15 rabbits.

**Example 2**

A school buys 45 circus tickets for 954 000 dong. An adult ticket costs 30 000 dong and a child ticket 18 000 dong. How many tickets of each kind does the school buy?

**Guided solution.**

Suppose all 45 tickets were child tickets: $18\,000 \times 45 = 810\,000$ dong.
That is $954\,000 - 810\,000 = 144\,000$ dong too little.
Swapping a child ticket for an adult ticket adds $30\,000 - 18\,000 = 12\,000$ dong.
Adult tickets: $144\,000 : 12\,000 = 12$.
Child tickets: $45 - 12 = 33$.
**Answer:** 12 adult tickets, 33 child tickets.

## Example 3

A maths quiz has 25 questions. Each right answer earns 4 points and each wrong answer loses 2 points. Phương answers every question and scores 64 points. How many questions does Phương get right?

### Guided solution.

Suppose all 25 answers were right: $4 \times 25 = 100$ points.
Phương has $100 - 64 = 36$ points fewer.
Each wrong answer (instead of a right one) costs $4 + 2 = 6$ points.
Wrong answers: $36 : 6 = 6$.
Right answers: $25 - 6 = 19$.
*Check:* $19 \times 4 - 6 \times 2 = 76 - 12 = 64$.
**Answer:** 19 questions.

## Example 4

There are 30 chickens and rabbits. The chickens have 6 more legs than the rabbits. How many of each are there?

### Guided solution.

Suppose all 30 animals were chickens: the chickens would have $30 \times 2 = 60$ legs and the rabbits 0, a difference of 60 legs.
The real difference is 6, so it must fall by $60 - 6 = 54$ legs.
Swapping one chicken for one rabbit: the chickens lose 2 legs and the rabbits gain 4, so the difference falls by $2 + 4 = 6$ legs.
Rabbits: $54 : 6 = 9$.
Chickens: $30 - 9 = 21$.
*Check:* $21 \times 2 - 9 \times 4 = 42 - 36 = 6$.
**Answer:** 21 chickens, 9 rabbits.

## Example 5

A bicycle park holds 36 vehicles: bicycles (2 wheels), tricycles (3 wheels) and cars (4 wheels), with 114 wheels in all. There are as many bicycles as tricycles. How many vehicles of each kind are there?

### Guided solution.

Put one bicycle and one tricycle in a *group*: a group has 2 vehicles and $2 + 3 = 5$ wheels.
Suppose there were no cars at all: $36 : 2 = 18$ groups, with $5 \times 18 = 90$ wheels.
That is $114 - 90 = 24$ wheels too few.
Swapping one group for 2 cars keeps the number of vehicles and adds $2 \times 4 - 5 = 3$ wheels.
Swaps: $24 : 3 = 8$, so there are $2 \times 8 = 16$ cars.
Groups left: $18 - 8 = 10$, i.e. 10 bicycles and 10 tricycles.
*Check:* $10 + 10 + 16 = 36$ vehicles; $20 + 30 + 64 = 114$ wheels.
**Answer:** 10 bicycles, 10 tricycles, 16 cars.

## Practice

**Problem 1**

A yard has chickens and dogs: 25 heads and 70 legs. How many chickens and how many dogs are there?

**Solution**

Suppose all were chickens: $25 \times 2 = 50$ legs, which is $70 - 50 = 20$ too few.
Each dog has $4 - 2 = 2$ more legs.
Dogs: $20 : 2 = 10$; chickens: $25 - 10 = 15$.
**Answer:** 15 chickens, 10 dogs.

**Problem 2**

A car park has 40 vehicles, motorbikes (2 wheels) and cars (4 wheels), with 104 wheels in all. How many cars are there?

**Solution**

Suppose all were motorbikes: $40 \times 2 = 80$ wheels, which is $104 - 80 = 24$ too few.
Each car has 2 more wheels.
Cars: $24 : 2 = 12$; motorbikes: $40 - 12 = 28$.
**Answer:** 12 cars (28 motorbikes).

**Problem 3**

Grandma gives Khôi 18 notes of 10 000 dong and 20 000 dong, worth 250 000 dong in all. How many notes of each kind are there?

**Solution**

Suppose all were 10 000 notes: $10\,000 \times 18 = 180\,000$ dong.
Too little by $250\,000 - 180\,000 = 70\,000$ dong.
A 20 000 note is worth 10 000 dong more.
20 000 notes: $70\,000 : 10\,000 = 7$; 10 000 notes: $18 - 7 = 11$.
**Answer:** 11 notes of 10 000, 7 notes of 20 000.

**Problem 4**

12 boats take 58 people across a lake, every boat full. A big boat seats 6 people and a small boat 4. How many boats of each kind are there?

**Solution**

Suppose all were small: $4 \times 12 = 48$ people, $58 - 48 = 10$ too few.
A big boat seats 2 more.
Big boats: $10 : 2 = 5$; small boats: $12 - 5 = 7$.
**Answer:** 5 big boats, 7 small boats.

**Problem 5**

The 36 pupils of class 5A plant 120 trees. Each boy plants 4 trees and each girl 3 trees. How many boys and girls are there?

**Solution**

Suppose all were girls: $3 \times 36 = 108$ trees, $120 - 108 = 12$ too few.
Each boy plants 1 more tree.
Boys: $12 : 1 = 12$; girls: $36 - 12 = 24$.
**Answer:** 12 boys, 24 girls.

**Problem 6**

15 lorries carry 59 tonnes of rice in one trip, every lorry full. Some carry 3 tonnes and the others 5 tonnes. How many lorries of each kind are there?

**Solution**

Suppose all carried 3 tonnes: $3 \times 15 = 45$ tonnes, $59 - 45 = 14$ too few.
A 5-tonne lorry carries 2 more.
5-tonne lorries: $14 : 2 = 7$; 3-tonne lorries: $15 - 7 = 8$.
**Answer:** 8 lorries of 3 tonnes, 7 lorries of 5 tonnes.

**Problem 7**

Ducks and goats have 48 heads and 136 legs. How many of each are there?

**Solution**

Suppose all were ducks: $48 \times 2 = 96$ legs, $136 - 96 = 40$ too few.
A goat has 2 more legs.
Goats: $40 : 2 = 20$; ducks: $48 - 20 = 28$.
**Answer:** 28 ducks, 20 goats.

**Problem 8**

A quiz has 20 questions: 5 points for each right answer and 1 point off for each wrong one. Nga answers all of them and scores 76. How many does she get right?

**Solution**

All right: $5 \times 20 = 100$ points; she is $100 - 76 = 24$ points down.
Each wrong answer costs $5 + 1 = 6$ points.
Wrong: $24 : 6 = 4$; right: $20 - 4 = 16$.
**Answer:** 16 questions.

**Problem 9**

A crate holds 30 bottles of juice, some of 0.5 litre and some of 1.5 litres, 27 litres in all. How many bottles of each size are there?

**Solution**

Suppose all were 0.5-litre bottles: $0.5 \times 30 = 15$ litres, $27 - 15 = 12$ litres too few.
A big bottle holds $1.5 - 0.5 = 1$ litre more.
1.5-litre bottles: $12 : 1 = 12$; 0.5-litre bottles: $30 - 12 = 18$.
**Answer:** 18 bottles of 0.5 l, 12 bottles of 1.5 l.

**Problem 10**

Tuấn buys 24 stamps, some at 3 000 dong and some at 5 000 dong, and pays 92 000 dong. How many of each does he buy?

**Solution**

Suppose all cost 3 000: $3\,000 \times 24 = 72\,000$ dong, 20 000 dong too little.
A 5 000 stamp costs 2 000 more.
5 000 stamps: $20\,000 : 2\,000 = 10$; 3 000 stamps: $24 - 10 = 14$.
**Answer:** 14 stamps at 3 000, 10 stamps at 5 000.

**Problem 11**

A classroom has 21 desks: some seat 2 pupils and some seat 3. When 50 pupils sit down, every seat is taken. How many desks of each kind are there?

**Solution**

Suppose all desks seated 2: $2 \times 21 = 42$ pupils, $50 - 42 = 8$ too few.
A 3-seat desk seats 1 more.
3-seat desks: 8; 2-seat desks: $21 - 8 = 13$.

**Answer:** 13 desks for 2, 8 desks for 3.

**Problem 12**

A box holds 22 spiders (8 legs each) and ants (6 legs each), with 152 legs in all. How many spiders are there?

**Solution**

Suppose all were ants: $6 \times 22 = 132$ legs, $152 - 132 = 20$ too few.
A spider has 2 more legs.
Spiders: $20 : 2 = 10$ (ants: 12).
**Answer:** 10 spiders.

**Problem 13**

A piggy bank holds 30 coins of 500 dong and 1 000 dong, worth 21 000 dong. How many coins of each kind are there?

**Solution**

Suppose all were 500-dong coins: $500 \times 30 = 15\,000$ dong, 6 000 dong too little.
A 1 000 coin is worth 500 more.
1 000 coins: $6\,000 : 500 = 12$; 500 coins: $30 - 12 = 18$.
**Answer:** 18 coins of 500, 12 coins of 1 000.

**Problem 14**

In a basketball match a team scores 26 baskets, some worth 2 points and some worth 3 points, 60 points in all. How many 3-point baskets does it score?

**Solution**

Suppose all were 2-pointers: $2 \times 26 = 52$ points, 8 too few.
A 3-pointer is worth 1 more.
3-pointers: $8 : 1 = 8$ (2-pointers: 18).
**Answer:** 8 baskets.

**Problem 15**

A worker is hired for 24 days. Each working day pays 250 000 dong and each day off costs a fine of 50 000 dong. At the end he receives 5 100 000 dong. How many days off did he take?

**Solution**

Working all 24 days: $250\,000 \times 24 = 6\,000\,000$ dong.
Shortfall: $6\,000\,000 - 5\,100\,000 = 900\,000$ dong.
Each day off costs $250\,000 + 50\,000 = 300\,000$ dong.
Days off: $900\,000 : 300\,000 = 3$.
**Answer:** 3 days.

## Problem 16

There are 40 chickens and rabbits. The chickens have 32 more legs than the rabbits. How many of each are there?

**Solution**

If all were chickens, the difference would be $40 \times 2 = 80$ legs.
It must fall by $80 - 32 = 48$.
Each swap (chicken $\to$ rabbit) lowers it by $2 + 4 = 6$.
Rabbits: $48 : 6 = 8$; chickens: $40 - 8 = 32$.
*Check:* $64 - 32 = 32$.
**Answer:** 32 chickens, 8 rabbits.

## Problem 17

A firm delivers 500 glasses. It is paid 2 000 dong for each glass that arrives safely, but for each broken glass it is paid nothing and must pay back 8 000 dong. It receives 940 000 dong. How many glasses were broken?

**Solution**

If none broke: $2\,000 \times 500 = 1\,000\,000$ dong.
Shortfall: $1\,000\,000 - 940\,000 = 60\,000$ dong.
Each broken glass costs $2\,000 + 8\,000 = 10\,000$ dong.
Broken: $60\,000 : 10\,000 = 6$.
**Answer:** 6 glasses.

## Problem 18

In 15 days Linh reads 236 pages. On some days she reads 12 pages and on the other days 20 pages. On how many days does she read 20 pages?

**Solution**

Suppose she read 12 pages every day: $12 \times 15 = 180$ pages, 56 too few.
A 20-page day gives 8 more.

20-page days: $56 : 8 = 7$ (12-page days: 8).
**Answer:** 7 days.

**Problem 19**

Minh travels for 2.5 hours and covers 14 km. Part of the time he runs at 8 km per hour and the rest he walks at 4 km per hour. For how long does he run?

**Solution**

Suppose he walked the whole time: $4 \times 2.5 = 10$ km, $14 - 10 = 4$ km too few.
Each hour of running instead of walking adds $8 - 4 = 4$ km.
Running time: $4 : 4 = 1$ hour (walking: 1.5 hours).
**Answer:** 1 hour.

**Problem 20**

Mum buys 8 kg of apples and oranges for 315 000 dong. Apples cost 45 000 dong per kg and oranges 30 000 dong per kg. How many kilograms of each does she buy?

**Solution**

Suppose all were oranges: $30\,000 \times 8 = 240\,000$ dong, 75 000 dong too little.
1 kg of apples costs 15 000 dong more.
Apples: $75\,000 : 15\,000 = 5$ kg; oranges: $8 - 5 = 3$ kg.
**Answer:** 5 kg of apples, 3 kg of oranges.

**Problem 21**

A tour group of 99 people travels in 9 minibuses, every seat taken. Some buses have 7 seats and some have 16 seats. How many buses of each kind are there?

**Solution**

Suppose all had 7 seats: $7 \times 9 = 63$ people, $99 - 63 = 36$ too few.
A 16-seat bus carries 9 more.
16-seat buses: $36 : 9 = 4$; 7-seat buses: $9 - 4 = 5$.
**Answer:** 5 buses with 7 seats, 4 buses with 16 seats.

**Problem 22**

A test has 30 questions: 4 points for a right answer, 2 points off for a wrong answer, 0 for a question left blank. Hà leaves 4 questions blank and scores 74. How many does she get right?

**Solution**

Hà answers $30 - 4 = 26$ questions.
If all 26 were right: $4 \times 26 = 104$ points; she is $104 - 74 = 30$ down.
Each wrong answer costs $4 + 2 = 6$.
Wrong: $30 : 6 = 5$; right: $26 - 5 = 21$.
**Answer:** 21 questions.

**Problem 23**

A drawer holds 34 notes of 2 000 dong and 5 000 dong. The 2 000 notes are worth 12 000 dong more than the 5 000 notes. How many notes of each kind are there?

**Solution**

If all were 2 000 notes, the difference would be $2\,000 \times 34 = 68\,000$ dong.
It must fall by $68\,000 - 12\,000 = 56\,000$.
Each swap lowers it by $2\,000 + 5\,000 = 7\,000$.
5 000 notes: $56\,000 : 7\,000 = 8$; 2 000 notes: $34 - 8 = 26$.
*Check:* $52\,000 - 40\,000 = 12\,000$.
**Answer:** 26 notes of 2 000, 8 notes of 5 000.

**Problem 24**

Using 52 matchsticks, An makes 15 separate shapes, some triangles and some squares (one matchstick per side). How many squares does he make?

**Solution**

If all were triangles: $3 \times 15 = 45$ sticks, 7 too few.
A square uses 1 more stick.
Squares: 7 (triangles: 8).
**Answer:** 7 squares.

**Problem 25**

Hoa buys 12 pencils and pens for 64 000 dong. A pencil costs 3 000 dong and a pen 7 000 dong. How many pens does she buy?

**Solution**

If all were pencils: $3\,000 \times 12 = 36\,000$ dong, 28 000 too little.
A pen costs 4 000 more.
Pens: $28\,000 : 4\,000 = 7$ (pencils: 5).
**Answer:** 7 pens.

**Problem 26**

In a football league a win earns 3 points, a draw 1 point and a loss 0. A team plays 12 matches, loses none and has 28 points. How many matches does it draw?

**Solution**

If it won all 12: $3 \times 12 = 36$ points; it has $36 - 28 = 8$ fewer.
A draw instead of a win gives $3 - 1 = 2$ points fewer.
Draws: $8 : 2 = 4$ (wins: 8).
**Answer:** 4 matches.

**Problem 27**

20 boxes hold 196 apples. Some boxes hold 8 apples and the others 12. How many boxes of each kind are there?

**Solution**

If all held 8: $8 \times 20 = 160$, 36 too few.
A box of 12 holds 4 more.
Boxes of 12: $36 : 4 = 9$; boxes of 8: $20 - 9 = 11$.
**Answer:** 11 boxes of 8, 9 boxes of 12.

**Problem 28**

A group buys 50 zoo tickets for 1 520 000 dong. An adult ticket costs 40 000 dong and a child ticket 25 000 dong. How many adults are there?

**Solution**

If all were children: $25\,000 \times 50 = 1\,250\,000$ dong, 270 000 too little.
An adult ticket costs 15 000 more.
Adults: $270\,000 : 15\,000 = 18$ (children: 32).
**Answer:** 18 adults.

**Problem 29**

In a shooting game each player fires 30 shots. A hit scores 6 points and a miss loses 3 points. Bảo scores 126 points. How many hits does he make?

**Solution**

All hits: $6 \times 30 = 180$ points; he is $180 - 126 = 54$ down.
Each miss costs $6 + 3 = 9$.
Misses: $54 : 9 = 6$; hits: $30 - 6 = 24$.
**Answer:** 24 hits.

### Problem 30

A car park holds 26 cars and motorbikes. The motorbikes have 4 more wheels than the cars. How many of each are there?

### Solution

If all were motorbikes, the difference would be $2 \times 26 = 52$ wheels.
It must fall by $52 - 4 = 48$; each swap (motorbike $\to$ car) lowers it by $2 + 4 = 6$.
Cars: $48 : 6 = 8$; motorbikes: $26 - 8 = 18$.
*Check:* $36 - 32 = 4$.
**Answer:** 8 cars, 18 motorbikes.

## Challenge – Extension

### Problem 31 ⋆

24 vehicles carry 133 people, every seat taken: motorbikes (2 wheels, 2 seats), 7-seat cars and 16-seat cars (4 wheels each). There are 64 wheels in all. How many vehicles of each kind are there?

### Solution

*Wheels.* If all were motorbikes: 48 wheels, 16 too few; a car has 2 more.
Cars: $16 : 2 = 8$; motorbikes: $24 - 8 = 16$.
*Seats.* Motorbikes carry $2 \times 16 = 32$ people, so the cars carry $133 - 32 = 101$.
If all 8 cars had 7 seats: 56, which is 45 too few; a 16-seat car has 9 more.
16-seat cars: $45 : 9 = 5$; 7-seat cars: $8 - 5 = 3$.
**Answer:** 16 motorbikes, 3 cars with 7 seats, 5 cars with 16 seats.

### Problem 32 ⋆

A company has 20 lorries of three kinds: kind A (4 wheels, carries 3 tonnes), kind B (6 wheels, 6 tonnes) and kind C (10 wheels, 6 tonnes). Together they carry 96 tonnes in one trip and have 124 wheels. How many lorries of each kind are there?

### Solution

*Tonnes.* B and C both carry 6 t. If all 20 carried 3 t: 60 t, 36 too few; each B or C carries 3 more.
B and C together: $36 : 3 = 12$; A: $20 - 12 = 8$.
*Wheels.* A has $4 \times 8 = 32$ wheels, so B and C have $124 - 32 = 92$.
If all 12 were B: 72 wheels, 20 too few; a C has 4 more.
C: $20 : 4 = 5$; B: $12 - 5 = 7$.
**Answer:** 8 of kind A, 7 of kind B, 5 of kind C.

**Problem 33** ⁎⁎

A contest has 40 questions: 5 points for right, 2 points off for wrong, 0 for a blank. Quang scores 128 points, and he has as many wrong answers as blanks. How many does he get right?

**Solution**

Group 1 wrong answer with 1 blank.
All right: $5 \times 40 = 200$; Quang is $200 - 128 = 72$ points down.
A group (instead of 2 right answers) costs $(5 + 2) + 5 = 12$ points.
Groups: $72 : 12 = 6$, i.e. 6 wrong and 6 blank.
Right: $40 - 12 = 28$.
**Answer:** 28 questions.

**Problem 34** ⁎

Lan buys 30 notebooks, some at 5 000 dong and some at 8 000 dong. She spends 45 000 dong more on the 8 000 notebooks than on the 5 000 ones. How many of each does she buy?

**Solution**

If all were 8 000 notebooks, the difference would be $8\,000 \times 30 = 240\,000$ dong.
It must fall by $240\,000 - 45\,000 = 195\,000$; each swap lowers it by $8\,000 + 5\,000 = 13\,000$.
5 000 notebooks: $195\,000 : 13\,000 = 15$; 8 000 notebooks: 15.
*Check:* $120\,000 - 75\,000 = 45\,000$.
**Answer:** 15 of each.

**Problem 35** ⁎

Chickens and rabbits have 100 legs. If every chicken had 4 legs and every rabbit 2, there would be 80 legs. How many of each are there?

**Solution**

Add the two counts: each animal is counted with $2 + 4 = 6$ legs, so $6 \times$ (animals) $= 180$.
Animals: $180 : 6 = 30$.
If all 30 were chickens: 60 legs, 40 too few; a rabbit has 2 more.
Rabbits: 20; chickens: 10.
**Answer:** 10 chickens, 20 rabbits.

**Problem 36** ⁎

100 cakes are shared by 100 people: each adult gets 3 cakes and every 3 children share 1 cake. How many adults and how many children are there?

**Solution**

Group 1 adult with 3 children: 4 people eat $3 + 1 = 4$ cakes.
Groups: $100 : 4 = 25$.
Adults: 25; children: $3 \times 25 = 75$.
*Supposition:* if all 100 were adults they would need 300 cakes, 200 too many; swapping an adult for a child saves $3 - \frac{1}{3} = \frac{8}{3}$ cake, so children: $200 : \frac{8}{3} = 75$.
**Answer:** 25 adults, 75 children.

**Problem 37** *

A car travels for 5 hours and covers 230 km: on the highway at 50 km per hour and on a country road at 30 km per hour. How many hours does it drive on the highway?

**Solution**

If it drove on the country road all the time: $30 \times 5 = 150$ km, 80 km too few.
Each highway hour adds 20 km.
Highway: $80 : 20 = 4$ hours (country road: 1 hour).
**Answer:** 4 hours.

**Problem 38** *

A shop sends 1 200 eggs to market. It is paid 300 dong for every whole egg; for each broken egg it gets nothing and pays a fine of 2 700 dong. It receives 330 000 dong. How many eggs were broken?

**Solution**

No breakages: $300 \times 1\,200 = 360\,000$ dong; shortfall 30 000 dong.
Each broken egg costs $300 + 2\,700 = 3\,000$ dong.
Broken: $30\,000 : 3\,000 = 10$.
**Answer:** 10 eggs.

**Problem 39** *

For 30 days Nam puts either 2 000 dong or 5 000 dong into his piggy bank each day, 117 000 dong in all. On how many days does he save 5 000 dong?

**Solution**

If he saved 2 000 every day: 60 000 dong, 57 000 too little.
A 5 000 day gives 3 000 more.
Days: $57\,000 : 3\,000 = 19$.
**Answer:** 19 days.

**Problem 40** ★★

Six teams play a round-robin tournament (each pair plays once). A win earns 3 points, a draw 1 point, a loss 0. The teams score 41 points altogether. How many matches end in a draw?

**Solution**

Each team plays 5 others; the number of matches is $6 \times 5 : 2 = 15$.
A match with a winner gives $3 + 0 = 3$ points in total; a draw gives $1 + 1 = 2$.
If no match were drawn: $3 \times 15 = 45$ points, 4 too many.
Each draw gives 1 point fewer, so draws: $4 : 1 = 4$.
**Answer:** 4 matches.

**Problem 41** ★★

A test has 25 questions: 4 points for right, 1 point off for wrong, 0 for a blank. An answers all and scores 65. Bình gets as many right as An, but leaves blank 3 of the questions An got wrong and gets the rest wrong. What is Bình's score?

**Solution**

An: all right would give 100; he is 35 down; each wrong costs 5.
An's wrong answers: $35 : 5 = 7$; right: 18.
Bình: 18 right, 3 blank, $7 - 3 = 4$ wrong.
Score: $18 \times 4 - 4 \times 1 = 68$.
**Answer:** 68 points.

**Problem 42** ★

Class 5C buys 25 gifts for 380 000 dong: some at 12 000 dong and some at 20 000 dong. If they had bought the numbers the other way round, how much would they have paid?

**Solution**

If all cost 12 000: 300 000 dong, 80 000 too little; a dearer gift costs 8 000 more.
20 000 gifts: $80\,000 : 8\,000 = 10$; 12 000 gifts: 15.
The other way round: $10 \times 12\,000 + 15 \times 20\,000 = 420\,000$ dong.
**Answer:** 420 000 dong.

**Problem 43** ★

There are 6 more chickens than rabbits, and 84 legs altogether. How many of each are there?

**Solution**

Take away the 6 extra chickens: $84 - 6 \times 2 = 72$ legs, and now there are as many chickens as rabbits.
A pair (one chicken, one rabbit) has 6 legs, so pairs: $72 : 6 = 12$.
Rabbits: 12; chickens: $12 + 6 = 18$.
**Answer:** 18 chickens, 12 rabbits.

## Problem 44 *

A shop sells 40 shirts for 5 520 000 dong. Some are sold at the full price of 150 000 dong and the rest at a 20% discount. How many shirts are sold at the discount?

**Solution**

Discounted price: $150\,000 \times 80\% = 120\,000$ dong.
If all were at full price: 6 000 000 dong, 480 000 too much.
Each discounted shirt brings 30 000 less.
Discounted: $480\,000 : 30\,000 = 16$.
**Answer:** 16 shirts.

## Problem 45 *

Mr Thành travels 120 km in 4 hours, partly by bicycle at 15 km per hour and partly by coach at 45 km per hour. How far does he cycle?

**Solution**

If he cycled all the time: $15 \times 4 = 60$ km, 60 too few; each coach hour adds 30 km.
Coach: $60 : 30 = 2$ hours; bicycle: 2 hours.
Distance cycled: $15 \times 2 = 30$ km.
**Answer:** 30 km.

## Problem 46 **

There are 18 small creatures: spiders (8 legs, no wings), dragonflies (6 legs, 2 pairs of wings) and cicadas (6 legs, 1 pair of wings). They have 118 legs and 20 pairs of wings. How many of each are there?

**Solution**

*Legs.* If all had 6 legs: 108, 10 too few; a spider has 2 more.
Spiders: 5; insects: 13.
*Wings.* If all 13 insects were cicadas: 13 pairs, 7 too few; a dragonfly has 1 pair more.
Dragonflies: 7; cicadas: $13 - 7 = 6$.

**Answer:** 5 spiders, 7 dragonflies, 6 cicadas.

### Problem 47 *

In an archery game each player shoots 20 arrows: a hit scores 10 points and a miss loses 5 points. Đức scores 45 points more than Long. How many more hits does Đức make?

### Solution

Both shoot 20 arrows. One more hit means one fewer miss, so the score grows by $10 + 5 = 15$ points.

Extra hits: $45 : 15 = 3$.

**Answer:** 3 hits.

### Problem 48 *

A 300-litre tank is filled using 22 buckets of water, some of 10 litres and some of 15 litres. How many 15-litre buckets are used?

### Solution

If all held 10 litres: 220 l, 80 l too few; a big bucket holds 5 l more.

15-litre buckets: $80 : 5 = 16$ (10-litre: 6).

**Answer:** 16 buckets.

### Problem 49 **

A bag has 48 blue and red marbles. $\frac{1}{3}$ of the blue marbles and $\frac{1}{4}$ of the red marbles make 14 marbles altogether. How many marbles of each colour are there?

### Solution

Suppose all 48 were blue: $\frac{1}{3}$ of 48 is 16, which is 2 too many.

Swapping a blue marble for a red one changes the count by $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.

Red: $2 : \frac{1}{12} = 24$; blue: $48 - 24 = 24$.

*Check:* $8 + 6 = 14$.

**Answer:** 24 blue, 24 red.

## Problem 50 ★★

A wallet holds 23 notes worth 600 000 dong: notes of 10 000, 20 000 and 50 000 dong. There are twice as many 10 000 notes as 20 000 notes. How many notes of each kind are there?

## Solution

Group two 10 000 notes with one 20 000 note: 3 notes worth 40 000 dong.

If all 23 notes were 50 000: 1 150 000 dong, 550 000 too much.

Swapping 3 notes of 50 000 for one group lowers the value by $150\,000 - 40\,000 = 110\,000$.

Groups: $550\,000 : 110\,000 = 5$.

10 000 notes: 10; 20 000 notes: 5; 50 000 notes: $23 - 15 = 8$.

**Answer:** 10 notes of 10 000, 5 of 20 000, 8 of 50 000.

# Mixed Review – Paper 1

**Covers:** Chapters 1 – 13 (percentages, cuboids and cubes, volume, painted cubes, uniform motion, statistics and probability, Venn diagrams, working backwards, supposition). **Suggested time:** 60 minutes.

## Problem 1

**a)** Find 35% of 240.

**b)** 27 is what percentage of 45?

**c)** 15% of a number is 27. Find the number.

## Solution

**a)** $240 \times 35 : 100 = 84$.

**b)** $27 : 45 = 0.6 = 60\%$.

**c)** $27 : 15 \times 100 = 180$.

**Answer: a)** 84; **b)** 60%; **c)** 180.

## Problem 2

A shop sells a bicycle for 1 800 000 dong and makes a profit of 20% of the cost price. Find the cost price and the profit.

## Solution

The selling price is $100\% + 20\% = 120\%$ of the cost price.

Cost price: $1\,800\,000 : 120 \times 100 = 1\,500\,000$ dong.

Profit: $1\,800\,000 - 1\,500\,000 = 300\,000$ dong.

**Answer:** Cost price 1 500 000 dong; profit 300 000 dong.

## Problem 3

A jacket costs 500 000 dong. Its price is cut by 10%, and later the new price is raised by 10%. What is the final price? By what percentage has the price changed compared with the start?

## Solution

After the cut: $500\,000 \times 90\% = 450\,000$ dong.

After the rise: $450\,000 \times 110\% = 495\,000$ dong.

The price falls by 5 000 dong, and $5\,000 : 500\,000 = 1\%$.
**Answer:** 495 000 dong; it is 1% lower.

**Problem 4**

A cuboid is 8 dm long, 5 dm wide and 4 dm high. Find its lateral surface area and its total surface area.

**Solution**

Lateral area: $(8 + 5) \times 2 \times 4 = 104$ dm$^2$.
Area of one base: $8 \times 5 = 40$ dm$^2$.
Total area: $104 + 40 \times 2 = 184$ dm$^2$.
**Answer:** 104 dm$^2$; 184 dm$^2$.

**Problem 5**

The total surface area of a cube is 294 cm$^2$. Find the edge and the volume of the cube.

**Solution**

Area of one face: $294 : 6 = 49$ cm$^2$.
Since $7 \times 7 = 49$, the edge is 7 cm.
Volume: $7 \times 7 \times 7 = 343$ cm$^3$.
**Answer:** Edge 7 cm; volume 343 cm$^3$.

**Problem 6**

A cuboid water tank is 1.5 m long, 0.8 m wide and 1 m high. The water reaches 75% of the height. How many litres of water are in the tank?

**Solution**

Water height: $1 \times 75\% = 0.75$ m.
Volume of water: $1.5 \times 0.8 \times 0.75 = 0.9$ m$^3$ $= 900$ dm$^3$.
1 dm$^3$ = 1 litre, so there are 900 litres.
**Answer:** 900 litres.

**Problem 7**

A glass tank has a base 50 cm long and 40 cm wide. When a stone is put in (and is completely under water), the water rises by 3 cm. Find the volume of the stone.

**Solution**

The stone's volume equals the volume of the water "pushed up":
$50 \times 40 \times 3 = 6\,000$ cm$^3$ $= 6$ dm$^3$.
**Answer:** $6\,000$ cm$^3$ ($6$ dm$^3$).

**Problem 8**

A wooden cube of edge 4 cm is painted on all its faces and then cut into 1 cm cubes. How many small cubes have **a)** 3 painted faces; **b)** 2 painted faces; **c)** 1 painted face; **d)** no painted face?

**Solution**

There are $4 \times 4 \times 4 = 64$ small cubes.
**a)** The 8 corners have 3 painted faces.
**b)** Each of the 12 edges has $4 - 2 = 2$ middle cubes: $12 \times 2 = 24$.
**c)** Each of the 6 faces has $2 \times 2 = 4$ inner cubes: $6 \times 4 = 24$.
**d)** The hidden core is $2 \times 2 \times 2 = 8$ cubes. (*Check:* $8 + 24 + 24 + 8 = 64$.)
**Answer:** 8; 24; 24; 8.

**Problem 9**

A car leaves at 7:15 and drives at 45 km per hour for 2 hours 20 minutes. **a)** How far does it travel? **b)** At what time does it arrive?

**Solution**

**a)** 2 hours 20 minutes $= 2\frac{1}{3}$ hours $= \frac{7}{3}$ hours.

Distance: $45 \times \frac{7}{3} = 105$ km.

**b)** 7 h 15 min + 2 h 20 min = 9 h 35 min.
**Answer: a)** 105 km; **b)** at 9:35.

**Problem 10**

Towns A and B are 150 km apart. At 7:00 a car leaves A for B at 40 km per hour, and at the same time a motorbike leaves B for A at 35 km per hour. When do they meet, and how far from A?

**Solution**

Each hour they come $40 + 35 = 75$ km closer.
Time to meet: $150 : 75 = 2$ hours, so they meet at $7 + 2 = 9$ o'clock.
Distance from A: $40 \times 2 = 80$ km.

**Answer:** At 9:00, 80 km from A.

### Problem 11

At 6:00 a cyclist leaves home at 12 km per hour. At 7:00 a motorbike sets off from the same place along the same road at 36 km per hour. When does the motorbike catch up, and how far from home?

### Solution

By 7:00 the cyclist is $12 \times 1 = 12$ km ahead.
The motorbike gains $36 - 12 = 24$ km each hour.
Catching-up time: $12 : 24 = 0.5$ hour $= 30$ minutes, i.e. at 7:30.
Distance from home: $36 \times 0.5 = 18$ km.
**Answer:** At 7:30, 18 km from home.

### Problem 12 ⋆

Hưng cycles up a hill at 12 km per hour and comes back down the same road at 24 km per hour. Find his average speed for the whole trip.

### Solution

Take the road to be 24 km (any length gives the same answer).
Up: $24 : 12 = 2$ hours; down: $24 : 24 = 1$ hour.
Average speed: $(24 + 24) : (2 + 1) = 16$ km per hour.
**Answer:** 16 km per hour.

### Problem 13

At 40 km per hour a bus needs 3 hours for a journey. At what speed must it go to make the same journey in 2.5 hours?

### Solution

Distance: $40 \times 3 = 120$ km.
Speed needed: $120 : 2.5 = 48$ km per hour.
**Answer:** 48 km per hour.

### Problem 14

A boat goes downstream at 18 km per hour and upstream at 12 km per hour. Find the speed of the current and the speed of the boat in still water.

**Solution**

Downstream − upstream = 2× current speed.
Current: $(18 - 12) : 2 = 3$ km per hour.
Still water: $18 - 3 = 15$ km per hour.
**Answer:** Current 3 km/h; still water 15 km/h.

**Problem 15** *

A train 180 m long passes a signal post in 12 seconds. **a)** Find its speed in m/s and in km/h. **b)** How long does it take to cross a bridge 420 m long?

**Solution**

**a)** Passing the post, the train moves its own length: $180 : 12 = 15$ m/s.
$15$ m/s $= 15 \times 3\,600 : 1\,000 = 54$ km/h.
**b)** To cross the bridge it moves $180 + 420 = 600$ m: $600 : 15 = 40$ seconds.
**Answer: a)** 15 m/s = 54 km/h; **b)** 40 seconds.

**Problem 16** *

A circular track is 400 m long. Two runners start from the same point at 3 m/s and 5 m/s. **a)** If they run in opposite directions, after how many seconds do they first meet? **b)** If they run in the same direction, after how many seconds does the faster one first catch the slower one?

**Solution**

**a)** Together they cover one lap: $400 : (3 + 5) = 50$ seconds.
**b)** The faster one must gain one full lap: $400 : (5 - 3) = 200$ seconds.
**Answer: a)** 50 seconds; **b)** 200 seconds.

**Problem 17**

**a)** In 5 rounds of a quiz, Ngân scores 8, 7, 9, 10 and one more score. Her average is 8.4. What was the missing score?

**b)** Tú tosses a coin 40 times and gets heads 18 times. Write the ratio of the number of heads to the number of tosses as a fraction and as a percentage.

**Solution**

**a)** Total of the 5 scores: $8.4 \times 5 = 42$.
Missing score: $42 - (8 + 7 + 9 + 10) = 42 - 34 = 8$.
**b)** $\frac{18}{40} = \frac{9}{20} = 45\%$.

**Answer: a)** 8; **b)** $\frac{9}{20} = 45\%$.

## Problem 18

Class 5D has 40 pupils. 25 like football, 18 like chess and 7 like both. How many pupils like neither?

### Solution

Pupils who like at least one: $25 + 18 - 7 = 36$ (the 7 were counted twice).
Neither: $40 - 36 = 4$.
**Answer:** 4 pupils.

## Problem 19 *

Mai spends half her money on books, then $\frac{1}{3}$ of what is left on pens. She has 40 000 dong left. How much money did she have at first?

### Solution

Working backwards: 40 000 dong is $1 - \frac{1}{3} = \frac{2}{3}$ of what was left after the books.

After the books: $40\,000 : \frac{2}{3} = 60\,000$ dong.

That is half her money, so at first: $60\,000 \times 2 = 120\,000$ dong.
**Answer:** 120 000 dong.

## Problem 20 *

A farm has chickens and rabbits: 35 heads and 94 legs. How many of each are there?

### Solution

Suppose all were chickens: $35 \times 2 = 70$ legs, $94 - 70 = 24$ too few.
A rabbit has 2 more legs: rabbits $24 : 2 = 12$; chickens $35 - 12 = 23$.
**Answer:** 23 chickens, 12 rabbits.

# Mixed Review – Paper 2

**Covers:** Chapters 1 – 13 (percentages, cuboids and cubes, volume, painted cubes, uniform motion, statistics and probability, Venn diagrams, working backwards, supposition). **Suggested time:** 60 minutes.

## Problem 1

A class has 25 girls and 15 boys. **a)** What percentage of the class are girls? **b)** The number of boys is what percentage of the number of girls?

### Solution

The class has $25 + 15 = 40$ pupils.

**a)** $25 : 40 = 0.625 = 62.5\%$.

**b)** $15 : 25 = 0.6 = 60\%$.

**Answer: a)** 62.5%; **b)** 60%.

## Problem 2

Mrs Hạnh puts 5 000 000 dong into a savings account that pays 0.5% interest per month. How much interest does she get after one month, and how much is then in the account?

### Solution

Interest: $5\,000\,000 \times 0.5 : 100 = 25\,000$ dong.

In the account: $5\,000\,000 + 25\,000 = 5\,025\,000$ dong.

**Answer:** 25 000 dong; 5 025 000 dong.

## Problem 3

On the first day Khánh reads 40% of a book, and on the second day 35% of the book. There are 75 pages left. How many pages does the book have?

### Solution

Pages left: $100\% - 40\% - 35\% = 25\%$ of the book.

The book: $75 : 25 \times 100 = 300$ pages.

**Answer:** 300 pages.

## Problem 4

200 g of salt water contains 5% salt. How many grams of pure water must be added to make salt water with 4% salt?

### Solution

Salt: $200 \times 5 : 100 = 10$ g; adding water does not change it.
In the new salt water, 10 g is 4%, so it weighs $10 : 4 \times 100 = 250$ g.
Water to add: $250 - 200 = 50$ g.
**Answer:** 50 g.

## Problem 5

An open box (no lid) is a cuboid 30 cm long, 20 cm wide and 15 cm high. Find the area to be painted on the outside.

### Solution

Lateral area: $(30 + 20) \times 2 \times 15 = 1\,500$ cm$^2$.
Bottom: $30 \times 20 = 600$ cm$^2$.
Area to paint: $1\,500 + 600 = 2\,100$ cm$^2$.
**Answer:** $2\,100$ cm$^2$.

## Problem 6

A cuboid has a lateral surface area of 200 cm$^2$ and a height of 5 cm. Its length is 4 cm more than its width. Find its volume.

### Solution

Perimeter of the base: $200 : 5 = 40$ cm, so length + width = 20 cm.
Length: $(20 + 4) : 2 = 12$ cm; width: $12 - 4 = 8$ cm.
Volume: $12 \times 8 \times 5 = 480$ cm$^3$.
**Answer:** 480 cm$^3$.

## Problem 7

A swimming pool is 25 m long, 10 m wide and 1.6 m deep. A pump puts in 50 m$^3$ of water each hour. Starting empty, how long does it take to fill the pool to 80% of its depth?

### Solution

Water depth: $1.6 \times 80\% = 1.28$ m.
Volume: $25 \times 10 \times 1.28 = 320$ m$^3$.
Time: $320 : 50 = 6.4$ hours $= 6$ hours 24 minutes.

**Answer:** 6 hours 24 minutes.

### Problem 8

A wooden cuboid 3 cm × 4 cm × 5 cm is painted on all its faces and cut into 1 cm cubes. How many small cubes have **a)** 3 painted faces; **b)** 2 painted faces; **c)** 1 painted face; **d)** no painted face?

### Solution

There are $3 \times 4 \times 5 = 60$ small cubes.

**a)** The 8 corners.

**b)** Middle cubes on the edges: each length gives 4 edges, so $4 \times (1 + 2 + 3) = 24$.

**c)** Inner parts of the faces: $2 \times (1 \times 2 + 2 \times 3 + 1 \times 3) = 2 \times 11 = 22$.

**d)** The core: $1 \times 2 \times 3 = 6$. (*Check:* $8 + 24 + 22 + 6 = 60$.)

**Answer:** 8; 24; 22; 6.

### Problem 9

How many 1 cm cubes are needed to build a solid cube of edge 5 cm? How many are needed if the cube is hollow, so that only its outer layer (one cube thick) is built?

### Solution

Solid cube: $5 \times 5 \times 5 = 125$ small cubes.

The hollow inside is a cube of edge $5 - 2 = 3$ cm: $3 \times 3 \times 3 = 27$ cubes.

Outer layer: $125 - 27 = 98$ small cubes.

**Answer:** 125; 98.

### Problem 10

Lan leaves home at 6:45 and walks 4.5 km to her grandparents' house at 5 km per hour. At what time does she arrive?

### Solution

Time: $4.5 : 5 = 0.9$ hour $= 54$ minutes.

Arrival: 6 h 45 min + 54 min = 7 h 39 min.

**Answer:** At 7:39.

### Problem 11 *

Two people 27 km apart walk towards each other at 4 km per hour and 5 km per hour. A dog starts with the first person and runs back and forth between them at 10 km per hour until they

meet. How far does the dog run?

**Solution**

Time until they meet: $27 : (4 + 5) = 3$ hours.
The dog runs all that time: $10 \times 3 = 30$ km.
**Answer:** 30 km.

**Problem 12**

Car A is 60 km ahead of car B, and both drive in the same direction. Car A goes at 40 km per hour and car B at 55 km per hour. After how many hours does B catch up with A?

**Solution**

B gains $55 - 40 = 15$ km each hour.
Time: $60 : 15 = 4$ hours.
**Answer:** 4 hours.

**Problem 13** ★

A van travels 120 km: the first 60 km at 30 km per hour and the next 60 km at 60 km per hour. Find its average speed for the whole journey.

**Solution**

Time: $60 : 30 + 60 : 60 = 2 + 1 = 3$ hours.
Average speed: $120 : 3 = 40$ km per hour (not $(30 + 60) : 2 = 45$).
**Answer:** 40 km per hour.

**Problem 14**

A canoe moves at 14 km per hour in still water, and the river flows at 2 km per hour. Landing B is 48 km downstream of landing A. How long does the canoe take to go from A to B and back to A?

**Solution**

Downstream: $14 + 2 = 16$ km/h, taking $48 : 16 = 3$ hours.
Upstream: $14 - 2 = 12$ km/h, taking $48 : 12 = 4$ hours.
Total: $3 + 4 = 7$ hours.
**Answer:** 7 hours.

## Problem 15

Two trains run towards each other on parallel tracks: one is 150 m long at 20 m/s, the other 130 m long at 15 m/s. How many seconds pass from the moment their fronts meet until their backs pass each other?

### Solution

Together they must cover both lengths: $150 + 130 = 280$ m.

They approach at $20 + 15 = 35$ m/s.

Time: $280 : 35 = 8$ seconds.

**Answer:** 8 seconds.

## Problem 16 *

On a circular track, Hòa runs one lap in 6 minutes and Dũng in 10 minutes. They start together from the same point in the same direction. After how many minutes does Hòa first catch up with Dũng? How many laps has Hòa run by then?

### Solution

Each minute Hòa runs $\frac{1}{6}$ lap and Dũng $\frac{1}{10}$ lap, so Hòa gains $\frac{1}{6} - \frac{1}{10} = \frac{1}{15}$ lap.

To gain one full lap: $1 : \frac{1}{15} = 15$ minutes.

Hòa's laps: $15 : 6 = 2.5$ laps.

**Answer:** 15 minutes; 2.5 laps.

## Problem 17

**a)** In a test, 3 pupils score 6, 5 pupils score 7, 8 pupils score 8 and 4 pupils score 9. Find the average score.

**b)** A dice is rolled 50 times and shows a six 8 times. What percentage of the rolls show a six?

### Solution

**a)** Pupils: $3 + 5 + 8 + 4 = 20$.

Total: $3 \times 6 + 5 \times 7 + 8 \times 8 + 4 \times 9 = 18 + 35 + 64 + 36 = 153$.

Average: $153 : 20 = 7.65$.

**b)** $8 : 50 = 0.16 = 16\%$.

**Answer: a)** 7.65; **b)** 16%.

## Problem 18

In a group of 50 pupils, 30 learn English, 26 learn computing and 10 learn neither. How many learn both?

### Solution

Pupils who learn at least one subject: $50 - 10 = 40$.

Both: $30 + 26 - 40 = 16$.

**Answer:** 16 pupils.

## Problem 19 *

On the first day a family uses half the water in a tank and 20 litres more. On the second day it uses half of what is left and 10 litres more. 50 litres remain. How much water was in the tank at first?

### Solution

Work backwards.

Before the second day: $(50 + 10) \times 2 = 120$ litres.

At first: $(120 + 20) \times 2 = 280$ litres.

**Answer:** 280 litres.

## Problem 20 *

A test has 30 questions: 4 points for a right answer and 1 point off for a wrong one. Sơn answers all of them and scores 95. How many does he get right?

### Solution

All right: $4 \times 30 = 120$ points; he is 25 points down.

Each wrong answer costs $4 + 1 = 5$ points: wrong $25 : 5 = 5$; right $30 - 5 = 25$.

**Answer:** 25 questions.

# End-of-Book 2 Test

**Time:** 60 minutes. **Total:** 10 marks. **Questions:** 20.

*Part A* – 0.4 *mark each.* *Part B* – 0.5 *mark each.* *Part C* – 0.75 *mark each.*

## Part A – Quick answers (4 marks)

### Problem 1

What is 12.5% of 80?

**Solution**

$80 \times 12.5 : 100 = 10$.

**Answer:** 10.

### Problem 2

Write the ratio of 3 to 8 as a percentage.

**Solution**

$3 : 8 = 0.375 = 37.5\%$.

**Answer:** 37.5%.

### Problem 3

What is the volume of a cube with edge 4 cm?

**Solution**

$4 \times 4 \times 4 = 64$ cm$^3$.

**Answer:** 64 cm$^3$.

### Problem 4

2.5 m$^3$ = … dm$^3$.

**Solution**

1 m$^3$ = 1 000 dm$^3$, so 2.5 m$^3$ = 2 500 dm$^3$.

**Answer:** 2 500 dm$^3$.

**Problem 5**

What is the total surface area of a cuboid 6 cm long, 4 cm wide and 3 cm high?

**Solution**

$(6 \times 4 + 6 \times 3 + 4 \times 3) \times 2 = (24 + 18 + 12) \times 2 = 108$ cm$^2$.

**Answer:** 108 cm$^2$.

**Problem 6**

A car drives at 48 km per hour for 45 minutes. How far does it go?

**Solution**

45 minutes $= 0.75$ hour; $48 \times 0.75 = 36$ km.

**Answer:** 36 km.

**Problem 7**

How long does it take to walk 9 km at 4 km per hour?

**Solution**

$9 : 4 = 2.25$ hours $= 2$ hours 15 minutes.

**Answer:** 2 hours 15 minutes.

**Problem 8**

A boat goes at 10 km per hour in still water; the current is 2 km per hour. What is its speed upstream?

**Solution**

$10 - 2 = 8$ km per hour.

**Answer:** 8 km per hour.

**Problem 9**

A coin is tossed 25 times and shows heads 12 times. What percentage of the tosses show heads?

**Solution**

$12 : 25 = 0.48 = 48\%$.

**Answer:** 48%.

**Problem 10**

Chickens and rabbits have 10 heads and 28 legs. How many rabbits are there?

**Solution**

All chickens: 20 legs, 8 too few; each rabbit adds 2: rabbits $8 : 2 = 4$.

**Answer:** 4 rabbits.

## Part B – Working out (3 marks)

**Problem 11**

After a 20% discount a backpack costs 240 000 dong. What was its price before the discount?

**Solution**

240 000 dong is $100\% - 20\% = 80\%$ of the old price.

Old price: $240\,000 : 80 \times 100 = 300\,000$ dong.

**Answer:** 300 000 dong.

**Problem 12**

A cuboid tank is 1.2 m long, 0.5 m wide and 0.8 m high. How many litres does it hold when full?

**Solution**

$1.2 \times 0.5 \times 0.8 = 0.48$ m$^3$ $= 480$ dm$^3$ $= 480$ litres.

**Answer:** 480 litres.

**Problem 13**

Find the lateral surface area and the total surface area of a cube with edge 7 cm.

**Solution**

One face: $7 \times 7 = 49$ cm$^2$.

Lateral: $49 \times 4 = 196$ cm$^2$; total: $49 \times 6 = 294$ cm$^2$.

**Answer:** 196 cm$^2$; 294 cm$^2$.

**Problem 14**

Towns A and B are 210 km apart. At 8:00 two cars leave A and B and drive towards each other at 45 km per hour and 60 km per hour. At what time do they meet?

**Solution**

They close up $45 + 60 = 105$ km each hour.
Time: $210 : 105 = 2$ hours; they meet at $8 + 2 = 10$ o'clock.
**Answer:** At 10:00.

**Problem 15**

A train 200 m long goes through a tunnel 300 m long in 25 seconds (from the front entering to the back leaving). Find its speed in km per hour.

**Solution**

Distance: $200 + 300 = 500$ m; speed: $500 : 25 = 20$ m/s.
$20$ m/s $= 20 \times 3\,600 : 1\,000 = 72$ km/h.
**Answer:** 72 km per hour.

**Problem 16**

Of 35 pupils, 20 go swimming, 17 go dancing and 5 do neither. How many do both?

**Solution**

At least one activity: $35 - 5 = 30$.
Both: $20 + 17 - 30 = 7$.
**Answer:** 7 pupils.

## Part C – Problem solving (3 marks)

**Problem 17** *

A cube of edge 5 cm is painted on all its faces and cut into 1 cm cubes. How many small cubes have **a)** exactly 2 painted faces; **b)** exactly 1 painted face; **c)** no painted face?

**Solution**

**a)** Each of the 12 edges has $5 - 2 = 3$ middle cubes: $12 \times 3 = 36$.
**b)** Each of the 6 faces has $3 \times 3 = 9$ inner cubes: $6 \times 9 = 54$.
**c)** Core: $3 \times 3 \times 3 = 27$. (*Check:* $8 + 36 + 54 + 27 = 125$.)
**Answer: a)** 36; **b)** 54; **c)** 27.

## Problem 18 *

On the first day a shop sells $\frac{1}{3}$ of its rice and 4 kg more. On the second day it sells half of what is left and 3 kg more. 22 kg remain. How much rice did the shop have at first?

### Solution

Before the second day: $(22 + 3) \times 2 = 50$ kg.

$50 + 4 = 54$ kg is $1 - \frac{1}{3} = \frac{2}{3}$ of the rice at first.

At first: $54 : \frac{2}{3} = 81$ kg.

*Check:* $81 - 27 - 4 = 50$; $50 - 25 - 3 = 22$.

**Answer:** 81 kg.

## Problem 19 *

A boat goes downstream from A to B in 2 hours and comes back upstream in 3 hours. The current flows at 3 km per hour. Find the speed of the boat in still water and the distance AB.

### Solution

Downstream speed is 6 km/h more than upstream speed.

In 2 hours downstream the boat gains $6 \times 2 = 12$ km over 2 hours upstream; the third upstream hour covers these 12 km.

Upstream speed: 12 km/h; still water: $12 + 3 = 15$ km/h.

AB: $12 \times 3 = 36$ km. (*Check:* $(15 + 3) \times 2 = 36$.)

**Answer:** 15 km per hour; 36 km.

## Problem 20 **

A contest has 40 questions: 5 points for right, 2 points off for wrong, 0 for a blank. Thu leaves 4 questions blank and scores 124 points. How many does she get right?

### Solution

Thu answers $40 - 4 = 36$ questions.

If all were right: $5 \times 36 = 180$ points; she is $180 - 124 = 56$ points down.

Each wrong answer costs $5 + 2 = 7$ points: wrong $56 : 7 = 8$; right $36 - 8 = 28$.

**Answer:** 28 questions.